Short-Reach Optics for AI Compute

An Optical Systems Engineering Handbook: Design, Validation, Qualification, Manufacturing, and Fleet Debug

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July 25, 2026

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From First Principles to the State of the Art

A concise technical overview of the optics that wire together large-scale AI systems. Compiled from public sources (IEEE, OIF, MSA specifications, vendor disclosures, and industry literature). Figures and third-party announcements are cited to their sources; any errors are the author's own.

Preface

Artificial intelligence has become an infrastructure problem. Training and serving frontier models at scale is no longer limited only by the accelerator at the center of the rack, but by how many accelerators can be wired together efficiently, reliably, and within a fixed power envelope. That wiring is increasingly optical, and the optics (especially the lasers inside them) have become a first-order lever on the cost, power, and reliability of the whole system.

This book is a concise technical overview of that layer: the short-reach optical interconnects that stitch together AI datacenters, from in-package optical I/O out to intra-rack links, deliberately setting aside the 2–10 km campus links that belong to coherent optics (Section 3.3). It concentrates on the subjects that decide whether these links work at scale: IM/DD physics and vocabulary; lasers and external light sources; WDM and wavelength locking; quantitative noise and sensitivity models; validation from bench to fleet; and reliability and manufacturing at volume. Two chapters bracket those fundamentals: why inference-scale computing puts the interconnect on the critical path (Chapter 1), and how AI datacenter networks are built where optics dominate cost and power (Chapter 9).

How to use this handbook. Start from the requirement, not the component. Move downward from system requirements to architecture, subsystem, component, and only then to the physics needed to make or test the decision. A choice at one layer constrains the next. A VCSEL points toward 850 nm, multimode fiber, and direct modulation. A DFB at 1310 nm points toward single-mode fiber and leaves a choice among direct modulation, an EML, or an external modulator. Neither path is best in isolation. Each closes a different reach, power, cost, manufacturing, and service model.

Every major chapter asks four questions: How does it work? How is it measured, and what uncertainty does the measurement remove? How does it fail? How is it debugged? The aim is operational judgment. The purpose of engineering is not to find certainty. It is to reduce uncertainty enough to make the next decision. Validation, measurement, debugging, qualification, supplier choices, and production are all that same work under different names. Debugging narrows hypotheses with measurements rather than matching a symptom to a memorized list. Throughout the book, watch how power, noise, timing, spectral, and control margins erode together before a link crosses its BER limit.

On sources. The industry moves quickly. Where the text cites specific products or figures (co-packaged-optics programs, per-lane roadmaps, energy-per-bit trends) it draws on public disclosures current as of early 2026, cited in the references. Where a claim is an inference rather than an established fact, the text says so. History and trend notes are included where they help explain why today's defaults exist, not as a full chronology of the field.

Key idea. The goal of an optical systems engineer is not to know every component. It is to make good engineering decisions under uncertainty using measurements, physics, and evidence. At gigawatt, multi-generation scale, the optical interconnect and its lasers are a first-order lever on power, cost, and reliability. Everything here serves that argument.

Contents

Chapter 1

Why the interconnect matters

1.1 The two themes of this book

Large AI systems fail or succeed as much on wiring as on FLOPs. The short-reach optics that connect accelerators, switches, and memory sit on the critical path for latency, power, and uptime, and two parts of that layer decide whether the system scales: the *lasers* that generate the light, and the discipline of *IM/DD validation*[1] that proves those links work from the first bench eye through production and fleet life. This book keeps returning to those two themes. Modulation formats, WDM, reliability, and network architecture matter because they serve them.

[1] IM/DD = intensity modulation with direct detection: the short-reach datacenter transceiver style. See Chapter 3.

Key idea. Two things dominate short-reach optics at scale: *lasers* and *IM/DD validation*. Everything else orbits those two.

1.2 The context: purpose-built inference silicon

The pressure on the interconnect is not abstract. It comes from a shift in how large AI systems are built: vertically integrated, purpose-built silicon deployed at gigawatt scale, with networking named as a first-order design axis beside compute and memory. A representative example is *Jalapeño*, a purpose-built LLM *inference* accelerator Broadcom and a hyperscaler partner announced in 2026 as a blank-slate design, the first chip in a multi-generation compute platform.[2] Its public features are typical of the class:

[2] Source: Broadcom and partner, "LLM-optimized inference chip," 24 June 2026.

- **Partners and integration.** Broadcom provides silicon implementation and networking (including *Tomahawk* switch silicon); Celestica provides board, rack, and system integration.

- **Cadence.** A roughly nine-month design-to-tape-out cycle, with the designers' own models used to accelerate parts of the flow.
- **Scale.** Deployment at gigawatt scale, over multiple generations, beginning in late 2026.
- **Design thesis.** "Reduce data movement and balance compute, memory, and *networking* resources" to reach realized utilization close to theoretical peak.

That last point is the hook for this book. Purpose-built inference silicon does not treat networking as a commodity NIC bolted on after the die is done; it budgets interconnect power and latency alongside FLOPs and HBM. Partners and cadence matter because they set who owns the switch ASIC and how fast generations turn: Broadcom for silicon and networking (including Tomahawk), Celestica for board and rack, a roughly nine-month design-to-tape-out cycle, and gigawatt-scale deployment planned across multiple generations from late 2026. Once networking sits on that line, the optical interconnect is how the system scales past a single package to a gigawatt cluster, and laser quality plus IM/DD validation become infrastructure problems rather than module afterthoughts.

1.3 Why inference makes the interconnect matter

Inference is not training. Once a model is trained, serving it is dominated by two phases with very different bottlenecks (developed fully in Chapter 9):

Prefill processing the prompt: highly parallel and compute-bound, much like training.

Decode generating tokens one at a time: autoregressive and *memory-bandwidth-bound*, because every token streams the model weights through the compute units again.

Frontier models do not fit on one chip. They are sharded across many accelerators, so every generated token triggers *collective communication* across the fabric: all-reduce for tensor parallelism, all-to-all for mixture-of-experts routing, point-to-point for pipeline stages. The interconnect therefore sits on the *latency critical path* of inference, not merely the plumbing between training runs.

Key idea. Reliable, low-power IM/DD links directly set how large and how dependable an inference fabric can be. That is why the optical layer has become central to AI system design.

1.4 The shifting bottleneck

Each generation of AI infrastructure has been limited by a different resource. The progression explains why optics moved from a commodity NIC accessory to a first-order design axis:

Compute-limited Early deep learning (pre-2016). GPUs were scarce and slow; models fit on one card; the network barely mattered.

Memory-limited Larger models and larger batches (2016–2020). HBM bandwidth set training throughput; the network carried gradients but was rarely the gate.

Network-limited Sharded frontier models (2020–present). Tensor, pipeline, and expert parallelism put collectives on the critical path; fabric bandwidth and tail latency now limit realized compute use (Sections 9.6 and 9.7).

Power-limited Gigawatt-class deployments (emerging). Site megawatts cap total capacity; every pJ/bit the interconnect saves is a watt returned to compute (Section 9.13).

The bottleneck did not replace the previous one; it stacked on top. A modern cluster is simultaneously memory-bandwidth-bound in decode, network-bound in collectives, and power-bound at the site. The interconnect sits at the intersection of the last two, which is why this book treats optics as infrastructure rather than as a module datasheet exercise.

1.5 AI clusters are communication machines

An accelerator does useful work only while its operands arrive on time. Model parallelism spreads weights, activations, and expert routing across many devices, so the fabric repeatedly moves partial results between otherwise fast compute engines. Three traffic patterns set the pressure:

All-reduce combines partial results across a group and returns the result to every member. The slowest path can hold the whole group at the synchronization point (Section 9.7).

All-to-all moves different payloads between every pair, as in mixture-of-experts routing. It tests bisection bandwidth, queueing, and path balance rather than one peak link rate.

Point-to-point carries pipeline stages and storage or memory traffic. Tail latency and retries can matter more than average throughput.

More compute increases the amount of traffic the system can inject. It does not increase fabric capacity. Once links or switches saturate, accelerators wait at collectives and realized compute use falls. Public large-model training data show why this is also an availability problem: network faults are only one failure bucket, but a synchronous job pays for every interruption across the whole allocation [23]. Compute scales faster than communication unless the network, optics, and software are designed as one system.

1.6 The systems engineering loop

Work from the system downward:

Requirements
↓
Architecture
↓
Subsystem
↓
Component
↓
Needed physics

A component choice is never isolated. A VCSEL usually commits the link to an 850 nm multimode path, direct modulation, and a short reach. A 1310 nm DFB points toward single-mode fiber and can feed a DML, EML, Mach–Zehnder modulator, or ring. Those paths then set detector material, fiber plant, thermal control, test coverage, and service policy. Start with reach, lane rate, power, cost, lifetime, and manufacturing volume. Choose the component only after those constraints rule out the other paths.

Validation reduces uncertainty. Characterization asks what the design does across its operating range. Margin testing asks how close it is to a limit. Environmental and life testing ask which mechanisms move with temperature, stress, and time. Production testing asks whether process variation and assembly escapes can be caught at useful test cost. A passing result is useful only when the test answers one of those questions (Chapters 7 and 8).

Margin erodes before the link fails. Power, noise, timing, and spectral margins usually move a little at a time. A connector adds loss, a laser loses slope efficiency, a bias point drifts, and a ring moves toward the edge of its lock range. No single change must be large. Their sum pushes normal unit and temperature variation across the BER threshold. Track margin as a ledger over temperature, lot, age, and workload rather than as one room-temperature pass/fail point.

Debug by eliminating hypotheses. First scope the failure: one unit, one lot, one vendor, one site, or the fleet. Then classify its pattern: sudden or gradual, constant or temperature-dependent, power-related or signal-quality-related. Choose the next measurement for its ability to separate competing causes. The debugging pyramid below gives the order; the failure-analysis handbook in Chapter 10 gives the symptom-led procedures.

1.7 Engineering lens

1.7.1 How it works

The interconnect carries partial results between accelerators, and collectives make the slowest link set the pace for the whole job. At AI scale the optical layer is the dominant medium between racks, so its bandwidth, latency, and uptime become the cluster's bandwidth, latency, and uptime.

1.7.2 How it is measured

At the system level: step time, collective latency, accelerator idle fraction, and tail behavior. At the link level: pre-FEC BER, FEC error distribution, optical power, module temperature, and flap count. At the architecture level: delivered bandwidth per watt, link count, and FIT-weighted availability (Sections 5.13 and 7.12).

1.7.3 How it fails

Fabric capacity can fall behind injected traffic. A single marginal link can stall a synchronous collective. Connector contamination, laser aging, thermal excursions, and firmware mismatches all reduce margin invisibly until the workload crosses the cliff. Service errors during expansion (polarity, cleaning, fiber routing) are a common source of day-one failures.

1.7.4 How it is debugged

Use the debugging pyramid: start at the system symptom, narrow to signal quality, walk the link budget, bisect the subsystem, then identify the physical root cause. Do not skip layers.

1.8 The debugging pyramid

When a link fails, work from the top down. Each layer narrows the search before you open a connector or reseat a module. This framework reappears in every chapter.

Layer 1: System What is failing at the workload level? BER, throughput, latency, training instability, collective stall. Start here because the symptom often rules out entire subsystems.

Layer 2: Signal quality What changed in the signal? Eye opening, jitter, noise, equalization margin, FEC error distribution. These are what the host and module already report (Sections 7.8 and 7.12).

Layer 3: Link budget Where did margin disappear? Optical power, receiver sensitivity, insertion loss, extinction ratio, ORL. Walk the ledger from transmitter to receiver (Section 7.7).

Layer 4: Subsystem Which block is responsible? Laser, modulator, driver, photodiode, TIA, DSP, connector, fiber, or host SerDes. Bisect with loopbacks and golden swaps (Sections 7.9 and 7.10).

Layer 5: Physical root cause What mechanism explains the failure? Aging, contamination, thermal stress, process variation, assembly defect, calibration error, firmware bug. This is where you open FA or 8D (Section 8.10).

Do not skip layers. A direct jump to root cause without first confirming the system symptom and localizing the subsystem wastes weeks on the wrong part. The pyramid is a discipline, not a checklist: each layer produces a measurement that either confirms or falsifies the hypothesis before you descend.

1.9 Interview and design review questions

Concept.

- Why does optics become favorable over copper at higher data rates?
- What distinguishes a communication-limited system from a compute-limited one?
- Why does a synchronous collective amplify single-link failures?

Design.

- What traffic pattern sets the requirement: all-reduce, all-to-all, or a latency-sensitive point-to-point path?
- What is the measured bottleneck: SerDes reach, switch radix, fiber count, module power, cooling, or collective efficiency?
- Why does passive copper, a direct-attach cable, or an active electrical cable not close the required reach and rate?
- Does a pluggable, linear pluggable, or co-packaged engine move risk into a domain the team can test and service?

Debug.

- Training throughput dropped after scaling. Where in the debugging pyramid (Section 1.8) do you start?
- How do you distinguish a network bottleneck from a compute or memory bottleneck using only host-visible telemetry?
- What data would show that added optical bandwidth does not improve job time?

Manufacturing and operations.

- What failure rate per link is acceptable given the total link count and the target job uptime?
- How does the service model (hot-swap vs board pull) change the architecture?
- What is the fleet cost of a wrong triage classification?

1.10 How to read this book

The chapters build from requirements to fleet operation:

1. Chapters 2 and 3: energy, IM/DD vocabulary, architecture, modulators, FEC, and equalization.
2. Chapter 4: quantitative noise, RIN, sensitivity (use with Section 7.7).
3. Chapter 5: requirements-led source and modulation decisions, measurement, aging, and service architecture.
4. Chapter 6: wavelength locking, thermal crosstalk, CW-WDM, on-chip MUX.
5. Chapters 7 and 8: measurement ladder, link budgets, qual, packaging.

6. Chapter 9: scale-up/out, pluggables, CPO/XPO, inference collectives.
7. Chapter 10: symptom-led root-cause isolation and corrective action.

To use the book as a design drill, pick one link style (retimed 800G DR, LPO, or CPO WDM) and trace it end to end through Sections 3.2, 9.3 and 9.10.

Key idea. Start from the system requirement and work downward. Validate to reduce uncertainty, track how several small losses consume margin, and debug by choosing measurements that eliminate hypotheses.

Chapter 2

First principles: the energy of moving a bit

Before any device, modulation format, or standard, one quantity governs short-reach interconnect design: the energy required to move a bit from one place to another. David Miller's work on optical interconnects to silicon lays out the clearest first-principles framework for this,[1] and although the specific numbers are years old, the scaling arguments are what matter, and they still decide where the optics go.

[1] D. A. B. Miller, "Device requirements for optical interconnects to silicon chips," *Proc. IEEE*, 2009; and "Attojoule optoelectronics for low-energy information processing and communications," *JLT*, 2017.

2.1 The electrical baseline: charging a wire

To send a bit down an electrical line you charge and discharge its capacitance through a voltage swing. To order of magnitude,

$$E_{\mathrm{elec}} \approx \tfrac{1}{2}\, C\, V^2,$$

and the line capacitance grows with length, $C \approx c' L$ (with c' on the order of a hundred-odd femtofarads per millimeter, depending on the medium).[2] So the energy to move a bit electrically *rises with distance*, and the resistive-capacitive delay and the equalization needed to fight it rise with both distance and data rate.

[2] The exact prefactor depends on the signaling scheme (voltage-mode, current-mode, terminated transmission line) but the *length dependence* is the first principle that matters here.

2.2 The optical alternative: energy at the ends

An optical link spends its energy differently. The dominant costs are the *conversions at the two ends* (driving the laser or modulator, and the receiver that turns light back into charge) plus the wall-plug inefficiency of the laser. Over short reaches, waveguide and fiber loss are small, so the energy per bit is, to first order, *independent of* *distance*.

Key idea. Electrical energy per bit grows with length; optical energy per bit is set at the endpoints and is roughly flat with length. That single contrast is the seed of everything else.

2.3 The break-even distance, and why optics moves toward the chip

Put the two together and there is a cross-over length: below it, electrical wins; above it, optical wins. The decisive observation is what happens as data rates climb: the electrical cost at a given length rises (more charging events per second, worse loss at higher frequency, more equalization), so the *break-even distance shrinks*.

That is the whole history of the field in one sentence. As rates went from gigabits to hundreds of gigabits per lane, the cross-over marched inward: campus, then rack, then board, then package, and now die-to-die. It is the first-principles reason co-packaged optics and in-package optical I/O exist (Sections 6.6 and 9.10), and the reason the scope of this book is the *shortest* links (Section 3.3).

2.4 The receiver, capacitance, and attojoule targets

Miller's sharpest point concerns the receiver. To register a bit, the photocurrent must develop a detectable voltage on the receiver's input node, so the detection energy again looks like a CV^2 on that node's capacitance. Minimize the photodetector and input capacitance (by integrating the detector tightly with the first transistor, eliminating parasitic pads and wires) and you can detect a bit with *fewer photons and less energy*. This is the argument for close electronic–photonic integration, and it is what makes sub-100 fJ/bit (and, in principle, attojoule-class) devices conceivable.

[3]

[3] This is why integration is not a packaging convenience but an energy strategy: shortening the electrical path between detector and amplifier lowers capacitance, which lowers both the energy and the optical power a link needs.

2.5 The floors: photons per bit and noise

Two physical floors bound how far this can go. A real receiver needs enough photons per bit for adequate signal-to-noise: today hundreds, with ideal detection in the handful-of-photons range. That sets a floor on received optical power, and hence on laser energy. Separately, the kT/C noise on the receiver node (not the Landauer $kT \ln 2$ limit of logic, which is far smaller) is the practical noise floor for communication, and it too rewards small capacitance.

Miller is careful to separate the energy of *logic* from the energy of *communication*; short-reach interconnect is overwhelmingly a communication-energy problem, dominated by the endpoints described above.

2.6 Recent progress toward the limits

Miller's numbers are years old, but the framework has aged well: the last few years have been a steady march of experiments toward the floors it predicts, and they validate rather than overturn it. Three threads are worth knowing.

First, **low-capacitance 3D integration is delivering the receiver energy Miller argued for**. A 2025 demonstration integrated an 80-channel transceiver in three dimensions, stacking the photonics directly on the CMOS to minimize the receiver-node capacitance, and reported roughly 120 fJ$/$bit.[4] Tellingly, it reaches that number not by pushing per-lane rate but by using *many slow channels* (about 10 Gb$/$s each) so each receiver stays in its most sensitive, lowest-energy regime, with WDM providing the aggregate bandwidth. That is Miller's prescription almost verbatim.

[4] S. Daudlin *et al.*, "Three-dimensional photonic integration for ultra-low-energy, high-bandwidth interchip data links," *Nature Photonics* 19:502–509, 2025.

Second, **the capacitance argument is directly measurable**. A co-designed 12 nm-FinFET-on-silicon-photonics transceiver using direct-bond interconnect cut input parasitic capacitance by about 75 %, which bought ~ 6 dB of receiver sensitivity and pushed link energy to a few hundred fJ/bit.[5] The CV^2 story is not a metaphor; you can watch the sensitivity move as the capacitance drops.

[5] P.-H. Chang *et al.*, "A 3D integrated energy-efficient transceiver … co-designed 12 nm FinFET and silicon photonic ICs," *JLT* 41(21):6741–6755, 2023.

Third, **WDM receivers with near-zero-power wavelength control are approaching sub-pJ/bit at terabit scale**: a single-chip 32-channel WDM PAM4 receiver reached 1.024 Tb$/$s on one fiber at under 0.38 pJ$/$bit with no DSP or equalization.[6] And on the device side, alternative emitters are being pushed into the same regime, a 2025 microLED link demonstrated 200 fJ$/$bit transmitter energy at BER < 10^{-12} with no FEC,[7] exactly the LED branch of Miller's device-by-device comparison.

[6] A. Pirmoradi *et al.*, "A single-chip 1.024 Tb$/$s silicon photonics PAM4 receiver," arXiv:2507.12452, 2025.

[7] Avicena, ECOC 2025 demonstration of the LightBundle microLED interconnect.

Key idea. The 2009/2017 framework has not been superseded; it has been confirmed in hardware. The winning recipe in every recent result (low-capacitance co-/3D-integration plus many WDM channels at modest per-lane rate) is precisely what Miller's energy accounting recommends. What has changed is only the vertical axis: from theoretical attojoule targets toward demonstrated hundreds-of-fJ/bit links.

2.7 The energy ladder

A useful system model orders operations by how far information must move:

$$\begin{array}{c}\text{arithmetic operation}\\ \downarrow \\ \text{local SRAM access}\\ \downarrow \\ \text{HBM access}\\ \downarrow \\ \text{package crossing}\\ \downarrow \\ \text{board crossing}\\ \downarrow \\ \text{rack crossing}\\ \downarrow \\ \text{datacenter crossing.}\end{array}$$

The exact energy at each rung depends on process, signaling, utilization, and reach. The ordering is the point: capacitance, equalization, conversion, and cooling costs grow as the bit leaves the compute core. Miller's endpoint model explains why the optical crossover moves inward as electrical rate rises [2, 3]. A design review should therefore count bytes moved and distance crossed, not only arithmetic throughput.

2.8 Debug discussion: why more compute can make training slower

When a larger accelerator count produces worse step time, treat it as a scaling failure, not proof that the compute is slow.

1. **Observe.** Compare useful compute time, collective time, queueing, retransmits, and idle time at each scale point.
2. **Hypothesize.** Check for a saturated fabric tier, uneven routing, stragglers, memory pressure, or a synchronization barrier that grows with group size.
3. **Measure.** Plot collective latency and delivered bandwidth against accelerator count. Correlate link error and retry counters with the tail of the step-time distribution.
4. **Bisect.** Hold the model fixed and vary topology, group size, message size, or route. A roofline-style split between compute and data movement keeps the diagnosis tied to measured limits [71].

5. **Act.** Fix the limiting rung: reduce communication, rebalance the collective, add fabric capacity, shorten the electrical reach, or move the optical conversion closer to the package.

Debug story

Observed. Training throughput fell when the accelerator count doubled.
Investigation. Compute kernels held the same efficiency, while all-reduce tail latency and link retries rose on one fabric tier.
Finding. The added workers spent more time waiting at the collective barrier than computing.
Root cause. The network tier was oversubscribed for the new group size, and one marginal link amplified the tail.
Resolution. The team rebalanced routes, removed the weak link, and set the next scale gate on measured collective latency rather than peak FLOPs.

2.9 Engineering lens

2.9.1 How it works

Electrical energy per bit grows with distance; optical energy is spent at the endpoints and is flat with distance. So as lane rates rise and the electrical cost climbs, the crossover where optics wins marches inward toward the chip. Tight electronic-photonic integration lowers receiver capacitance, cutting both energy and the optical power a link needs.

2.9.2 How it is measured

Energy per bit (pJ/bit) is the headline metric, measured end to end from host SerDes through the optical link. Break it into electrical (SerDes, driver, TIA), electro-optic (modulator, laser wall-plug), and thermal (cooling overhead, PUE). Compare against the roofline: bytes moved times energy per byte against the power envelope.

2.9.3 How it fails

A link that meets its energy target on a quiet bench can exceed it in the field when equalization depth increases, retransmits rise, or thermal control draws extra current. A poor co-packaging yield or high coupling loss forces higher laser power and erases the integration advantage. A firmware update that increases DSP activity raises module watts without changing the optics.

2.9.4 How it is debugged

Compare measured link power against the design budget, split by block. If the total exceeds the target, identify which block moved: SerDes, driver, laser bias, TEC, or DSP. Correlate power anomalies with BER, temperature, and equalizer state. A link that draws more power than expected is often also running at reduced margin.

2.10 Interview and design review questions

Concept.

- Why does electrical energy per bit grow with distance while optical does not?
- What physical quantity sets the receiver's minimum detectable energy?
- Why does co-packaging reduce energy per bit even without changing the laser?

Design.

- At what reach does the optical crossover occur for your target lane rate, and what assumptions drive that number?
- How does doubling the lane rate change the crossover distance?
- Which rung of the energy ladder dominates total system power for this workload?

Debug.

- More accelerators were added but training throughput fell. Where on the energy ladder is the bottleneck?
- A link's power consumption rose after a firmware update. What changed: equalization depth, retransmits, or thermal control?

Manufacturing and operations.

- What determines the power budget at the faceplate: module watts, cooling capacity, or rack-level allocation?
- How do you verify that a deployed link actually achieves the energy per bit the design predicted?

2.11 Why this framework anchors the book

Everything that follows is an effort to approach these floors at the required data rate and reliability. Laser wall-plug efficiency (Chapter 5) sets how much optical power you can afford. Modulation and FEC (Chapter 3 and sections 3.6 and 3.12) trade SNR for reach and rate. WDM (Chapter 6) amortizes one laser source across many wavelengths. Receiver noise and sensitivity (Chapter 4, section 4.4.3, and section 7.7) decide whether that power is enough. Co-packaging and energy-per-bit trends (Section 9.13) show the same first principle playing out as copper reach collapses and optics move onto the interposer.

Key idea. Short-reach optics is, at bottom, an energy-per-bit optimization. Optical energy is spent at the endpoints and is flat with distance; electrical energy grows with distance and rate; so rising data rates push optics ever closer to the chip. Miller's framework is the lens the rest of this book looks through.

Chapter 3

Intensity modulation, direct detection

3.1 What IM/DD means

Almost every short-reach datacenter link still uses the same basic deal with physics: put data on optical *power*, and recover it with a photodiode. That is *IM/DD*: *intensity modulation, direct detection*. The transmitter encodes bits as changes in optical intensity; the receiver is a photodiode plus a transimpedance amplifier (TIA) that turns photocurrent back into voltage. There is no local oscillator and no coherent mixing.[1] Detection is square-law (photocurrent proportional to optical power), so phase is discarded. That limitation is acceptable inside the rack and across a few hundred meters of fiber, where cost, power, and latency matter more than spectral efficiency. It is why IM/DD, not coherent, wires AI compute fabrics today.

[1] Contrast with *coherent* detection, which mixes the signal against a local-oscillator laser to recover amplitude *and* phase. Coherent wins for long-haul spectral efficiency; IM/DD wins on cost and power for short reach.

3.2 The IM/DD transceiver chain

Every pluggable module and every co-packaged engine is a rearrangement of the same chain. Once you can name the blocks, you can place equalization, FEC, and validation measurements without getting lost in form-factor jargon.

Transmit laser or CW source → modulator (EML, MZM, ring) → driver → fiber coupling (Chapter 5 and section 3.14.3).

Channel fiber, connectors, MUX (Section 7.2.2 and section 6.3).

Receive photodiode → TIA (optional CTLE) → SerDes/CDR or linear ADC to host (Sections 3.6 and 4.5).

Digital KP4 FEC in host or retimer (Section 3.12); module DSP optional (Section 9.3, section 9.5.1, and table 9.4).

The rest of this chapter fills in modulation physics, equalization, FEC, and the modulator platforms. Noise math and measurement practice live in Chapters 4 and 7.

3.2.1 A pluggable link, end to end

The block list above is one module. A working rack-to-rack link chains two of them back to back across a fiber, and it helps to trace a single bit through the whole path once. Figure 3.1 follows that path as a folded loop: the transmit side runs left to right across the top, the fiber turns the corner on the right, and the receive side runs back along the bottom into the far switch.

Start at the switch ASIC in rack A. It builds Ethernet frames, runs the reconciliation and coding sublayers, and encodes KP4 FEC (Sections 3.8 and 3.12), then hands parallel bit streams to the host *SerDes*. The SerDes serializes each stream to a 112 GBd PAM4 lane, applies transmit FFE, and drives the host PCB and the module cage connector. That electrical hop, host to module, is the AUI (an OIF VSR-class channel; Sections 3.3 and 3.14). Inside the module, an optional DSP or retimer reshapes the lane, or for a linear pluggable (LPO) nothing does and the host SerDes owns the whole electrical budget (Section 3.6 and section 3.14.2). The driver then swings a modulator, an EML, Mach-Zehnder, or ring (Section 3.14.3), and electrons become photons. That is the first domain crossing, marked E→O in the figure.

The fiber plant is the quiet middle: duplex or parallel single-mode fiber, connectors, and patch panels carrying the light a few meters to a few hundred meters (Section 7.2.2 and section 3.3). At the far module the light hits a photodiode and TIA and becomes current again (O→E, the second crossing; Section 4.5). An optional module DSP cleans it up, and the rack B host SerDes recovers timing, equalizes the lane, and decodes KP4. The switch ASIC reassembles frames and forwards them into the fabric, out a NIC, and over the in-node link (PCIe or an NVLink-class fabric) into the destination GPU or CPU. Every pluggable link is this shape; form factors and CPO only rearrange where the boundaries fall.

This is the *scale-out* path, the rack-to-rack Ethernet fabric where optics already dominate. The *scale-up* fabric that ties GPUs together (NVLink- or UALink-class; Section 9.1) is a different link. Its reach is much shorter, so most of it is still copper: direct-attach cable or backplane under roughly a meter, with no optics in the path at all. Optics only appears once rack densification pushes those links past the copper wall, and even then it borrows the same IM/DD building blocks over a different link and protocol layer. So the chain in Figure 3.1 is the one that carries the optical volume today; scale-up is where optics is arriving next.

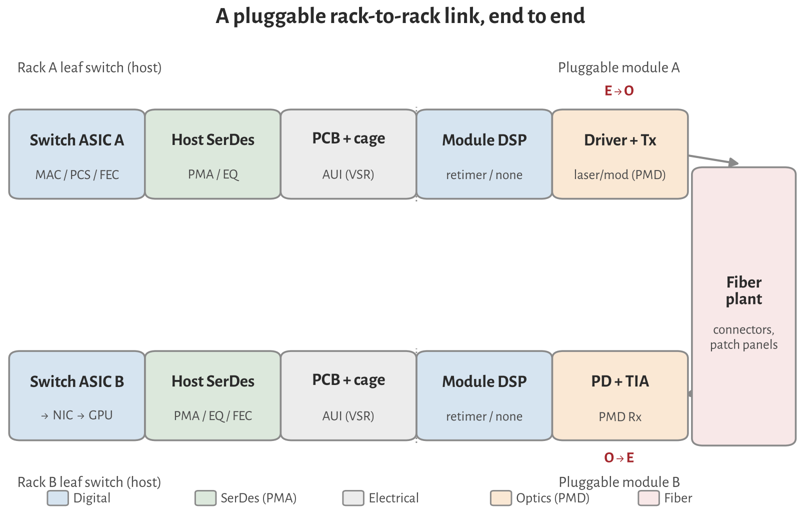

Figure 3.1: A pluggable rack-to-rack link, drawn as a folded loop. Transmit runs left to right (switch ASIC ⟶ host SerDes ⟶ cage/AUI ⟶ module DSP ⟶ driver and optics), the fiber turns the corner, and receive runs back into the far switch, then on to the NIC and GPU. Colors mark the domain of each block; the two E↔O crossings sit at the optics.

3.3 Reach regimes, and the scope of this book

"Optical interconnect" is a wide phrase. A die-to-die link of a few millimeters and a 10 km campus span both move bits on light, but they solve different problems with different physics. AI compute fabrics live at the short end of that spectrum, where lasers, IM/DD, and validation dominate cost, power, and reliability. This book stays there and largely sets aside the 2–10 km campus and data-center-interconnect links that have moved toward coherent optics.

Regime	Distance	Notes
In-package / die-to-die	mm–cm	optical I/O chiplets (e.g. Section 6.6)
Chip-to-chip / co-packaged	cm	CPO engines on the switch/XPU substrate
Scale-up (rack scale)	∼1–10 m	XPU-to-XPU; copper below ∼1–2 m, optics beyond
Intra-rack / intra-row	up to ∼100–500 m	SR over MMF; DR single-mode
Campus / DCI *(out of scope)*	2 km (FR), 10 km (LR), and beyond	increasingly coherent, not IM/DD

Table 3.1. Reach regimes. This book focuses on the top four.

The dividing line is roughly where a link leaves the compute fabric. Inside that boundary, from millimeters in-package out to a few hundred meters, IM/DD is the workhorse and the laser is the component that gates scale. That is the territory this book treats.

3.4 Where coherent takes over

3.4.1 The two detection schemes, in one paragraph

IM/DD encodes bits as changes in optical intensity and recovers them with a square-law photodiode. Phase and polarization are discarded at detection. Coherent detection mixes the received field against a narrow-linewidth local-oscillator (LO) laser on a balanced photodiode pair, recovering amplitude and phase on both polarizations. The receiver digitizes in-phase and quadrature samples and runs heavy DSP: carrier recovery, polarization demux, chromatic dispersion compensation, and soft-decision FEC. The coherent transmitter typically needs an I/Q modulator (or equivalent) under the same linewidth discipline. That stack buys spectral efficiency and reach tolerance. It costs extra photonic parts, ADC/DSP die area, power, and latency.

3.4.2 Why short reach stays IM/DD

Inside the compute fabric, from millimeters in-package out to a few hundred meters of single-mode fiber, loss and dispersion are modest: standard G.652.D fiber runs about 0.2 dB/km at 1550 nm with a zero-dispersion wavelength near 1310 nm, so a few hundred meters costs well under a decibel [91]. A PAM4 IM/DD link with equalization, FEC, and a reasonable laser closes without coherent mixing. The energy argument from Chapter 2 applies directly: IM/DD keeps the expensive digital work at the electrical ends where process scaling helps, and the optical path is passive fiber. Rack and row runs also lean on bend-insensitive G.657 fiber in patch panels and shelves, and the SR/DR reach classes below map onto the OM4/OM5 multimode and OS1/OS2 single-mode cabling classes that ISO/IEC 11801 and TIA-492 define for the datacenter plant [92]. Retimed 800G pluggables run roughly 14–18 W; LPO trims to roughly 7–9 W by deleting module DSP [72]. A 400ZR-class coherent pluggable is built around a coherent DSP that alone draws roughly 5–8 W inside a 15 W module budget [85], landing near 45 pJ/bit at 400 Gb/s. Over a 100 m rack link that extra DSP power does not buy enough margin to justify the BOM and service complexity (Section 9.13).

3.4.3 What coherent buys, and its price

Coherent wins where the channel is harder: kilometers of fiber, amplified DWDM, and dispersion that would crush a directly modulated PAM4 eye. Polarization multiplexing and higher-order QAM raise bits per symbol. Electronic dispersion compensation removes the need for tight wavelength and chirp control on the transmit laser. LO mixing gives a sensitivity boost that IM/DD only matches with more launched power. The price is upfront: linewidth requirements on both the transmit laser and the LO (often well below 100 kHz for long reach), I/Q modulators or integrated coherent PICs, high-speed ADCs, and a DSP that dominates module power. OIF standardized 400ZR coherent pluggables with a 15 W module budget for DCI spans to roughly 120 km [85]. ZR+ variants push reach further at similar or higher power. That trade is right for campus and metro DCI. It is not the default inside an AI rack.

3.4.4 The crossover, and where it is moving

Today the practical line sits near 2 km: intra-datacenter DR and shorter hops stay IM/DD (Section 3.3 and table 3.13); campus FR/LR and DCI have moved to coherent 400ZR/ZR+ (Table 3.1). Pressure pushes coherent inward on per-lane rates: at 224G and especially 448G the IM/DD SNR and TDECQ margin tighten, and some proposals explore coherent-lite or coherent receivers for the longest scale-out hops [10, 14]. Pressure keeps IM/DD dominant for AI compute: LPO and CPO attack the power wall (Section 9.13 and section 3.14.2) by stripping module DSP and shortening electrical reach, not by adding an LO and ADC chain. For inference fabrics at in-package to a few hundred meters, IM/DD remains the workhorse through the 224G generation. The open question is 448G and beyond: whether the last scale-out meters stay PAM4/PAM6 IM/DD or whether a stripped coherent option appears for links that already run retimed modules today (Section 3.14.3).

3.5 PAM4: more bits per symbol

Through the 10G and early 25G eras, most short-reach optics used two-level NRZ: one eye, one decision threshold, simple receivers. As per-lane rates climbed past about 50 Gb/s, keeping NRZ would have demanded more electrical bandwidth than connectors and SerDes could afford. The industry answer was *PAM4*: four amplitude levels carrying two bits per symbol. Line rates below are in gigabaud (*GBd*); host I/O is built in *SerDes* blocks.

GBd Gigabaud: 10^9 symbols per second. Line rate = baud $\times$ bits per symbol.

SerDes Serializer/deserializer: high-speed I/O that maps parallel data to PAM4 on the wire.

Table 3.2: Per-lane rates and symbol rates (PAM4).

Per-lane rate	Symbol rate	Context
100G/lane	$\approx$53.1 GBd	400G/800G today
200G/lane	$\approx$106–112 GBd	800G/1.6T ramp
224G/lane	$\approx$112 GBd (CEI-224G)	next SerDes generation

PAM4 halves the required bandwidth versus NRZ for a given bit rate, but it costs about 9.5 dB of SNR (three eyes instead of one, spaced one-third as far apart). That SNR hit is why modern links assume equalization, DSP, and forward error correction as part of the architecture rather than as optional polish (Section 3.6). Looking ahead, the 448G debate is partly about whether to stay on PAM4 at still higher baud or move to PAM6/PAM8 to ease the electrical channel (Section 3.14.3).

3.6 Equalization and clock recovery

Once PAM4 became the default, the electrical channel stopped looking like a wire and started looking like a filter. PCB traces, connectors, and cables low-pass the signal; the receiver must undo that inter-symbol interference (ISI) before slicing. Three equalizer classes appear in every short-reach link, plus clock recovery when the bit clock is rebuilt.

ISI Inter-symbol interference: energy from neighboring symbols overlapping the current unit interval.

CTLE Continuous-time linear equalizer: analog high-frequency boost in SerDes or TIA.

CTLE (continuous-time linear equalizer) provides analog high-frequency boost, often a zero-pole pair in the SerDes front-end, the TIA, or a redriver chip. CTLE is cheap and low latency but fixed: it cannot adapt tap-by-tap to an arbitrary channel.

FFE Feed-forward equalizer: FIR filter. TDECQ applies a bounded FFE as the reference equalizer.

FFE (feed-forward equalizer) is a finite-impulse-response filter with pre- and post-cursor taps. Host SerDes and module DSPs use FFE (often with CTLE ahead of it) to open the eye. Transmitter and dispersion eye closure quaternary (TDECQ) applies a *bounded* reference FFE when scoring transmitters (Section 7.4).

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

DFE Decision-feedback equalizer: cancels post-cursor ISI using past symbol decisions.

DFE (decision-feedback equalizer) cancels post-cursor ISI using past symbol decisions. CEI electrical eye budgets reference an 8-tap DFE at the slicer (Section 9.5.2). DFE adds latency and error propagation but recovers more ISI than FFE alone at high loss.

CDR Clock-data recovery: extracts bit clock from the data stream; present in retimed modules, absent in LPO.

CDR (clock-data recovery) extracts bit timing from the data stream and re-samples the eye. A *retimer* includes CDR plus equalization and regenerates a clean output; a *redriver* has CTLE/VGA but no CDR (Section 9.5.1).

Where these blocks sit depends on the module style (Section 9.3 and chapter 9):

- **Fully retimed pluggable:** host SerDes → connector → module DSP (FFE/DFE/CDR) → optical engine. The module cleans a bad electrical channel.
- **Redriver / ACC:** CTLE (+ VGA) in the cable or mid-channel; host SerDes still owns CDR and heavy EQ.
- **LPO:** no module DSP/CDR. Host SerDes EQ and FEC must survive the full path; module may add only CTLE in the TIA and a linear driver (Section 9.5.1). Optical TDECQ and electrical margin both tighten.

Figure 3.2 shows the correct SerDes equalizer order, then where those blocks live in a retimed module versus LPO.

Key idea. CTLE boosts analog bandwidth; FFE/DFE cancel ISI digitally; CDR rebuilds timing in retimers. LPO deletes the module-side safety net, so host equalization and KP4 margin (Section 3.12) become the whole electrical story.

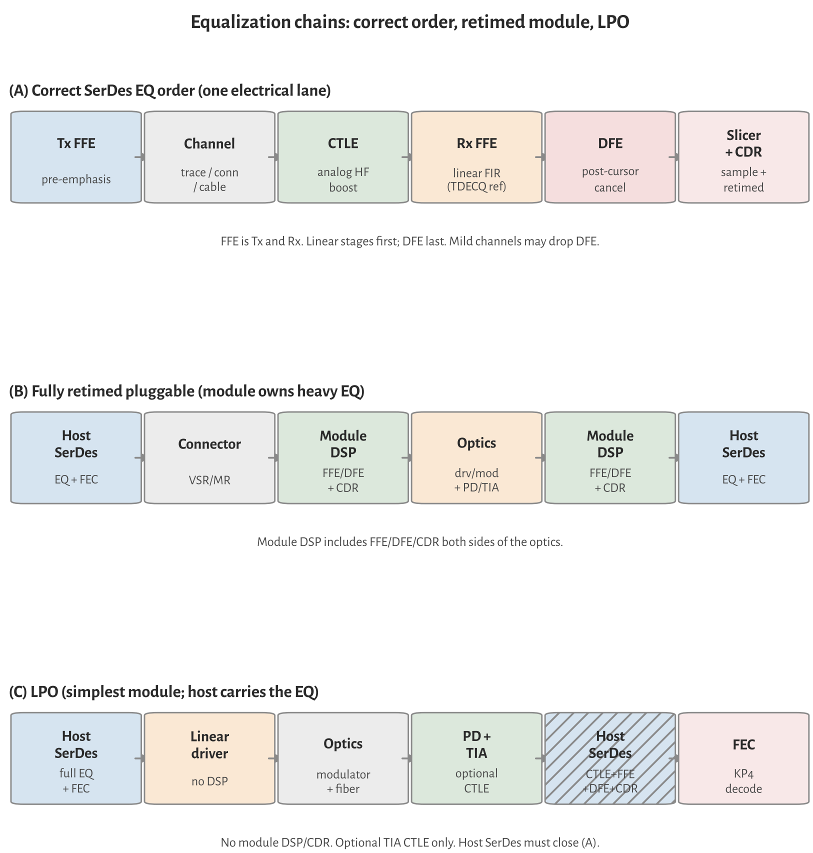

Figure 3.2: Equalization chains. (A) Correct order on one electrical lane: Tx FFE, channel, CTLE, Rx FFE, DFE, then slicer/CDR. (B) Fully retimed pluggable: module DSP owns FFE/DFE/CDR on both sides of the optics. (C) LPO: no module DSP; the host SerDes must close the full EQ and FEC path.

3.7 SerDes and DSP: who does the work

The equalizers above have to run somewhere, and “SerDes” and “DSP” get used loosely for that somewhere. They are not the same thing, and at 112 GBd the line between them has mostly dissolved. Pinning down the vocabulary makes the retimed versus LPO versus CPO argument (Section 9.5.1 and chapter 9) concrete instead of a wall of acronyms.

3.7.1 What the SerDes is

SerDes is the high-speed electrical I/O macro on an ASIC: switch silicon, a NIC, an accelerator, or a module chip. On transmit it serializes wide, slow parallel data from the MAC/PCS into one PAM4 lane on the wire; on receive it deserializes the sampled lane back to parallel words. That is the literal serializer/deserializer job. Everything else the SerDes does is in service of getting a clean symbol stream across a lossy channel.

3.7.2 Analog SerDes versus DSP-based SerDes

How the SerDes conditions the signal splits into two design styles. An *analog* (mixed-signal) SerDes equalizes in the analog domain (CTLE, a few analog FFE/DFE taps) and slices directly. It is low power and low latency but limited in how much loss it can undo and how flexibly it adapts. A *DSP-based* SerDes puts a high-speed ADC on the receive lane, digitizes the eye, and does FFE, DFE, and timing recovery numerically, with a DAC and digital pre-distortion on transmit. It burns more power and adds latency, but it closes far more channel loss and adapts tap-by-tap. Above roughly 50 GBd the DSP-based architecture won: 112 GBd (224G) host SerDes and module retimers are ADC/DSP designs. So the "DSP" that cancels ISI at 224G usually lives *inside* the SerDes, not in a separate chip.

DSP SerDes DSP-based SerDes: ADC on the Rx lane, digital FFE/DFE and timing recovery, DAC on Tx. Standard above $\sim$50 GBd; the "DSP" that cancels ISI at 224G lives inside the SerDes.

3.7.3 Two things people call "the DSP"

That is the first meaning of DSP: the digital equalization and clock recovery inside a modern SerDes. The second meaning is *the module DSP*, a distinct retimer chip in a pluggable that has its own ADC/DSP SerDes on the host side, an FFE/DFE/CDR core, a gearbox (Section 3.14.3), and often the FEC engine, then drives the optics. When the book says a retimed module "has a DSP" and LPO "deletes the DSP" (Section 9.5.1), it means this second chip. Deleting it does not delete DSP from the link; it moves the equalization burden back onto the *host* SerDes DSP and onto FEC.

3.7.4 Where FEC sits

FEC is a third block, and it is usually not part of the SerDes proper. KP4 encode and decode live in the PCS/FEC layer on the host (or in the module DSP when one is present), operating on the recovered symbol stream (Section 3.12). This is why a link can have a healthy SerDes eye and still fail on FEC, or ride an ugly eye that KP4 cleans: the SerDes recovers symbols, FEC fixes what is left. Validation reads pre-FEC BER exactly at that SerDes-to-FEC boundary.

3.7.5 Mapping it back to module styles

Table 3.3 sorts the blocks by where they physically run for the three module styles. Figure 3.2 shows the same split as signal chains. Read the LPO column as the design point that matters most for AI power budgets: the host SerDes DSP and host FEC carry the entire electrical channel, because the module has no retimer to hide behind (Section 3.14.2).

Block	Retimed	LPO	CPO (XSR)
Serialize / PAM4	Host SerDes	Host SerDes	Host SerDes
Rx EQ (FFE/DFE)	Module DSP	Host SerDes DSP	Host SerDes DSP
CDR / retiming	Module DSP	Host SerDes	Host SerDes
KP4 FEC	Host (or module)	Host	Host
Optical drive	Module DSP out	Linear driver	On-package engine

Table 3.3: Where each block runs, by module style.

Key idea. The SerDes is the ASIC's electrical PHY; at 224G its equalization and clock recovery are done in DSP inside the SerDes. "The module DSP" is a separate retimer chip. LPO removes that retimer, so the host SerDes DSP plus KP4 FEC (Section 3.12) own the whole electrical channel. Optics begin only after that handoff.

3.8 The Ethernet PHY stack: where optics attaches

The SerDes, the FEC, and the optics are not loose parts. IEEE 802.3 stacks them into named sublayers, and knowing the stack tells you exactly where the optical transceiver plugs in and which block owns each impairment. Ethernet lives at the bottom two OSI layers: the *MAC* (data link) builds frames and carries the addressing, and the *PHY* (physical) turns those frames into signals on the wire. Everything in this book sits inside the PHY.

MAC Media access control: Ethernet data-link sublayer; builds frames, addresses, CRC. MAC rate is payload throughput.

PHY Physical layer: PCS plus PMD; maps MAC frames to the line code.

3.8.1 The sublayers, top to bottom

The PHY is itself a stack. Table 3.4 lists the sublayers a 400G/800G port walks through on transmit (receive runs in reverse).

Sublayer	Function	Domain
MAC	Frame, address, CRC	Digital, host
RS	Reconciliation: adapt MAC to the PHY across the xMII	Digital, host
PCS	64B/66B, 256B/257B transcode, scramble, RS-FEC	Digital
PMA	Serialize, lane mux, clock recovery (SerDes)	Mixed-signal
PMD	Modulate/detect light: laser, driver, PD, TIA	Optical

Table 3.4: IEEE 802.3 PHY sublayers, transmit order.

Two interface names sit between these blocks and cause most of the confusion. The *xMII* (for example 400GMII) is a wide parallel bus inside the chip between MAC and PCS. The *AUI* (for example 400GAUI-4) is the electrical serial lane set that leaves the host and crosses the faceplate connector to the module. The AUI is the CEI electrical channel your SerDes has to survive (Section 3.7 and chapter 9); the *PMD* is the optical transceiver on the far side of it.

xMII Media-independent interface: a wide parallel bus between MAC and PCS inside the chip (e.g. 400GMII).

AUI Attachment unit interface: the electrical serial lanes between host and module (e.g. 400GAUI-4, at the faceplate cage).

PMD Physical medium dependent: the optical transceiver (Tx/Rx) in IEEE 802.3.

3.8.2 Where the optics attaches

Optics is the PMD, and only the PMD. Everything above it is electrical or logical and is medium-independent: the same MAC, RS, PCS, and PMA feed a copper DAC, a multimode SR module, or a single-mode DR module. That is the whole point of the layering. When IEEE names `400GBASE-DR4` or `800GBASE-DR8`, it is naming a PMD (Section 3.13); the digital stack above is shared. It is also why the retimed-versus-LPO-versus-CPO argument is really about *where the PMA and PCS/FEC physically sit* relative to the AUI, not about changing the optics (Section 3.7).

3.8.3 Line-rate accounting

The layering is where the MAC-rate-versus-line-rate gap (Table 3.6) actually comes from. Take one 400G port. The MAC delivers 400 Gb/s of payload. The PCS transcodes it (256B/257B adds about 0.4%) and then RS(544,514) FEC adds 30 parity symbols per 514 data symbols, a $544/514$ expansion of roughly 5.8% relative to payload (equivalently 5.5% of the coded line, Section 3.12). The two multiply out to $257/256 \times 544/514 = 1.0625$, so the line carries $400 \times 1.0625 = 425$ Gb/s, split as four AUI lanes at 106.25 Gb/s (53.125 GBd PAM4). The optical PMD carries that same 425 Gb/s. So "400G Ethernet" is a 400 Gb/s MAC rate riding a 425 Gb/s line, and the extra 25 Gb/s is transcode plus FEC, not wasted bandwidth.

3.8.4 What the PCS actually does

Inside the PCS, the MAC's 64-bit words are first wrapped in 64B/66B blocks: a 2-bit sync header (`01` for data, `10` for control) lets the receiver find block boundaries in the serial stream. Four 66-bit blocks are then transcoded into one 257-bit block, which compresses the four 2-bit sync headers to a single bit and, crucially, produces a 256-bit payload that aligns cleanly with 10-bit RS-FEC symbols (lcm$(256, 10) = 1280$, so five blocks make exactly 128 symbols with no remainder). The RS(544,514) encoder then adds parity (Section 3.12), and the PMA distributes the coded symbols across logical lanes, bit-multiplexes them down to the physical AUI lanes, and serializes each to PAM4. The payload bits themselves are never altered by transcoding; only the framing overhead is squeezed.

PCS Physical coding sublayer: 64B/66B line coding, 256B/257B transcoding, scrambling, and RS-FEC. Maps MAC data to the coded line stream.

Where these blocks physically live is an implementation choice, not a layer rule. The PCS/FEC can sit in the host switch ASIC or in a module retimer DSP; LPO deletes the module DSP and forces the host to own PCS/FEC and all of the PMA equalization (Section 3.7 and section 9.5.1). The layer names stay the same regardless of which chip runs them.

Key idea. Ethernet is MAC (frames) plus PHY, and the PHY is RS, PCS, PMA, PMD. Optics is only the PMD; everything above is medium-independent, which is why one digital stack feeds copper, multimode, or single-mode. The MAC-to-line-rate gap is transcode $(257/256)$ times FEC $(544/514)$, giving 425 Gb/s on a 400G port. The module-style debate is about which side of the AUI runs the PMA and PCS/FEC.

3.9 Test points: where the link is measured

Every metric in this book is taken *somewhere*. A datasheet line like "TDECQ $\leq$ 3.4 dB" or "stressed sensitivity -3.1 dBm" means nothing until you name the reference plane it was measured at. The standards name those planes TP0 through TP5, plus the host-referenced pair TP1a and TP4a, and they run in order from the transmit silicon to the receive silicon. Learning them turns a scattered pile of numbers into a map: each spec belongs to exactly one plane, and each plane is owned by one document. Figure 3.3 walks the chain; Table 3.5 is the lookup card.

The mistake to avoid is treating TP2 and TP3 as electrical points just past a connector. They are *optical* planes at the fiber. TP2 is the light launched into the fiber, measured after the module's electrical-to-optical conversion (driver plus laser or modulator). TP3 is the light arriving at the far module, before its optical-to-electrical conversion (photodiode plus TIA). The electrical planes are the ones on either side of those two crossings.

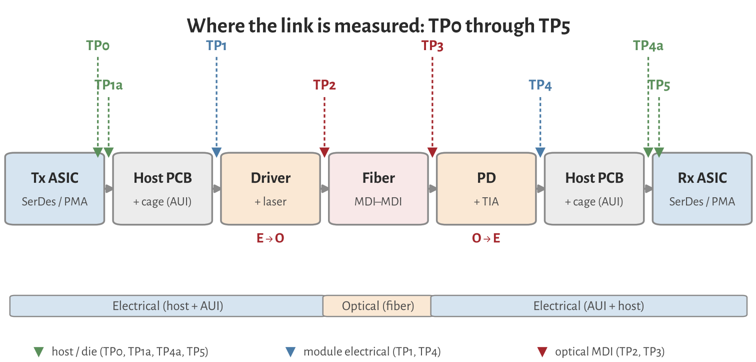

Figure 3.3: Test points on a one-way IM/DD link. Green marks the host and die-pad electrical planes (TP0, TP1a, TP4a, TP5); blue the module's electrical connector (TP1, TP4); red the optical planes at the fiber *MDI* (TP2, TP3). The two domain crossings, E$\rightarrow$O in the module transmitter and O$\rightarrow$E in the module receiver, sit between the electrical and optical planes.

READ THE CHAIN IN ORDER. At the transmit end, **TP0** is the transmit SerDes output at the die pad, the silicon designer's reference. **TP1a** is the same host signal referenced at the module cage, the *AUI* the module must accept; its electrical eye quality is *EECQ*, the electrical analog of TDECQ. **TP1** is the module's electrical input on the far side of the mated connector. The module then converts electrical to optical, and **TP2** is the optical launch at the transmitter *MDI*, where TDECQ, TECQ,

EECQ Electrical eye closure quaternary: the electrical analog of TDECQ, quoted at the host-referenced points TP1a and TP4a.

MDI Medium dependent interface: the optical connector where a module meets the fiber. TP2 (Tx launch) and TP3 (Rx input) sit here.

OMA_{outer}, ER, and RIN are specified (Chapter 7 and section 7.4). After the fiber, **TP3** is the optical input at the receiver MDI, where stressed receiver sensitivity (SECQ) is specified. The module converts back to electrical at **TP4** (its electrical output), **TP4a** is the host-referenced input near the receive SerDes under worst-case module output, and **TP5** is the receive die pad.

Point	Domain	Location	Principal measurement	Owning spec
TP0	Electrical	Tx ASIC/SerDes die pad (host)	Silicon Tx quality; design reference, hard to probe once packaged	OIF CEI C2M
TP1a	Electrical	Host output at the cage, host-referenced (SerDes output via a host compliance board)	Host Tx eye, EECQ; the AUI the module must accept	OIF CEI C2M; LPO MSA
TP1	Electrical	Module electrical input (module side of the mated connector)	Module input stressor calibration	IEEE 802.3 AUI; OIF CEI
TP2	Optical	Transmitter MDI, fiber launch (after E→O)	TDECQ, TECQ, OMA_{outer}, ER, $RIN_x OMA$	IEEE 802.3 PMD; LPO MSA
TP3	Optical	Receiver MDI, fiber input (before O→E)	Stressed receiver sensitivity, SECQ	IEEE 802.3 PMD; LPO MSA
TP4	Electrical	Module electrical output (module side of the connector)	Module Rx electrical output, EECQ	IEEE 802.3 AUI; OIF CEI
TP4a	Electrical	Stressed host input, host-referenced (near the Rx SerDes)	Host Rx under worst-case module output	OIF CEI C2M; LPO MSA
TP5	Electrical	Rx ASIC/SerDes die pad (host)	Silicon Rx recovery; design reference, hard to probe	OIF CEI C2M

Table 3.5. Test-point reference planes on a short-reach IM/DD link, transmit to receive. The optical planes TP2 and TP3 carry the transmitter- and receiver-quality specs; the electrical planes carry the host and module eye specs. See Table 9.3 for the concrete LPO MSA assignments.

Who owns which plane. The split is not arbitrary. IEEE 802.3 optical PMD clauses define the transceiver planes: the optical TP2 and TP3, and the module's electrical PMD input and output at TP1 and TP4 [8]. The OIF Common Electrical I/O chip-to-module work owns the electrical planes deeper into the host: the die pads TP0 and TP5, and the host-referenced compliance points TP1a and TP4a, which are measured through host and module compliance boards to de-embed the fixture [66]. The LPO MSA then binds both sides into a single product contract for a DSP-less module, with the six-point ladder (TP1a, TP1, TP2, TP3, TP4, TP4a) tabulated in Table 9.3 [79].

The numbering is a family of conventions, not one universal set. IEEE optical PMD clauses commonly define only TP1 through TP4, with the optical measurements at TP2 and TP3; the die pads TP0/TP5 and the host-referenced TP1a/TP4a come from the chip-to-module electrical world. Fibre Channel historically used its own reference points. Match the labels to the document you are reading rather than assuming a shared TP0-to-TP5 spine.

Where compliance testing concentrates. Optical transmitter compliance lives at TP2 (TDECQ, OMA, ER); optical receiver compliance lives at TP3 (stressed sensitivity). Electrical host and module compliance concentrate at TP1a and TP4a, which is why a module that passes optical TDECQ at TP2 can still fail interop if it misses EECQ at TP1a. TP0 and TP5 sit inside the package and are informative for silicon design but hard to probe in a finished system. A retimed module recovers the signal before TP4; a linear (LPO) module does not, so at 224G its host EQ and FEC carry the whole electrical channel (Section 9.5.1 and section 3.13).

Key idea. The plane makes the number. TP2 and TP3 are optical, at the fiber MDI, and carry transmitter and receiver quality (TDECQ, stressed sensitivity). TP1 and TP4 are the module's electrical connector; TP1a and TP4a are host-referenced near the SerDes; TP0 and TP5 are the die pads. IEEE 802.3 owns the optical planes, OIF CEI owns the die and host-referenced electrical planes, and the LPO MSA binds all of them into one DSP-less product contract (Table 9.3).

3.10 The link-budget vocabulary

These terms are the language of both datasheets and debug. Learn them as a ledger, not as a glossary quiz: when a link fails, you are almost always arguing about one of them.

OMA Optical modulation amplitude: outer PAM4 level swing $P_1 - P_0$.

ER Extinction ratio: P_1/P_0 in dB. Higher ER widens the OMA for a given average power; trades against chirp and driver swing.

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed $\mathrm{RIN_x OMA}$ under fixed ORL.

OMA (optical modulation amplitude) $P_1 - P_0$: the real signal swing. For PAM4 the "outer OMA" spans the outermost levels.

Extinction ratio (ER) P_1/P_0, in dB. Trades against insertion loss and against *TDECQ* (Chapter 7).

RIN (relative intensity noise) the laser's own amplitude noise; sets a noise floor that matters more as OMA shrinks.

Chirp the unwanted frequency modulation that accompanies intensity modulation. Large in directly modulated lasers; small and controllable in externally modulated lasers. See Section 3.11.

Chromatic dispersion penalty dispersion × chirp × distance produces inter-symbol interference. This drives the wavelength plan and the choice between directly and externally modulated lasers (Section 3.11).

Receiver sensitivity the minimum OMA needed to hit a target bit-error ratio; "stressed" sensitivity adds a calibrated impairment for margin.

[2]

[2] A clean mental model: start with transmitter OMA, subtract connector and fiber loss, subtract penalties (dispersion, TDECQ, reflection), and compare the result to receiver sensitivity (Sections 4.4 and 7.7 and chapter 4). What remains is margin.

3.11 Chirp, dispersion, and the IM/DD penalty

IM/DD detects power, not phase, but phase still matters on the fiber. Intensity modulation couples to frequency through the laser or modulator *chirp* parameter

α: a change in output power shifts the instantaneous optical frequency. Single-mode fiber then converts that frequency wander into intensity distortion via chromatic dispersion (different λ travel at slightly different speeds).

The penalty grows with chirp $\times$ dispersion $\times$ distance. At O-band (~1310 nm) dispersion is near zero ($|D| \lesssim 3$ ps/(nm·km)), which is why silicon photonics and many datacenter SMF links use 1310 nm class wavelengths. At C-band (~1550 nm) dispersion is larger; chirpy sources pay more per kilometer.

SMF Single-mode fiber: 9 μm core; O-band (1310 nm) or C-band (1550 nm). G.652.D and G.657 for datacenter plant.

Source choice sets chirp (Chapter 5, table 3.12, and section 3.14.3):

- **DML:** large α; fine for tens of meters of MMF or uncooled SR, poor for FR-class SMF unless rate and reach stay short.
- **EML / external MZM / ring:** low chirp; default for DR/FR and CPO.
- **Reflections:** light reflected back into the laser raises effective chirp and RIN; optical return loss (ORL) specs exist for this reason (Section 7.2.2).

ORL Optical return loss: reflected power back toward the laser. Low ORL raises RIN and can seed burst errors.

For the distances this book treats (in-package through a few hundred meters), dispersion is often secondary to TDECQ, connector loss, and receiver noise. It becomes decisive when a low-chirp assumption fails (aging EAM bias, DML on SMF, or FR edge cases). TDECQ embeds dispersion in its name because the reference receiver includes a dispersion penalty model for the standardized test channel.

3.12 Forward error correction

Modern PAM4 links do not meet raw bit-error-ratio targets on their own; they lean on *FEC*. The datacenter workhorse is *KP4*, a Reed–Solomon code RS(544,514) operating on 10-bit symbols. The name comes from `100GBASE-KP4` in IEEE 802.3bj: K for backplane, P for PAM4, 4 for four lanes [9]. Today "KP4" means that FEC code on any PAM4 Ethernet link (optical DR, 224G SerDes, and backplane), not only the original backplane PHY. The link is specified to a *pre-FEC* BER threshold, on the order of 2.4×10^{-4}, which FEC then drives down to an effectively error-free *post-FEC* BER.

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

KP4 IEEE 802.3 Clause 91 FEC for PAM4: Reed–Solomon RS(544,514) on 10-bit symbols, 5.5% overhead. Pre-FEC BER target 2.4×10^{-4}.

RS(544,514) Reed–Solomon block code: 544 symbols out, 514 payload, 30 parity. Corrects up to 15 symbol errors per codeword.

pre-FEC BER measured before the FEC decoder. Standards and validation sweeps use this number, not post-FEC.

post-FEC BER after FEC correction. Target is effectively error-free (< 10^{-12} class).

Two consequences matter in practice:

1. Validation targets are stated in pre-FEC BER, and FEC symbol-error histograms are a rich debug signal (they reveal *how* a link is failing, not just that it is).
2. Emerging *linear-drive* optics (*LPO/LRO*, Chapter 9) remove the DSP/retimer to save power, which tightens the link budget and leans even harder on FEC and on transmitter quality (Section 3.6).

LPO Linear pluggable optics: no retimer/DSP in the module; host SerDes drives the modulator directly.

LRO Linear retimed optics: retimer in the module but no heavy DSP; lighter than fully retimed pluggables.

KP4 in practice. *KP4* maps to Reed–Solomon RS(544,514) on 10-bit symbols: 514 payload symbols, 30 parity, up to 15 correctable symbol errors per codeword. Coding overhead is $30/544 \approx 5.5\%$, so the *line rate* on the wire exceeds the MAC/info rate (for example, 802.3dj delivers ≈211 Gb/s payload on a 224 Gb/s PAM4 lane). Table 3.6 is the decoder for the three rate names you will see in 802.3dj and CEI-224G docs (example: one 200G Ethernet lane).

Term	Meaning	Example
MAC rate	User Ethernet throughput at MAC	200 Gb/s per lane
Line rate	Bits on wire incl. FEC overhead	∼211–224 Gb/s (context-dependent)
Symbol rate	Baud on the electrical/optical lane	∼112 GBd PAM4

Table 3.6: Three rate names on one 802.3dj lane (200G Ethernet).

The optical PHY is qualified to a *pre-FEC* BER threshold, typically 2.4×10^{-4} for KP4-class links. That corresponds to $Q \approx 3.5$ on the Gaussian model of Section 4.1. Post-FEC the target is effectively error-free (10^{-12} to 10^{-15} class). Validation always reports pre-FEC BER during bring-up; post-FEC confirms the decoder is working.

FEC symbol-error *histograms* are a debug tool: clustered errors point to burst impairments (reflections, power droop); sparse errors point to Gaussian noise margin. At 448G/lane, CEI notes that KP4 at 10^{-4} pre-FEC may not close on 40 dB-class channels without stronger codes (Table 3.10): MLC plus higher overhead RS, or hard-decision FEC in demos (Section 3.14.3).

3.13 Ethernet optical PMDs and reach classes

IEEE 802.3 names the optical transceiver (*PMD*) separately from the MAC rate. Short-reach IM/DD classes that matter for AI fabrics:

SR (short reach) multimode fiber, VCSEL at 850 nm or evolving MMF solutions; rack and row distances; modal noise and bandwidth limit legacy SR.

DR (datacenter reach) single-mode, typically 500 m class at 1310 nm; PAM4, KP4, TDECQ limits; the mainstream 400G/800G/1.6T module reach.

FR (far reach) single-mode, 2 km class; same optics family as DR with tighter transmitter specs; chirp and dispersion matter more at the margin.

SR Short reach: multimode fiber class (typically 100 m over OM4); VCSEL at 850 nm. Legacy rack optics.

MMF Multimode fiber: 850–940 nm VCSEL links (OM3/OM4/OM5); reach and modal BW limit to SR distances.

DR Datacenter reach: IEEE 802.3 single-mode class, typically 500 m at 1310 nm. Default for AI scale-out optics.

FR Far reach: IEEE 802.3 single-mode class, 2 km. Tighter chirp and dispersion budget than DR.

Naming examples: 400G-DR4 (four lanes), 800G-DR8, 800G-FR8 (eight λ, 2 km). 802.3dj adds 200G/lane Ethernet; the 400G/lane study group targets the next generation (Table 3.13 and section 3.14.3). Campus LR and coherent DCI sit beyond this book's scope (Table 3.1).

3.14 The per-lane roadmap: 224G and beyond

Per-lane rate is the axis the whole industry advances along, because doubling it roughly doubles switch and module capacity for the same lane count. The electrical I/O roadmap is set by the OIF's Common Electrical I/O (CEI) projects, and optics tracks it closely.

Table 3.7: The CEI per-lane roadmap.

Generation	Per lane	Modulation	Ethernet rates
CEI-112G	112 Gb/s	PAM4	100/200/400G
CEI-224G	224 Gb/s	PAM4 (≈112 GBd)	200/400/800/1600G
CEI-448G	448 Gb/s	TBD (PAM4/6/8)	400/800/1600/3200G

Figure 3.4 puts that ladder on a longer clock: OIF CEI has roughly doubled the rate per differential pair every four to five years since the early 2000s. Open markers for 224G and 448G follow the demo deck's "202X" placement; 224G is shipping now, 448G is still pathfinding [66].

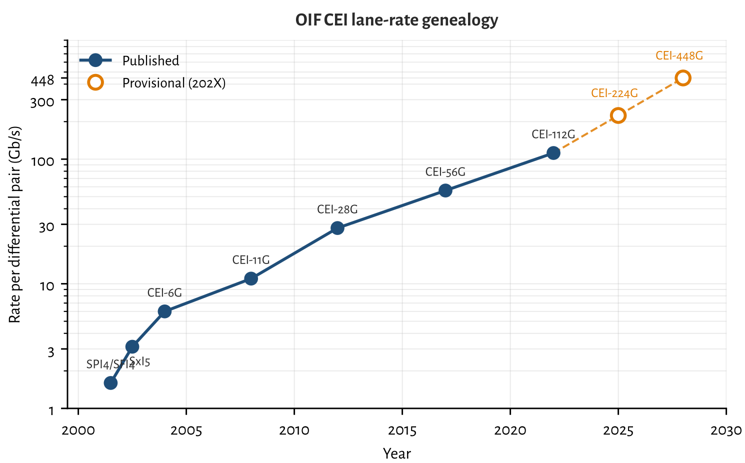

Figure 3.4: OIF CEI rate per differential pair versus year (log scale). Data from the OFC 2025 CEI interoperability demo genealogy; 224G/448G years are provisional placements for the deck's 202X entries, and 448G's rate is the naming target (PAM order still open).

3.14.1 224G is settled; the frontier is deployment

CEI-224G kicked off in 2022 as the electrical follow-on to CEI-112G. It kept PAM4 and roughly doubled the baud, to about 112 GBd per lane, with reach projects from XSR through LR now maturing. That choice is no longer the open debate; it is the shipping baseline. Eight 224G lanes make a 1.6 Tb/s port, and the same SerDes generation feeds 102.4 Tb/s-class switch silicon (Section 9.10). Copper reach has collapsed to about a meter, DSP-based SerDes are assumed, and a *CEI-224G-Linear* variant defines linear operation without a DSP/retimer in the optical module: the electrical foundation for LPO (Chapter 9). The remaining work is deployment: closing LPO margins, qualifying ELSFP banks for CPO, and holding yield at volume, not inventing a new modulation alphabet.

Figure 3.5 is the reach map those projects sit on: *XSR*/XSR+ inside the package (die-to-die and die-to-optical-engine), *VSR* from ASIC to a faceplate pluggable, *MR* chip-to-chip on the board (PCB or twinax), and *LR* for backplane and longer copper cables [66]. The media labels (host PCB traces, twinax, optical fiber, optical module) are where the loss budget actually lives; the CEI class name is the electrical recipe for that hop. Table 3.8 is the matching lookup card from the CEI-224G project map (OFC 2025 demo framing) [67][66].

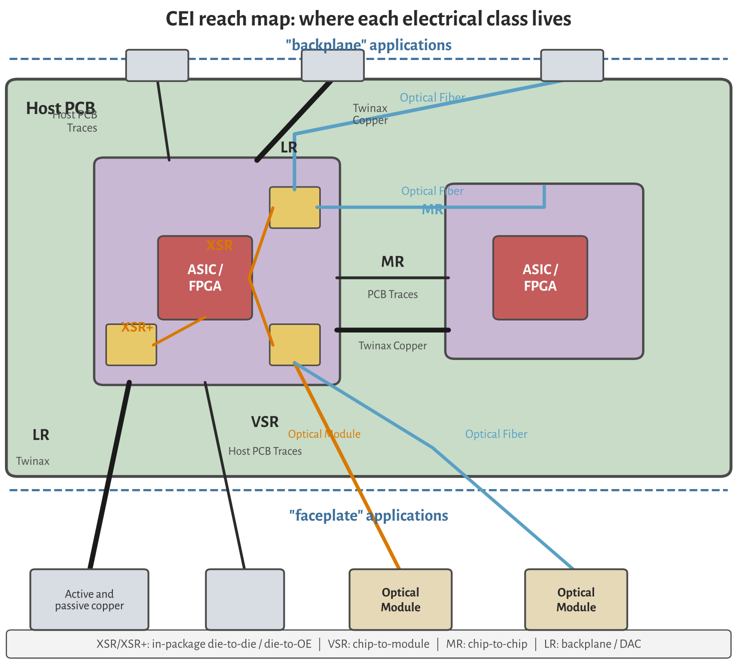

Figure 3.5: CEI reach map on a host PCB (redrawn). Backplane applications sit above the board; faceplate cages and modules sit below. XSR/XSR+ are in-package; VSR is chip-to-module; MR is chip-to-chip; LR covers backplane and active/passive copper.

Class	Hop	Nominal reach / channel	Notes
XSR	Die-to-die or die-to-OE	$\lesssim$50 mm package substrate	CPO / chiplet optics; light EQ
VSR	Chip-to-pluggable module	$\sim$200 mm host + $\sim$20 mm module, 1 connector	Classic faceplate retimed path
MR	Chip-to-chip / midplane	$\sim$500 mm, 1 connector	Board-scale copper
LR	Backplane or copper cable	$\sim$1000 mm host+daughter, 2 connectors	DAC/ACC/AEC territory
Linear	Chip-to-linear module	Same cages as VSR-class ports; no module DSP	LPO foundation; host EQ + FEC

Table 3.8. CEI-224G electrical classes (project map). BER target on the reach classes is 10^{-15} or better with FEC allowed. Draft LR/MR IAs were member-review as of OFC 2025; Linear is the separate full-linear module track.

One SerDes core may not cover XSR through LR efficiently: short reaches want simple, low-power equalization, while LR burns DSP to close tens of dB of loss. That is why Linear is its own project rather than "VSR with the DSP deleted," and why CPO (XSR) and LPO (Linear) show up as different power/serviceability bets (Section 3.14.2 and sections 9.3 and 9.10).

3.14.2 Deploying 224G: LPO, COM, and TDECQ corners

At 224G the alphabet is settled (PAM4 + KP4). What fails in the field is the *margin stack*: electrical channel operating margin (COM) on the host side, transmitter and dispersion eye closure quaternary (TDECQ) on the optical side, and the production corners that couple them (Section 7.9). Retimed modules still dominate 1.6T faceplate ports because their DSP absorbs host-channel sin. LPO and LRO exist to delete that DSP for power and latency; they only ship when both ledgers close without it [3][72][66][4][37].

[3] **??**IEEE EPS, Linear Pluggable Optics overview (2026).

[4] Semtech 224G/lane MZM drivers + TIAs (GN1834 family) for LPO/LRO/CPO (OFC 2026).

The two ledgers that must close together. A retimed module can hide a bad host PCB behind module EQ. An LPO module cannot. You therefore run two go/no-go tests as one program:

Electrical COM After the CEI reference TX, channel, and Rx (CTLE + 8-tap DFE), COM ≥ 3 dB is the usual pass line at the slicer (Section 9.5.2). At 112 GBd the residual eye after that equalizer is only a few millivolts tall; KP4 then cleans pre-FEC BER near 2.4×10^{-4} down to 10^{-15} class (Section 3.12).

COM Channel operating margin: statistical SNR of the fully equalized link. CEI go/no-go is typically COM $\geq$3 dB.

Optical TDECQ / TECQ Outer OMA, RLM, and TDECQ (or *TECQ* without the test fiber) score the transmitter after a bounded reference FFE (Section 7.4). LPO MSA language for 100G/lane DR already couples OMA to max(TECQ, TDECQ) and caps TDECQ near 3.4 dB; 224G-class linear modules inherit the same philosophy even while CEI-224G-Linear and IEEE 802.3dj freeze the exact limits [5][79][6][8][66].

TECQ Transmitter eye closure quaternary: same measurement as TDECQ without the test fiber. Used with TDECQ in LPO MSA OMA tables.

[5] LPO MSA 100G-DR-LPO v1.0: 53.125 GBd PAM4 SMF to 500 m; TDECQ/TECQ $\leq$3.4 dB; builds on 802.3 + CEI-112G-LINEAR.

[6] IEEE P802.3dj: 200/400/800/1600G Ethernet (200G/lane; SA ballot 2026).

CEI-224G-Linear defines the host/module electrical test points (TP1/TP1a, TP4/TP4a) so a linear module can sit between a DSP host SerDes and the fiber without its own CDR [66]. Commercial 224G driver/TIA families advertise that interface explicitly (tunable swing, on-chip CTLE, CEI-224G-Linear host EQ) Semtech 224G. If either ledger is soft, prefer retimed or LRO until the host PCB and module linearity improve; do not "fix" LPO with more FEC alone.

Where 224G LPO programs actually break. The failure modes are familiar once you stop treating LPO as a cheaper OSFP:

- **Host FIR / CTLE mistuned.** Taps pegged or CTLE boost too aggressive raises COM loss and looks like a bad module. Golden-swap the module first (Section 7.9).
- **Connector and package return loss.** VSR/MR channels already sit near the edge at 112 GBd; a resonant stub or long bondwire eats the few mV of slicer margin (Section 9.5 and section 9.5.1).

- **Module nonlinearity.** Driver or TIA compression wrecks RLM; TDECQ climbs even when average power looks fine. Linear-optics parts exist because retimed DSP no longer hides this (Section 3.14.3 and section 4.5).
- **ORL / RIN feedback.** Dirty fiber or a bad isolator raises effective RIN and floors pre-FEC BER while LIV still looks healthy (Section 4.3 and section 7.2.2).
- **Chassis thermal + neighbor load.** Faceplate case temperature and adjacent lanes move bias, TEC, and (for rings) lock; TDECQ and unlock show up together (Table 7.4 and section 6.5).

Half-retimed LRO as the pragmatic middle. When full LPO will not close on the target host, *LRO/TRO* (retimed TX, linear RX) keeps roughly half the DSP power and still relaxes the Tx eye into the fiber [72]. Many AI clusters take that path first: spend watts on the harder electrical→optical direction, keep the receive path linear into the host. The validation split is the same: COM and stressed host input on the linear side, TDECQ on the retimed Tx side (Sections 7.5 and 9.3).

CPO at the same SerDes generation. Co-packaged engines shipping in 2025–26 typically run *200 Gb/s per optical channel* on 100G/200G SerDes into microring banks with external lasers (Section 9.10): same CEI-224G-class shoreline as faceplate 224G, but the lossy pluggable connector is gone and the laser service model moves to ELSFP (Section 5.14). Deployment corners shift from cage thermals to FAU mate, lock hold under neighbor heaters, and ELS hot-swap (Table 7.4 and chapter 6). The electrical alphabet is still 224G PAM4; the hard part is packaging and wavelength control.

Key idea. 224G deployment is a margin problem, not a modulation problem. Close COM and TDECQ together under production-representative corners; use LRO when full LPO will not; treat CPO as the same SerDes generation with a shorter electrical path and a harder laser/lock service model.

3.14.3 448G is where the modulation debate lives

The next step, *CEI-448G* (framework published in late 2025, targeting 3.2 Tb/s systems from 2026 onward), is where the long PAM4 consensus finally comes under pressure [7] [14]. A full CEI-448G glossary of terms is in Section A.7. The debate is not whether 448 Gb/s *per lane* is useful (eight lanes make a 3.2 Tb/s OSFP-class port), but what *symbol rate* and *modulation order* each part of the link must run at to get there.

[7] OIF, *Next Generation CEI-448G Framework* (OIF-FD-CEI-448G-01.0, 2025).

OSFP Octal Small Form-factor Pluggable: high-power faceplate cage for 800G/1.6T/3.2T modules; larger thermal envelope than QSFP-DD.

Line rate, symbol rate, and where each applies. *448G/lane* in CEI means roughly 448 Gb/s of payload on one differential electrical pair between host and module (or between dies in a package). Ethernet maps the same lane count to aggregate rates (400G, 800G, 1.6T, 3.2T). FEC and coding overhead sit on top of the 448 Gb/s target, as they did at 112G and 224G.

For PAM-M, the relationship is

$$R_{\text{line}} \approx \log_2(M)\, R_{\text{sym}},$$

with R_{sym} the baud rate and the Nyquist frequency $f_N \approx R_{\text{sym}}/2$. At 448 Gb/s the OIF framework tabulates the options in Table 3.9 [14]. PAM4 needs 224 GBd and $f_N \approx 112$ GHz. PAM6 drops to 173 GBd and $f_N \approx 87$ GHz. PAM8 drops further to 149 GBd and $f_N \approx 75$ GHz. Each step down in baud buys channel bandwidth at the cost of finer amplitude levels, lower SNR margin, and heavier FEC/DSP.

The critical split is *where* on the board those numbers apply:

- **In-package (*XSR*/XSR+, die-to-die, die-to-OE):** trace lengths of millimeters to a few centimeters. Package bandwidth can reach $\gtrsim$115 GHz with $\leq$0.5 mm BGA pitch and skip-layer routing. 448G-PAM4 at 224 GBd is the working assumption here, including linear drive from host SerDes straight into a co-packaged optical engine [14].
- **Host PCB plus pluggable connector (*VSR*/*MR*/*LR*):** meters of PCB, a module connector, and often a cable. Measured end-to-end channel bandwidth is limited to about 90 GHz today, mostly by connector pin stubs and resonance notches above that frequency [14]. At 90 GHz, PAM4 at 112 GHz Nyquist does not close; PAM6 or PAM8 is the fallback unless connector technology moves the roll-off past 112 GHz.
- **Optical lane (fiber):** IM/DD at 448 Gb/s still targets PAM4 at ≈224 GBd per wavelength. The fiber channel is not connector-limited in the same way; TDECQ, chirp, dispersion, and receiver bandwidth dominate instead of copper insertion loss [14][8] [30].

XSR CEI extra-short reach: die-to-die or die-to-OE, mm-scale traces.

VSR CEI very short reach: host PCB plus one module connector (∼200 mm).

MR CEI medium reach: longer host plus module channel than VSR.

LR CEI long reach: longest CEI copper class before optics take over.

[8] St-Arnault *et al.*, 3.2 Tb/s IM/DD with TFLN modulators (arXiv:2503.24147, preprint).

Figure 3.6 is the packaging map behind that split: co-packaged copper (*CPC*), co-packaged optics (CPO), and faceplate pluggables (retimed, LPO/LRO, AEC/DAC) all leave the host IC at different places, which is why connector bandwidth and host EQ budgets change together [14].

CPC Co-packaged copper: copper I/O at the package edge (OIF map); feeds faceplate cages or short copper without an on-package optical engine.

What ships today on that map (224G / 200G Ethernet). Before the 448G fork, every branch of Figure 3.6 already has a shipping answer at the CEI-224G / IEEE 802.3dj lane rate. Faceplate ports run eight 224 Gb/s PAM4 lanes into 1.6 Tb/s OSFP-class modules (retimed DSP modules still dominant; LPO and LRO gaining share where host COM and TDECQ close, Section 3.14.2). Copper inside the rack uses DAC, ACC, and AEC at the same SerDes generation. CPO engines shipping with Tomahawk 6 and Quantum-X class switches are typically *200 Gb/s per optical channel* on

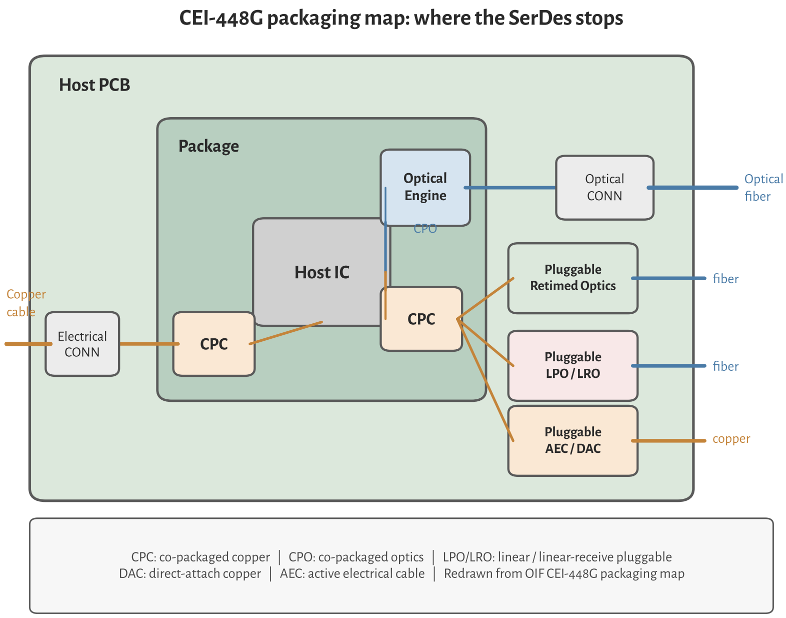

Figure 3.6: CEI-448G packaging map (redrawn). CPC brings copper to the package edge; CPO puts an optical engine in the package; face-plate cages take retimed optics, LPO/LRO, or AEC/DAC. Same host IC, different SerDes reach and EQ burden.

100G/200G SerDes into microring banks with external lasers (Section 9.10): same packaging map, one generation behind the 448G electrical debate. KP4 FEC, CEI-224G-Linear for LPO, and KP4 pre-FEC BER near 2.4×10^{-4} are the settled electrical/optical contract (Sections 3.12 and 3.14 and section 9.5.2). The rest of this section is about what breaks when you try to double that lane rate on the same shoreline.

So the "448G debate" is really two linked questions: can the *electrical* shoreline run PAM4 at 224 GBd, and can the *optical PMD* modulate and detect at the same symbol rate? Figure 3.7 maps the three dominant architectures and the baud rate at each hop.

Table 3.9: PAM-M options at 448 Gb/s/lane (OIF CEI-448G framework, pre-FEC line-rate target).

Scheme	Symbol rate	f_N	UI	SNR penalty vs. PAM4
PAM4	224 GBd	112 GHz	4.5 ps	0 dB
PAM6	173 GBd	87 GHz	5.8 ps	≈2 dB
PAM8	149 GBd	75 GHz	6.7 ps	≈4–5 dB

FEC overhead and the SNR budget. At 224G, IEEE and CEI both lean on *KP4* FEC: Reed–Solomon RS(544,514) from 802.3 Clause 91, with coding overhead $30/544 \approx 5.5\%$ and a pre-FEC BER target near 2.4×10^{-4} for optical PHYs [8][14]. Post-FEC the corrected BER target is 10^{-15} class. The line rate on the wire is higher than the MAC/info rate: a 200 Gb/s Ethernet lane (802.3dj) runs PAM4 at 112 GBd (224 Gb/s raw) and delivers ≈211 Gb/s of payload after KP4, rounded to 200 Gb/s at the MAC.

448G pushes FEC harder. CEI-448G lists pre-FEC BER 10^{-4} as the working target

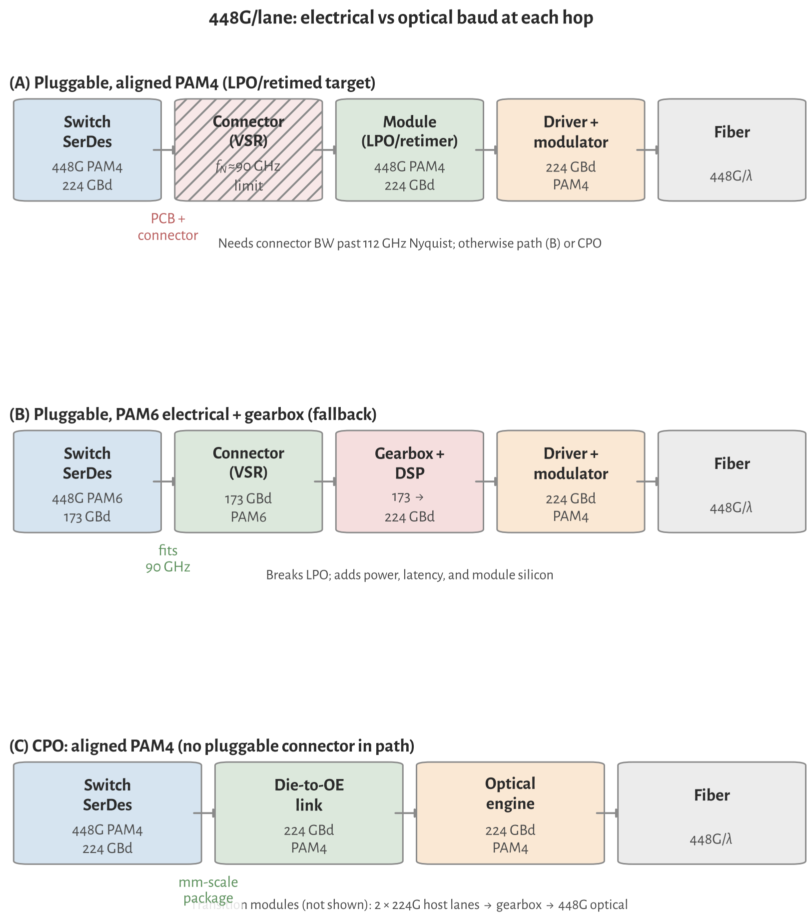

Figure 3.7: 448G/lane signal paths: aligned pluggable PAM4 (connector-limited), PAM6 electrical with module gearbox (LPO incompatible), and CPO or gearboxed 224G transition.

(same as prior CEI generations) but notes that 10^{-4} at $\geq$40 dB channel loss may not close without a *stronger* code than KP4 [14]. PAM6 adds $\sim$2 dB SNR penalty on top, so workshop assumptions often pair PAM6 with *MLC* and higher-overhead FEC ($\sim$12%) for a fair comparison to PAM4 with KP4 [9] [28]. Table 3.10 summarizes the landscape (OH = coding overhead); 448G codes are not finalized (see caption for sources).

Sources: 802.3dj/KP4 baseline [8]; CEI-448G electrical targets [14]; PAM6 FEC assumption [28]; optical demo [30].

Higher FEC overhead feeds back into the modulation debate: every extra parity bit raises the line rate and Nyquist frequency, which hurts a bandwidth-limited copper channel. That is one reason PAM6 (lower baud) and stronger FEC get paired, and why CPO (shorter electrical path) keeps PAM4 attractive.

MLC Multi-level coding: coded modulation paired with PAM6/PAM8. OIF 448G workshop models often add a strong outer RS code; $\sim$12% overhead vs. KP4 at 5.5%.

[9] OIF 448G AI Workshop: native modulation vs. gearboxed 224G (2025).

OH Coding overhead: fraction of line rate spent on parity symbols (KP4 $\approx$5.5%).

HDFEC Hard-decision FEC: high-overhead codes used in optical demos and long-reach products (10–15% OH), not the host KP4 budget.

Context	FEC	OH	Role at 448G/lane
802.3dj / 224G	KP4 RS(544,514)	5.5%	Baseline; pre-FEC BER near 2.4e-4.
CEI-448G, PAM4 elec.	KP4 or stronger	5.5%+	Target pre-FEC 1e-4; KP4 may fail on 40 dB-class channels.
CEI-448G, PAM6 elec.	MLC + strong RS	12%	Offsets PAM6 SNR penalty; adds latency (workshop assumption).
448G optical demos	HD-FEC / product	10–15%	3.2T TFLN demo threshold; not necessarily host FEC.
FEC architecture	end-to-end vs. terminated	n/a	448G may need dedicated electrical inner code.

Table 3.10. FEC at 448G/lane (448G codes provisional).

Electrical SerDes: feasible, but the channel is the gate. 448G SerDes themselves are widely treated as feasible on advanced CMOS ($\leq$N3/N2): ADC/DAC DSP transmitters and receivers demonstrated at 224 Gb/s are the template, and coherent-optics SerDes with $\sim$100 GHz class AFE at 200 GBd is cited as encouraging precedent for doubling to 448G-PAM4 [14]. Synopsys and others have published channel simulations showing margin for both PAM4 and PAM6 on 224G-class channels when equalization and FEC scale with the modulation order[10] [27]. Power targets cluster around 0.5–2.5 pJ/bit at 448G, similar in spirit to 224G scaling.

[10] Synopsys, *Designing for 448G* white paper (2025).

The blocker is not the PLL or the DAC; it is the *passive channel*. Connector bandwidth near 90 GHz forces a fork:

1. **Fix the channel for PAM4:** new high-density connectors, shorter reach, skip-layer PCBs, cabled-host internal twinax. Simulations show passive CPC reach up to $\sim$1.2 m at 448G-PAM4 under optimistic connector assumptions [14]. This path preserves electrical/optical format alignment and keeps LPO/LRO architectures viable (Chapter 9).

2. **Keep the channel, change modulation:** PAM6 at 173 GBd fits a 90 GHz channel but adds $\sim$2 dB SNR penalty and pushes the host toward stronger FEC than KP4. PAM8 relaxes bandwidth further at $\sim$4–5 dB penalty [14].

3. **Bypass the bad channel:** co-packaged optics with millimeter-scale die-to-OE links. Host 448G-PAM4 SerDes drives the optical engine directly; the pluggable connector never sees 224 GBd [14].

If electrical PAM6 wins on the PCB but optics stay at PAM4, the module needs a *gearbox* (rate/format conversion) in the retimed path. That adds power, latency, and silicon area, and it breaks linear pluggable optics: LPO requires the host electrical waveform to match what the optical modulator expects [14][72]. First 448G optical modules may therefore ship with gearboxed 224G SerDes (two 224G lanes per 448G optical lane) before native 448G electrical I/O is ready [28].

Optical modulation at 224 GBd: hard, but no longer hypothetical. On the optics side, the industry assumption is still IM/DD PAM4 at $\approx$224 GBd for 448 Gb/s per λ [14][30]. That requires roughly 112 GHz class electro-optic bandwidth in the modulator (and comparable photodiode/TIA bandwidth in the receiver), plus a driver with enough linear swing and RF BW. By 2025–26 that stack exists in demos and, increasingly, in announced commercial parts: silicon rings and MZMs with peaking, TFLN MZMs past 100 GHz EO bandwidth, EMLs still owning volume at 200G/lane, and SiGe drivers / Ge receivers catching up. The snapshot below is the state of play; device sections later unpack each family.

- **Silicon microring modulators:** resonant, compact modulators for dense WDM and CPO (Section 3.14.3). OFC 2026 demos reach 224–416 Gb/s PAM4 with inductive peaking [11] [42] [12] [43].
- **Silicon Mach–Zehnder modulators:** the broadband, non-resonant counterpart to microrings (Section 3.14.3). OFC 2026 demos reach 400G/lane PAM4 with SiGe drivers [13] [45].
- **EML:** integrated DFB + EAM remains the default through 200G/lane for DR/FR pluggables: one chip, low chirp, mature supply chain.
- **TFLN MZM:** thin-film lithium niobate Mach–Zehnder modulators with CW lasers exceed 100 GHz EO bandwidth and are the leading path to native 224 GBd IM/DD (Section 3.14.3). A 3.2 Tb/s system (eight $\times$225 GBd PAM4 lanes) with 3 nm CMOS SerDes and TFLN modulators over 2 km *FR8*/*DR8* was reported in 2025 [30].
- **Drivers:** commercial modulator drivers with >120 GHz RF BW for 400G PAM4 EML/MZM/TFLN platforms appeared in 2026 [14] [36].
- **Receivers:** Ge/Si PIN and APD photodiodes above 100 GHz and 224G TIAs exist (Section 4.5); the receive side is not the long pole relative to modulator/driver bandwidth at 448G.

[11] Lin *et al.*, 90-GHz Si microring, 224-Gb/s PAM4, inductive tuning (OFC 2026 Th2A.14).

[12] Peng *et al.*, Si microring >400 Gbps PAM4, T-coil peaking (OFC 2026 M2A.2).

[13] Dong *et al.*, 400G/lane Si MZM PAM4 with SiGe driver (OFC 2026 postdeadline Th4A.4).

FR8/DR8 802.3 800G single-mode reaches: FR8 = 2 km, DR8 = 500 m (eight lanes per direction).

[14] MACOM 448G/lane PAM4 drivers (MAOM-025408/022404): >120 GHz RF BW (OFC 2026).

Silicon microring and microdisk modulators. A microring modulator (MRM) wraps a phase-shifter waveguide into a closed loop coupled to a straight *bus* waveguide. A *microdisk* is the same idea in disk form: a pillar cavity evanescently coupled to the bus, often with a wider free spectral range (FSR) in a smaller footprint. Both are resonant filters as well as modulators: when the input wavelength sits on resonance, drop-port power is high; off resonance it is rejected. Data modulation shifts the resonance (carrier depletion or injection in an embedded pn junction) or detunes the laser relative to a fixed ring, mapping voltage to intensity at the through or drop port.

MRM Microring modulator: resonant Si modulator; compact, WDM-native, needs wavelength lock. CPO workhorse at 200G/ch.

FSR Free spectral range: wavelength span between adjacent resonances of a ring or etalon. Sets WDM channel packing density.

The central design trade is *photon lifetime versus bandwidth*. A high-Q ring stores photons longer, which improves modulator efficiency (V_π) but narrows the optical passband and creates an electrical bandwidth ceiling through the RC-limited junction. Coupling strength, ring radius, and whether the device operates in under-coupled or over-coupled regime set Q, extinction, and the electro-optic (EO) roll-off. That trade does not appear in broadband Mach–Zehnder modulators (Section 3.14.3).

Three constraints dominate ring modulator design at 100–400G per λ. First, EO bandwidth: the junction RC roll-off is often below the target Nyquist frequency, so inductive peaking (T-coils on the drive path), optimized detuning from resonance, and compact RLC co-design extend EO BW without widening the ring so much that Q collapses [42][43]. Production CPO rings target 50–90 GHz; conference demos report 90–110+ GHz with aggressive peaking [42][43][15][44]. Second, wavelength alignment: ring resonance drifts roughly 10 GHz/°C in silicon, so each λ in a WDM bank needs the laser or the ring tuned onto the modulator resonance (Chapter 6 and section 6.4). Thermal crosstalk from neighbors shifts resonances in dense arrays, which is why fleet validation must include corner temperature and adjacent-channel heating (Section 6.5 and chapter 7). Third, optical loss and FSR: ring radius sets FSR ($\Delta\lambda$ between adjacent resonances). Microdisk and *Euler* ring layouts widen FSR so more channels fit under one free spectral range without collisions [44], while residual coupling loss and on-resonance insertion loss (often 1–3 dB class per modulator) eat link budget.

[15] OFC 2026 M2A.1: Euler Si microring, 256-Gb/s PAM4, >67-GHz EO BW, 3-THz FSR.

Integration is where rings win. A single SOI die can pack dozens of ring modulators and filters for CW-WDM (Chapter 6 and section 6.6), each fed by one wavelength from an external comb or multi-wavelength ELS (Chapter 5). The photonic engine co-packaged with a switch ASIC (Broadcom Bailly/Davisson, NVIDIA Quantum-X/Spectrum-X, Section 9.10) uses microring modulators at 200 Gb/s per channel today. Fiber count stays low because many λ share one waveguide; area and modulator count scale with WDM order rather than with faceplate port count.

State of the art in 2025–26 shows how far that peaking path has been pushed. Inductive and wavelength tuning carried silicon MRMs to 224 Gb/s PAM4 at 90 GHz EO BW [42]. T-coil-peaked designs report 416 Gb/s PAM4 ($\approx$208 GBd) with >110 GHz 1-dB EO BW and TDECQ 2.88 dB at 1 Vpp [43]. Euler microdisk rings show 256 Gb/s PAM4 with >67 GHz EO BW and 3 THz FSR in O-band [44]. These are lab and conference results, but they match the 200G/lane CPO shipping point and probe 448G-class lane rates when paired with sufficient electrical drive.

The platform choice is a packaging and control decision as much as a bandwidth one. Prefer rings when many λ must fit on one PIC and modulator area dominates: CPO WDM engines, scale-up optical I/O, and any architecture that already budgets for wavelength locking (Chapters 6 and 9). Prefer a silicon MZM for single-λ DR/FR where a flat passband avoids lock loops (Section 3.14.3). Prefer TFLN when you need native 224 GBd in a pluggable and ring thermal control at fleet scale looks harder than hybrid assembly (Section 3.14.3).

Silicon Mach–Zehnder modulators. Silicon photonics builds the Mach–Zehnder modulator (MZM) as a push-pull interferometer on a silicon-on-insulator (SOI) rib or strip waveguide. Phase shifters in each arm use *carrier depletion* in an embedded pn junction: reverse bias pulls carriers out of the waveguide core, lowering refractive index and shifting optical phase. Intensity modulation comes from recombining the arms at a 3-dB coupler, so chirp stays low compared with directly modulated lasers (Chapter 5).

MZM Mach–Zehnder modulator: push-pull interferometer; broadband, low chirp. Si MZM, TFLN MZM, or hybrid platforms.

SOI Silicon on insulator: substrate for silicon photonics; 220 nm Si layer on buried oxide confines the optical mode.

Three constraints mirror every high-speed MZM, but silicon's weak electro-optic coefficient (Δn per volt is far smaller than lithium niobate) sets the numbers. The optical S_{21} response must span the Nyquist frequency ($\approx$56 GHz at 112 GBd, 112 GHz at 224 GBd); junction capacitance, series resistance, and traveling-wave electrode (TWE) microwave loss set the roll-off. Production Si MZMs for 100–200G/lane modules typically quote 70–100+ GHz 3-dB BW; differential-drive layouts and compact 300-mm platforms report 80–95 GHz class results [16][47][17][46]. $V_\pi L$ for carrier-depletion Si is often $\approx$1.5–2.5 V·cm, so millimeter-scale devices need 2–4 V peak-to-peak drive at the target baud inside the linear range of a SiGe or BiCMOS modulator driver (448G-class drivers exceed 120 GHz RF BW) [36]. Unlike resonant rings, an MZM is broadband, so WDM channels do not fight thermal lock (Chapter 6); the cost is length: mm-scale shifters and splitters add 2–4 dB on chip, and the device is far larger than a ring, which matters in dense CPO tiles.

[16] OFC 2026 W3E.5: differential-drive Si MZM, 81.8-GHz 3-dB BW, 100-GBd PAM8 eyes.

[17] OFC 2026 M2A.6: 500-μm Si MZM, 94.7-GHz median EO BW, 2.4-dB IL (300-mm platform).

Integration is the main reason Si MZM survives competition from EML and TFLN. The modulator shares a die with Ge photodiodes, fiber grating couplers or edge couplers, monitors, and (in WDM products) MUX/de-MUX filters, all in a CMOS foundry-compatible flow. An external CW DFB or ELS (Chapter 5) couples in through a spot-size converter; the RF driver is usually a separate die wire-bonded or 2.5D packaged next to the PIC, the same assembly style as TFLN modules but without bonding a second optical material stack.

State of the art in 2025–26: Si MZMs are the default modulator in 100G/lane and 200G/lane DR/FR silicon-photonics transceivers. Pushing to 400G/lane IM/DD was open until OFC 2026, when Coherent reported 400 Gb/s PAM4 per lane with a Si MZM and commercial SiGe driver (2.5 V swing) [45]. The same conference cycle showed a 500 μm compact MZM on a 300-mm platform with 94.7 GHz median EO BW and 2.4 dB on-chip insertion loss [46], and a differential-drive MZM with 81.8 GHz 3-dB BW with eyes to 100 GBd PAM8 [47]. These are lab and conference demos, not shipping modules, but they close the headline lane-rate gap with ring modulators while keeping a flat passband that rings only match with tight wavelength control (Section 3.14.3).

SiPh Silicon photonics: waveguides and modulators on SOI; CMOS fab-compatible. Mainstream for DR/FR and CPO.

Thin-film lithium niobate Mach–Zehnder modulators. *TFLN* refers to a sub-micrometer slice of lithium niobate bonded to a silicon or silica handle wafer. Compared with bulk LN, the tight optical mode confinement shrinks electrode gap and interaction length, which cuts the half-wave voltage–length product

TFLN Thin-film lithium niobate: sub-μm LN on Si handle. >110 GHz EO BW; leading path to native 224 GBd PAM4.

$V_\pi L$ while keeping a wide *electro-optic* (EO) bandwidth. The modulator is a Mach–Zehnder interferometer (MZM): a *Pockels* phase shifter in each arm, driven in push-pull, converts RF voltage on a *traveling-wave electrode* (TWE) into intensity at the output coupler.

EO Electro-optic: conversion from voltage to optical phase or intensity. EO bandwidth sets modulator speed.

TWE Traveling-wave electrode: RF transmission line on a modulator that velocity-matches the optical mode for high BW.

Three design constraints set whether a TFLN MZM can run at 224 GBd PAM4 ($f_N \approx 112$ GHz). EO bandwidth, measured as the S_{21} roll-off of optical response versus RF drive, must clear that Nyquist: production-class devices quote $\gtrsim$110 GHz 3-dB BW, and research devices with low-*k* underfill or advanced TWE layouts extrapolate toward 200+ GHz [18][32][19][OFC 2026 W1A.3]. $V_\pi L \approx 1$–2 V·cm is typical, so a 5–7 mm device gives $V_\pi \approx 1.5$–2 V that must fit inside the linear swing of a SiGe or InP driver at 112 GHz class [36][20][31]. Along the TWE, microwave and optical group indices must match or bandwidth collapses; low-loss underfill, narrow-gap CPW or co-planar layouts, and transparent conductive oxide (TCO) electrodes trade insertion loss against efficiency [31][32].

[18] He *et al.*, dual-band TFLN MZM 390-Gb/s PAM8, 220-GHz extrapolated BW (Laser Photonics Rev. 2025; arXiv:2411.15037).

[19] OFC 2026 W1A.3: 200-mm TFLN platform, 110-GHz EO, $V_\pi L = 1.9$ V·cm.

[20] Liu *et al.*, TFLN MZM 224-Gb/s PAM4, 108-GHz EO BW (OFC 2025 M3K.2; arXiv:2311.05119).

Demonstrated IM/DD results on TFLN MZMs include 224 Gb/s PAM4 at 108 GHz EO BW (O-band, $V_\pi L = 1.02$ V·cm) [31] and 390 Gb/s PAM8 on the same dual-band chip (extrapolated 220 GHz BW, sub-fJ/bit in lab) [32]. System-level proof came with eight $\times$225 GBd PAM4 lanes over 2 km using 3 nm SerDes and packaged TFLN modulators [30]. Commercial suppliers (HyperLight, Lumiphase, and foundry lines on 200-mm silicon) now ship 110 GHz-class packaged MZMs aimed at 200–240 GBd signaling [21][35][OFC 2026 W1A.3].

[21] HyperLight, 110-GHz packaged TFLN IQ modulators for 240-GBaud class (Sept. 2025 announcement).

Integration looks unlike monolithic silicon photonics. A CW DFB or ELS feeds the TFLN chip through a Si/SiN coupler; the RF driver sits on a separate die, wire-bonded or flip-chip mounted with matched 50 Ω lines. Hybrid Si–LN platforms (silicon waveguides, LN overlay) were demonstrated early for 100G/lane and remain a template for co-packaged assemblies [22][34]. The laser is not on the TFLN chip, so alignment, fiber attach, and thermal bias of the MZM quadrature point become validation items (Chapter 7).

[22] He *et al.*, hybrid Si–LN integration (Nature Photonics 2019).

EML and the electro-absorption modulator. An *EML* integrates a DFB laser with an *EAM* (electro-absorption modulator) on one InP chip. The EAM is a reverse-biased absorption region: voltage shifts the band edge (Franz-Keldysh or quantum-confined Stark effect), attenuating light with far less chirp than direct current modulation (Section 3.11).

EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

EMLs dominate 100G/lane and 200G/lane DR/FR pluggables because they are single-chip, mature in supply chain, and match $\sim$70–100 GHz class EO bandwidth (Table 3.12). The design limits that show up in validation are EAM bias and aging (bias sets extinction and chirp; aging drifts the curve and shows up as TDECQ/RLM creep, Chapters 7 and 8), driver swing (a few volts inside the linear absorption region; 448G-class EML drivers appeared alongside MZM drivers in 2026 [36]), and thermal headroom (uncooled datacom is standard; slope efficiency and bias must stay in range across case temperature).

EML wins on cost and integration through 200G/lane. Above that, external modulators (Si MZM, TFLN, rings with CW laser) chase bandwidth and chirp headroom (Section 3.14.3).

Modulator drivers: requirements, records, outlook. Every external modulator (Si MZM, TFLN, ring, EAM) needs an RF path that delivers enough *linear* swing at the target baud. That path is either a dedicated SiGe/BiCMOS *modulator driver* die, or (for LPO) the host SerDes itself. Laser *bias* drivers are a different circuit and a different noise budget (Section 5.8). Do not share the modulator driver's switching returns with the CW bias rail.

SiGe Silicon-germanium BiCMOS: high-f_T process for TIAs and modulator drivers at 100–200+ GHz bandwidth.
BiCMOS Bipolar CMOS: process combining bipolar (high f_T) and CMOS on one die; used for high-speed TIAs and drivers.

What the driver must deliver. At 224 GBd PAM4 the symbol Nyquist frequency is 112 GHz, so a usable driver is not just "fast enough on paper." It needs RF bandwidth roughly $\gtrsim$100–120 GHz (3 dB) with flat gain and low group-delay ripple, output swing matched to V_π or EAM drive (often $\sim$1.5–3 V peak-to-peak, differential or single-ended, part- and modulator-dependent), linearity good enough that RLM and TDECQ stay inside the PMD budget when the optical path adds its own compression (Section 7.4), and return loss / matching into 50 Ω (or the designed modulator impedance) so bondwire and package reflections do not eat the bandwidth you paid for on the die. Distributed (traveling-wave) SiGe drivers dominate the high-BW niche: many cascoded gain cells along an artificial transmission line. Lumped drivers win on area and power at lower baud. Flip-chip bumped die and short wirebonds matter as much as f_T: a 120 GHz die behind a long bond loop is a 70 GHz module.

Record and commercial snapshot (2025–26). Table 3.11 separates research demos (often with offline DSP) from shipping or announced commercial parts aimed at 1.6T/3.2T modules. Commercial anchors: MACOM's MAOM-025408 MZM and MAOM-022404 EML drivers (>120 GHz RF BW, OFC 2026) [36]; Semtech's GN1877/GN1887 224G quad/octal family for LPO/LRO/CPO across SiPh, InP, and TFLN [37]. Research anchors: a 130 nm SiGe distributed driver at 105.7 GHz BW and 2.25 V swing running 232 GBd PAM4 into TFLN (offline DSP) [23] [38]; OFC 2026 co-designed engines at 210 GBd / 420 Gb/s PAM4 (TFLN TOSA) [24] [39], 180 GBd PAM4 (InP MZM + EML-class driver) [25] [40], and differential SiGe+TFLN at 140 GBd PAM8 / 1.4 pJ/bit [26] [41]; plus the Si MZM postdeadline with a commercial SiGe driver at 400 Gb/s/lane and 2.5 V swing [45].

[23] RFIC 2026: 105.7 GHz SiGe distributed driver; 232 GBd PAM4 into TFLN (research).
[24] OFC 2026 W4J.4: 210 GBd / 420 Gb/s PAM4 TFLN TOSA (co-designed driver).
[25] OFC 2026 W3E.6: 180 GBd PAM4 InP MZM + 224G-class EML driver.
[26] OFC 2026 M4D.4: differential SiGe+TFLN; 140 GBd PAM8 at 1.4 pJ/bit.

LPO and the host-as-driver path. For *LPO*, the host SerDes (or a linear driver in the module with no retimer) is the modulator driver. Waveform fidelity is end to end: host FFE, connector ISI, driver/modulator nonlinearity, and TIA all land in TDECQ

Part / paper	Process	RF BW	Rate / swing	Note
MACOM MAOM-025408 (MZM)	SiGe	>120 GHz	448G-class PAM4	Commercial; SiPh MZM
MACOM MAOM-022404 (EML)	SiGe	>120 GHz	448G-class PAM4	Commercial; EML / TFLN
Semtech GN1877 / GN1887	—	224G-class	224 Gb/s/lane	Quad/octal; LPO path
RFIC 2026 distributed drv	130 nm SiGe	105.7 GHz	232 GBd; 2.25 V	Research; TFLN; offline DSP
OFC 2026 Si MZM + SiGe	commercial SiGe	(drv-limited)	400 Gb/s/lane; 2.5 V	Postdeadline Th4A.4
OFC 2026 TFLN TOSA	co-designed	—	210 GBd / 420 Gb/s	Driver+TFLN engine
OFC 2026 InP MZM + EML drv	InP + SiGe-class	76 GHz MZM	180 GBd PAM4	Co-packaged engine
Nokia/Bell Labs hybrid	SiGe + TFLN	—	140 GBd PAM8; 1.4 pJ/bit	Diff. drive; low $V_\pi L$

Table 3.11. Modulator-driver snapshot (c. 2026). Commercial rows are vendor announcements (bandwidth/swing as published); research rows often use offline DSP and short RF interconnects. "448G-class" means aimed at ∼400–448 Gb/s/lane PAM4 modules, not a CEI compliance claim. Sources cited in the paragraph above.

and pre-FEC BER (Sections 3.6 and 7.4 and section 9.5.1). That is why linear-optics driver families (e.g. Semtech's 224G set) advertise CEI-224G-Linear host EQ and tunable swing: the module cannot clean up what the host launches Semtech 224G. Retimed modules hide some of this behind DSP; they also burn the watts LPO was meant to save.

Outlook. Driver roadmaps are no longer waiting on papers alone. Commercial 448G-class parts are shipping as dies: MACOM's >120 GHz MZM and EML drivers (OFC 2026) are the clearest public 400G/lane announcement MACOM, while research benches already run past 200 GBd on short RF paths. Expect a short period where optics and drivers lead host SerDes, so gearboxed 224G electrical into 448G optical remains common (Section 3.14.3). The hard problems are packaging (bondwire, FAU, faceplate connectors), co-design of peaking with modulator S_{21}, and LPO cases where driver linearity and host COM sit beside TDECQ as first-order validation items (Section 9.5.2 and section 7.9). After the driver, the next ceilings are modulator EO bandwidth and the PD/TIA noise stack (Section 3.14.3 and section 4.5).

Key idea. A 448G-class modulator driver is a >100 GHz linear SiGe amp with swing matched to V_π, not a SerDes pin. Commercial dies now claim >120 GHz for MZM/EML/TFLN; research shows ∼230 GBd on co-designed benches. Package and modulator match set the real ceiling; LPO makes the host the driver.

Gearboxes: when electrical and optical rates diverge. A *gearbox* converts lane rate or modulation format between electrical and optical domains. The 448G transition often uses *two* 224G electrical lanes into one 448G optical lane (Figure 3.7 and section 3.14.3) when host SerDes or connectors lag optics.

Gearbox Rate/format converter between electrical and optical lanes; common in 448G transition; incompatible with strict LPO.

Gearboxes add power, latency, and silicon area. They also break strict LPO: the host waveform no longer matches what the optical lane carries, so a retimed or half-retimed module is required [14][72]. Expect gearboxed 3.2T modules before native 448G electrical I/O and matched VSR connectors land in volume.

Putting the platforms side by side, Table 3.12 summarizes the trade space at 100–400G per λ; Section 3.14.3 expand each path. Through 200G/lane DR, EML still leads on cost and single-chip integration. Silicon rings and MZMs lead in CPO where area and CMOS fab matter, with Si MZM preferred for flat single-λ DR/FR and rings preferred for dense WDM. TFLN leads when you need $\gtrsim$100 GHz EO BW with low chirp for native 224 GBd PAM4 in a pluggable or NPO module and silicon cannot close the 112 GHz Nyquist without heavy peaking.

Platform	EO BW	Per-λ demo	Chirp	Integration	Typical use
Si microring	90–110 GHz	224–416 Gb/s PAM4	low	monolithic SiPh	CPO, WDM
Si MZM	70–100+ GHz	400 Gb/s PAM4	low	monolithic SiPh	DR/FR, CPO
TFLN MZM	108–110+ GHz	224–390 Gb/s PAM4/8	very low	hybrid + driver die	400G/lane pluggable, FR
EML	70–100 GHz	200 Gb/s PAM4	low	single InP chip	DR/FR to 200G/lane

Table 3.12. IM/DD transmitter platforms at 100–400G per λ (2026 snapshot).

The honest summary: *modulating* at 224 GBd PAM4 is demonstrated in multiple material systems when the electrical drive path is short and the driver is dedicated (not a lossy meter of PCB plus OSFP connector). *Modulating* at that rate from a switch ASIC through a pluggable module is the harder system problem. Silicon photonics rings without peaking still sit below the 112 GHz Nyquist needed for 448G-PAM4; TFLN, EML, and peaked Si rings are the near-term paths. WDM (Chapter 6) and CPO remain the architectural escape hatches: more aggregate bits without forcing every electrical lane to 224 GBd on a lossy shoreline.

Where the standards conversation stands. Electrical and Ethernet standards are climbing the same ladder on slightly different clocks. OIF CEI owns the electrical I/O recipe; IEEE 802.3 owns how Ethernet names the lane and the optical PMD. Neither has locked 448G modulation yet.

OIF CEI (electrical I/O). No modulation order is locked yet in CEI-448G. PAM4 remains attractive if connectors reach $\gtrsim$112 GHz and if optics stay on PAM4 (alignment for LPO, lower FEC overhead, backward compatibility with 224G gear). PAM6 is the leading compromise for bandwidth-starved host channels if connector progress stalls [14][27][29]. Electrical IAs target a 2026–28 window [14].

[27] Cignal AI on OIF 448G workshop (April 2025).

IEEE 802.3 (Ethernet PHY/MAC). Ethernet standards trail CEI slightly but follow the same lane-doubling cadence. Table 3.13 is the map as of mid-2026.

Project	Status	Eth. lane	PHY line
802.3df	Feb 2024	100 Gb/s	≈112G PAM4
802.3dj	SA ballot 2026	200 Gb/s	≈224G PAM4
802.3 400G/lane SG	Mar 2026	400 Gb/s	≈448G

Table 3.13: IEEE 802.3 lane-rate generations.

802.3df (approved February 2024) defined 200/400/800 Gb/s Ethernet using **100 Gb/s lanes** (112 GBd PAM4 class). **802.3dj** is the active 200 Gb/s-lane amendment: 200/400/800/1600 Gb/s Ethernet with PAM4 at 112 GBd, KP4 FEC, and a large PHY portfolio (copper and IM/DD optics). The draft cleared 802.3 working-group recirculation ballot in February 2026 and entered IEEE Standards Association ballot; approval is expected around late 2026 [8][28][10].

[28] IEEE 802.3 Ethernet for AI ad hoc (400G/lane CFI, 2025–26).

400 Gb/s per lane is the next Ethernet milestone. The *Ethernet for AI* (E4AI) ad hoc incubated demand for 3.2 Tb/s ports and 400G/lane signaling [10]. A call for interest in March 2026 led to an IEEE 802.3 **400 Gb/s/lane Signaling Study Group** (chartered 13 March 2026) to draft a PAR for electrical interconnects and single-mode fiber reaches up to 500 m [29][11]. The proposed scope has two tracks: a fast track for 400/800/1600 Gb/s on 400G/lane copper and IM/DD optics (scale-up, intra/inter-rack), and a follow-on phase for **3.2 Tb/s** and additional PHYs [10][11]. That aligns with OIF CEI-448G and industry 3.2T module work, with one naming difference: IEEE quotes **400 Gb/s per lane** (MAC/info rate); CEI quotes **448 Gb/s** (line rate including FEC overhead). They refer to the same generation.

[29] IEEE 802.3 400 Gb/s/lane Signaling Study Group (chartered March 2026).

Until native 448G host SerDes and matched connectors land, expect a transition generation that gearboxes 224G electrical lanes into 448G optical lanes (Figure 3.7), then a consolidation generation with 400G/lane Ethernet PHYs where electrical and optical both run PAM4 at 224 GBd end to end.

3.15 Engineering lens

3.15.1 How it works

IM/DD is the whole game in one line: put data on optical power, read it with a square-law photodiode, and let FEC and equalization buy back the SNR that PAM4 spends. Everything else in this chapter is where each block in that chain lives and how the standards name it.

3.15.2 How it is measured

Name the reference plane before naming the metric. At TP2, use a calibrated power meter and digital communications analyzer (DCA) for average power, outer optical modulation amplitude (OMA), extinction ratio (ER), relative level mismatch (RLM), and transmitter and dispersion eye closure quaternary (TDECQ). At TP3, use a stressed source, attenuator, and bit-error-ratio tester (BERT) for receiver sensitivity. Use a vector network analyzer (VNA) for electrical and electro-optic *S*-parameters when bandwidth or reflections are suspect. Acceptance limits come from the optical physical-medium-dependent (PMD) specification and the program acceptance test plan (ATP), with the test planes in Section 3.9 and methods in Chapter 7 [8, 79].

3.15.3 How it fails

The common escapes are low ER from a bad bias point, low OMA from loss or limited drive, high TDECQ from bandwidth or nonlinearity, a receiver driven into saturation, and reflection-induced burst errors. Multi-lane links add power imbalance and electrical lane mismatch. A clean average-power reading does not clear the transmitter because ER, RLM, and noise can still close the eye.

Failure mode: Low extinction ratio

Symptoms. Average power passes, but OMA is low and receiver sensitivity degrades.
Likely causes. Incorrect laser or modulator bias, a compressed driver, a detuned ring, or an aged electro-absorption modulator.
Measurements. DCA level histograms, ER and RLM, bias sweeps, wavelength, and pre-FEC BER versus received power.
Mitigations. Restore the bias point, remove compression, retune the resonance, and add the failing corner to calibration and ATP.

3.15.4 How it is debugged

For degraded receiver sensitivity, first verify the power meter and reference plane. Then measure OMA, ER, RLM, and TDECQ at TP2. If the transmitter passes, inspect insertion loss and optical return loss (ORL), then run stressed sensitivity at TP3. A power sweep separates a noise-limited waterfall from saturation or a BER floor. A golden transmitter and receiver split the last ambiguity. Change one block at a time and preserve the failing state before cleaning, reseating, or retuning it.

Debug story

Observed. Receiver sensitivity was several dB worse than the qualification baseline.
Investigation. Average power passed, but the DCA showed low ER. A bias sweep restored both ER and the BER waterfall.
Finding. The receiver was healthy; the transmitted zero level was too high.
Root cause. A laser-bias calibration table selected the wrong temperature bin.
Resolution. The calibration was corrected, limits were added across temperature, and the escaped firmware revision was quarantined.

3.16 Interview and design review questions

Concept.

- Why is PAM4 preferred over NRZ at 200G/lane despite its SNR penalty?
- What does KP4 FEC buy, and what does it cost in overhead and latency?
- Why does IM/DD dominate short reach while coherent wins at longer distances?

Design.

- Which plane owns each limit, and can production reach that plane without a lab-only fixture?
- Which margin dominates: OMA, TDECQ, ORL, receiver sensitivity, or host channel operating margin?
- What prevents saturation at maximum launch power and minimum channel loss?
- Which bias or equalizer settings are calibrated, and how are bad tables detected before ship?

Debug.

- Average power is in spec but BER fails. What do you measure next?
- The equalizer taps are saturated. What does that tell you about the channel?
- FEC errors are clustered rather than random. What failure mechanisms produce bursts?
- What production escape leaves power in range while BER fails?

Manufacturing and operations.

- What is the fastest production test that catches a bad bias calibration?
- How do you correlate ATE TDECQ to a reference DCA?
- What changes in the ATP when you move from retimed to LPO modules?

Key idea. IM/DD is intensity in, power out, with FEC and DSP closing the gap that PAM4's SNR penalty opens. Know OMA, ER, chirp and dispersion, pre-FEC BER, and the reference plane for every number. Then measure, bisect, and correct the failing block.

Chapter 4

Quantitative models: noise, RIN, and BER

The previous chapters argued mostly in architecture and measurement vocabulary. This one puts numbers behind the two questions every link engineer eventually asks: *what bit-error ratio (BER) will this receiver deliver?* and *what is the minimum signal it needs?* The answers follow a short chain of physics that has been stable for decades of IM/DD design: Gaussian statistics at the decision circuit, a handful of noise sources, and the way relative intensity noise (RIN) turns into an error floor. Understanding that chain is what lets you read a sensitivity number or a RIN floor without treating the datasheet as magic.

Everything here is backed by short, reproducible Python (in `sims/`), so the figures are computed curves rather than sketches. The models follow the treatment in Säckinger's *Analysis and Design of Transimpedance Amplifiers for Optical Receivers*,[1] and the sanity checks in the code reproduce that book's worked numbers.

[1] E. Säckinger, *Analysis and Design of Transimpedance Amplifiers for Optical Receivers*, Wiley, 2018, ch. 4.

4.1 The decision: from Q to BER

A binary receiver samples a noisy voltage and compares it to a threshold. If the noise is Gaussian and the threshold is optimally placed, the BER depends on a single quality factor Q, the separation between the one and zero levels measured in units of their combined noise:

$$Q = \frac{I_1 - I_0}{\sigma_1 + \sigma_0}, \qquad \mathrm{BER} = \tfrac{1}{2}\,\mathrm{erfc}\left(\frac{Q}{\sqrt{2}}\right).$$

This is the workhorse relation of link design. Its power is that it needs no assumption about pulse shape or spectrum, only that the sampled noise is Gaussian.[2] Two reference points anchor everything that follows: the classic uncoded target

[2] Q here is the electrical decision quality, not to be confused with a laser-cavity or resonator Q.

BER $= 10^{-12}$ needs Q $= 7.03$, while the *KP4* Reed–Solomon FEC (Chapter 3) corrects a pre-FEC BER $\approx 2.4 \times 10^{-4}$, needing only Q ≈ 3.5.

```
from scipy.special import erfc, erfcinv
import numpy as np

def q_to_ber(q):
    return 0.5 * erfc(q / np.sqrt(2))

def ber_to_q(ber):
    return np.sqrt(2) * erfcinv(2 * ber)
```

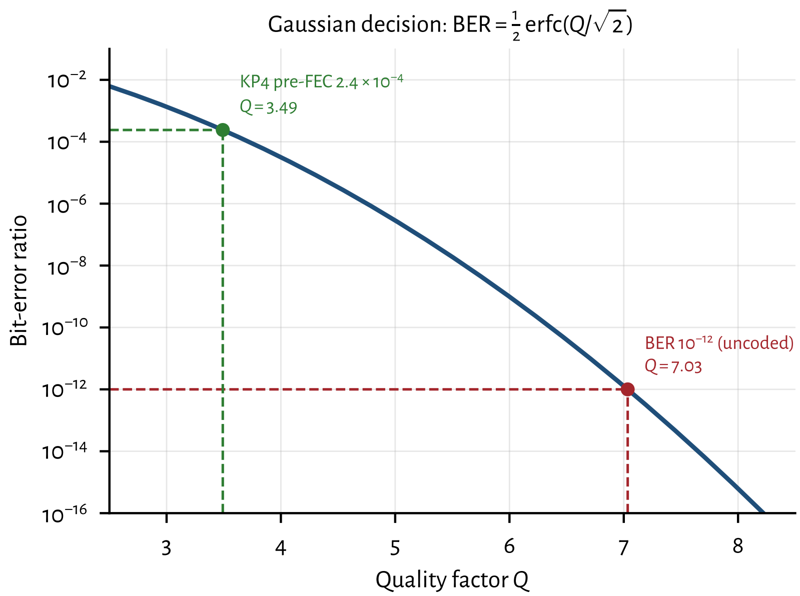

Figure 4.1: The Gaussian decision curve. Every dB of Q buys orders of magnitude of BER near the knee, which is why FEC (trading a modest Q for a huge BER improvement) is decisive.

Figure 4.1 shows why the curve is so steep near the operating point: a small change in Q (equivalently, in received power) moves the BER by orders of magnitude. This steepness is exactly what FEC exploits: nudging the required Q from 7.03 down to 3.5 relaxes the optical power budget by several dB.

4.2 The receiver noise budget

Q is only as good as our estimate of σ. Three noise sources dominate a short-reach IM/DD receiver, and they add in quadrature (they are independent):

Thermal / circuit noise the input-referred noise of the TIA and following stages, roughly white, so its variance scales with the noise bandwidth: $\sigma_{\text{th}}^2 = i_n'^2\,\text{BW}$, where i_n' is a noise-current density (A/$\sqrt{\text{Hz}}$). *Signal-independent.*

Shot noise from the discreteness of photocurrent, $\sigma^2_{\text{shot}} = 2q\,I\,\text{BW}$. *Grows with signal*, so it is larger on the ones than the zeros.

RIN (relative intensity noise) the laser's own intensity fluctuations, $\sigma^2_{\text{RIN}} = \text{RIN}_{\text{lin}}\,I^2\,\text{BW}$, with $\text{RIN}_{\text{lin}} = 10^{\text{RIN[dB/Hz]}/10}$. *Grows with the square of signal*, the key fact of the next section.

Because shot and RIN noise are signal dependent, we evaluate σ_1 and σ_0 separately and form Q with their sum. The core of the model is just this:

```
def nrz_q(p_avg_w, responsivity, i_thermal_rms, bw,
          er_db=np.inf, rin_db_hz=-np.inf):
    p1, p0 = er_levels(p_avg_w, er_db)      # one / zero
        ↪ optical powers
    i1, i0 = responsivity * p1, responsivity * p0
    var1 = i_thermal_rms**2 + 2*Q_E*i1*bw   # thermal +
        ↪ shot
    var0 = i_thermal_rms**2 + 2*Q_E*i0*bw
    if np.isfinite(rin_db_hz):
        rin = 10**(rin_db_hz / 10)
        var1 += rin * i1**2 * bw            # RIN grows
            ↪ as I^2
        var0 += rin * i0**2 * bw
    return (i1 - i0) / (np.sqrt(var1) + np.sqrt(var0))
```

4.3 RIN and the BER floor

Here is the consequence that makes RIN worth its own section. Thermal noise is fixed, so pouring on more optical power raises $I_1 - I_0$ while σ_{th} stays put, so Q improves without limit. But RIN noise scales *with the signal itself*: $\sigma_{\text{RIN}} \propto I$. Once RIN dominates, the signal and its noise grow together and Q stops improving. Taking the high-power, high-extinction limit (thermal and shot negligible, $I_0 \to 0$):

$$Q_{\text{max}} = \frac{I_1}{\sigma_{\text{RIN},1}} = \frac{1}{\sqrt{\text{RIN}_{\text{lin}}\,\text{BW}}}.$$

This is a hard ceiling: no amount of transmit power or receiver sensitivity can push Q past it, so there is a BER floor set entirely by the laser and the bandwidth. Equivalently, the power penalty to hold a target Q is

$$\text{PP} = \frac{1}{\sqrt{1 - Q^2\,\text{RIN}_{\text{lin}}\,\text{BW}}},$$

which diverges as $Q \to Q_{\text{max}}$.

Figure 4.2 shows the floor directly; Figure 4.3 plots the ceiling versus RIN for three lane rates. The bandwidth dependence matters: doubling the lane rate doubles the noise bandwidth and drops Q_{max} by $\sqrt{2}$ (≈ 1.5 dB of margin), so RIN that is harmless at 25G can bite at 200G. This is the quantitative reason the laser chapter (Chapter 5) lists RIN among the parameters that decide pass/fail.

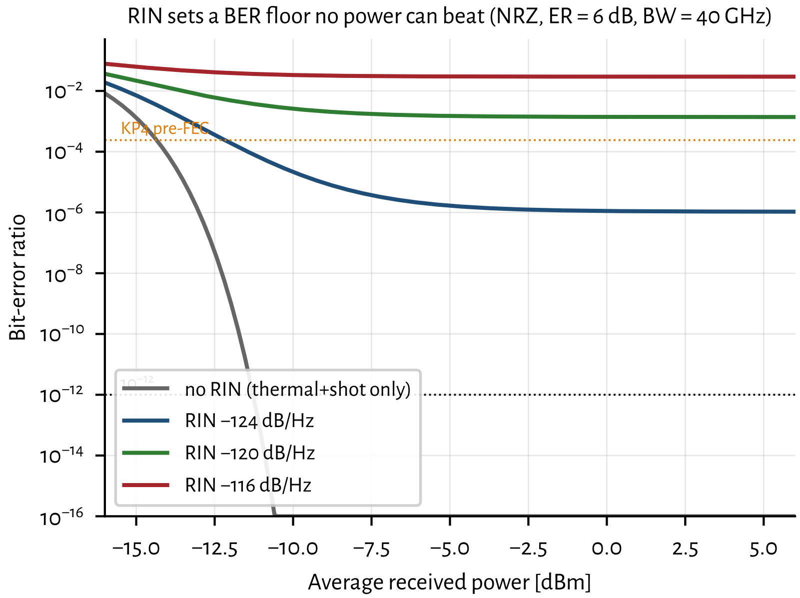

Figure 4.2: With RIN present, the BER stops falling no matter how much power is added. The thermal/shot-only curve dives; each RIN level flattens into a floor. (RIN values here are deliberately high to make the floor visible in-frame; good DFBs at < -150 dB/Hz have no floor at these rates; see text.)

4.3.1 Typical RIN values (2026)

How much RIN headroom do real sources have? Table 4.1 collects representative figures. Two cautions on reading them. First, standards quote RIN_xOMA, RIN referenced to the OMA and measured under a specified optical return loss (ORL) x, because back-reflections into the laser raise its noise, so a spec limit is a stressed, worst-case number, not the device's quiet intrinsic RIN.[3] Second, RIN degrades with optical feedback, so isolator-free and co-packaged designs care as much about *feedback tolerance* as about the isolated number, one reason quantum-dot lasers (near-zero linewidth-enhancement factor) are attractive for CPO [56].

A RIN number in dB/Hz is, by itself, incomplete: because RIN is *relative*, it only becomes an absolute noise current once the photocurrent $I = \mathcal{R}P$ is fixed. The intensity-noise current density is

$$i_{\text{RIN}} = \sqrt{\text{RIN}_{\text{lin}}}\, I \quad [\text{A}/\sqrt{\text{Hz}}], \qquad S_{\text{RIN}} = \text{RIN}_{\text{lin}}\, I^2 \quad [\text{A}^2/\text{Hz}],$$

so it scales linearly with received power. Table 4.1 therefore lists both the RIN and the current density it produces at a common reference operating point ($\mathcal{R} = 0.8$ A/W, $P_{\text{rx}} = 0$ dBm, i.e. $I = 0.8$ mA), the units a receiver designer actually compares against.

For scale, at that same 0.8 mA the *shot*-noise density is $\sqrt{2qI} = 16$ pA/$\sqrt{\text{Hz}}$ ($S = 2.6 \times 10^{-22}$ A^2/Hz) and a good high-speed TIA adds roughly 25 pA/$\sqrt{\text{Hz}}$ of thermal noise. So a VCSEL at -140 dB/Hz (80 pA/$\sqrt{\text{Hz}}$) already dominates both, while a heterogeneous source at -160 dB/Hz (8 pA/$\sqrt{\text{Hz}}$) is a minor term. The key

[3] For example IEEE 802.3 / 100G Lambda MSA short-reach PAM4 links cap $\text{RIN}_{17.1}\text{OMA}$ at -136 dB/Hz with 17.1 dB ORL .

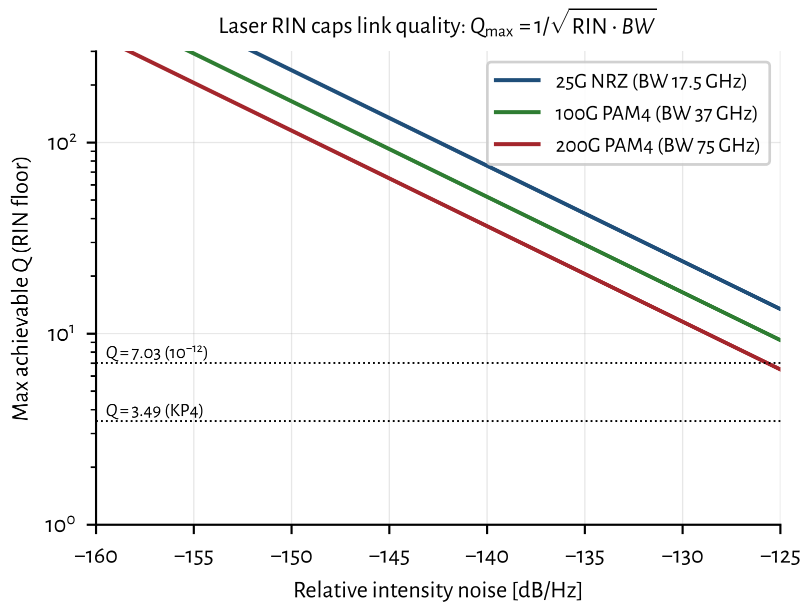

Figure 4.3: The RIN ceiling $Q_{max} = 1/\sqrt{RIN \cdot BW}$. Wider receiver bandwidth (higher lane rate) integrates more RIN, lowering the ceiling. Where a curve dips below the dotted anchors, that link can no longer reach the corresponding BER.

asymmetry: thermal noise is fixed and shot grows only as $\sqrt{I}$, but RIN grows as I, so at low received power thermal wins and RIN is irrelevant, and only above a break-in power (Figure 4.4) does RIN take over. That is why quoting a RIN figure without an operating power says little.

Put these against the ceiling. At 200G-PAM4 bandwidths (BW $\approx$ 75 GHz), even the worst spec-compliant number (–136 dB/Hz) gives $Q_{max} \approx 23$ (far above the $Q = 7$ needed for 10^{-12}), so for well-behaved sources RIN is *not* the limiter; thermal noise is. RIN becomes the story only when feedback, aging, or a marginal source pushes the effective figure toward –125 dB/Hz, where Q_{max} falls through the uncoded target. That is why the practical spec is written against a stressed ORL, and why feedback-tolerant sources matter for dense, isolator-free integration. A third path to excess intensity noise is electrical: laser bias-driver current noise converts to equivalent RIN (Section 5.8) and must be budgeted separately from intrinsic laser RIN.

Key idea. Thermal noise is beaten by power; RIN is not. Because $\sigma_{RIN} \propto I$, intensity noise imposes a floor $Q_{max} = 1/\sqrt{RIN \cdot BW}$ that no link budget can climb past, and it worsens as lane rates (and thus bandwidths) rise. Good 2026 sources (–145 to –165 dB/Hz) sit comfortably below that floor; feedback and aging are what erode the margin.

Source	RIN (dB/Hz)	i_{RIN} @ 0 dBm (pA/$\sqrt{\mathrm{Hz}}$)	Note
Standards spec limit (400G-FR4, 100GBASE-BR)	≤ -136	≥ 127	stressed, w/ ORL
Datacom VCSEL, 850 nm (MMF)	-135 to -145	45 to 142	quiet parts ≤ -145
Good datacom DFB / EML	-145 to -155	14 to 45	CPO ELS targets ≤ -145
Quantum-dot laser on Si	-140 to -150	25 to 80	temp-stable, isolator-free
Heterogeneous / self-injection-locked Si laser	-155 to -165	4.5 to 14	high-Q feedback
Lab record (QD, quiet pump + injection lock)	down to ~ -168	~ 3.2	research

Table 4.1. Representative RIN by source type (c. 2026) and the intensity-noise *current* it produces at a reference operating point: $\mathcal{R} = 0.8$ A/W, $P_{\mathrm{rx}} = 0$ dBm, so $I = 0.8$ mA. The density is $i_{\mathrm{RIN}} = \sqrt{\mathrm{RIN}_{\mathrm{lin}}}\, I$; the PSD in A^2/Hz is its square. Spec limits are stressed $\mathrm{RIN}_x\mathrm{OMA}$ values; the rest are typical intrinsic RIN.

4.4 Sensitivity and OMA

4.4.1 Common misconception: sensitivity does not guarantee a working link

Received power exceeding the datasheet sensitivity number does not guarantee low BER. Sensitivity is measured under specific, controlled assumptions: a clean transmitter, known extinction ratio, specified ORL, and a particular equalization state. In the field, BER depends on all of these simultaneously:

- received optical power;
- extinction ratio and OMA (not just average power);
- RIN under the actual ORL;
- jitter (random and deterministic);
- ISI from bandwidth limits and dispersion;
- receiver thermal noise and any TIA compression;
- equalization state and FEC margin.

Sensitivity is therefore a shorthand for receiver noise performance under reference conditions, not a complete description of whether the link closes. A link that "has enough power" can still fail if ER is poor, RIN is elevated, or the equalizer is saturated. Always close the full link budget (Section 7.7), not just the power line.

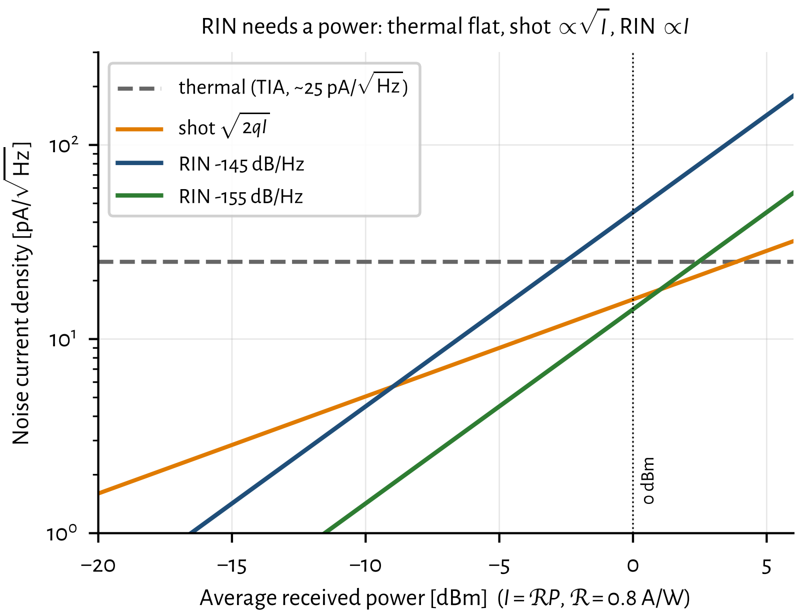

Figure 4.4: Noise current densities versus received power. Thermal is flat, shot $\propto \sqrt{I}$, RIN $\propto I$; the RIN curves cross the fixed thermal floor only above a break-in power, which is why RIN only "makes sense" once the optical power (hence $I = \mathcal{R}P$) is stated.

4.4.2 OMA versus average power

Two links may have identical average optical power but different BER. The reason is OMA and extinction ratio. At fixed average power P_{avg}, the OMA depends on ER through

$$\frac{\text{OMA}}{P_{\text{avg}}} = \frac{2(\text{ER} - 1)}{\text{ER} + 1},$$

where ER is the linear extinction ratio P_1/P_0. A link with 10 dB ER ($\text{ER}_{\text{lin}} = 10$) delivers $\text{OMA}/P_{\text{avg}} = 18/11 \approx 1.636$, while 6 dB ER ($\text{ER}_{\text{lin}} \approx 4$) delivers $6/5 = 1.200$. The difference is about 1.36 dB of OMA at the same average power. Whether that 1.36 dB matters depends on where the receiver sits on its BER waterfall: near the sensitivity cliff a 1 dB OMA change moves BER by orders of magnitude (Section 4.1 and figure 4.2), while well above sensitivity the same change is invisible. The impact also depends on RIN, bandwidth, equalization state, and TDECQ; ER alone does not set BER.

For PAM4, the relevant quantity is the outer OMA ($P_3 - P_0$), and level spacing (RLM) matters independently. A transmitter that launches adequate average power but has compressed inner eyes (poor RLM) will fail TDECQ even though a power meter reads in-spec. This is why TDECQ, OMA, and RLM are specified together, not average power alone (Section 7.4).

4.4.3 The sensitivity formula

Turning the question around (*what is the least power that meets a target BER?*) gives the sensitivity. Referring the receiver's input noise current i_n back to the optical input through the responsivity $\mathcal{R}$:

$$P_{\text{sens}} = \frac{Q\,i_n}{\mathcal{R}} \qquad \text{(average power)}, \qquad P_{\text{sens}}^{\text{OMA}} = \frac{2\,Q\,i_n}{\mathcal{R}}.$$

Modern short-reach standards specify the *OMA* (optical modulation amplitude, $P_1 - P_0$) rather than average power, because it decouples the sensitivity spec from the transmitter's extinction ratio. As a check, the textbook example ($i_n = 1\ \mu$A, $\mathcal{R} = 0.8$ A/W, BER $= 10^{-12}$) gives $P_{\text{sens}} = 7.03 \times 1\ \mu\text{A}/0.8 = 8.8\ \mu$W, or -20.6 dBm, which the code reproduces. A finite extinction ratio costs a further PP $= (\text{ER}+1)/(\text{ER}-1)$: 0.87 dB at 10 dB ER, 2.2 dB at 6 dB ER. These penalties feed directly into the link budgets of Chapter 3 and the transmitter and dispersion eye closure quaternary (TDECQ) discussion of Chapter 7.

Worked example: DR4-class budget check. Take a 200G/lane DR link with Ge-on-Si PIN ($\mathcal{R} = 0.9$ A/W), TIA $i_n = 13$ pA/$\sqrt{\text{Hz}}$, bandwidth ≈ 60 GHz, target pre-FEC BER 2.4×10^{-4} (Q ≈ 3.5, Sections 3.12 and 4.1).

Integrated noise: $i_n\sqrt{\text{BW}} \approx 13 \times 10^{-12} \times \sqrt{60 \times 10^9} \approx 3.2\,\mu$A rms. Required OMA:

$$P_{\text{OMA,sens}} = \frac{2Qi_n\sqrt{\text{BW}}}{\mathcal{R}} \approx \frac{2 \times 3.5 \times 3.2\,\mu\text{A}}{0.9} \approx 25\,\mu\text{W} \approx -16\,\text{dBm}.$$

Add $\sim$3 dB TDECQ penalty, $\sim$2 dB connector/fiber, $\sim$2 dB system margin: need Tx OMA $\approx -16 + 7 \approx -9$ dBm class at the receiver input, i.e. roughly -6 to -4 dBm launched depending on reach. If measured sensitivity is worse, bisect RIN, reflections, and ER (Section 7.7 and section 7.2.2).

4.5 Receiver technologies and their noise (2026)

The sensitivity formula $P_{\text{sens}} = Q\,i_n/\mathcal{R}$ has exactly two device inputs: the photodiode responsivity $\mathcal{R}$ and the amplifier's input-referred noise current i_n. So "receiver noise performance" is really a statement about the detector–TIA pair, and the winning short-reach recipe is the one the question anticipates: a *waveguide germanium-on-silicon PIN* feeding a *tightly integrated CMOS or SiGe-BiCMOS TIA*. The reasons are all in the two parameters above plus a parasitic (Section 4.5 details the TIA side).

Photodiodes and transimpedance amplifiers. The photodiode converts photons to photocurrent; the *TIA* (transimpedance amplifier) converts current to voltage with low input-referred noise. For PAM4 at 100–224G/lane:

TIA Transimpedance amplifier: converts PD current to voltage; input capacitance sets input-referred noise.

- **PIN (Ge-on-Si):** no internal gain; lowest excess noise; mainstream (Table 4.4). Capacitance at the TIA input dominates i_n.
- **APD**: internal multiplication gives 5–9 dB sensitivity gain at the cost of excess noise factor and bias voltage; Ge/Si APDs now reach >100 GHz class [61].
- **UTC/MUTC:** electron-only transport for >200 GHz BW and high saturation; used when linearity and speed beat raw sensitivity [65].

PIN PIN photodiode: p-intrinsic-n structure; no internal gain. Ge-on-Si waveguide PIN is the mainstream short-reach detector.

APD Avalanche photodiode: internal gain improves sensitivity; excess noise factor and bias complexity vs PIN.

UTC Uni-traveling-carrier photodiode: electron-only transport; >200 GHz BW and high saturation current. Linear/LPO niche.

The TIA often embeds *CTLE* for LPO (Section 3.6 and section 9.5.1): a fixed high-frequency boost before the host SerDes ADC. Co-packaging PD and TIA (sub-mm interconnect) is the noise win repeated throughout this book (Chapter 2).

Why Ge-on-Si wins the mainstream. Germanium grown on silicon absorbs the O- and C-bands, is fully CMOS-process-compatible, and is built *on the same PIC* as the modulators and couplers. Modern devices reach $\mathcal{R} \approx 0.8$–1.0 A/W with dark currents of single-digit to tens of nA and, in 2025 research, –3 dB bandwidths beyond 100 GHz (e.g. a recessed Ge/Si PIN at 106 GHz, 0.93 A/W, < 10 nA; SiN-coupled lateral Ge > 110 GHz at 1 mA) [58]. Crucially, monolithic or 3D/flip-chip co-integration keeps the PD-to-TIA interconnect sub-millimetre, so the input node capacitance stays tens of fF, and since a TIA's input-referred noise rises with that capacitance ($i_n \propto C f^{3/2}$ in the front-end limit), *short is quiet*. This is the same capacitance argument that drives co-packaging in Chapter 2 (Miller's receiver point), now cashed out as noise current.

What the TIA must deliver. The TIA is the receive twin of the modulator driver (Section 3.14.3). At 224 GBd PAM4 you need bandwidth $\gtrsim$50–70 GHz for a 112 GBd Nyquist-class front-end (often less than the Tx driver BW because the optical channel and reference receiver already band-limit; LPO pushes for flatter, more linear TIAs), input-referred noise in the low teens of pA/$\sqrt{\text{Hz}}$ once co-packaged with a low-C PD, linearity / overload so large OMA and reflections do not crush PAM4 levels (RLM) or trip AGC into a bad corner, and optional CTLE for LPO/LRO so the host SerDes sees a usable eye without module DSP (Section 3.6 and section 9.5.1). Noise scales with input capacitance: $i_n \propto C f^{3/2}$ in the front-end limit. That is why co-packaging (or monolithic PD+TIA) is not optional at 200G+: every millimetre of bondwire is noise and BW you cannot recover with FEC.

Noise levels you actually budget. Table 4.2 puts the model numbers next to published front-ends. Shot noise at 0 dBm into $\mathcal{R} = 0.8$ A/W is $\sqrt{2qI} \approx 16$ pA/$\sqrt{\text{Hz}}$; a good TIA sits near that floor. Worse i_n or higher C burns sensitivity linearly via $P_{\text{sens}} = Q\, i_n/\mathcal{R}$ (Section 4.4).

2026 linear PAM4 front-ends land around 10–17 pA/$\sqrt{\text{Hz}}$: e.g. 16.9 pA/$\sqrt{\text{Hz}}$ at 112G in 16-nm CMOS (–8.2 dBm sensitivity) [4] [59] and 13.2 pA/$\sqrt{\text{Hz}}$ at 224G in

[4] **??**112G PAM4 linear TIA in 16-nm CMOS: 16.9 pA/$\sqrt{\text{Hz}}$; –8.2 dBm.

Front-end	i_n (pA/$\sqrt{\text{Hz}}$)	BW / rate	Sensitivity note
Typical "good" short-reach TIA (book model)	$\sim$25	—	older rule-of-thumb floor
16-nm CMOS + co-pkg PD	16.9	32 GHz / 112G PAM4	-8.2 dBm class
55-nm SiGe $4\times$112 GBd linear TIA	13.2	65 GHz / 224G	$\sim$1.2 pJ/bit
Shot noise @ 0 dBm, $\mathcal{R} = 0.8$ A/W	$\approx$16	—	physics floor at that power

Table 4.2. Input-referred TIA noise densities used in link budgets (c. 2023–26 published front-ends). Integrate $i_n\sqrt{\text{BW}}$ for rms noise before applying Q (Section 4.4). Sources in text.

55-nm SiGe BiCMOS (65 GHz BW, 1.2 pJ/bit) [5][60]. Put these in the model: $i_n \approx 13$ pA/$\sqrt{\text{Hz}}$ integrated over ~ 60 GHz is ≈ 3.2 μA rms, giving an NRZ average sensitivity near -15 dBm; the PAM4 level penalty lands OMA sensitivities in the -8 to -10 dBm range these front-ends report.

[5] **??**224G SiGe TIA: 65 GHz BW, 13.2 pA/$\sqrt{\text{Hz}}$, $\sim$1.2 pJ/bit.

Record and commercial snapshot (2025–26). Table 4.3 pairs detectors and TIAs. Commercial linear-optics TIAs (Semtech GN1834L/DL, GN1838DL) target LPO/LRO/CPO at 224G/lane with on-chip EQ; Semtech has also shown 448G-class PMD ICs (TN14740 TIA) at OFC 2026 demos (vendor demonstration; not a volume datasheet claim) [37][6][64]. On the detector side, recessed Ge/Si PINs at 106 GHz / 0.93 A/W [58], Ge/Si UMC-APDs at 105 GHz with $\sim$9 dB sensitivity gain over PIN at 224/260G PAM4 [7][61], waveguide Ge/Si APDs toward 100 GHz at 2 A/W [8][62], and OFC 2026 Ge-on-Si APDs at 180 GBd PAM4 [9][63] mark the research edge. UTC/MUTC PDs remain the high-saturation / >200 GHz niche [65].

[6] Semtech OFC 2026: 448G/lane TN14740 TIA + TN622 driver demos (provisional).

[7] Ge/Si UMC-APD: 105 GHz BW; 224/260G PAM4 at -10.9/-10.1 dBm.

[8] JLT 2026: waveguide Ge/Si APD >100 GHz; 2 A/W at 70 GHz.

[9] OFC 2026 Th3F.2: Ge-on-Si APD to 180 GBd PAM4 (O/C-band).

Part / paper	Type	BW / i_n or $\mathcal{R}$	Rate / sens.	Note
Semtech GN1834L/DL / GN1838DL	TIA	224G-class; linear+EQ	224 Gb/s/lane	Commercial LPO/CPO family
Semtech TN14740 (demo)	TIA	448G-class	448G/lane demo	OFC 2026 booth; provisional
SiGe $4\times$112 GBd TIA	TIA	65 GHz; 13.2 pA/$\sqrt{\text{Hz}}$	224G PAM4	Research / product-class paper
16-nm CMOS + co-pkg PD	TIA+PD	16.9 pA/$\sqrt{\text{Hz}}$	112G; -8.2 dBm	Co-packaged win
Recessed Ge/Si PIN	PD	106 GHz; 0.93 A/W	200 GBd-class	<10 nA dark
Ge/Si UMC-APD	APD	105 GHz @ $M \approx 7$	224/260G; -10.9/-10.1 dBm	$\sim$9 dB over PIN
Ge/Si WG APD (300 mm)	APD	>100 GHz; 2 A/W @ 70 GHz	400G/lane target	7 V bias class
OFC 2026 Ge-on-Si APD	APD	70–100 GHz; 1.5–2 A/W	180 GBd PAM4	O- and C-band
UTC / MUTC-PD	PD	>110–200 GHz	high I_{sat}	Linear / LPO niche

Table 4.3. Receiver snapshot (c. 2025–26). Commercial TIA rows are vendor announcements; APD/PIN rows mix production-intent SiPh with research demos. Sensitivities are as published (FEC threshold varies).

Reasonable alternatives to Ge PIN + quiet TIA. Table 4.4 lays the detector menu out. III-V InGaAs PINs (flip-chipped) trade monolithic integration for higher power handling and remain common in discrete modules. Avalanche photodiodes add internal gain for ~ 5–9 dB of sensitivity (attractive for power-starved or high-split links) at the cost of excess noise, bias complexity, and, historically, bandwidth;

that bandwidth excuse is fading fast above 100 GHz [61–63]. Uni-traveling-carrier (UTC/MUTC) PDs use electron-only transport for very high saturation current, linearity, and bandwidth (> 200 GHz) but modest responsivity, a fit for linear/LPO and > 200 GBd analog optics more than for raw sensitivity [65]. SOA-preamplified receivers bolt optical gain ahead of the PD for large effective responsivity and reach, but pay in ASE noise figure, power, and complexity.

SOA Semiconductor optical amplifier: inline optical gain (III-V); adds ASE noise. Used as preamplifier for sensitivity.

Detector	$\mathcal{R}$ (A/W)	–3 dB BW	Integration	Where it fits
Ge-on-Si waveguide PIN	0.8–1.0	60–> 100 GHz	monolithic on SiPh	mainstream short-reach / CPO
III-V InGaAs PIN (flip-chip)	0.6–0.9	60–> 100 GHz	hybrid / flip-chip	discrete modules, high power
APD (Ge/Si, InP, UMC)	effective ↑ (gain)	up to ~ 100 GHz+	hybrid / emerging	power-starved links; $+5$–9 dB sens.
UTC / MUTC-PD	0.1–0.8	> 110–200 GHz	III-V	linear/LPO, > 200 GBd, high saturation
SOA-preamplified PD	effective $\gg 1$	~ 50 GHz	III-V PIC	tight power budgets; adds ASE noise

Table 4.4. Short-reach receiver detector options, c. 2026. Ranges span production to recent research; APD/UTC/SOA figures are effective (with gain) or device-record.

Outlook. Volume short-reach receive stays on PIN + SiGe/CMOS TIA: noise in the low teens of pA/$\sqrt{\text{Hz}}$ with > 100 GHz Ge PINs is enough for DR/FR PAM4 when co-packaged, so the fight is capacitance and yield, not a new detector physics. Linear optics raises the bar: LPO/LRO need high linearity, on-chip EQ, and multi-lane density (Semtech's 224G family is the public commercial marker; 448G TIA demos are still provisional, Section 3.14.3). APDs are back in the 200G conversation when several dB of sensitivity gain changes a power-limited plant; UTC/MUTC matter when the impairment is saturation or > 200 GBd analog fidelity rather than photons per bit. Bondwire and FAU still set C and BW, which is why CPO/NPO win on noise for the same reason they win on energy (Chapter 2).

Key idea. Receiver performance is $\mathcal{R}$ and i_n, with i_n set by PD+TIA capacitance. Budget 10–17 pA/$\sqrt{\text{Hz}}$ co-packaged TIAs and > 100 GHz Ge PIN/APD detectors; use APD gain or UTC saturation when the link demands it. LPO makes TIA linearity as important as raw noise.

4.6 NRZ versus PAM4, at equal bit rate

The 224G-per-lane roadmap (Chapter 3) rides on PAM4, so it is worth seeing the trade quantitatively. PAM4 sends two bits per symbol using four levels, so at a fixed bit rate its symbol rate (and thus noise bandwidth) is halved, collecting less noise. But its three eyes each span only a third of the OMA, costing roughly $20 \log_{10} 3 \approx 9.5$ dB of vertical separation. Figure 4.5 pits the two at a common 100 Gb/s: PAM4's narrower bandwidth partly offsets its level penalty, but it still needs several dB more received power for the same BER, repaid by halving the electrical bandwidth the SerDes and optics must support. That balance is exactly why 100G/lane went NRZ and 200G/lane went PAM4.

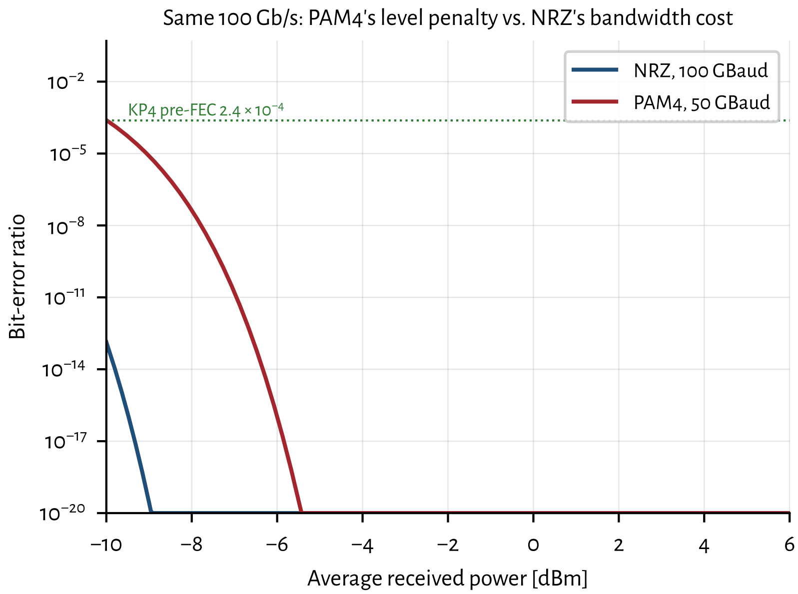

Figure 4.5: NRZ (100 GBaud) versus PAM4 (50 GBaud) at the same 100 Gb/s. PAM4 pays a level penalty but relaxes bandwidth; the crossover with the KP4 pre-FEC threshold sets the required operating power.

4.7 Engineering lens

4.7.1 How it works

Receiver performance reduces to one number, the quality factor Q: signal separation divided by combined noise. Thermal noise yields to more power, but RIN does not, which is why a BER floor exists. The rest of this chapter computes, measures, and defends Q.

4.7.2 How it is measured

A BER model earns trust only when every term has a bench measurement. Use a BERT and calibrated optical attenuator for BER versus received OMA. Measure receiver input-referred noise with the photodiode dark and illuminated. Use a DCA for OMA, ER, and eye quality, and a photodiode plus electrical spectrum analyzer for relative intensity noise (RIN). Repeat power sweeps at temperature and per lane. Acceptance is a curve, not one point: the required waterfall must cross the program's pre-FEC threshold with margin and without a floor (Sections 4.1 to 4.4).

4.7.3 How it fails

Thermal noise dominates at low optical power, shot noise grows with photocurrent, and RIN grows with signal power. Reflections, crosstalk, jitter, and compression break a simple Gaussian fit. The signatures differ: a horizontal BER floor points to signal-proportional noise or bursts; a uniform horizontal shift points to lost power or added receiver noise; one lane moving alone points to its optical or electrical path.

Failure mode: BER floor

Symptoms. BER improves with received power, then stops improving.
Likely causes. RIN, reflection-driven multipath interference, laser-bias noise, crosstalk, or periodic jitter.
Measurements. BER versus OMA, RIN spectrum, ORL sweep, FEC error distribution, and a quiet laser-bias source.
Mitigations. Remove the correlated noise source, improve ORL, isolate supplies, or reject the laser if its intrinsic RIN misses the ATP.

4.7.4 How it is debugged

When BER moves from 10^{-12} class to 10^{-6} class, save the failing condition and work in this order: received OMA, transmitter OMA and ER, receiver noise, RIN and ORL, electrical jitter, and lane crosstalk. Sweep power before changing settings. If the curve recovers with power, quantify the sensitivity shift. If it floors, split intrinsic laser noise from board noise and reflection feedback. If only temperature moves it, repeat the same terms hot and cold instead of assigning the change to "thermal margin."

Debug story

Observed. One lane developed a BER floor after the product board replaced the bench supply.
Investigation. The optical power sweep flattened. Intrinsic RIN on the quiet source passed, but discrete tones appeared with the product rails active.
Finding. The laser was not the noisy element.
Root cause. A switching regulator coupled into the laser bias path.
Resolution. The bias filter and return path were changed, and a powered-board RIN check was added to validation.

4.8 The debugging fork: did received power change?

When BER degrades, ask one question first:

Did received optical power change? If yes, the failure is in the power path:

- laser degradation (threshold rise, slope loss, COD);
- connector contamination or damage;
- coupling loss (fiber attach, FAU drift);
- fiber break or bend loss;
- MUX/de-MUX imbalance or grid misalignment.

Or did the required power increase? If received power is stable but BER worsened, the failure is in signal quality:

- jitter (random or deterministic);
- RIN (intrinsic, feedback-driven, or bias-driver);
- ISI (bandwidth degradation, equalizer saturation);
- extinction ratio degradation (bias drift, EAM aging);
- receiver sensitivity degradation (TIA noise, overload);
- timing recovery issues (CDR, clock path).

This fork often narrows an investigation in minutes. Power-path failures show up on a meter; signal-quality failures require a DCA, BERT, or spectrum analyzer. Apply it before opening the package, changing settings, or blaming a supplier (Section 7.12).

4.9 Interview and design review questions

Concept.

- Why does RIN impose a BER floor that no amount of optical power can overcome?
- What is the physical difference between thermal noise, shot noise, and RIN?
- Why is OMA a better receiver metric than average power?

Design.

- Which measured curve supports each noise term in the model?
- Over what power range does the Gaussian assumption fit the data?
- What evidence separates thermal noise, shot noise, RIN, jitter, and crosstalk?
- Which corner would create a floor that a room-temperature sensitivity point misses?

Debug.

- Optical power is unchanged but BER worsened from 10^{-12} to 10^{-6}. Where do you start? (Section 4.8)
- The BER curve flattens at high power. What mechanisms produce a floor?
- A bench supply is replaced with the product board and RIN rises. What changed?
- What result would disprove the current root-cause hypothesis?

Manufacturing and operations.

- How do you screen a laser whose quiet RIN passes but stressed RIN fails?
- What ATP row catches a bias-driver noise contribution that only appears with host rails active?
- At what received power does RIN begin to dominate over thermal noise for this receiver?

Key idea. Four relations carry most of short-reach link design: the Gaussian BER(Q), the quadrature noise budget (thermal + shot + RIN), the RIN floor $Q_{\mathrm{max}} = 1/\sqrt{\mathrm{RIN} \cdot \mathrm{BW}}$, and the sensitivity $P_{\mathrm{sens}} = Q\, i_n/\mathcal{R}$. Close each relation with a measured curve, then use the curve shape to debug the link.

Chapter 5

Choosing light sources and modulation

Do not choose a laser by comparing data sheets in isolation. Start with reach, lane rate, fiber plant, power, cost, lifetime, manufacturing volume, and service policy. Those requirements select an architecture. The architecture then limits the useful source, modulation, detector, packaging, and validation choices.

5.1 System requirements and architecture paths

Freeze the system problem before asking a supplier for samples:

Reach and fiber decide whether multimode loss and modal bandwidth are acceptable or whether the link needs single-mode fiber.

Lane rate and bandwidth decide whether direct modulation closes the eye or whether the link needs an EML, MZM, or ring.

Power and cooling include laser wall-plug power, modulator loss, driver power, TEC power, and control overhead.

Cost and volume include fiber plant, assembly alignment, burn-in, test time, yield, and field replacement, not only die price.

Reliability and service decide whether the source can stay inside the package or must be a replaceable ELSFP.

The choices are coupled. A cost-driven, short multimode link points toward an 850 nm VCSEL, multimode fiber, silicon photodiodes, and direct modulation. A longer single-mode path points toward a 1310 nm DFB or EML and germanium or

III-V detection. Dense WDM and co-packaged optics often move the CW source away from the modulator, which adds wavelength control, fiber attach, and service requirements. Section 5.6 and table 5.3 turns these paths into supplier specifications.

5.2 Light-source and modulation choices

The short-reach market uses a small set of source families. Each fits a different constraint set:

DFB (distributed feedback) a grating along the active region gives single-mode output; the workhorse continuous-wave (CW) or directly modulated source for CWDM and LAN-WDM (Section 5.4).

CW Continuous wave: unmodulated laser output. External modulators (MZM, ring, EAM) encode data onto a CW source.

DBR Distributed Bragg reflector: laser grating outside the gain region; enables tunable single-mode output.

DBR (distributed Bragg reflector) the grating sits outside the gain region. Choose it when tunability or separate control of gain and wavelength is worth added control and qualification work.

External-cavity laser a gain element and an external wavelength-selective cavity provide narrow linewidth and tunability. Choose it when spectral purity or lock range matters more than package size, cost, and control-loop simplicity. Most short-reach IM/DD links do not need it [RIO PLANEX]; the cited product is vendor orientation, not a datacenter source recommendation.

DML Directly modulated laser: modulate bias current; high chirp, low cost.

DML (directly modulated laser) modulate the bias current directly: cheap and low-power, but chirp-limited over dispersive fiber (Section 5.3).

EML Electro-absorption modulated laser: DFB plus EAM on one InP chip; dominant 100–200G/lane pluggable Tx.

EML (externally modulated laser) *EML*: a DFB integrated with an *EAM*. Low chirp and high bandwidth make it the dominant 100–200G/lane transmitter for single-mode links at DR (500 m) and shorter (Section 5.4 and section 3.14.3).

CW laser + TFLN MZM an external CW source feeds a thin-film lithium niobate Mach–Zehnder modulator on a separate chip. Very low chirp and $\gtrsim$100 GHz EO bandwidth make this the leading path to 400G/lane pluggables and high-baud FR links; see Section 3.14.3 and table 3.12.

CW laser + Si MZM an external CW source feeds a silicon Mach–Zehnder modulator on the same PIC (Section 3.14.3). Low chirp, flat passband, and CMOS fab integration make this the default for 100–200G/lane DR/FR SiPh modules; 400G/lane demos appeared in 2026.

CW laser + Si ring same laser architecture, but a microring or microdisk modulator on the PIC (Section 3.14.3). Smaller footprint and strong WDM/CPO fit; wavelength lock and thermal crosstalk dominate validation (Chapter 6).

CW-WDM / multi-wavelength sources high-power, multi-wavelength CW lasers (per the CW-WDM MSA) that feed comb-like WDM architectures (Sections 5.16 and 6.6).

VCSEL 850–940 nm multimode sources for short-reach links over multimode fiber; cheap but reach-limited and less relevant at 200G/lane.

VCSEL Vertical-cavity surface-emitting laser: 850 nm class, MMF SR links.

External laser source (ELS/ELSFP) a pluggable laser module supplying CW light to a co-packaged switch, so a failed laser is field-replaceable (Section 5.14).

Attribute	VCSEL direct	DFB direct	DFB + EAM	CW DFB/DBR + MZM	CW DFB/DBR + ring	External cavity + modulator
Wavelength / fiber	850–940 nm / MMF	1310 nm / SMF	1310 nm / SMF	1310 or 1550 nm / SMF	WDM grid / SMF	Tunable grid / SMF
Modulation fit	Direct only	Direct	Integrated EML	Si or TFLN MZM	Resonant Si ring	External MZM or ring
Bandwidth / reach	Short, modal limit	Chirp-limited	High BW, low chirp	High BW, broad passband	High BW, lock-limited	High BW, architecture-specific
RIN / linewidth	RIN and modal noise	RIN and chirp	RIN plus EAM bias	RIN; linewidth usually secondary	RIN plus spectral alignment	Low linewidth; verify feedback response
Power / efficiency	Low Tx complexity	Low Tx complexity	Driver and EAM loss	Laser, driver, and MZM loss	Laser, heater, and lock power	Laser plus control overhead
Reliability	Junction and temperature wear	Facet and active-region wear	Laser plus EAM aging	Source, attach, and bias drift	Source, heater, and lock faults	Cavity, package, and lock faults
Manufacturing	Array-friendly, MMF plant	Simple Tx, SMF attach	Mature integrated Tx	Multi-die attach and RF match	Dense PIC, tight thermal control	Tight optical assembly and control
Validation burden	Modal, temperature, aging	Chirp, LIV, RIN	LIV, RIN, EAM sweep, TDECQ	Source plus bias and RF path	Source plus resonance and crosstalk	Spectrum, lock, feedback, environment

Table 5.1. Decision matrix for common source and modulation paths. Entries show the dominant engineering concern, not fixed rankings. Program limits come from Table 5.4.

5.3 Directly modulated lasers and VCSELs

Before EMLs and silicon photonics took over single-mode datacenter ports, most volume optics were either a cheap *DML* on single-mode fiber or a *VCSEL* array into multimode fiber. Both still matter at the low-cost, short-reach edge of the market, and both show why chirp, modal bandwidth, and temperature push AI fabrics toward externally modulated single-mode sources.

A DML modulates laser bias current directly. The transmitter is simple and efficient, but the same carrier dynamics that make modulation easy also produce chirp: intensity changes drag the optical frequency along (Section 3.11). Over multimode or very short single-mode runs that is often acceptable. Over dispersive single-mode fiber at tens of GBd, the chirp turns into inter-symbol interference and closes the eye. Validation therefore focuses on extinction ratio, pattern-dependent chirp, and RIN, not just average power.

VCSELs took a different path. They emit from a vertical cavity at 850–940 nm straight into multimode fiber, so parallel arrays are easy to assemble and cheap to ship. That combination made VCSEL SR optics the default for early 40G/100G Ethernet inside the rack (100G-SR4 and its cousins): short ribbons of MMF, high lane count, low dollars per gigabit. The same physics that made them attractive also capped their future. Multimode fiber has modal bandwidth and modal noise limits; VCSEL bandwidth and reliability both degrade with temperature; and as lane rates climb toward 100 G and 200 G, those limits arrive sooner. The industry response has been incremental (better OM4/OM5 fiber, tighter specs, sometimes PAM4 on MMF) rather than a clean leap to 400G/lane SMF DR. In practice, MMF reach and modal dispersion keep VCSEL links in the SR box (Section 3.13), while hyperscale AI fabrics standardize on single-mode DR/FR and CPO.

Neither family is the path to 400G/lane SMF DR. EMLs and external modulators (Section 5.4, section 3.14.3, and table 3.12) own that space. Pattern-aware chirp linearization can stretch a DML a little farther, but it does not change the physics at FR distances: if you need low chirp and high EO bandwidth at fleet scale, you leave direct modulation behind.

5.4 DFB and EML: the workhorse transmitters

Once single-mode DR/FR became the hyperscale default, most short-reach ports started with an InP laser chip. Two configurations still dominate production: the CW or directly modulated DFB, and the EML that adds an electro-absorption modulator on the same die.

InP Indium phosphide: III-V semiconductor for lasers, EAMs, SOAs, and high-speed photodiodes in the 1310/1550 nm bands.

DFB. A distributed-feedback laser has a grating along the active region that selects one longitudinal mode. Spec-sheet metrics that matter in bring-up are threshold current, slope efficiency, SMSR (typically many tens of dB on a clean part), RIN, and wavelength vs. temperature/current. Used as a CW source for SiPh or TFLN modulators, or as a DML when chirp is acceptable (Section 5.3). Uncooled datacom DFBs ride case temperature with a known $d\lambda/dT$; cooled parts add a TEC and lock to a grid.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

EML. An electro-absorption modulated laser integrates a DFB with an *EAM* on one chip (Section 3.14.3). Reverse bias on the EAM sets absorption and extinction; chirp stays far below a DML. That combination, not marketing, is why EMLs became the volume answer for 100G/lane and then 200G/lane DR/FR pluggables: one chip, low chirp, mature supply chain. Validation adds EAM bias sweeps, aging of the absorption curve, and driver-match checks on top of the DFB LIV/SMSR/RIN suite (Sections 5.7 and 5.13).

When to pick which. Through 200G/lane DR, EML usually wins on cost and integration. A CW DFB (or ELSFP/CW-WDM bank) plus Si MZM, ring, or TFLN wins when the modulator must sit on silicon or needs $\gtrsim$100 GHz EO bandwidth (Table 3.12 and section 3.14.3). At CPO scale the laser often leaves the optical engine entirely so it can be replaced without pulling the ASIC package (Section 5.14). Looking forward, 400G/lane pluggables are pushing harder toward external CW plus TFLN or high-BW silicon modulators, while EMLs remain the workhorse of the installed 100–200G base.

Source	Typical use	Top risks
DML	short reach, cost-driven	chirp/dispersion, extinction ratio
EML	$\leq$DR, 100–200G/lane	EAM bias/aging, thermal
CW + TFLN MZM	400G/lane FR/DR, NPO	MZM bias drift, fiber attach, driver match
CW + Si MZM	DR/FR SiPh, 100–400G/lane	driver match, bias drift, fiber coupling
CW + Si ring	CPO, WDM transceivers	wavelength lock, thermal crosstalk, coupling
VCSEL	SR over MMF	modal noise, reach, temperature
ELS / ELSFP	co-packaged optics	connectorization, fleet serviceability

Table 5.2. When each source is used, and its top validation risks.

5.5 Choosing the modulation path

The source decision and modulation decision must close together. Direct modulation minimizes parts and power but carries laser chirp into the link. An EML adds an EAM on the laser die and is a mature low-chirp path for 100–200G/lane. A silicon MZM uses more area and drive but gives a broad optical passband. A ring is compact and fits dense WDM, but adds resonance control and thermal-crosstalk tests. TFLN offers high bandwidth and low chirp with a separate material platform and assembly flow.

Table 5.1 compares the system consequences. The device operation, bandwidth, insertion loss, and driver interfaces live in Section 3.14.3 and table 3.12. Keep that physics in one place. Here the decision is whether the link can carry the added power, control, assembly, and validation burden.

5.6 Laser requirements: from roadmap to specs

Laser requirements only work when they are numbers a supplier can fail and a link budget can close. Start from the interconnect roadmap choice, then fill a short requirements slice; the ATP in Section 8.10 is how that slice is enforced on every lot.

Roadmap forks that set the laser. Each architecture decision forces a different requirements set (Table 5.3):

Roadmap choice	Laser implication	Specs you must freeze early
Pluggable EML vs CW+Si/TFLN	Integrated EAM vs external CW + modulator	EAM bias/aging and TDECQ vs CW power class, RIN, and modulator V_π match
On-package laser vs ELSFP/CW-WDM	Field replace vs FIT inside the package	Connector/ORL/mate cycles and hot-swap CMIS vs COD/aging inside ASIC thermal
Isolator vs isolator-free (CPO)	Feedback tolerance vs quiet RIN only	Stressed RIN_xOMA at stated ORL; monitor PD / lock policy
Single-λ vs CW-WDM / comb	One line vs N lines into rings/filters	Per-line power flatness, SMSR, grid, crosstalk (Section 5.16)
Retimed vs LPO	Module DSP hides Tx vs host sees raw eye	Laser+modulator TDECQ/RLM floor vs host COM budget (Sections 3.14.3 and 9.5.2)
Derate policy	Operating I, T, power below abs-max	Bias window, thermal class, FIT/E_a assumptions (Section 5.13)

Table 5.3. Architecture forks and the laser specs each one forces. Freeze these before DVT samples are built (Section 8.10).

One-page requirements slice. Table 5.4 is the PRD-sized list. Fill every row with a number (or an explicit "N/A for this architecture") before you negotiate ATP limits. Do not leave RIN without an ORL, or power without a case-temperature class.

Parameter	How to set the number	Measure / ATP	Reject if	Derate / ops note
Launch power / class	Link budget + connector loss + aging margin (Section 7.7)	Power meter; ELSFP class	Below min at rated T	Cap max power for COD
Wavelength / grid	PMD or ring FSR plan; $d\lambda/dT$ headroom (Chapter 6)	OSA / wavemeter	Off-grid at case T	TEC setpoints
SMSR floor	Datasheet + modal-noise budget	OSA	Below floor at T	Watch aging
RIN (quiet + stressed)	BER floor vs BW (Section 4.3); ORL from plant	PD+ESA; stated ORL	Above limit at ORL	Bias-driver noise budget (Section 5.8)
Bias window	LIV kink-free range at max case T	LIV	Kink in window	Run below abs-max I
EAM / MZM (if any)	ER, RLM, TDECQ at baud (Section 7.4)	DCA + bias sweep	TDECQ/RLM fail	Bias aging policy
ORL / isolator	Architecture: isolator-free needs tighter RIN	ORL meter; mate cycles	ORL out of range	Cleaning / ELS mate life
CMIS monitors	What fleet triage will read (Section 7.12)	CMIS dump	Missing alarms / bad state machine	Enable sequence (Section 7.9)
FIT / life	Fleet failures/day target (Section 5.13)	GR-468 + E_a	Screen escape	Burn-in depth; ELS replace

Table 5.4. Laser requirements one-pager. Every cell needs a program number; this table is the structure, not the limits.

How to fill numbers (method, not invention). Work backward from the link, not forward from a marketing slide. The four steps below turn an architecture choice into ATP limits:

1. Close the optical ledger at target pre-FEC BER (Sections 3.12 and 7.7). That sets minimum launch OMA/power and maximum allowed penalties (transmitter and dispersion eye closure quaternary, TDECQ; ORL/RIN).
2. From receiver BW and the RIN ceiling $Q_{\text{max}} = 1/\sqrt{\text{RIN} \cdot \text{BW}}$ (Section 4.3), set a stressed RIN limit with margin under the plant ORL you will actually see (not only a quiet bench).
3. From case-temperature and derating policy, set the LIV bias window and thermal class so the laser never sits on a kink or at abs-max in the fleet (Sections 5.13

and 7.9).

4. From service model, choose ELSFP mate-cycle / hot-swap requirements or accept on-package FIT and write COD/aging screens accordingly (Section 5.14).

Hand the filled slice to the supplier with the ATP checklist (Table 8.3). If a roadmap slide cannot point to a row in Table 5.4, the requirement is not real yet.

Key idea. Laser leadership is a requirements sheet: architecture forks force specific specs (power, grid, RIN@ORL, SMSR, bias window, CMIS, FIT). Fill Table 5.4 from the link budget and fleet model, then enforce it with the ATP (Section 8.10).

5.7 LIV, SMSR, and RIN: the measurement playbook

These three measurements decide whether a laser chip or module is usable. The instruments are standard; the skill is knowing which failure each one catches.

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

LIV (light–current–voltage). The LIV curve plots optical power and forward voltage versus bias current. Read off threshold I_{th}, slope efficiency (mW/mA above threshold), kink-free operating range, and thermal rollover at high current or high case temperature. Figure 5.1 is a labeled schematic (not measured data).

High-temp LIV failures look like: I_{th} rise, slope collapse, early rollover, or a kink that moves into the bias window. Those map to aging, TEC saturation, or package thermal resistance (Section 5.13).

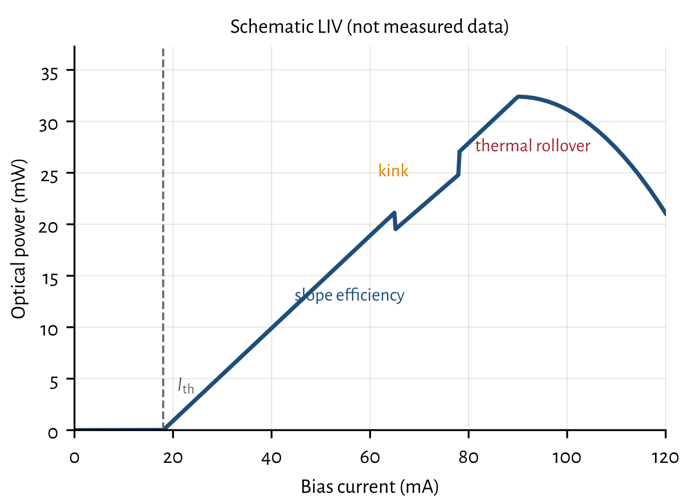

Figure 5.1: Schematic LIV curve with threshold, slope, kink, and thermal rollover labeled. Idealized for teaching; use measured LIV for pass/fail.

SMSR Side-mode suppression ratio: power difference (dB) between the lasing mode and the strongest side mode (OSA).

SMSR (side-mode suppression ratio). SMSR is the power difference (dB) between the lasing mode and the strongest side mode on an optical spectrum analyzer (OSA). Datacom single-mode parts require high SMSR so side modes do not steal power or seed modal noise. Spec-sheet floors are part-specific; treat the datasheet or ATP limit as authoritative. SMSR collapse under temperature or aging is a reject: the laser is leaving single-mode operation.

RIN (relative intensity noise). Measure RIN with a calibrated photodetector and RF spectrum analyzer (or a dedicated RIN analyzer), under a controlled optical return loss. Distinguish *intrinsic* RIN (quiet bench, high ORL) from stressed RIN_xOMA used in Ethernet/MSA specs. IEEE 802.3 / 100G Lambda class links cap $\mathrm{RIN}_{17.1}$OMA at -136 dB/Hz with 17.1 dB ORL[1] [100G Lambda MSA / IEEE 802.3]. Quiet datacom DFB/EML parts typically sit well below that when feedback is controlled; CPO ELS designs care as much about feedback tolerance as about the quiet number (Section 4.3.1 and section 4.3).

[1] IEEE 802.3 / 100G Lambda MSA: $\mathrm{RIN}_{17.1}$OMA ≤ -136 dB/Hz.

Parameter	Instrument	Pass/fail intent	Failure signature
LIV	SMU + power meter / integrating sphere	I_{th}, slope, kink-free bias window	high-temp rollover; kink in bias range
SMSR	OSA	single-mode purity vs. datasheet/ATP	side modes rise with T or age
RIN	PD + ESA / RIN analyzer	intrinsic and stressed RIN_xOMA	RIN rises with ORL; BER floor (Section 4.3)
Bias-driver noise	SMU vs. product bias board	$\mathrm{RIN}_{\mathrm{eq}}$ from i_n (Section 5.8)	RIN rises with rails on, flat vs. ORL
Wavelength	OSA / wavemeter	grid placement, $d\lambda/dT$, $d\lambda/dI$	walk off ring or MSA grid
EAM bias (EML)	bias sweep + DCA/TDECQ	extinction, chirp, RLM	aging shifts absorption curve

Table 5.5. Laser measurement playbook: what to measure, with what, and what failure looks like.

5.8 Laser drivers and the RIN budget

Modulator RF drivers (Section 3.14.3) deliver swing and bandwidth into an EAM or MZM. Laser *bias* drivers are a different circuit: they set a quiet constant current into the diode. Current noise on that path becomes optical intensity noise and adds in the RIN budget of Section 4.3. Confusing the two is a common debug miss: a great SiGe PAM4 driver can still ruin a CW laser if its supply or ground couples into the bias rail.

From current noise to equivalent RIN. Above threshold, optical power tracks bias approximately as $P \propto (I - I_{\mathrm{th}})$. Relative intensity fluctuations then track relative current fluctuations:

$$\mathrm{RIN}_{\mathrm{eq,lin}} \approx \left(\frac{i_n}{I - I_{\mathrm{th}}}\right)^2, \qquad \mathrm{RIN}_{\mathrm{eq}}[\mathrm{dB/Hz}] = 20\log_{10}\left(\frac{i_n}{I - I_{\mathrm{th}}}\right),$$

where i_n is the one-sided current-noise density in A/$\sqrt{\text{Hz}}$ at the laser terminals (driver plus board pickup). The approximation assumes linear slope efficiency and ignores intrinsic laser dynamics; it is a budget tool, not a device model.

Worked numbers at $I - I_{\text{th}} = 50$ mA (typical CW DFB window): $i_n = 500$ pA/$\sqrt{\text{Hz}}$ maps to $\text{RIN}_{\text{eq}} \approx -160$ dB/Hz; 270 pA/$\sqrt{\text{Hz}}$ maps to about -165 dB/Hz. Commercial low-noise laser drivers quote roughly 50–500 pA/$\sqrt{\text{Hz}}$ at 1 kHz depending on current range (Table 5.6); the Koheron DRV200 family is a concrete example [2][Koheron DRV200]. Against a good datacom intrinsic RIN of -145 to -155 dB/Hz (Section 4.3.1), those 1 kHz densities look comfortable. The budget tightens when $(I - I_{\text{th}})$ is small (near threshold, derated CW, or low-current VCSELs), when you integrate broadband switching noise rather than a 1 kHz spot, or when SerDes/DSP rails dump discrete tones onto the bias network.

[2] Koheron DRV200: low-noise CW laser driver; i_n at 1 kHz by current range.

Driver class (example)	i_n @ 1 kHz	RIN_{eq} @ 50 mA	What it means
Ultra-low-noise CW (DRV200-A-40)	55 pA/$\sqrt{\text{Hz}}$	≈ -179 dB/Hz	Bench / metrology floor
Low-noise CW (DRV200-A-200)	270 pA/$\sqrt{\text{Hz}}$	≈ -165 dB/Hz	Typical quiet CW source
Higher-current CW (DRV200-A-400)	480 pA/$\sqrt{\text{Hz}}$	≈ -160 dB/Hz	Still below -155 intrinsic
Shared digital LDO, poor PSRR	often $\gg$1 nA/$\sqrt{\text{Hz}}$ + tones	can exceed -145	False "RIN" on ESA

Table 5.6. Bias-driver current noise converted to equivalent RIN at $I - I_{\text{th}} = 50$ mA using $\text{RIN}_{\text{eq}} = 20\log_{10}(i_n/(I - I_{\text{th}}))$. Densities for the DRV200 rows are from the Koheron datasheet at 1 kHz; the last row is qualitative (board-dependent).

CW / ELSFP / CW-WDM paths. For external CW sources feeding Si or TFLN modulators, design the bias path as a low-noise current source with high supply rejection, local decoupling at the diode, and a star ground that does not share return with SerDes switching currents. Automatic power control () loops that close through a monitor PD suppress slow drift; keep the loop bandwidth well below the RIN measurement band and quiet enough that the loop itself does not inject intensity noise. ELSFP and CW-WDM modules hide this circuitry inside the pluggable (Sections 5.14 and 5.16); acceptance still needs module-level RIN with the host bias and management rails connected, not only a quiet SMU on the bare die.

APC Automatic power control: feedback loop from a monitor PD that holds average optical power; slow loop BW keeps it out of the RIN band.

DML and EML. A *DML* shares one diode for bias and RF: a bias tee (or on-chip bias network) combines a quiet DC source with the RF driver. Excess RF driver broadband noise, poor tee isolation, or supply ripple on the bias arm all raise measured RIN and chirp-related penalties. An *EML* splits the problem: keep the DFB bias as quiet as a CW source, and treat the EAM RF driver under Section 3.14.3. EAM drive amplitude sets extinction and chirp; DFB bias noise still lands in optical intensity before the modulator.

What to measure on the bench. Bisect electrical vs. optical RIN:

1. Measure intrinsic RIN with a quiet SMU or known low-noise driver and high ORL

(Section 5.7).

2. Repeat with the product bias board / module rails connected. Any rise is driver or supply contribution, not laser physics.

3. Sweep ORL. Rise with reflection is feedback-driven laser RIN (Section 4.3.1); rise independent of ORL points at the electrical path.

4. Look for discrete spurs on the ESA (switching frequencies, CMIS clocks). Spurs fail stressed $\mathrm{RIN_xOMA}$ even when the broadband floor looks fine (Section 4.3.1).

Key idea. Treat laser bias noise as a RIN term: $\mathrm{RIN_{eq}} \approx (i_n/(I - I_{\mathrm{th}}))^2$. Quiet CW drivers at tens to hundreds of pA/$\sqrt{\mathrm{Hz}}$ usually sit under a -145 dB/Hz intrinsic floor at 50 mA; digital supply pickup, near-threshold bias, and DML bias-tee leakage are what actually burn the budget.

5.9 How lasers fail

Six mechanisms account for most laser field returns. Each has a distinct telemetry signature, so classify before you open FA.

Threshold current increase I_{th} rises from its ship value at fixed temperature, usually with slope efficiency dropping in step. Points to active-region or facet degradation (Section 5.13).

Slope efficiency degradation Output power per unit bias current falls even when I_{th} is stable. A separate wear-out track from threshold rise; both show up on the same LIV sweep.

Wavelength drift The lasing line walks off its grid slot or ring resonance. Distinguish laser drift from TEC or ring drift by holding one actuator fixed and moving the other (Section 6.4 and chapter 6).

Aging (SMSR collapse, mode hopping) Side modes grow relative to the main mode, or the laser hops between modes under temperature or current. An OSA trend over time is the tell.

Thermal runaway A positive feedback loop where higher junction temperature raises threshold current and cuts slope efficiency, so more drive power turns to heat for the same optical output, raising temperature further until the TEC saturates and the laser rolls over. Triggered by a failed or saturated TEC, a blocked heat path, or operation above the rated thermal class. Distinct from ordinary wear-out because it is fast (minutes, not months) once it starts; the failure-analysis handbook has the full symptom-to-cause breakdown (Section 10.8).

Monitor photodiode failure The control loop's own sensor drifts or fails, so the laser looks unstable when the real fault is in the feedback path, not the gain medium (Section 5.20.3).

5.10 Separate thermal behavior from long-term aging

Thermal response is reversible on the time scale of a temperature sweep or cycle. It changes threshold current, slope efficiency, wavelength, EAM bias, TEC current, and ring alignment. Measure it with controlled case-temperature sweeps, loaded thermal corners, heater sweeps, and thermal cycling. Repeat the measurement after returning to the starting temperature. Recovery points toward an operating-point or control problem.

Long-term aging is cumulative. Threshold current rises, slope efficiency falls, contacts degrade, defects grow, and an absorption or spectral curve can move permanently. Measure those changes with HTOL, accelerated life testing, and periodic LIV, spectrum, and modulation readouts. A temperature cycle can expose a weak attach or calibration error, but it does not by itself establish a lifetime acceleration model.

Do not merge the data sets. A high-temperature BER failure that clears at room temperature needs thermal-margin work. A room-temperature baseline that keeps moving after each stress interval needs an aging or damage hypothesis.

5.11 Calibration: what drifts and what triggers retuning

Calibration exists because no transmitter runs at a datasheet point. Every unit has its own threshold, slope, absorption curve, quadrature point, and resonance, and each of those moves with temperature and age. The operating points a product actually stores are:

Laser bias / APC target the bias current or monitor-PD power setpoint that holds launch power. Drifts as threshold rises and slope falls with age (Section 5.13); a drifting monitor PD corrupts it silently (Section 5.20.3).

EAM bias (EML) the reverse-bias point that sets extinction and chirp. Moves with case temperature and with absorption-curve aging, so production parts store bias versus temperature, not one number.

MZM quadrature the phase bias that holds the modulator at its linear point. Drifts with temperature, stress, and age; a bias-control loop tracks it, and a railed loop is a telemetry alarm, not a retune request.

Ring heater / lock point the tuner power that aligns resonance to the laser line. Consumes headroom as ambient rises and neighbors heat; a railed DAC means the tuning range is exhausted (Sections 6.4 and 6.5).

Tables are usually segmented by temperature. Segment boundaries are a real failure mode: the debug story in Section 5.20.4 is a healthy laser reading the wrong temperature segment after thermal cycling. Keep calibration tables under change control and record the table version with every test result, or failures cannot be replayed.

Recalibration should be triggered by evidence, not habit: a control-loop error residual that no longer converges, an actuator (TEC, heater, bias DAC) that approaches its rail, telemetry that disagrees with an external reference, a temperature excursion beyond the table range, or a repair, rework, or firmware change that invalidates stored coefficients. ATP must verify calibration at the temperature corners the fleet will see, not only at the station ambient (Section 7.9).

5.12 How lasers are qualified

Qualification projects the failure mechanisms of Section 5.9 forward from a short bench test to years of field life. Three stress classes do the work:

HTOL (high-temperature operating life) Run a sample lot at elevated temperature and bias for a fixed duration (often 1000 hours) and track LIV, SMSR, and wavelength drift. HTOL is the primary input to the Arrhenius life projection below.

Burn-in A shorter, sometimes 100%-screen stress that removes infant-mortality units before ship, rather than projecting life. Burn-in trades test time for escape rate (Section 8.9).

Environmental stress Temperature cycling, damp heat, vibration, and shock catch packaging, attach, and mechanical failure modes that HTOL does not. They qualify different risks and should not be treated as substitutes for long-term aging data (Section 8.2).

Together with the Arrhenius acceleration factor, these three stresses turn a qualification lot into a defensible FIT number.

Observable aging signatures. Watch LIV and spectrum over HTOL or field life:

- threshold rise and slope drop (active-region / facet degradation);
- SMSR collapse (mode competition);

- EAM bias creep on EMLs (absorption curve shift $\rightarrow$ TDECQ/RLM drift);
- RIN rise under feedback (ORL or isolator failure);
- COD (catastrophic optical damage) at the facet under overstress.

COD Catastrophic optical damage: sudden, irreversible facet failure in a laser under thermal or optical overstress.

Each signature should appear in the ATP and in field telemetry triage (Sections 7.12 and 8.2).

5.13 Aging curves, derating, and fleet FIT

Lasers wear out. At fleet scale that is not a footnote; it sets architecture (ELSFP vs. integrated laser) and operating policy (derating, burn-in).

Arrhenius / E_a Life-acceleration model: temperature stress multiplies wear-out by $\exp[(E_a/k)(1/T_{\text{use}} - 1/T_{\text{stress}})]$.

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

[3] Telcordia GR-468-CORE: reliability assurance for optoelectronic devices.

Arrhenius life projection. Telcordia GR-468-CORE qualifies optoelectronic parts with accelerated stress (HTOL, temperature cycle, damp heat) and projects field life with Arrhenius acceleration [3] [80]:

$$\text{AF} = \exp\left[\frac{E_a}{k_B}\left(\frac{1}{T_{\text{use}}} - \frac{1}{T_{\text{stress}}}\right)\right],$$

where E_a is the activation energy for the wear-out mechanism under test, k_B is Boltzmann's constant, and temperatures are absolute. Document E_a, sample size, and confidence bounds when converting a 1000-hour HTOL lot into field-year FIT. Activation energies are mechanism-specific; use the value justified in the qual plan, not a generic number copied from another product.

When the projection is valid. Acceleration assumes the stress speeds up the *same* physical mechanism the fleet will see. The projection fails in two ways: the stress activates a mechanism the field never sees (solder creep or moisture ingress at a stress temperature the product never reaches), or the field sees a mechanism the stress never exercises (connector wear, bias-rail transients, thermal cycling from traffic load). So a qual number is a hypothesis, not a fact: compare field-return Pareto and failure signatures against the qual projection, and treat divergence as evidence that E_a or the mechanism model is wrong, not that the fleet is unlucky (Section 7.12).

Derating. Run below absolute-max current, case temperature, and optical power. Derating extends wear-out life and reduces COD risk. Uncooled datacom parts already sit near thermal limits at high case temperature; cooled or faceplate ELSFP modules (Section 5.14) buy headroom by moving heat off the ASIC package.

Worked FIT example (assumptions labeled). FIT is failures per 10^9 device-hours. For illustration only, assume 50 FIT per laser (confirm against your supplier qual; do not treat 50 as a measured claim) and a fabric with 5×10^5 lasers (order-of-magnitude for a large AI cluster with several optical links per accelerator). Expected failures per day:

$$\frac{5 \times 10^5 \times 50 \times 24}{10^9} \approx 0.6 \text{ laser failures/day.}$$

That is why field-replaceable ELSFP modules, burn-in screens, and derating are design inputs, not afterthoughts (Chapter 8).

5.14 ELS and ELSFP: architecture, pinout, qual

ELSFP (External Laser Small Form-Factor Pluggable) is the OIF form factor for faceplate-pluggable CW laser modules that feed co-packaged optical engines [4][49]. The lasers sit at the coolest part of the system (front panel), hot-swap when they fail, and keep thermal load off the ASIC and photonic engine.

ELSFP External Laser Small Form-Factor Pluggable (OIF): faceplate CW laser module for CPO; 24-pin card edge, CMIS, blind-mate MT.

[4] OIF ELSFP Implementation Agreement (OIF-ELSFP-02.0, 2023): 24-pin card edge, CMIS, blind-mate MT.

Mechanical and optical. The module uses a card-edge electrical interface and a blind-mate multi-fiber optical connector at the rear (MT-class ferrules), which improves eye safety for high CW power by keeping live fiber inside the chassis [49]. One ELSFP can feed more than one optical engine. OIF defines optical power classes, thermal classes, and wavelength assignments (e.g. DR-type 1311 nm and FR-type CWDM4 grids) so hosts and modules interoperate.

Management and hot-swap. ELSFP uses CMIS and the CMIS module state machine over TWI. On plug-in the module resets, initializes management, and stays in low-power mode with lasers *off* until the host transitions it to ModuleReady and explicitly enables lasers [49]. `ModPrsL` and `IntL` support presence detect and asynchronous alarms for safe hot-swap.

Electrical pinout (OIF-ELSFP-02.0 Table 7). Twenty-four contacts: multiple 3.3 V VCC and GND pins, module reset (`ResetL`), low-power mode (`LPModeL`), two-wire serial management (`SCL`/`SDA`), presence (`ModPrsL`), and interrupt (`IntL`), plus reserved pins for future power/ground [49]. Table 5.7 summarizes the published map.

Qual hooks for suppliers. Acceptance test plans should cover the checklist in Table 8.3 and section 8.10: laser LIV/SMSR/RIN inside the module; optical power-class compliance; connector mating cycles and contamination/ORL; burn-in before ship; CMIS register sanity; and thermal class at rated case temperature. Module bring-up must also prove the CMIS enable sequence and ModuleReady laser policy

Pin	Function	Requirements	Notes
1–3	VCC	1.5 A, 3.3 V	with noise filtering
4	TBD	reserved	future power
5	ResetL	pull-up 10 kΩ	reset module, LVTTL
6	LPModeL	MMC on only	low-power mode (low), LVTTL
7	TBD	reserved	future ground
8–10	GND	1.5 A, 3.3 V	with noise filtering
11	TBD	reserved	—
12	SCL	TWI clock	host 4.7 kΩ pull-up; module $\geq$10 kΩ
13	SDA	TWI data	same pull-ups as SCL
14	TBD	reserved	—
15–17	GND	1.5 A, 3.3 V	with noise filtering
18	TBD	reserved	future ground
19	ModPrsL	shorted to GND in module	presence (low), LVTTL
20	IntL	pull-up 10 kΩ	interrupt, LVTTL
21	TBD	reserved	future power
22–24	VCC	1.5 A, 3.3 V	with noise filtering

Table 5.7. ELSFP electrical pinout (adapted from OIF-ELSFP-02.0 Table 7). Lasers power only in ModuleReady after host command; default on plug-in is lasers off [49].

(Section 7.9). Field returns split between laser wear-out and connector/fiber-attach faults; keep both in the triage tree (Section 7.12).

5.15 Optical safety and laser classes

5.15.1 Hazard and laser classes

Laser safety for interconnects is governed by IEC 60825-1 (laser product classification) and IEC 60825-2 (optical-fiber communication systems, OFCS) [88]. Classes run from Class 1 (safe under normal use) through Class 1M (safe unless the beam is collected by optics), Class 3R/3B, and Class 4. At 1310 nm and 1550 nm the beam is invisible, which raises the operational risk: technicians cannot see exposure. The retinal-hazard band ends near 1400 nm, but corneal and skin hazards remain, and single-mode power confined to a $\sim$9 µm core is high radiance even at modest milliwatt levels.

Short-reach datacom modules are usually engineered so each fiber port stays Class 1 or Class 1M under rated launch power. That is a design constraint on EML/DFB bias and on how much power each lane launches, not a label you add after the fact.

5.15.2 Hazard level = aggregate, not per-lane

The safety case scales with *total* launched power at an accessible location, not with a single DFB data sheet. CW-WDM and ELS banks concentrate many lines on one MT or MPO ferrule (Section 5.14). A connector that breaks out eight or sixteen fibers can exceed a per-lane Class 1 budget even when each lane is modest. IEC 60825-2 assigns hazard levels (1 through 4) to each accessible port in the OFCS based on the radiant power that could escape during service [88]. That is why ELS architecture and fiber count drive classification, not the laser chip alone.

5.15.3 Open-fiber protection: APR and ALS

When fiber continuity is lost, open connectors and broken fiber can expose hazardous power. *APR* (automatic power reduction) holds output at or below Hazard Level 1M and probes for re-mate with safe low-power pulses. *ALS* (automatic laser shutdown) cuts power entirely and was common on older SDH links; for modern high-power systems APR with automatic restart is the preferred pattern because restart probes stay within the hazard limit [89]. ITU-T G.664 requires power reduction to Hazard Level 1M within about 3 s of a continuity break, a restart inhibit window, and restart only at safe power.

APR Automatic power reduction: holds laser output at safe level on fiber break; probes for re-mate at Hazard Level 1M.
ALS Automatic laser shutdown: cuts laser power on fiber break. APR with restart preferred in modern systems (ITU-T G.664).

These mechanisms tie directly to CMIS and bring-up policy: lasers enable only when the host commands ModuleReady (Sections 7.8 and 7.9). APR/ALS is what makes a live ELSFP hot-swap survivable in a running rack (Section 7.9).

5.15.4 What validation and ops owe

Optical safety is a validation deliverable, not a compliance sticker. ATP should verify APR/ALS trip threshold and timing on representative open-fiber faults; label modules and cages with the rated class; document max launched power per port and per MPO breakout; and write service procedures for multi-fiber connectors. At fleet scale, a hot-swap runbook that assumes ALS works but was never tested in ATP is a real hazard. Fold the APR/ALS check into the ELS hot-swap corner in Section 7.9 alongside mate-cycle and ORL tests.

5.16 CW-WDM source validation

Multi-wavelength CW sources (CW-WDM MSA) feed dense ring or filter banks on a PIC (Section 6.6 and chapter 6). Validation is per-channel plus cross-channel:

- power flatness across λ (uneven OMA after the modulator bank);
- per-channel SMSR and wavelength grid placement;
- channel crosstalk and residual ASE between lines;
- lock to microring resonances under temperature and neighbor heating (Sections 6.4 and 6.5 and section 3.14.3);

- RIN and ORL sensitivity for each line (Section 5.7 and section 4.3.1).

Examples: Ayar Labs SuperNova (CW-WDM MSA-compliant, feeds TeraPHY) [5][50][6][48]; Broadcom ELSFP banks on Tomahawk CPO (Sections 5.14 and 9.10); quantum-dot comb lasers (Ranovus, Quintessent) aimed at many λ from one chip. Source tests live here; locking and on-chip MUX live in Chapter 6.

[5] Ayar Labs SuperNova CW-WDM source and TeraPHY optical I/O.

[6] CW-WDM MSA Rev 1.0 (2021): O-band 8/16/32-λ CW grids (9/18/36 nm spans); modular vs integrated ports; RIN/SMSR/linewidth methods.

5.17 Light-source supply strategy

The sourcing decision follows the same architecture fork as the optical design: buy a merchant source, buy a serviceable external module, or bind the source to the photonic package. Evaluate each path by qualification ownership, second-source portability, lot traceability, test access, field replacement, and change-control rights. A vendor list ages quickly and does not answer those questions.

Merchant DFB, EML, or CW die preserve module-level design freedom and can support a second source, but the integrator owns attach, driver match, screening, and package reliability.

External CW-WDM or ELSFP module moves source qualification and management into a replaceable unit. The system still owns connector, ORL, hot-swap, and host interoperability (Sections 5.14 and 5.16).

Multi-wavelength source reduces source count and can simplify WDM fan-out, but couples channel yield, power flatness, control, and replacement into one unit (Section 5.16).

Source integrated with the PIC reduces optical interfaces and can improve density, but makes laser yield and wear-out part of package yield and service life.

Approach	Qualification ownership and risk
Merchant DFB/EML/CW die	Integrator owns attach, driver match, screen, and module qual
External CW-WDM / ELSFP module	Supplier owns source module; system owns interface and service qual
Multi-wavelength source	Shared yield, power-flatness, and replacement risk across channels
Source integrated with PIC	Highest density; laser yield and life become package risks

Table 5.8. Light-source sourcing paths and the qualification ownership each one creates.

[7]

[7] The fork is a reliability and ownership decision. An external source makes the source replaceable but adds a managed optical interface. An integrated source removes that interface but places source yield and wear-out inside the package. Qualify the architecture you will service, not only the laser die.

5.18 Why lasers are the reliability bottleneck

At the scale of a large optical fleet the laser is usually the reliability-limiting component. It is an active device with wear-out physics that passive optics and even photodiodes largely lack:

- *Catastrophic optical damage* (COD) at the facet.
- Gradual facet and active-region degradation (accelerated by temperature, following Arrhenius kinetics; Section 5.13).
- EAM aging in EMLs; coupling and solder drift in packaged assemblies.

Because failures scale with the number of lasers, a fleet of $100{,}000$+ links turns a modest per-laser FIT rate into a steady stream of field failures (Sections 5.13 and 8.2). The mitigations shape architecture: field-replaceable external laser sources (ELSFP, CW-WDM), redundancy, burn-in screening to weed out infant mortality, and derating (running lasers below their maximum to extend life).

5.19 Margin erosion over temperature, lot, and life

A link rarely loses all margin in one event. The source can lose launch power as slope efficiency falls. Connector loss and ORL can rise after service. EAM or MZM bias can move. A ring can consume spectral headroom as its heater approaches range. Driver noise can raise the BER floor while none of these changes violates its stand-alone limit.

Track five ledgers:

Power margin launch power, coupling, connector and MUX loss, receiver sensitivity, and aging reserve.

Noise margin intrinsic and feedback-driven RIN, bias-rail noise, receiver noise, and crosstalk.

Timing margin source and modulator bandwidth, dispersion, driver and host jitter, and equalization reserve.

Spectral margin laser wavelength, SMSR, filter or ring passband, thermal drift, and lock range.

Control margin headroom in APC, TEC, heaters, ring lock, bias DACs, and calibration tables. A railed loop can fail the link while the diode is still healthy.

Recompute the link at combined production corners. A nominal part at nominal temperature says little about whether a slow loss in two ledgers will push a tail unit across the pre-FEC BER limit. Section 7.9 and table 7.5 carry the same ledgers into validation and fleet triage. The interview review compresses this checklist in Section A.3.4.

5.20 Engineering lens

5.20.1 How it works

A laser is an active device with wear-out physics, which makes it both the first line of the link budget and the fleet's reliability bottleneck. The chapter's device families and LIV/SMSR/RIN measurements all serve one question: will this source stay in spec for years at temperature?

5.20.2 How it is measured

Qualify the laser as a set of curves across temperature, bias, ORL, and age, not a room-temperature data-sheet point. The measurement playbook (LIV, SMSR, RIN, wavelength, and EAM checks with their instruments and pass/fail intent) is in Section 5.7 and table 5.5; the stress classes that project field life are in Sections 5.12, 5.13 and 8.2 [80].

5.20.3 How it fails

The six field-return mechanisms are catalogued in Section 5.9: threshold rise, slope droop, wavelength drift, aging (SMSR collapse and mode hopping), thermal runaway, and monitor-photodiode failure. Manufacturing adds die, wafer, lot, and assembly spread to every one. The mechanism that most often misleads triage is a healthy laser behind a bad feedback sensor, so it gets the worked callout below.

Failure mode: Monitor photodiode drift

Symptoms. Reported power falls or the bias loop moves, but an external power meter does not show the same change.
Likely causes. Monitor-PD responsivity drift, transimpedance gain error, contamination in the monitor path, or a bad calibration coefficient.
Measurements. External power meter, monitor current, bias current, LIV, and loop error versus temperature.
Mitigations. Repair the monitor path or calibration, add disagreement alarms, and do not raise laser bias to compensate for a false reading.

5.20.4 How it is debugged

For power degradation, compare external optical power, monitor current, bias, and case temperature before changing the setpoint. Rerun LIV at the failing temperature and compare it with ship data. If LIV moved, inspect SMSR, wavelength, and RIN to classify active-region, facet, or modal change. If LIV is stable, move to coupling, connector, monitor-PD, and control-loop checks. For a wavelength excursion, inspect OSA data and TEC current together. For a bias anomaly, replace the product driver with a quiet source before blaming the diode.

Debug story

Observed. BER worsened after thermal cycling while average optical power stayed in range.
Investigation. The DCA showed that extinction ratio had collapsed. LIV and SMSR were unchanged, and an EAM bias sweep restored the eye.
Finding. The light source was healthy, but its modulator operating point was wrong.
Root cause. A calibration table used the wrong temperature segment after the cycle.
Resolution. The table and screening limits were fixed, and EAM bias sweep data became part of the thermal-cycle readout.

5.21 Engineering checklist

Decision or test	Question it answers	Evidence to retain
Architecture	Does the source and modulation path close reach, rate, power, cost, and service?	Requirement allocation and rejected alternatives
LIV	Is the operating window clear of threshold, kinks, and rollover?	Curves by unit, lot, temperature, and age
Spectrum	Does wavelength and SMSR stay inside the assigned grid and filter passband?	OSA or wavemeter data across corners
RIN and ORL	Does noise margin survive the reflection environment?	Quiet and stressed RIN with stated ORL and bandwidth
Modulation	Does bias, drive, chirp, and bandwidth close the eye?	Bias sweeps, TDECQ or equivalent, and driver conditions
Thermal behavior	Are reversible shifts within control and actuator range?	Temperature and heater sweeps, TEC current, recovery data
Long-term aging	Which parameters drift permanently, and at what rate?	HTOL intervals, LIV, spectrum, and modulation trends
Manufacturing	Can the ATP catch bad units and lot drift at useful test cost?	Limits, guard bands, GR&R, yield, and reaction plan
Fleet operation	Which monitors distinguish source, modulator, cooler, and optical path?	Telemetry map, alarm thresholds, and golden baselines

Table 5.9. Source and modulation engineering checklist. Each row ties a decision to evidence, not only a test name.

5.22 Interview and design review questions

Concept.

- Why is the laser typically the reliability-limiting component in an optical link?
- What physical mechanism causes a DFB laser's threshold current to rise with age?
- Why does an isolator-free design tighten the RIN requirement?

Design.

- Why would you choose an EML over direct modulation for this reach and lane rate?
- When does a ring's density justify its heater and wavelength-control burden?
- Which aging model and activation energy match the observed failure mechanism?
- What are the distributions of threshold, slope, wavelength, SMSR, and RIN across wafer, lot, assembly site, and temperature?
- Which screens remove infant mortality, and what good units do those screens consume?
- Does the architecture make the laser field-replaceable, and what failure rate justifies that choice?

Debug.

- Optical power fell but the monitor photodiode reports no change. What do you check?
- BER worsened after thermal cycling while average power stayed in range. Which measurement would separate source aging, EAM bias, wavelength, and connector hypotheses?
- Which telemetry distinguishes laser wear from monitor, TEC, connector, and bias-driver faults in the fleet?

Manufacturing and operations.

- How would you qualify a second laser supplier without assuming the first supplier's failure distribution?

- What are the process control limits and the supplier's reaction plan when a trend crosses them?
- How many HTOL hours at what temperature project 10 years of field life at your operating conditions?
- What lot-level data must the supplier provide at each NPI gate?
- What is the expected laser replacement rate for this fleet size and FIT?

Key idea. Measure LIV, SMSR, wavelength, and RIN as distributions across temperature, lot, and age. Tie each requirement to an ATP row, each life claim to a physical mechanism, and each field alarm to a measurement that separates the laser from its driver, monitor, cooler, and optical path.

Chapter 6

WDM and wavelength-locked lasers

Early datacenter optics mostly ran one wavelength per fiber. That worked while port counts were modest. At AI scale, fiber count itself becomes a first-order cost and cable-plant problem, so the industry packed more channels onto each strand. The price of that packing is control: once channel spacing tightens, or once the modulator is a wavelength-selective ring, someone must keep laser and filter locked together. Few phrases carry as much architectural information as "wavelength-locked laser," because locking only appears under those interconnect choices. Ring and MZM device physics stay in Section 3.14.3; per-λ laser ATP and aging stay in Chapter 5. This chapter covers grids, lock loops, thermal crosstalk, MUX budget, and CW-WDM architecture.

6.1 Why multiplex wavelengths at all

At $100{,}000$+ accelerator scale, every extra fiber is another connector, another patch, and another failure mode. *Wavelength-division multiplexing* (WDM) puts many independent channels on a single fiber, each on its own wavelength, so bandwidth per fiber rises without adding fiber. Each wavelength can still be an ordinary IM/DD channel. WDM and IM/DD are orthogonal; you simply run IM/DD *per wavelength*.

WDM Wavelength division multiplexing: multiple λ on one fiber. CWDM (20 nm spacing) or DWDM ($\leq$100 GHz).

Historically the industry climbed a ladder of spacing. Coarse CWDM4 used $\approx$20 nm slots and uncooled lasers. LAN-WDM tightened that for 2 km-class FR4. Dense grids and then CW-WDM O-band combs for CPO pushed spacing into the 100–800 GHz class and made active locking mandatory. Those spacings are standardized grids, not vendor choices: the 20 nm CWDM slots follow the ITU-T G.694.2 wavelength grid (18 channels, 1271–1611 nm), and the 50/100/200 GHz datacom DWDM spacings follow the ITU-T G.694.1 frequency grid anchored at 193.1 THz [90]. CWDM4 uses the four O-band lines of that CWDM grid; the CW-WDM combs in Section 6.6 define their own O-band grids for dense integration. Table 6.1 is that ladder as you will meet it in short-reach AI optics today.

Grid family	Spacing (class)	Channels / fiber	Cooling / lock	Typical short-reach use
CWDM4	$\approx$20 nm	4	Uncooled; loose control	FR-class pluggables; faceplate WDM
LAN-WDM	$\approx$800 GHz ($\approx$4–5 nm @ 1310 nm)	4	Cooled or tight open-loop	2 km-class FR4 (edge of book scope)
Datacom DWDM	200/100/50 GHz	many	Locked to grid	Discrete DFB/EML DWDM modules
CW-WDM / CPO O-band	100–800 GHz class (MSA spans)	8 / 16 / 32	Locked; often ring-tuned	CPO engines, optical I/O chiplets

Table 6.1. WDM grids for short-reach AI interconnects. CW-WDM MSA normative grids sit in O-band with 9/18/36 nm spans and 8/16/32-line sets (Section 6.6); spacing is set by the chosen span and channel count, not by Ethernet CWDM4.

6.2 Why "locked" is the operative word

WDM alone does not force active locking. CWDM4 packs four wavelengths with enough spacing that uncooled lasers can wander and still stay in their slots. Locking becomes the operative word only when either the channel spacing is tight or the modulator itself is wavelength-selective. Those two situations drive nearly every modern CPO and dense optical-I/O control loop.

6.2.1 Tight DWDM grids

To pack channels at 50–200 GHz class spacing, each laser must sit on its grid slot or adjacent channels collide. Emphasizing "locking" therefore hints at spacing tighter than CWDM. You do not stress locking for coarse, uncooled CWDM4; you do as soon as the grid looks like LAN-WDM, datacom DWDM, or a CW-WDM comb.

DWDM Dense wavelength division multiplexing: channel spacing $\leq$100 GHz; used in metro/long-haul and dense CPO WDM.

6.2.2 Microring modulators and WDM locking

Resonant ring and microdisk modulators are the dense-WDM workhorse (Section 3.14.3). Their resonance drifts by roughly 10 GHz/°C in silicon, so even a stable laser is not enough: the laser and the ring must be wavelength-locked. Either lock the laser to the ring, or thermally tune the ring onto the laser with a feedback loop. That laser–ring co-locking is the central control problem in ring-based WDM links and in co-packaged optics, and it is why neighbor heat and case-temperature ramps belong in validation, not only in the thermal section of a datasheet.

6.3 WDM filters, grids, and on-chip multiplexing

WDM is not only lasers and locking (Section 6.6): the PIC needs wavelength selective routing. The MUX/demux stage is a first-class link-budget line item (Section 7.7), not a packaging footnote.

PIC Photonic integrated circuit: waveguides, modulators, detectors, and couplers on one chip (typically SOI at 220 nm Si).

Hardware choices.

AWG / echelle gratings multiplex and demultiplex on silicon or glass. Insertion loss is often 2–5 dB per MUX stage; adjacent-channel crosstalk and passband ripple land in OMA and transmitter and dispersion eye closure quaternary (TDECQ).

Ring filter banks drop/port routing in microring banks sets how many λ share a bus waveguide. Thermal tuning per ring is common (Section 3.14.3 and section 6.5).

Hybrid some engines use a coarse AWG plus fine ring filters; count every stage in the ledger.

MZMs trade area for calm wavelength behavior. When each lane carries its own laser (DR/FR SiPh modules), silicon Mach–Zehnder modulators sidestep ring locking (Section 3.14.3). Rings remain the default when many λ share one PIC and area dominates (Section 3.14.3 and chapter 9).

Where MUX defects land. Table 6.2 maps common MUX faults to the measurement that catches them.

Fault	Optical symptom	Hits	Catch with
Stage insertion loss	Lower launch OMA on all λ	Link budget OMA	Power meter / OMA
Passband ripple / tilt	Uneven OMA across bank	Weakest λ	Per-λ OMA map
Adjacent-channel crosstalk	Closed eyes, RLM/TDECQ up	Tx quality / BER	Isolation sweep + DCA
MUX imbalance	One λ weak, neighbors OK	Single-lane BER	Per-lane power + BER
Grid misalignment	Filter edge clipping	TDECQ, unlock risk	OSA + lock status

Table 6.2. MUX/demux defects and where they appear in validation. Isolation and imbalance tests belong in ATP for any dense WDM engine (Sections 5.16 and 6.7).

Validation adds channel isolation sweeps, grid alignment across temperature, and MUX imbalance (uneven OMA per λ). Treat the weakest channel as the budget-limiting lane, not the average.

6.4 Lock-loop mechanics

Wavelength locking closes the loop between source and filter. Pick a technique based on whether the laser, the ring, or both must be steered (Section 3.14.3

and chapter 7).

Error-signal sources.

Wavelength locker Etalon + dual-PD error signal that trims laser TEC or current onto a grid slot (common on discrete DFB/EML).

Etalon-based wavelength locker A fixed reference etalon plus a pair of photodiodes produces an error signal proportional to wavelength offset; feedback trims laser temperature or current onto the grid. Common on discrete DFB/EML modules.

Laser-to-ring thermal feedback Monitor the ring's through/drop power (or a dither tone) and heat the ring, or trim laser current, to park the carrier on resonance. Default for dense microring WDM banks in CPO and optical I/O.

Injection / external-cavity locking Stabilize a laser's wavelength and linewidth against an external reference cavity; higher performance, more parts. Rare in short-reach volume products.

Athermal design Engineer the device so its resonance barely moves with temperature, reducing the control burden. Athermal does not remove MUX grid alignment; it shrinks the loop authority you need.

Digital supervisory loop CMIS-exposed monitors and firmware on modern modules; link training at 224G/448G may iterate EQ and wavelength trim together (Section 9.3).

What you trim. Three actuators show up repeatedly, and the bring-up order usually starts with the slowest, highest-authority knob. Laser TEC / temperature moves the whole comb or a single DFB on the frequency axis. Laser bias current is the fine wavelength trim (and also changes power), so watch RIN and SMSR when you use it as a locker (Section 5.8). Ring heaters park each microring onto its assigned λ and are the per-channel control in dense banks (Section 6.5).

Capture versus hold. *Capture* is acquiring lock from a cold or unlocked state: coarse scan of heater or TEC until the monitor error signal crosses zero, then close the loop. *Hold* is rejecting thermal drift and neighbor crosstalk once locked. Loop bandwidth must be fast enough to track case-temperature ramps and adjacent-heater steps, but slow enough not to fight the data path or inject RIN through bias modulation. Silicon rings at $\sim$10 GHz/°C set the disturbance scale: a 1 °C neighbor step is tens of GHz of resonance walk, which is a large fraction of a 100–200 GHz grid slot (Section 3.14.3).

Failure signatures. When a loop loses lock, bisect laser versus ring by toggling the TEC setpoint and the ring heater independently (Sections 6.7, 7.6 and 7.12). The one-lane and intermittent signatures this produces, and their causes, are catalogued in Section 6.9.3.

6.5 Thermal crosstalk and heater budget

Dense ring banks share a substrate. Heating one ring to stay on resonance shifts neighbors. That is , and it is why a single-lane lock test at room temperature is not a product test.

Thermal crosstalk Heater or self-heating on one ring shifts neighbors' resonances; worst case is full-bank traffic at max case *T*.

Where heat comes from. Self-heating from the ring's own heater (and absorbed optical power) shifts its resonance, so the lock loop must settle with the lane at operating optical power, not dark. Adjacent heaters on nearest-neighbor and next-nearest rings in a WDM bank are the next disturbance; the worst case is all neighbors at max heater power while you hold lock. Package and ASIC load add a common-mode walk: switch or XPU case-temperature ramps and local hotspots move the whole bank, and a shared TEC or cold plate sets how much of that walk the lock loop must reject (Section 7.9).

Design and validation implications. Budget heater range with headroom for crosstalk, not just for the coldest and hottest case alone. Layout (heater placement, thermal isolation trenches, ring pitch) is a reliability and yield problem as much as a control problem. In validation, simultaneous full-traffic on neighbors plus max case T is a *lock* test: unlock, BER walk, or TDECQ rise on one λ under neighbor load points at thermal design, not at a bad laser die (Sections 7.9 and 7.12).

6.6 External multi-wavelength sources (CW-WDM)

Dense WDM with ring modulators needs a source of many clean, stable wavelengths. The industry answer is a *disaggregated* external laser: a single multi-wavelength continuous-wave (CW) module supplies a comb of wavelengths over fiber to the photonic engine, where microrings imprint data onto each one. The *CW-WDM MSA* standardizes those sources for AI, HPC, and high-density optics [48]. Source-side measurement detail (per-channel power, SMSR, RIN, lock under neighbor heat) is in Section 5.16; this section is the architecture contract.

What the MSA specifies (and what it does not). Rev 1.0 (June 2021) defines O-band wavelength grids, port configurations, and measurement methods. It does *not* standardize mechanical form factors, management pins, or full link parameters; those stay application-specific or move to form-factor MSAs such as ELSFP [49][48].

Core normative content:

- **Grid sets:** 8+1 and 16+1 lines in a 9 nm span; 8+1 / 16+1 / 32+1 in 18 nm and 36 nm spans (shortest line optional in each set).

- **Spacings (class):** for the 18 nm span, channel spacing is 400 / 200 / 100 GHz for 8 / 16 / 32-line sets; the 9 nm and 36 nm spans scale spacing with span width (100–800 GHz class). Normative MSA grids are denser than 5 nm; coarser CWDM-like spacings are informative only.
- **Two physical configs:** *modular* (each fiber carries one λ) and *integrated* (each fiber carries the full comb).
- **Power classes and AS parameters:** output power classes span low to high launch; SMSR, RIN, and linewidth floors are defined with measurement methods, with many limits marked application-specific (AS) in the normative tables.

Informative appendix examples (not universal product guarantees) often quote $\approx$30 dB SMSR, ≈ -135 dB/Hz RIN, $\approx$20 MHz linewidth, $\pm$1 dB per-line power variation, and -20 dB ORL tolerance for 18 nm-span examples. Treat those as negotiation anchors; write the ATP to your link budget (Section 5.16 and table 5.4).

Why disaggregate the laser. External CW-WDM / ELSFP sources are field-replaceable. Lasers dominate wear-out FIT (Chapters 5 and 8); keeping them off the ASIC package turns a COD or facet failure into a hot-swap, not a board pull. The photonic engine still needs per-λ lock to rings or filters.

Exemplar: SuperNova + TeraPHY. Ayar Labs' optical-I/O stack is aimed at *scale-up* (XPU-to-XPU) rather than switch fabric [50]:

TeraPHY an optical-I/O chiplet co-packaged with the host XPU, carrying the microring modulators and receivers.

SuperNova the external CW light source, positioned as the first CW-WDM-MSA-compliant 16-wavelength source, delivering up to 16 wavelengths into each of 16 fibers. That is light for 256 data channels (vendor claim: about 16 Tb/s bidirectional), and roughly $64\times$ the wavelength count of CWDM4 pluggables.

Vendor performance claims versus pluggables plus electrical SerDes (5–10$\times$ bandwidth, 10$\times$ lower latency, 4–8$\times$ better power efficiency) are marketing numbers; use them as orientation, not as ATP limits.[1]

System requirements the source must meet. For the PIC to close every lane, the comb must deliver, across temperature and with all ports active, a small set of properties at once: per-line power flatness (else MUX + modulator bank makes uneven OMA), grid placement and SMSR per line, RIN under the specified ORL, and absolute wavelength stable enough that ring heaters stay in range (Sections 5.16 and 6.5). Miss any one and a single λ will look like a modulator or lock-loop failure when the source is the real cause.

[1] Source: Ayar Labs, SuperNova light source, ayarlabs.com. Positioning: Ayar targets scale-up optical I/O between accelerators; Broadcom and NVIDIA CPO (Section 9.10) put the same building blocks on the *switch*.

6.6.1 Comb sources: one device, many lines

The SuperNova approach builds its comb from an array of discrete lasers, one distributed-feedback (DFB) die per wavelength, combined onto the output fiber. That is the shipping answer, and it scales cleanly to the 8 and 16 lines the MSA grids call for. Past a few dozen lines the die count, the combining loss, and the per-die wavelength trimming start to hurt, which is why a single device that emits a whole comb is attractive. Three device classes compete.

Quantum-dot mode-locked lasers (*QD-MLLs*) are the front-runner for a monolithic O-band comb. Mode locking in a single cavity produces evenly spaced lines at the cavity round-trip rate; quantum-dot gain adds low RIN, a near-zero linewidth-enhancement factor, and strong optical-feedback tolerance, the same properties that make quantum-dot lasers attractive for isolator-free co-packaging [2][102][56]. Reported O-band demos carry 14×100 Gb/s PAM4 over 10 km at ∼284 fJ/bit, and isolator-free variants target interconnect capacity beyond 3.2 Tb/s. These are research results, not qualified products, so treat the line counts and efficiencies as provisional.

[2] QD mode-locked laser combs: best single-device efficiency/size for DWDM; O-band demos to 14×100G PAM4 and beyond 3.2 Tb/s. Provisional.

Kerr microcombs take the opposite route: pump one high-*Q* microresonator and let four-wave mixing fill in many evenly spaced lines on a chip [3][103]. The line count and the spacing uniformity are excellent, and a 2025 demonstration added a monolithic demultiplexer that autonomously locks to and tracks the comb lines. The catch is power. Pump-to-comb conversion efficiency is modest, so each line leaves the chip weak and usually needs a booster or per-line amplifier before it reaches the modulator bank (Section 6.6.2). Microcombs also need a clean pump laser and careful thermal control to hold the soliton state.

[3] Integrated Kerr/soliton microcomb source: hundreds of chip-scale lines with autonomous locking, but low per-line power. Provisional.

Gain-switched and quantum-dash combs sit between the two: a directly driven laser produces a flatter, lower-line-count comb with simple electronics. They have reached multi-terabit aggregate rates in the lab but see less datacom traction than QD-MLLs.

For any of them the contract from the MSA does not change: the source must hold per-line power flatness, SMSR, RIN, and grid placement across temperature with every port active (Section 5.16). A comb that delivers 32 lines but drops 6 dB across the band, or whose edge lines miss the grid, buys nothing over an array of DFBs the PIC already knows how to drive.

6.6.2 Gain and power distribution across the bank

Whatever generates the comb, the light still has to survive the trip to the modulators. One source feeds a multiplexer, a splitter tree, and per-line routing before it reaches a ring, and each stage takes its cut. When the source is a chip-scale comb with weak lines (Section 6.6.1), or when the fan-out is large, a *semiconductor optical amplifier* (SOA) restores the budget.

WHERE THE GAIN SITS. A booster SOA placed right after the comb lifts every line before the split, so one device pays for the whole fan-out. A per-line SOA after the demultiplexer instead corrects line-to-line imbalance, at the cost of one amplifier per wavelength. The receiver-side SOA preamplifier (Table 4.4) is a separate job: there the goal is sensitivity, here it is launch power.

THE NOISE-FIGURE COST. An SOA adds amplified spontaneous emission (*ASE*), and its noise figure sets how much. The quantum floor is 3 dB; commercial O-band SOAs land near 6–7 dB with roughly 15 dB of gain and about 1.5 dB polarization-dependent gain [4][104]. Every dB of noise figure eats into the signal-to-noise ratio the receiver eventually sees, so an SOA that rescues launch power can cost link margin if it runs deep into saturation or amplifies an already-noisy comb. Quantum-dot SOAs grown on silicon are attractive for the same reason QD lasers are: low noise, wide O-band gain, and CMOS-compatible integration.

[4] O-band SOA: $\sim$15 dB gain, NF $\leq$7 dB (3 dB quantum floor), $\leq$1.5 dB PDG; QD-SOA on Si suits CPO. Booster/distribution gain.

HOLDING THE BANK FLAT. Gain is not uniform across the comb span, and SOAs compress near saturation, so the line that starts strongest is not the line that ends strongest. Per-line power flatness is a system spec (Section 6.6), held with some mix of source-side pre-emphasis, gain-tilt control, and per-line trimming. The alternative to any distribution-side gain is a higher-power source, which is why array-of-DFB designs that already meet the launch budget often skip the SOA entirely.

6.6.3 Polarization on the CW distribution path

IM/DD is forgiving about polarization where it counts most. The photodiode is a square-law detector, so the received state of polarization does not affect the recovered bits (Section 3.4). A standard single-mode drop to the receiver therefore needs no polarization control. The external-source and CW-WDM architectures move the sensitive part upstream, onto the path between the laser and the modulator.

WHERE IT MATTERS. Three elements on the CW feed care about the launched state. A *TFLN* Mach–Zehnder drives the electro-optic effect through TE-polarized light on one crystal axis (Section 3.14.3); light on the wrong axis sees little modulation. Silicon grating couplers are polarization-selective by construction, so coupling loss swings with the input state, a *polarization-dependent loss* (PDL) that comes straight off the launch budget. And a booster or per-line SOA adds polarization-dependent gain (Section 6.6.2), so a drifting state becomes a drifting per-line power. None of these sit after the photodiode, so none show up in a receiver-side budget: they act on light that has not yet been modulated.

HOW IT IS HELD. Keep the CW feed on *polarization-maintaining* (PM) fiber and PM connectors from the external laser (Chapter 5) to the modulator input, launched on the coupler's preferred axis. On-chip the light is already single-polarization once it is in a TE waveguide, so the discipline is really about the fiber run and the mate at the package. A co-packaged design with the laser on the same substrate shortens that run to millimeters and sidesteps most of the problem; an external-laser design pays for a PM path and its alignment tolerance instead.

6.7 Lock validation playbook

Instruments and BER methods live in Chapter 7. What is special to WDM is the order: you cannot trust a BER number on a ring bank until the comb is identified, the resonances are parked, and the lock loops hold under neighbor heat. The sequence below is the usual bring-up; skip a step and you will debug the wrong domain.

Bring-up order.

1. **Grid ID:** confirm each CW line (or DFB) is on the assigned channel with an OSA / wavemeter (Section 5.16).
2. **Coarse align:** park rings near resonance with open-loop heater sweeps; check through/drop monitors.
3. **Close lock:** enable the feedback loop; verify capture on every λ at operating optical power.
4. **Stress neighbors / temperature:** max case T, neighbor heaters and traffic on (Sections 6.5 and 7.9). Confirm hold, not just capture.
5. **Close the link:** BER / TDECQ / sensitivity on the weakest lane first (Sections 7.4 and 7.7).

Bisect laser versus ring. If one λ unlocks or walks, change one actuator at a time:

- change laser TEC / current with ring heater fixed: if the error follows the laser, the source or its locker is wrong;
- change ring heater with laser fixed: if the error follows the heater, the ring, monitor PD, or thermal crosstalk is wrong;
- if both look fine but OMA is low, inspect MUX imbalance and connector/ORL (Table 6.2 and section 7.2.2).

Fleet telltales. Slow BER creep with rising bias on one line is often laser wear-out (Section 8.4). Sudden unlock under neighbor load with healthy LIV is thermal crosstalk or lock firmware. One dark lane with neighbors up is COD, FAU, or a single ring/heater fail; classify with Section 7.12 before you open an 8D on the wrong supplier.

6.8 What wavelength locking implies about an architecture

Because locking is only worth its complexity under specific conditions, its presence narrows the design space considerably.

Implication	Strength
Some form of WDM is in use	Near-certain: locking only matters with WDM.
Dense (D)WDM rather than coarse CWDM	Likely: locking implies tight spacing.
Microring-based silicon photonics with external multi-wavelength (CW-WDM) sources, often trending toward co-packaged optics	Common in modern AI interconnects (e.g. Broadcom and NVIDIA Quantum-X programs), driven by fiber-count pressure at scale.
Discrete DFB/EML DWDM (no rings)	Also possible; locking alone does not prove rings.

Table 6.3. What a wavelength-locking requirement implies.

The solid conclusion is that **WDM is present and wavelength control is central**; whether the implementation is ring-based silicon photonics with external multi-wavelength sources or discrete DFB/EML DWDM depends on the specific system.

6.9 Engineering lens

6.9.1 How it works

WDM buys fiber count back by stacking wavelengths, and the price is control: once the grid is tight or the modulator is resonant, laser and filter must stay locked. The chapter's lock loops, thermal budgets, and source specs all exist to hold that lock across temperature and neighbor load.

6.9.2 How it is measured

Start with an OSA or wavemeter to identify every source line, then sweep each ring or filter while recording through-port, drop-port, heater current, monitor-PD signal, and lock error. Measure insertion loss and adjacent-channel leakage per lane. Close the loop and repeat BER, OMA, and TDECQ under case-temperature ramps, neighbor heater activity, source-power spread, and restart. Capture range and hold range are different acceptance tests; Section 6.7 gives the order, and the source measurements follow the CW-WDM MSA methods [48].

6.9.3 How it fails

One degraded lane points toward its laser line, ring, filter channel, heater, monitor, or local fiber attach. All lanes moving together points toward the shared CW source, thermal controller, supply, polarization path, or common MUX. Capture failures occur during startup or a large temperature step. Hold failures occur after lock, often under neighbor heat, power droop, or control-loop interaction. Treat those as separate signatures.

One lane fails. Possible causes, roughly in order of how often each shows up in the field:

- **Wavelength locker failure:** the etalon-based error signal drifts or the locker photodiodes degrade, so the loop steers the laser to the wrong setpoint even though the laser itself is healthy (Section 6.4).
- **Ring drift:** the microring's own resonance walks off the assigned λ under local heating or process aging, independent of the laser (Section 3.14.3).
- **AWG issues:** an arrayed-waveguide grating channel shifts passband center or picks up excess adjacent-channel crosstalk, clipping one lane at the filter edge while neighboring channels stay clean (Section 6.3 and table 6.2). Distinguish from a laser or ring fault by sweeping the source across the passband and watching for a grid-alignment signature rather than a lock-error signature.
- **Thermal crosstalk:** an adjacent heater's disturbance exceeds this lane's hold-range budget while its own actuators read nominal (Section 6.5).

Intermittent failure. Failures that appear and clear on their own point to control-loop dynamics rather than a broken part:

- **Lock acquisition issues:** the loop fails to capture reliably from a cold or power-cycled state, so failures correlate with restarts or ELS hot-swaps rather than with steady-state operation (Section 6.7).
- **Heater control instability:** the thermal feedback loop oscillates or overshoots, often triggered by a control-loop bandwidth that fights the data path or a gain that was tuned for a different neighbor-load condition. Shows up as a heater current that hunts rather than settles.
- **Temperature excursions:** a transient case-temperature ramp (fan failure, load step, HVAC event) pushes the loop past its hold range momentarily; the lane recovers once the transient passes, which is the distinguishing signature versus a genuine hardware fault.

Failure mode: Wavelength drift

Symptoms. BER rises with temperature, one WDM lane moves, and lock error or heater demand grows.
Likely causes. Laser wavelength drift, ring resonance drift, thermal coupling, a saturated TEC or heater, or a weak monitor signal.
Measurements. OSA or wavemeter, heater and TEC current, lock error, neighbor activity, and per-lane BER.
Mitigations. Restore thermal headroom, widen capture only after noise is checked, improve calibration, and reduce shared thermal or supply coupling.

6.9.4 How it is debugged

First decide whether the fault affects one lane or all lanes. Freeze heater and laser actuators long enough to measure the open-loop spectrum. Move the source while holding the ring fixed, then move the ring while holding the source fixed. If neither follows the failing BER, inspect MUX loss, polarization, connector ORL, and the electrical lane. For intermittent unlock, align lock-error, heater, temperature, supply, and neighbor-traffic traces on one time axis. A control loop cannot be debugged from final register values alone.

Debug story

Observed. One lane failed only after adjacent lanes began traffic.
Investigation. The source line stayed on grid, while the suspect ring heater railed and lock error grew with neighbor temperature.
Finding. The optical source and receiver passed; the resonance was being pushed out of hold range.
Root cause. Thermal coupling from the adjacent heater exceeded the control-loop budget.
Resolution. The heater map and feed-forward terms were corrected, and neighbor-load hold testing became an ATP corner.

6.10 Interview and design review questions

Concept.

- Why is WDM preferable to adding fibers at scale?
- Why does a microring modulator require wavelength locking while a Mach–Zehnder does not?
- What happens if laser wavelength drifts by 2 nm in a 200 GHz-spaced WDM system?

Design.

- What are capture and hold ranges across process, voltage, temperature, and source-power spread?
- Which resource is shared across lanes, and what does its failure signature look like?
- How much heater and TEC headroom remains at the worst thermal corner?
- Can telemetry distinguish laser drift, ring drift, MUX loss, and monitor-PD error without opening the package?

Debug.

- One WDM lane fails while neighbors are healthy. How do you determine whether the cause is the laser, ring, filter, heater, or connector?
- All lanes unlock after a temperature step. What shared resource do you check first?
- Which calibration or startup error can escape production and fail only after a field restart?

Manufacturing and operations.

- What production test proves lock hold under worst-case neighbor heat?
- How do you qualify a new CW-WDM source lot against the ATP?
- What field telemetry distinguishes a source failure from a ring-bank failure?

Key idea. Wavelength locking is a control-system problem wrapped around an optical link. Measure the grid, capture, hold, shared thermal paths, and per-lane margin. Then use one-lane versus all-lane behavior to split the source, ring bank, MUX, thermal controller, and common supplies.

Chapter 7

Optical validation

A datasheet that closes on a quiet bench is not a product. *Validation* reduces uncertainty about whether a link meets its requirements across the temperatures, hosts, connectors, production spread, and lifetime the fleet will actually see. Passing tests is an output, not the purpose. This chapter walks the ladder from a single device to a deployed fleet, the engineering question at each stage, module and system bring-up under production-like corners, and the hypothesis-driven debug method the work demands.

7.1 The validation ladder

Optical programs fail in the same places again and again: a part that looks good in characterization but cannot bring up on a production host, or a module that passes ATP and then unlocks under neighbor heat. The practical way to avoid those gaps is a ladder of five stages, each answering a sharper question than the last. Skipping one creates escapes that surface later at higher cost.

	Stage	Question	Activity and instruments
1	Bring-up	Does it link?	Power-on, CMIS state machine, first light, CDR lock, pre-FEC BER on golden host (Section 7.9)
2	Characterization	How much margin, across corners?	Sweep T, V, ORL, aging; TDECQ, sensitivity, link budget; component LIV/RIN/SMSR (Sections 7.4 and 7.7)
3	Margin and interop	Does margin survive the fleet?	Production-representative corners: chassis thermal, host rails, neighbor load, dirty fiber, ELS hot-swap, target host/NIC/ASIC (Section 7.9)
4	Stress qualification	Does it survive its projected life?	HTOL, temperature cycle, humidity, ESD, mating cycles per GR-468, GR-1221, JESD47 (Sections 8.2 and 8.3)
5	Production readiness	Can the supplier build it at volume?	Multi-lot yield, SPC stability, ATP coverage, ATE correlation, first-article, DVT/PVT gates (Sections 8.9 and 8.10)

Table 7.1. The validation ladder: five stages from bench to fleet. A sixth concern, scaled deployment and fleet triage, runs continuously once material ships (Section 7.12). Field telemetry, RMA tracking, and FIT burn-down feed back into stages 3–5.

For every metric at every stage, name the instrument, the reference plane (Section 3.9), the pass criterion, and the failure signature. A number without a plane and a method is not a measurement.

Each stage removes a different uncertainty. Bring-up removes basic integration unknowns. Characterization maps response and variation. Margin and interop tests find combinations that consume power, noise, timing, spectral, or control headroom (Section 5.19). Qualification separates life and environmental risks. Production readiness proves that the chosen screens can control variation and escapes at volume. Do not use one stage as evidence for another.

7.2 The core IM/DD measurements

Once the ladder is clear, the measurement list is organized around isolation: transmitter, channel, and receiver. That split is older than PAM4. Long before TDECQ, field engineers learned that a dark link can be a dead laser, a dirty connector, or a dead TIA, and that guessing which one burns hours. Bisecting those three domains is still how you keep debug from turning into simultaneous retunes of everything.

7.2.1 Transmitter

Start with the light leaving the faceplate or the CPO fiber array. For PAM4, the headline metric is *TDECQ* (transmitter and dispersion eye closure quaternary): a reference equalizer is applied to the captured eye and the residual penalty is reported in dB (Section 7.4). Alongside it you read *OMA* (outer), extinction ratio, and *RLM* (level linearity), plus wavelength, spectral width, and RIN with a bias-driver versus feedback bisect (Sections 5.7 and 5.8 and section 4.3.1).

RLM Relative level mismatch: PAM4 level spacing uniformity. 1.0 is perfectly even; CEI often requires $\geq$0.95.

What else you add depends on the transmitter style. Laser-bearing modules need LIV, threshold, slope, SMSR, and chirp checks for DMLs (Sections 5.4 and 5.7). External MZMs (TFLN or silicon) need EO S_{21}, V_π, quadrature bias versus temperature, and driver-path eye symmetry at baud (Section 3.14.3 and section 7.4). Microring banks need resonance alignment, thermal tuning, neighbor crosstalk, and peaking-network EO S_{21} (Section 3.14.3 and chapter 6). The point of the list is not completeness for its own sake: it is knowing which instrument answers which hypothesis when the eye closes.

7.2.2 Channel

If the transmitter looks clean into a golden receiver and the link still fails, the channel is next. Insertion loss from fiber, connectors, MUX/de-MUX (Section 6.3), and on-chip coupling (Section 3.14.3) is the first ledger line; plan about 1–3 dB per fiber interface. Chromatic dispersion (Section 3.11) matters more on FR-class SMF sweeps than on short DR links. Optical return loss (ORL) is the quiet killer: reflections back into the laser raise RIN and seed burst errors, which is why many DR/FR modules still carry isolators while some CPO engines rely on design margin and monitor photodiodes instead (Section 4.3.1 and chapter 5). Fiber attach (MPO/MTP, FAU, grating couplers) shows up as both yield and reliability (Section 8.8).

7.2.3 Receiver

Receiver work asks whether the front-end can still decide bits at the OMA that survives the channel. Measure sensitivity (minimum OMA for the target BER) and stressed-receiver sensitivity with a calibrated stressor for margin (Section 7.5), plus overload before the TIA saturates. Underneath those system numbers sit the photodiode/TIA pair: responsivity, bandwidth, and input-referred noise (Section 4.5 and chapter 4).

7.2.4 Link level

Only after Tx, channel, and Rx each look sane do you trust a full-link verdict: pre-FEC BER against the KP4 threshold (Section 3.12), post-FEC BER, FEC symbol-error histograms, and a signed link-budget ledger from transmitter OMA to receiver sensitivity with penalties and remaining margin. That ledger is the document you argue from in DVT; the BER alone is not.

7.3 Measurement mapping

The metrics above are scattered across Tx, channel, Rx, and link level because that is how you debug them. Table 7.2 collects the same metrics into one reference: what is measured, the instrument, why it matters, and the failure signature that points back to it. Use the chapter subsections for the debug logic; use this table to look up an instrument fast.

7.4 Transmitter and dispersion eye closure quaternary (TDECQ)

TDECQ (transmitter and dispersion eye closure quaternary) deserves a closer look because it is the metric that governs PAM4 transmitter acceptance. It answers a specific question: *how much worse is this transmitter than an ideal one, after a realistic receiver has done what it can to clean up the signal?*

Metric	Instrument	Why it matters	Failure signature
OMA / TDECQ	DCA + reference equalizer	Scores transmitter quality against an ideal source; governs PAM4 acceptance (Section 7.4)	TDECQ rises with no average-power change; points to bandwidth, RLM, or bias
Extinction ratio / RLM	DCA level histograms	Sets OMA at fixed average power (Section 4.4); poor RLM inflates TDECQ	Compressed inner eyes with passing average power
Wavelength / SMSR	OSA or wavemeter	Confirms grid placement and single-mode purity (Section 5.7)	Side modes rise with T or age; line walks off grid
RIN	PD + electrical spectrum analyzer	Sets the BER floor $Q_{\max} = 1/\sqrt{\text{RIN} \cdot \text{BW}}$ (Section 4.3)	BER improves with power then flattens (a floor)
Insertion loss / ORL	Power meter + ORL meter	First ledger line; reflections raise RIN and seed bursts (Section 7.2.2)	Burst errors with stable average power; RIN rises with ORL
Receiver sensitivity	BERT + calibrated attenuator	Minimum OMA at target BER, the budget's bottom line (Sections 4.4 and 7.5)	Waterfall shifts uniformly right without flooring
Pre-FEC BER / FEC histogram	BERT + FEC counters	The single number every other metric feeds; histogram shape reveals mechanism (Section 3.12)	Clustered errors point to bursts; sparse errors point to Gaussian noise margin
CMIS state / DDM	Host or CMIS tool	Confirms management layer before blaming optics (Section 7.8)	Module never reaches ModuleReady; DDM disagrees with bench truth

Table 7.2. Measurement mapping: metric, instrument, rationale, and failure signature in one reference. Cross-references point to the full treatment of each metric elsewhere in this chapter.

7.4.1 How it is measured

1. **Capture.** The optical waveform is acquired on a sampling oscilloscope (a DCA) through a standardized reference receiver (a fourth-order Bessel–Thomson filter at roughly half the baud rate) so every lab measures the same bandwidth.

2. **Equalize.** A defined *reference equalizer*, a *feed-forward equalizer* (FFE) with a small, bounded number of taps (commonly up to five), is applied. This models the modest equalization a real receiver would perform, so the transmitter is not penalized for *ISI* the system can remove anyway.

3. **Histogram.** Two narrow vertical histogram windows are placed inside the symbol (near 0.45 and 0.55 of the unit interval). The noise distribution is evaluated at the three PAM4 decision thresholds.

4. **Compute.** The algorithm finds the RMS Gaussian noise σ that, added to the equalized signal, would just reach a target symbol error ratio of 4.8×10^{-4} (the SER consistent with the KP4 pre-FEC budget). TDECQ is the ratio, in dB, of the noise an *ideal* transmitter could tolerate to the noise *this* transmitter can tolerate:

$$\text{TDECQ} = 10 \log_{10} \left(\frac{\sigma_{\text{ideal}}}{\sigma_{\text{measured}}} \right).$$

A worse transmitter tolerates less added noise before failing, so σ_{measured} shrinks and TDECQ rises. Lower is better; typical 100–200G/lane specifications cap it in the low single-digit dB range.

7.4.2 Related quantities and failure signatures

SECQ Stressed eye closure quaternary: receiver-side stressed test, analog of TDECQ. Applies calibrated optical stress before BER test.

SECQ the stressed-eye counterpart used on the receiver side, adding a calibrated stressor to test margin rather than transmitter quality alone. See Section 7.5.

RLM (relative level mismatch) measures how evenly the four PAM4 levels are spaced; poor RLM (uneven levels) inflates TDECQ.

Because TDECQ folds several impairments into one number, the way it fails is diagnostic: uneven levels point to modulator or driver linearity (RLM); residual eye closure the equalizer cannot fix points to excess ISI or limited bandwidth; a noise-limited result points to low OMA, RIN, or reflections. For external MZMs (TFLN or silicon), also check EO S_{21} bandwidth, V_π and bias quadrature drift with temperature, and RF return loss on the driver-to-modulator path (Section 3.14.3). This is why *LPO*, which removes the module's own DSP, raises the stakes on transmitter quality: there is less downstream equalization to hide behind, so TDECQ-class metrics become even more central.

7.5 SECQ and stressed-receiver testing

SECQ (stressed eye closure quaternary) mirrors TDECQ on the *receiver*: instead of scoring transmitter quality with a reference equalizer, the test applies a calibrated optical stressor (attenuation, ISI template, optional RIN) and asks how much margin remains before the receiver hits the target pre-FEC BER.

Stressed-receiver sensitivity and overload tests (Section 4.4) use the same philosophy: bracket the operating OMA range with impairments the link will see in the field. For LPO, where the module DSP is gone, SECQ-style margin on the host-side receiver (Section 3.6 and section 9.5.1) is as important as TDECQ on the transmitter.

7.6 Instruments

A failing PAM4 link rarely announces which block is wrong. The bench is how you force the answer: each instrument isolates one failure mode, and the loopback topology tells you which side of the optical connector owns it.

DCA Digital communication analyzer: sampling oscilloscope used for eye diagrams and TDECQ.

DCA (digital communication analyzer): sampling scope for PAM4 eyes, TDECQ, OMA, RLM (Section 7.4). Needs a reference receiver filter matched to the PHY under test.

BERT Bit-error-ratio tester: instrument or ASIC function that generates PRBS and counts errors for pre-/post-FEC BER.

BERT bit-error ratio at pre- and post-FEC; FEC symbol histograms (Section 3.12).

OSA Optical spectrum analyzer: measures wavelength, SMSR, and side-mode structure of a laser source.

OSA wavelength, SMSR, side modes, RIN estimates (Chapter 5 and section 4.3.1).

VOA / stressor assembly calibrated attenuation and optional ISI for SECQ and sensitivity sweeps.

VOA Variable optical attenuator: calibrated loss element for sensitivity sweeps and stressed-receiver testing.

Power meter average power; pair with DCA for OMA.

Thermal chamber + TEC controller corner validation; essential for rings (Section 3.14.3 and chapter 6) and laser grids.

Use electrical loopback (host SerDes), optical loopback (Tx$\rightarrow$Rx on module), and golden-host/golden-module interop to bisect faults (Section 7.2.2). If the fault follows the module under golden-host swap, stop blaming the SerDes; if it stays with the host, stop opening laser FA.

7.7 Building a link budget

A link budget is a signed dB (or power) ledger from transmitter to receiver. For IM/DD short reach, start from outer OMA at the Tx faceplate and subtract every loss and penalty until you compare against receiver sensitivity (with target BER and KP4 pre-FEC threshold, Sections 3.12 and 4.4).

Typical ledger (single-mode DR class). Start from Tx OMA on the DCA (or from average power and ER). Subtract connector/coupling loss (1–3 dB per mated pair; fiber attach in CPO), fiber loss ($\sim$0.3–0.4 dB/km at 1310 nm; often negligible at 500 m), and MUX/de-MUX if WDM (2–5 dB per stage, Section 6.3). Add penalties for TDECQ (already in the OMA spec for many PMDs), dispersion (Section 3.11), and ORL/RIN reflection (Sections 4.3.1 and 7.2.2). Compare the remainder to stressed sensitivity at pre-FEC BER 2.4×10^{-4}, and keep 1–3 dB+ of production margin (more for fleet corners). Electrical budgets parallel this for the host-to-module path: COM and pre-FEC BER (Section 9.5.2 and section 3.6). LPO requires *both* ledgers to close without module DSP help.

7.8 Module management: CMIS

7.8.1 What CMIS is, and why an optical engineer cares

CMIS (Common Management Interface Specification) is the vendor-neutral management layer between a host (switch ASIC, NIC, or test fixture) and a pluggable or on-board optical module. The host talks to the module over a two-wire bus (TWI, I2C-like) through a paged register map: identity, power mode, alarms, per-lane

CMIS Common Management Interface Specification: module management, monitors, link training at 224G/448G.

monitors, and (at 224G/448G) link-training and host signal- integrity tuning extensions [86]. CMIS covers QSFP-DD, OSFP, COBO, ELSFP, and CPO engines that expose the same management contract.

You touch CMIS on every bring-up and every field triage. It is how the host learns what module is seated, when lasers may turn on, what Tx/Rx power and temperature look like, and whether a link failed at the management layer or the optical layer. A module that passes BER on a bench with lasers forced on but cannot reach ModuleReady on a production host will fail in the fleet (Section 7.9).

7.8.2 The module state machine

CMIS defines a module state machine the host drives. After presence detect and power application, the module stays in low power until the host releases `LPModeL` (or the CMIS 5.x `LowPwr` equivalent). The host reads identifier pages, clears sticky interrupts, and steps the module toward ModuleReady. Only then should Tx lanes or ELS lasers enable. ELSFP modules that emit before ModuleReady are a reject: the host did not authorize light (Section 5.14).

Data paths have their own state machines in CMIS 5.x (data path states, and network path states for media-side links). For bring-up, map the sequence in Section 7.9 onto these transitions: presence and Vcc, CMIS init and ModuleReady, enable light, optical path check, electrical lock, traffic, snapshot. Skipping step 2 and jumping to BER is how interop failures hide until production.

7.8.3 The memory map: pages, monitors, control

The lower memory map holds module identity, status, interrupt flags, and alarm thresholds. Upper pages hold application descriptors, lane controls, tunable-laser support, versatile diagnostics (VDM), and command-data-block (CDB) firmware messaging [86]. Hosts select an application (lane count, host interface, media type) before bringing up traffic.

DDM (digital diagnostic monitoring) is the telemetry layer you read at scale: per-lane Tx and Rx optical power, laser bias current when exposed, module temperature, supply voltage, LOS/LOL flags, and alarm/warning bits. On WDM parts you also get wavelength or channel ID. This is exactly what Section 7.12 reads before anyone reaches for a DCA. On bring-up, dump the register map you will use in the field and treat that dump as the golden reference for later RMA comparisons.

DDM Digital diagnostic monitoring: per-lane telemetry in CMIS (Tx/Rx power, bias, temperature, LOS/LOL flags).

7.8.4 CMIS as a validation deliverable

CMIS correctness is part of production readiness, not a firmware afterthought. ATP should prove the state machine reaches ModuleReady across voltage and thermal corners; DDM monitors track bench truth (CMIS Tx power versus DCA, module temperature versus case T); alarms fire at the right thresholds; and firmware revision is ECO-controlled like laser die revision (Section 8.10). Multi-source interop failures are often CMIS, media-type, or firmware mismatches, not marginal TDECQ (Section 7.9). At fleet scale the register map is the only eyes you have on a module in the rack. If CMIS is wrong, triage starts blind.

7.9 Module and system bring-up

Characterization proves a sample can meet metrics on a quiet bench. Bring-up proves a module (then a system) can be powered, managed, and linked the way production and the fleet will actually run it. Lab-to-production programs fail in the gap between those two if you only ever test golden hosts, clean fiber, and room-temperature faceplates.

Module bring-up sequence. Run this order on every new module (pluggable, ELSFP, or CPO engine with CMIS). Do not skip ahead to BER: a link that "works" with lasers forced on and CMIS ignored will fail the first host that enforces the state machine (Section 5.14).

1. **Presence and power.** Detect module (`ModPrsL` or equivalent). Apply rails in the host power sequence. Confirm Vcc and module temperature in CMIS. Stay in low power (`LPModeL` asserted or ModuleLowPwr) until management is sane.

2. **CMIS init.** Read identifier, vendor, firmware rev, supported media. Clear sticky interrupts. Confirm the state machine can reach ModuleReady (or the pluggable equivalent) under host command. Dump the register map you will use in the field; that dump is your bring-up golden reference.

3. **Enable light.** Exit low power; enable Tx lanes / ELS lasers only after ModuleReady. Confirm Tx optical power and laser bias (if exposed) against the power class. Lasers that come up before the host asks are a reject for ELSFP (Section 5.14).

4. **Optical path.** Mate fiber (clean first). Check Rx power and LOS. Optical loopback first if the host path is unproven.

5. **Electrical lock.** Bring host SerDes / module CDR. Confirm LOL clear, equalizer taps not pegged (Section 3.6). For LPO, this is the host eye and COM path (Sections 3.14.3 and 9.5.2).

6. **Traffic.** PRBS or live FEC traffic. Pre-FEC BER vs. KP4 threshold (Section 3.12); glance at FEC symbol-error histogram shape.

LOS Loss of signal: receiver (or host) flag that optical input is below the detect threshold.

LOL Loss of lock: CDR or SerDes unlocked; electrical path not recovered even if light is present.

PRBS Pseudo-random binary sequence: deterministic test pattern that exercises all bit transitions for BER and eye measurements.

7. **Quality snapshot.** On a Tx-capable path: OMA/RLM/TDECQ or module diagnostics that proxy them (Section 7.4). Record CMIS + BER + case T together so later triage has a baseline (Section 7.12).

Table 7.3 is the short form you can put on a lab wall.

Step	Action	Pass signal	Fail → first look
1	Presence / Vcc / temp	CMIS alive, rails in range	cable, seat, PSU
2	CMIS state machine	ModuleReady (or equiv.)	firmware, TWI, LPMode
3	Enable Tx / ELS	Tx power in class; lasers on only when commanded	bias driver, enable pin, APC
4	Fiber / Rx power	Rx power up; LOS clear	dirty MT, polarity, break
5	CDR / SerDes lock	LOL clear; taps not saturated	host SI, LPO COM, retimer
6	Pre-FEC BER	below KP4 target with margin	Tx quality, ORL, Rx sensitivity
7	Snapshot	CMIS dump + BER + T logged	(needed for RMA later)

Table 7.3. Module bring-up checklist. LOS = loss of signal; LOL = loss of lock. Limits come from the ATP and PMD, not from this table.

Production-representative corners. Bench corners (T, V) are necessary and not sufficient. Before you call DVT or PVT done, run the corners that match how the fleet will abuse the link. Table 7.4 is the minimum set for IM/DD + laser programs.

Corner	What to run	Why it catches	Points to
Chassis thermal	Module in target rack/sled at airflow and power load; not only a quiet chamber on a bench fixture	Faceplate T and TEC load differ from chamber setpoints	derate, TEC, ring unlock
Host rails live	Bias / CMIS powered from host supplies with SerDes traffic on	Switching noise into laser bias looks like RIN (Section 5.8)	PSRR, ground, APC
Dirty fiber / ORL	Controlled contamination or ORL stress on MT/FAU; clean vs dirty BER	Field installs are not lab-clean; ORL raises RIN and bursts	connector, isolator, feedback
Cable plant	Production fiber length, MPO count, and bend radius	Extra loss and reflections eat margin the ledger assumed	link budget (Section 7.7)
ELS hot-swap	Pull/replace ELSFP under traffic (or under controlled traffic stop per CMIS)	Service action the architecture promised (Section 5.14)	state machine, mate cycles
Neighbor load	Adjacent modules/lanes at full traffic and max case T	Crosstalk, shared supply droop, thermal crosstalk on rings	WDM lock, SI, PSU
LPO / linear path	Host COM and pre-FEC BER without module DSP crutch	LPO fails here first (Sections 3.14.2, 3.14.3 and 9.5.2)	host FIR, module linearity
Voltage corners	Host Vcc min/max with traffic	Brown-out and CMIS glitches	power design, ATP

Table 7.4. Production-representative corners. A quiet BERT at 25 °C with pristine fiber is characterization, not production readiness.

System bring-up. A module that passes on a golden host can still fail in a real chassis:

- **Host path:** run the same sequence on the target NIC/switch ASIC SerDes, not only the lab BERT. LPO and half-retimed modules expose host FIR/CTLE mistakes that a retimed module hid (Section 9.5.1 and section 9.3).

- **Multi-lane / multi-module:** bring all lanes on a port, then neighbors in the same cage or tray. Watch thermal rise, supply droop, and CMIS temp alarms when the tray is loaded.
- **Golden swap:** known-good module in the suspect host slot, then suspect module in a known-good slot. That single swap splits host vs. module before you open FA (Section 7.12).
- **Interop:** at least one other vendor host or module if the program claims multi-source. Interop failures are usually CMIS, media type, or electrical eye, not laser physics.
- **ELS / CPO:** external laser modules add a second bring-up: ELSFP state machine and optical mate to the engine, then engine bring-up with light present (Sections 5.14 and 9.10). A dark engine with a healthy ELS is an optical connector or FAU problem until proven otherwise.

Exit criteria before "bring-up done." Call module bring-up done only when: CMIS state machine and enable sequence are correct; pre-FEC BER meets target on the *target* host with margin; a CMIS+BER+T snapshot is filed; and at least the chassis-thermal, host-rails, and ORL corners in Table 7.4 have been run on a representative unit. Call system bring-up done when golden-swap has split host vs. module issues and multi-lane / neighbor load has not opened a new failure mode. Everything after that is characterization depth, supplier gates (Section 8.10), or fleet triage (Section 7.12).

Key idea. Bring-up is a sequence (presence $\rightarrow$ CMIS $\rightarrow$ light $\rightarrow$ lock $\rightarrow$ BER $\rightarrow$ snapshot), then a system proof on the real host, then production-representative corners (chassis thermal, host-rail noise, ORL, ELS hot-swap, neighbor load). A quiet bench pass is not DVT.

7.10 The debug mindset

Debug at this level is data-driven, not opinion-driven. The method is disciplined bisection: change one domain at a time, and let the measurement tell you whether the transmitter, the channel, or the receiver moved.

1. Isolate transmitter versus channel versus receiver, using loopbacks.
2. Sweep temperature and voltage to expose corner-dependent failures.
3. Correlate failures to DSP equalizer tap values (Section 3.6) and FEC symbol-error statistics (Section 3.12); these tell you *how* the link fails.

The third step is where modern PAM4 links differ from older eye-mask work. Tap saturation and FEC histograms often reveal the failure mode before a single waveform screenshot does. Treat those as primary evidence, not as afterthoughts logged once BER already fails.

[1]

[1] Worked example. "The link fails only at high temperature." Candidate causes: laser wavelength drift off the grid or off a ring resonance; TEC saturation; EAM bias shift; elevated receiver thermal noise; LIV rollover (Sections 5.7 and 5.13). You bisect with measurements (spectrum for wavelength, LIV for the laser, receiver sensitivity at temperature) rather than guessing.

7.11 The debugging fork in validation

Apply the debugging fork (Section 4.8) before sweeping parameters or changing firmware: check the power meter or CMIS Rx power monitor first. If power moved, the fault is in the optical path (laser, coupling, connector, fiber, MUX); if power held but BER or TDECQ worsened, it is signal quality (bandwidth, noise, jitter, bias, equalization, reflection). This one check prevents the most common validation mistake: retuning an equalizer or laser bias when the real cause is a dirty connector. Then check which margin ledger moved (Section 5.19) before descending to component physics.

7.12 Fleet and field triage

Lab debug asks: *what is broken on this unit?* Fleet triage asks: *which bucket does this failure belong in, and who owns the fix?* Optical programs at fleet scale own that split across performance, reliability, and manufacturability. Wrong bucket wastes weeks (sending a contaminated connector to laser FA, or rewriting a SerDes FIR when the laser is rolling over).

Three buckets. Classify every field issue before deep root-cause work:

Performance the design or operating point does not close the budget under the conditions seen in the fleet. Examples: TDECQ/RLM marginal at case temperature, host COM tight on LPO, ring unlock under thermal crosstalk, ORL-driven RIN that the architecture assumed away. Fix is usually retune, derate, firmware, or a design/spec change (Section 7.4 and sections 3.14.3 and 9.5.2).

Reliability the unit met spec at ship and later degraded. Examples: LIV threshold rise, SMSR collapse, EAM bias creep, COD, TEC wear, epoxy creep on fiber attach. Fix is Arrhenius-backed life projection, burn-in/screen, derating, or field-replaceable lasers (Sections 5.13, 5.14, 8.2 and 8.4).

Manufacturability a subpopulation fails early or never met the ATP; the issue tracks lot, date code, supplier site, or assembly step. Examples: FAU misalign yield cliff, solder void on a driver die attach, incoming DPPM spike, CMIS register map mismatch on one firmware rev. Fix is SPC, ATP tighten, first-article, DPA, and 8D/CAPA with the supplier (Sections 8.8 and 8.10).

A single symptom can sit in more than one bucket until you bisect. The tree below forces the split with telemetry first, then a short bench confirm, then an RMA label. Chapter 10 expands the same method into symptom-led bench and fleet procedures.

Telemetry you actually read. At scale you rarely start with a DCA. Start with what the host and module already report:

- *CMIS* monitors and alarms: module temperature, supply rails, Tx/Rx optical power, laser bias (when exposed), wavelength or channel ID on WDM parts, LOS/LOL flags, and interrupt history (`IntL` on ELSFP; Section 5.14).
- Host link state: CDR lock, pre-FEC BER, FEC symbol-error histogram shape (Section 3.12), equalizer tap saturation (Section 3.6).
- Fleet context: rack position, case temperature, time since install, date code / lot, neighbor-link correlation (one bad fiber vs whole tray).

Decision tree (symptom $\rightarrow$ bucket). Table 7.5 is the working map. Read left to right: observe, check telemetry, pick a provisional bucket, then run the named confirm measurement before you open an RMA or change a design rule.

How to walk an incident (order of operations).

1. **Stabilize and capture.** Freeze CMIS dump, host BER/FEC counters, rack T, and install age before anyone reseats the module. Reseating destroys connector evidence.
2. **Localize.** One link vs tray vs rack. Tray-wide points at power, cooling, or a shared ELS. Single-link points at that module, fiber, or host lane.
3. **Classify** with Table 7.5. Write the bucket on the ticket before FA starts.
4. **Confirm** with the smallest measurement that can falsify the bucket (golden swap, clean/inspect, LIV, TDECQ, ORL). Do not skip to DPA.
5. **Act.**
    - Performance: change operating policy (derate, FIR, lock loop) or open a design/spec defect.

Symptom	First telemetry check	Bucket	Confirm on bench / FA	Typical fix owner
Link never comes up (fresh install)	CMIS presence, Vcc, Tx power flat-line, LOS	Mfg or install	Visual fiber/connector; golden module swap; CMIS dump	Ops install; supplier ATP if lot-correlated
Intermittent LOS / burst errors	Rx power dropouts; FEC bursts; ORL events	Perf (ORL) or mfg (contam.)	Clean/inspect MT; ORL meter; RIN vs ORL (Section 5.8 and section 4.3.1)	Ops cleaning; packaging if repeat RMA
Pre-FEC BER high, power OK	Tap saturation; RLM/TDECQ if logged; case T	Perf	DCA TDECQ/RLM; host COM; LPO vs retimed path (Section 7.4 and section 9.5.2)	Host SI / module Tx design
BER rises only at high case T	Module temp alarm; Tx power drop; λ walk	Perf or reliability	LIV at T; OSA grid; TEC current; EAM bias (Section 5.13)	Derate / TEC / laser supplier
Slow BER creep over weeks/months	Bias current up for same Tx power; SMSR if monitored	Reliability	LIV/SMSR vs ship ATP; Arrhenius lot history	Laser wear-out; ELS replace
Sudden hard fail, was healthy	Last good CMIS snapshot; neighbor links OK	Reliability (COD) or mfg (ESD)	Dark LIV; DPA on facet/solder; date-code cluster?	FA + supplier 8D
One date code / site fails early	Lot Pareto; burn-in escape rate	Mfg	Incoming SPC vs ATP; FA on sample of lot	Supplier CAPA; hold shipment
WDM / ring unlock, power OK	Channel ID; thermal of neighbors; lock-loop status	Perf	Resonance tune; crosstalk; CW-WDM line power (Sections 5.16, 6.5 and 6.7)	Lock firmware / thermal design
ELSFP swap restores link	Old module CMIS vs new; connector cycles	Reliability or mfg (connector)	Inspect MT; mating-cycle count; laser LIV in returned module (Section 5.14)	Laser vs connector split in FA

Table 7.5. Fleet triage map: symptom to provisional bucket to confirm measurement. Perf = performance (design/operating point); reliability = time-dependent wear; mfg = lot/process/install excursion.

- Reliability: replace (ELSFP hot-swap when available), update FIT burn-down, tighten burn-in or derate (Section 5.13).
- Manufacturability: quarantine lot, incoming hold, supplier 8D with DPA photos and ATP deltas (Section 8.10).

6. **Close the loop.** Feed the signature back into ATP and CMIS alarm thresholds so the next incident trips earlier.

Worked paths (three common tickets). "*High temp only.*" CMIS shows module near thermal limit and Tx power sagging. Bucket starts as performance (thermal design / derate) until LIV at temperature shows threshold rise matching an aged lot, which flips it to reliability. Measure OSA wavelength before blaming the laser: a ring unlock is still performance (Section 3.14.3 and chapter 6).

"*Random burst errors, average power fine.*" Check FEC histogram for clustered errors and CMIS for Rx power dropouts. Clean and measure ORL. If RIN rises with ORL, it is performance/architecture (feedback). If ORL is fine and bursts track a date code, it is mfg (intermittent fiber attach). If bursts grow over months at fixed ORL, suspect laser or driver aging (Sections 5.8 and 5.13).

"*ELSFP replace fixed it; returned module looks alive on the bench.*" Alive LIV with high ORL sensitivity or dirty MT face means connector/ORL (mfg/ops), not laser death. Dead or kinked LIV means reliability. Split those RMA codes explicitly or your FIT math will blame the wrong wear-out mode (Sections 5.14 and 8.8).

RMA labels that keep FIT honest. RMA codes should be distinct, not a single "optics fail":

RMA Return merchandise authorization: field-failed unit returned for FA. Distinct failure codes keep fleet FIT honest.

- laser wear-out (LIV/SMSR/EAM aging confirmed);
- COD / sudden dark;
- connector / contamination / ORL;
- fiber attach / FAU;
- driver / bias electronics;
- host / SerDes / LPO eye (not module);
- NFF (no-fault-found; track these; high NFF means bad triage).

NFF No fault found: RMA unit passes all tests on return. High NFF rate points at triage or intermittent connector faults.

NFF rate and lot Pareto are as important as FIT. A rising NFF with clean LIV points at install practice or intermittent connectors, not Arrhenius.

7.13 Engineering lens

7.13.1 How it works

Validation is a chain of evidence, not a single pass: a number means nothing without its reference plane, its corner, and its method. The chapter's ladder, instruments, and triage tree are that evidence chain from bench to fleet.

7.13.2 How it is measured

Use the least complex instrument that can falsify the current hypothesis. Table 7.2 maps every key metric to its instrument, rationale, and failure signature in one lookup; the bring-up sequence (Section 7.9) orders those instruments into a workflow.

7.13.3 How it fails

Validation fails when the setup, sample, or acceptance rule does not match the product. Common misses are a stale calibration, the wrong reference plane, a golden host that hides interop risk, pristine fiber that hides ORL sensitivity, short BER dwell, one lane tested without neighbors, and chamber temperature used as a substitute for measured case temperature. These are test escapes even when the device physics is sound.

Failure mode: Low optical power

Symptoms. A lane is dark or below its launch-power limit.
Likely causes. A laser or enable fault, coupling loss, connector contamination, fiber polarity, calibration error, or a power-meter setup mistake.
Measurements. Known source and meter, inspection scope, CMIS state and bias, power at successive planes, and a golden fiber or module.
Mitigations. Correct the setup first, then repair the failing source, attach, connector, or control path. Add the signature at the earliest production test that can catch it.

7.13.4 How it is debugged

Preserve the failing state and record software, firmware, calibration, fixture, cables, temperature, and supply. Verify the meter with a known source. Walk from power to spectrum to waveform to BER, moving one reference plane at a time. Use a golden swap to split host, module, and fiber. Only then stress temperature, voltage, ORL, and neighbors. Every corrective action needs a repeated failing test, a repeated passing test, and a guard against recurrence in ATP or telemetry.

Debug story

Observed. A new module lot showed low optical power on one station.
Investigation. The same units passed on a second station. A known source exposed an offset in the first power-meter path.
Finding. The lot was good, and the station was reading low.
Root cause. A reference jumper had been replaced without updating the path-loss calibration.
Resolution. The station was recalibrated, jumper identity was placed under change control, and a start-of- shift source check was added.

7.14 Interview and design review questions

Concept.

- Why is a passing BER on a golden bench not sufficient for production readiness?
- What is the difference between characterization and validation?
- Why does LPO raise the stakes on transmitter TDECQ?

Design.

- What requirement does each test prove, at which plane, and with which uncertainty and guardband?
- Which corner is represented only by a lab fixture rather than the target host or chassis?
- What is the fastest measurement that can falsify each top risk?
- Which setup error can make a bad unit pass or a good unit fail?

Debug.

- A new module passes on the bench but fails on the production host. What is your first measurement?
- BER is high but optical power looks fine. Apply the debugging fork (Section 7.11): what do you check next?
- A module works on host A but fails on host B. How do you determine ownership?
- Are raw data, calibration state, firmware, and sample identity stored well enough to replay a failure months later?

Manufacturing and operations.

- What is the minimum set of corners that proves production readiness?
- How do you detect tester drift before it becomes a yield cliff or a field escape?
- What exit criteria distinguish DVT from PVT?

Key idea. Validation is a chain of evidence. Start with calibrated power and management state, move through spectrum and waveform, then trust BER only after the blocks and reference planes are known. Run the target host, chassis, fiber, and neighbor corners before calling the product ready.

Chapter 8

Reliability and manufacturing at scale

A link that closes in the lab can still fail the business case if lasers die in the field or suppliers cannot hold yield. At gigawatt, multi-generation scale, reliability and manufacturability stop being afterthoughts and become design constraints: they decide whether you put the laser on the ASIC package or in a replaceable module, how hard you derate, and what ATP language you freeze with partners. This chapter covers the vocabulary of failure at scale, the qualification flows that project field life, and the supplier-execution work these systems demand.

8.1 The language of scale: FIT and DPPM

Fleet arguments need two different numbers. One is about life in the field; the other is about quality at the factory door.

FIT Failures in time: failures per 10^9 device-hours. Fleet FIT times link count sets expected failures per day.

FIT (failures in time) failures per 10^9 device-hours. Multiply a per-laser FIT by the number of lasers in a fleet and by hours to estimate failures per day.

DPPM (defective parts per million) the manufacturing-quality counterpart, measured at incoming or outgoing inspection.

[1]

[1] An order-of-magnitude exercise: a $100{,}000$-GPU cluster with several optical links per GPU has on the order of 10^5–10^6 lasers. Even a small per-laser FIT then implies a nontrivial number of link failures per day, which is exactly why field-replaceable external laser sources and redundancy are attractive.

8.2 Qualification flows

Optoelectronics inherited a common qualification language from telecom: *Telcordia GR-468-CORE*[80]. The core stress tests still show up on every laser and module program:

- HTOL (high-temperature operating life) and burn-in.
- Temperature cycling and damp heat.
- Electrostatic-discharge and mechanical stress.

Arrhenius acceleration underpins life projection: raising temperature accelerates wear-out by a factor set by the activation energy, so a bounded high-temperature test projects years of field life. Screening (burn-in) removes infant-mortality parts before deployment.

GR-468 in practice. Telcordia GR-468-CORE is the common qualification language for optoelectronic modules and discrete lasers. For optical engineers the actionable pieces are test-plan alignment (map your ATP to GR-468 stress sequences such as HTOL, temperature cycle, damp heat, and ESD so supplier and customer agree on pass/fail), activation energy (FIT projections use Arrhenius acceleration; document E_a and confidence bounds when converting 1000-hour HTOL to field years), sample-size humility (qualification lots are small; production SPC catches drift that qual missed, Table 8.2), and boundary clarity: qualify the laser die, the hermetic package, and the module assembly separately when failures split across those boundaries (Sections 5.13, 5.14 and 8.8).

GR-1221: the passive-component companion. GR-468 covers active optoelectronics. Its companion, *Telcordia GR-1221-CORE* (Generic Reliability Assurance Requirements for Passive Optical Components), covers the parts GR-468 does not: connectors, fiber couplers, WDM filters and MUX/DEMUX, splitters, and isolators [2] [81]. It uses the same style of stress sequence (damp heat, temperature cycling, mechanical, and aging tests) but scores pass/fail on insertion loss and return loss rather than on LIV. A short-reach link that leans on an on-package or blind-mate MUX and on external multi-wavelength sources carries a passive reliability budget that lives in GR-1221, not GR-468 (Section 8.8 and chapter 6). Split the qual the same way you split the FIT: active laser die under GR-468, silicon under JESD47, passive optics under GR-1221.

[2] Telcordia GR-1221-CORE: reliability assurance for passive optical components (connectors, couplers, WDM filters, isolators).

ATP sketch: EML module or ELSFP. A short acceptance sketch lives with the qual hooks in Section 5.14; the full ATP-as-contract, SPC, and 8D workflow is in Section 8.10. Failures that pass qual but fail field usually sit in derating policy or connector contamination (Sections 5.13 and 7.12).

8.3 Electronics reliability: driver, TIA, and DSP silicon

GR-468 covers the optoelectronic parts of the link: the laser die, the photodiode, and the hermetic or non-hermetic package around them. The modulator driver, TIA, retimer, and DSP (Section 3.14.3 and section 4.5) are ordinary CMOS or SiGe BiCMOS ICs, and they wear out and fail by a different, better-documented set of mechanisms. Treat them with the semiconductor industry's own qualification language, not with Arrhenius laser-aging math borrowed from Section 5.13.

JESD47: the silicon-side GR-468. JEDEC JESD47 is the baseline stress-test-driven qualification flow for a new IC, a device family, or a process change: temperature cycling, HTOL, HTSL (high-temperature storage life), autoclave or HAST (highly accelerated stress test) for moisture, and mechanical shock and vibration [3][93]. It plays the same role for driver and TIA silicon that GR-468 plays for the laser: a common list of stresses that a supplier runs once and a customer accepts instead of renegotiating a qual plan on every design win.

JESD47 JEDEC's baseline IC stress-test qualification standard: temperature cycle, HTOL, HTSL, autoclave/HAST, mechanical shock and vibration. The silicon-side counterpart to Telcordia GR-468 for optoelectronics.

[3] JEDEC JESD47M: baseline IC stress-test qualification flow (Aug 2025).

ESD and latch-up: failure modes GR-468 does not test. Two mechanisms are specific to ICs and absent from the laser-side wear-out map in Table 8.1:

ESD a discharge event during handling or assembly damages a gate oxide or junction. Component-level classification uses the human-body model (HBM) and charged-device model (CDM) test standards, *ANSI/ESDA/JEDEC JS-001* and *JS-002* [4][94]. A driver or TIA datasheet HBM/CDM rating is the number that protects the part on the factory floor, at fiber-attach and wire-bond stations where a laser die is also exposed.

[4] ANSI/ESDA/JEDEC JS-001 (HBM) and JS-002 (CDM) component ESD test standards.

Latch-up a parasitic thyristor structure in CMOS turns on under an overvoltage or current-injection event and holds a low-impedance path until power is cycled. *JESD78* defines the overvoltage and ± 100 mA current-injection test that classifies susceptibility by supply and signal pin [5][95]. A latched driver IC can look like a dead laser on the bench (no light, no LIV signature) until you check the supply current instead of the optical path.

JESD78 JEDEC IC latch-up test: overvoltage and ± 100 mA current-injection stress that classifies a CMOS/BiCMOS die's susceptibility to a parasitic-thyristor low-impedance latch.

[5] JEDEC JESD78F.02: IC latch-up test, overvoltage and ± 100 mA current injection.

Both mechanisms are 100%-screen or design-margin items, not something you project with an activation energy. If a driver fails ESD or latch-up in the field, that is a manufacturability or design-margin bucket item (Section 7.12), not a wear-out FIT argument.

AEC-Q100: a borrowed grade, not a requirement. *AEC-Q100* is the automotive industry's qualification standard for ICs, built on the same JEDEC JESD47/JESD22 stress methods with tighter ESD targets and named temperature grades from Grade 3

AEC-Q100 Automotive Electronics Council IC qualification standard: JEDEC stress methods plus tighter ESD targets and named temperature grades (Grade 3 to Grade 0). Not required for datacenter optics; useful as a borrowed signal when a driver/TIA die also ships an automotive part number.

(–40 to 85°C) up to Grade 0 (–40 to 150°C) [6][96]. Datacenter optics does not require Q100; the fleet lives in a controlled data hall, not an engine bay. It is still a useful signal: a driver, TIA, or retimer die that also ships in an automotive part number typically carries a published Q100 grade, and that grade is a fast proxy for the ESD/latch-up margin and temperature-cycle depth behind the datasheet, without re-running the qual plan yourself.

[6] AEC-Q100 Rev-J: automotive IC qualification on JEDEC methods, tighter ESD and thermal grades.

Where this lands in the ATP. Fold IC-level qual into the same acceptance and SPC structure used for the laser (Table 8.3 and section 8.10): require the supplier's JESD47 qual report and HBM/CDM/latch-up ratings for driver and TIA die at DVT, add an ESD handling audit to the incoming-QC checklist alongside laser LIV/SMSR sampling, and treat a driver/TIA silicon revision the same way you treat a laser die revision or a CMIS firmware rev: an ECO that needs first-article requalification, not a silent BOM swap.

8.4 Wear-out modes to know

Arrhenius math, derating, and the worked FIT example live in Section 5.13. This section is the mechanism catalog: how each failure shows up in ATP and telemetry, and which triage bucket owns it (Section 7.12). Do not run process CAPA on a wear-out part, and do not burn FIT math on a dirty connector.

Infant mortality versus wear-out versus packaging. Field failures come in three clocks, and mixing them up wastes CAPA. Infant mortality is early fails from latent defects; burn-in and HTOL screens remove them before ship (Section 8.2). Wear-out is gradual or sudden end-of-life in the laser or EAM under temperature, current, and optical power, projected with Arrhenius and derating (Section 5.13). Packaging and assembly faults (FAU align, epoxy creep, solder voids, connector wear) often dominate field returns once lasers are screened (Section 8.8). Destructive physical analysis (facet cross-section, EDX, FAU section) is required when the signature is ambiguous or when you need evidence for supplier 8D (Section 8.10).

Mechanism map. Table 8.1 is the working list for laser-bearing modules and CPO/ELS paths. Customize limits in the ATP; keep the classification discipline.

Catastrophic optical damage (COD) is the sudden facet failure under optical or electrical overstress. Gradual facet and active-region degradation move threshold and slope on a slower clock. EAM aging shifts the absorption curve and shows up as transmitter and dispersion eye closure quaternary (TDECQ) or RLM drift before the DFB LIV looks dead. TEC and lock failures mimic optical wear-out until you bisect heater versus laser (Section 6.7). Coupling and FAU faults belong with packaging, not with Arrhenius E_a arguments.

Mechanism	Observable	ATP / telemetry	Triage bucket
COD (facet)	Sudden dark or hard fail; was healthy	Dark LIV; DPA facet; date-code cluster?	Reliability (COD) or mfg (ESD)
Gradual facet / active region	I_{th} up, slope down over life	LIV trend vs ship ATP; HTOL lot history	Reliability (wear-out)
SMSR collapse	Side modes rise; modal noise / BER	OSA SMSR vs floor at T	Reliability; watch aging
EAM aging (EML)	TDECQ/RLM creep at fixed bias	EAM bias sweep + DCA; bias creep log	Reliability (EAM)
RIN rise	BER floor up; feedback sensitive	RIN @ ORL; isolator / connector check	Perf if ORL; reliability if isolator
TEC / thermal control	Unlock or λ walk; LIV may look fine	TEC current, case T, lock status	Perf (lock) or reliability (TEC)
Coupling / FAU / solder	Loss step, intermittent LOS, shock-related	ORL, mate cycles, DPA FAU/solder	Manufacturability / packaging
Driver/TIA latch-up (ESD)	Sudden hard fail; no light, no LIV signature; supply current spikes	Supply current vs bias; JESD78 rating; date-code cluster?	Mfg (ESD) or design margin
Connector wear / contamination	ORL creep after repeated mate cycles; RIN floor rise	Mate-cycle count vs IEC 61300-2-2 rating; endface grade (IEC 61300-3-35)	Manufacturability / packaging

Table 8.1. Wear-out and packaging mechanisms versus observables. Arrhenius projection and derating for the laser rows: Section 5.13. Electronics stress qualification: Section 8.3. Connector reliability: Section 8.8. Field classification workflow: Section 7.12.

8.5 The reliability bathtub: three failure regimes

Field failures follow three regimes with different clocks and different fixes. Mixing them wastes corrective action on the wrong mechanism.

Infant mortality (early life) Latent defects from manufacturing: weak solder joints, marginal laser die, contamination, firmware bugs that trip on first thermal cycle. Burns down rapidly with time. Fixed by burn-in, screening, tighter ATP, and first-article control (Section 8.2).

Useful life (constant rate) Random failures at a roughly steady FIT: cosmic-ray-induced single-event upsets, handling damage during unrelated service actions, and isolated material defects that passed all screens. Fixed by redundancy, field-replaceable modules, and design margin. Note: connector wear and contamination accumulate with mate cycles and are usage-driven degradation (Section 8.8), not constant-rate random failures; budget them separately from the flat-hazard FIT.

Wear-out (end of life) Physics-driven degradation: laser facet, active region, EAM absorption curve shift, TEC aging, epoxy creep. Onset depends on temperature, current, optical power, and time. Fixed by derating, Arrhenius-based life projection, and planned replacement intervals (Section 5.13).

A rising failure rate after years of service is wear-out and calls for replacement planning, not supplier 8D. A cluster of early failures on a new lot is infant mortality and calls for tighter screens. A steady trickle with no date-code correlation is useful-life random and calls for redundancy and fast repair. The triage tree in Section 7.12 forces this classification before corrective action starts.

8.6 Yield analysis

Yield is not one number. It splits by stage and by failure mode, and each split points at a different owner.

Wafer / die yield Process-limited: waveguide loss, ring resonance spread, heater shorts, photodiode dark current. Caught at wafer probe. Owner: foundry SPC.

Assembly yield Packaging-limited: fiber-array attach alignment, solder voids, wirebond pull, epoxy placement. Caught at module ATP. Owner: assembly supplier.

Test yield (first-pass) ATP-limited: units that fail one or more acceptance criteria on first pass. May include measurement-system false rejects (gauge R&R). Owner: test engineering.

Escaped DPPM (post-screen field failures) Units that passed all screens but fail in the fleet. Each escape is either a preventable coverage gap (the screen exists but missed the defect) or a residual latent failure that no cost-effective screen separates from good units. Owner: quality and reliability engineering.

Track yield by ATP row, lot, supplier site, tester, and date code. A yield drop that correlates with one tester is likely a measurement problem. A yield drop that correlates with one supplier lot is likely a process problem. A yield drop with no observed correlation requires further investigation: verify gauge repeatability, expand stratification (shift, fixture, material lot, firmware), and test guardband or specification mismatch as hypotheses before concluding. Do not open supplier corrective action until the measurement system is cleared (Section 8.9).

8.7 Escaped defect analysis

An escaped defect is a unit that passed every production screen and failed in the field. Post-screen field failures split into two categories with different corrective actions:

Preventable coverage escapes. A defect that a cost-effective screen could have caught but did not, because the screen was absent, miscalibrated, or insufficiently stressed. Each one requires an ATP or process-control change. Common mechanisms in optical modules:

- **Contamination:** particles trapped during assembly that only move under thermal cycling or vibration, shifting coupling or raising ORL intermittently.

- **Marginal alignment:** fiber-attach or FAU position within spec at room temperature but outside at the thermal corner, creating a temperature-dependent loss that the ATP chamber did not run.
- **Weak solder or wirebond:** passes pull test but fatigues under repeated thermal cycling, creating intermittent opens or increased resistance.
- **Firmware corner:** a state-machine path or calibration table entry that production never exercises but the fleet hits on a specific boot sequence, temperature bin, or host interaction.
- **Packaging stress:** residual mechanical stress from underfill cure or lid attach that relaxes over time, shifting alignment or birefringence.

For each preventable escape, trace the failure signature back to the earliest point in the production flow where it could have been caught, and add or tighten the screen there.

Residual latent failures. A defect that no cost-effective screen can separate from good units at the time of test. Examples include cosmic-ray-induced single-event upsets, rare material inclusions below detection limits, and process-marginal units that sit inside all guardbands but fail under a specific field combination of temperature, ORL, and neighbor load. These go into the residual FIT model, not into ATP. Document why no reasonable screen exists so the decision is auditable and revisitable as test technology improves.

8.8 Photonic packaging and module-level failures

Fleet FIT is not only laser wear-out. Once lasers are screened and derated, module and packaging failures often dominate field returns: the part that shipped with a clean LIV still loses light after shock, humidity, or a thousand ELSFP mate cycles. Fiber attach and FAU alignment fail from shock, humidity ingress, and epoxy creep; CPO fiber-array units add assembly steps that wafer test cannot catch (Section 9.10). Hybrid stacks (TFLN-on-Si, InP laser on Si, flip-chip drivers) introduce solder voids, underfill cracks, and RF return-loss drift (Section 3.14.3). Thermal paths matter too: uncooled datacom versus liquid-cooled XPO/CPO, and TEC failure that looks like wavelength drift off grid or off ring (Section 3.14.3 and chapter 6).

Connector reliability: MPO, mating cycles, and endface quality. Multi-fiber connectors are the highest-touch mechanical interface in the fleet: every ELSFP swap, every fiber-attach unit (FAU) rework, and every cable-plant install mates and unmates an MPO. The MPO/MT ferrule family (rectangular, 6.4 mm $\times$ 2.5 mm, guide-pin aligned, 8/12/16/24 fibers per row) is standardized in *IEC 61754-7*, split into

FAU Fiber array unit: a precision V-groove assembly that mates multiple fibers to a PIC edge or grating coupler array.

MPO Multi-fiber push-on: standard multi-fiber connector for parallel optics (8, 12, 16, 24, or 32 fibers per ferrule).

one-fibre-row and two-fibre-row parts [7][97]. That standard fixes geometry, not lifetime; lifetime comes from two companion test methods. *IEC 61300-2-2* specifies the mate/unmate cycling test connector datasheets are rated against, and *IEC 61300-3-35* grades endface scratches, pits, and debris into pass/fail zones on the fiber core and cladding [8][98]. TIA-568.3 sets 500 cycles as the structured-cabling mating-durability floor; MPO/MTP-class connectors in practice are commonly rated well above 1000 cycles, but that headroom erodes fast with the wrong cleaning discipline (Section 7.2.2).

[7] IEC 61754-7-1/-2: MPO/MT mechanical interface, one and two fibre rows.

[8] IEC 61300-2-2 mating durability and 61300-3-35 endface inspection grading.

Three practical consequences follow for an ELSFP or CPO fiber-attach program. First, ORL creep is a mating-cycle and cleaning problem before it is a laser problem: a rising RIN floor after repeated ELS swaps (Table 8.1) is diagnosed with an IEC 61300-3-35-style endface inspection, not a laser FA request. Second, mate-cycle count belongs in the same telemetry you already read for CMIS and DDM (Section 7.8); track it per connector, not per module, since a connector can outlive several module swaps or vice versa. Third, write the mating-cycle and endface-grade limits into the ATP explicitly (Table 8.3) rather than inheriting a generic MPO datasheet number: an ELS bank that hot-swaps weekly reaches a 500-cycle floor in under ten years, and a CPO fiber array that is field-serviced more aggressively reaches it faster still.

ELSFP cycling adds connector wear and contamination that raise ORL (Section 7.2.2 and section 5.14); the mating-cycle and endface-grade limits above are exactly the numbers that turn “the connector feels loose” into an ATP line item instead of a guess.

Destructive physical analysis (cross-section, EDX) and structured 8D/CAPA with suppliers close the loop from RMA to design rule (Sections 7.12 and 8.10). Without that loop, packaging FIT gets mis-attributed to laser Arrhenius models and the wrong part gets redesigned.

8.9 Production test at volume

8.9.1 Test time is a cost, coverage is a risk

Every second in the acceptance test plan (ATP) times millions of units is line capacity and real money. Every skipped measurement is escaped DPPM in the field (Section 8.1). The core tension in high-volume manufacturing is not “test or don’t test” but how much coverage you buy per second. The expensive optical steps are thermal soak and corner runs, TDECQ on a sampling scope, BER dwell long enough to trust a low pre-FEC target, laser burn-in, and mate-cycle stress on ELSFP connectors. Some screens are statistical (sample burn-in from a lot, audit TDECQ on a subset). Others must be 100%: CMIS state machine sanity, basic LIV/SMSR pass, and any test that catches a safety or enable-sequence fault (Sections 5.15 and 7.8).

8.9.2 Where the test happens: wafer, die, module, system

Push defect detection as far upstream as correlation allows. Wafer-level or PIC probe catches process shifts (waveguide loss, ring resonance drift, bad heaters) before fiber attach and packaging spend. Killing a bad die at probe is orders of magnitude cheaper than an RMA (Section 8.8). Module ATP is the full functional test: optical power class, TDECQ or proxy, sensitivity spot-check, CMIS bring-up, and connector/ORL on ELS parts. System or golden-host bring-up catches interop: media type, firmware rev, equalizer defaults, and the corners in Section 7.9. Wafer test cannot catch fiber attach, FAU alignment, epoxy creep, or connector wear. Those failures must survive to module ATP and, for some signatures, to fleet telemetry (Section 7.12).

8.9.3 ATE-to-bench correlation

Production testers are built for speed and cost, not lab fidelity. The number that matters is correlation: does the ATE TDECQ, OMA, or sensitivity track the DCA/BERT reference within a known offset and spread? Set ATP limits with guardbands derived from that spread. A drifting tester or a stale golden unit shows up as a yield cliff or a DPPM escape. Keep a golden module (and golden laser subassembly for ELS), run gauge R&R across testers and shifts, and correlate CMIS monitors to bench instruments the same way you correlate TDECQ (Section 7.8). If the ATE and the DCA disagree, fix the correlation before you argue with the supplier about spec.

8.9.4 Screens, guardbanding, and SPC

Burn-in and HTOL screens trade infant-mortality escape rate against test time and cost (Sections 8.2 and 8.4). Test limits are usually guardbanded tighter than the customer spec so field DPPM stays inside target under drift. SPC control charts on LIV, SMSR, RIN, TDECQ, and mate-cycle yield by lot, site, and date code catch a process shift before it becomes an 8D (Section 8.10). Production test is a yield, DPPM, and cost trade under a fixed reliability target. It is not a pass/fail checkbox after the optics already work on a golden bench.

8.10 Supplier execution playbook

The supplier path is milestones, performance targets, quality, and manufacturability triage. That is not a soft skill. It is a concrete contract: requirements, gates, acceptance tests, process control, and corrective action when a lot goes wrong.

NPI gates and exit criteria. Table 8.2 is the usual stage map. For lasers and IM/DD modules, write exit criteria that a supplier can fail clearly, not slogans.

Gate	Question	Laser / optics exit criteria (examples)	Who signs
EVT	Does it work at all?	First light; CMIS bring-up sequence (Section 7.9); LIV/SMSR/RIN on engineering samples; one link closes BER	Optical eng + supplier FAE
DVT	Spec across corners?	Full ATP at *T*/*V* corners; prod-rep corners (Section 7.9); TDECQ/OMA/sensitivity; GR-468 stress plan frozen; FIT model agreed	Optical eng + reliability
PVT	Buildable at yield?	Multi-lot yield vs ATP; SPC charts live; burn-in escape rate; FAIR on production tooling; bring-up on production host	Optical eng + SQE / mfg
MP	Sustained quality?	Steady DPPM; RMA Pareto owned; ECO control on CMIS/firmware and process	Program + supplier QM

Table 8.2. NPI gates with laser-relevant exit criteria. EVT/DVT/PVT/MP are stage names, not a schedule; dates and sample sizes belong in the program plan with the supplier.

Hold a gate if the exit data are missing. Shipping PVT material without frozen ATP limits is how field NFF and FIT arguments start (Section 7.12).

Requirements and ATP are the contract. ATP and the requirements doc are the contract. Write both and keep them versioned together:

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

1. **Requirements / PRD slice for the laser path:** fill Table 5.4 and section 5.6 (power class, grid, RIN@ORL, SMSR, derating, CMIS, FIT). Version it with the ATP.
2. **Acceptance test plan (ATP):** the measurable tests that prove those requirements on every ship lot (or on a defined sample). Map each ATP line to a GR-468 or design-validation stress where life is claimed (Section 8.2).

Table 8.3 is a working ATP checklist for an EML pluggable or an ELSFP CW module. Customize limits from the datasheet and the link budget; do not invent numbers in the ATP itself.

Incoming QC and SPC. Qual lots are small. Production catches drift that qual missed.

- **Incoming:** sample or 100% screen against a subset of the ATP (at least power, CMIS, and a laser LIV/SMSR sample). Track DPPM by date code and site.
- **SPC**: control charts on I_{th}, slope, SMSR, Tx power, and burn-in fallout. A process shift is a hold, not a hope.
- **First-article / FAIR:** when tooling, epi, or assembly site changes, rerun a defined FAIR package before open PO volume. Treat CMIS firmware revs the same way as process changes.

SPC Statistical process control: lot-by-lot control charts (I_{th}, SMSR, power) to catch drift between qual stress tests.

ATP item	Instrument / method	Pass intent	Ties to
LIV (I_{th}, slope, kink)	SMU + power meter	kink-free bias window at rated T	wear-out, derate (Section 5.7)
SMSR	OSA	single-mode vs. spec floor	modal noise, aging
RIN (intrinsic + stressed ORL)	PD + ESA	RIN_xOMA / quiet floor	BER floor (Section 4.3.1)
Wavelength / grid	OSA / wavemeter	channel ID; $d\lambda/dT$	WDM lock (Chapter 6)
Optical power class	power meter	ELSFP / MSA class met	link budget
EAM bias / chirp (EML)	bias sweep + DCA	ER, RLM, TDECQ at baud	Tx quality (Section 7.4)
CMIS / TWI bring-up	host or CMIS tool	registers, alarms, state machine	field telemetry
Connector / ORL	mate cycles + ORL meter	cycles + endface grade vs IEC 61300 limits	packaging (Sections 5.14 and 8.8)
Burn-in screen	HTOL sample or 100% screen	infant mortality culled	GR-468 (Section 8.2)
Driver/TIA ESD, latch-up	supplier JESD47/HBM/CDM report; sample audit	rating on file; no latch-up under current injection	IC reliability (Section 8.3)
Thermal class	chamber at case T	LIV/RIN/CMIS still pass	derate policy

Table 8.3. Acceptance checklist for laser-bearing modules (EML or ELSFP). Limits are program-specific; the structure is what you negotiate with the supplier.

Excursions: 8D / CAPA. When a lot fails ATP, incoming, or field triage lands in the manufacturability bucket (Section 7.12), run structured corrective action:

1. **Contain:** quarantine WIP and ship holds; identify suspect date codes in the fleet.
2. **Evidence pack:** failing ATP rows, CMIS dumps, LIV/SMSR/RIN plots, DPA photos (facet, solder, FAU cross-section) compared to a golden unit.
3. **8D / CAPA**: root cause with the supplier (process step, material lot, firmware), corrective action, and preventive control (ATP tighten, SPC limit, poke-yoke).
4. **Verify:** re-run FAIR on the corrected process; watch field RMA codes for that date-code family for a defined burn-in window.

8D Eight disciplines (8D) corrective action: contain, root-cause, correct, and prevent with the supplier; close only with verification evidence.

Do not close 8D on “operator error” without a control that would have caught it at ATP. If FA shows laser wear-out on a young unit, it may be a reliability screen gap, not a supplier process bug; reclassify with Section 7.12 before you argue FIT.

Milestone hygiene with partners. Align the partner calendar to gates, not slideware:

- freeze requirements before DVT samples are built;
- freeze ATP limits before PVT yield is claimed;
- freeze FIT/E_a assumptions before reliability marketing numbers ship;
- require ECO notice on laser die revision, TEC vendor, FAU epoxy, driver/TIA silicon revision (Section 8.3), and CMIS firmware.

Your job in those meetings is to name the measurement that would kill the gate. If nobody can point to an ATP row or a corner, the milestone is not real.

Key idea. Reliability at scale is mechanism discipline plus supplier gates. Classify failures with the wear-out map (Table 8.1) before you argue FIT or open 8D. Laser wear-out gets Arrhenius and GR-468; driver and TIA silicon gets JESD47, HBM/CDM ESD, and latch-up ratings (Section 8.3); MPO connectors get mating-cycle and endface-grade limits (Section 8.8). Supplier execution is a gated contract: requirements and ATP at DVT, multi-lot SPC and FAIR at PVT, then 8D with evidence when a lot or field Pareto says manufacturability. Do not run process CAPA on a wear-out failure, and do not burn FIT math on a dirty or worn connector.

8.11 From component FIT to fabric availability

The FIT arithmetic in Section 5.13 gives a rate: about 0.6 laser failures per day for a fleet of 5×10^5 lasers at 50 FIT. That number sizes the RMA pipeline and the ELS spares bin (Section 5.14), but it does not say what a failure costs or how a running job survives one. Two facts turn a per-component rate into a fabric problem.

First, a training or large inference job is synchronous. A collective (Section 9.7) waits for its slowest member, so a single dead or slow link stalls the whole group, not just one endpoint (Section 9.6). A link that flaps for a second is a stall for every accelerator in that collective. The optical FIT the earlier chapters budget therefore matters out of proportion to its share of the parts count.

Second, at cluster scale failures are continuous, not rare. Meta's published Llama 3 run is the clearest public data point: 16,384 H100 GPUs over 54 days logged 466 interruptions (419 unexpected), roughly one every three hours, while holding about 90% effective training time [9] [23]. GPU and HBM3 faults dominated at close to half; network switch and cable faults were 35 events, 8.4% of the total. The optical link is a minority of hard job stops, but 8.4% of a failure every three hours is still tens of network events per run, and the ELS, module, and connector FIT this chapter budgets (Sections 8.1 and 8.8) lands in exactly that bucket.

[9] Meta, *Llama 3 herd of models* (arXiv:2407.21783, 2024): 16,384 H100s, 54 days, 419 unexpected interruptions (∼1 per 3 h, ∼90% uptime); network switch/cable 8.4%, GPU/HBM3 ∼47%.

So the design question shifts. It is no longer "how reliable is one link" but "how does a fabric of 10^5 links keep a job running through a failure every few hours." The answers are architectural, and the optical engineer feeds each one.

Redundancy and rails. Rail-optimized topologies (Section 9.2) already give parallel planes; dual-plane and dual-ToR designs let a lost link degrade bandwidth instead of dropping an endpoint. Redundancy multiplies the link and laser count, which feeds straight back into the FIT budget: more resilience is more parts that can fail.

Detection and reroute. Transient faults stay below the job. KP4 FEC (Section 3.12) absorbs the error bursts a marginal link throws; link-level retry and sub-second link-flap detection plus adaptive routing steer traffic off a degraded link before the scheduler notices. Vendor fabrics (NVIDIA Spectrum-X and Quantum, Broadcom Tomahawk) advertise adaptive or "cognitive" routing and link-level retry for this. Treat the specifics as vendor orientation, but the mechanism is why transient optical faults rarely reach the hard-stop bucket above.

Topology reconfiguration. When a link or rack dies for good, an optical circuit switch re-wires the topology around it in milliseconds, so the scheduler routes around the dead node instead of stalling the pod (Section 9.9) [21]. Component FIT still applies; the fabric survives each failure by re-wiring optically.

Sparing and field service. Hot spare nodes and lanes cover the interval between failure and repair. Field-replaceable external lasers (Section 5.14) make a dead laser a faceplate swap rather than a fabric outage, which is the architectural reason ELS decouples laser FIT from switch FIT. The connector mating-cycle and endface budget (Section 8.8) sets how many of those swaps the plant survives.

The cost of a failure closes the loop. A hard interruption is lost compute plus the time to detect, reroute or reschedule, and restart from the last checkpoint. Fast detection and reroute shrink that lost time, which is the fabric-level reason the module work in this chapter pays off: derating (Section 5.13), burn-in and screens (Section 8.2), and a tight ATP (Section 8.10) lower the failure rate, and a resilient fabric lowers the cost of each failure that slips through. The two multiply.

8.12 Engineering lens

8.12.1 How it works

At fleet scale, reliability and manufacturability are design constraints: a modest per-part FIT times millions of parts is a steady stream of failures. The chapter's qualification, yield, and supplier discipline all aim at keeping that stream small and classifiable.

8.12.2 How it is measured

Reliability is measured as a distribution over stress and time. Qualification records failures by mechanism, stress, lot, and sample history. Production records yield, fallout by ATP row, measurement distributions, gauge repeatability, and escaped defects per million. Fleet data add install age, temperature, firmware, supplier lot, return code, and no-fault-found rate. Keep the chain from wafer or die measurement through module ATP to field return so a drift can be traced to its first observable point (Sections 7.12, 8.9 and 8.10).

8.12.3 How it fails

Programs fail at scale through wear, variation, and escapes. Wear shifts a unit after ship. Variation produces a weak tail across wafer, lot, site, or assembly line. An escape is a defect the current screen cannot see or a control that was not run. Yield can drop because the product changed, the process moved, incoming material moved, calibration changed, or the tester moved. Do not open supplier corrective action until the measurement system is cleared.

Failure mode: Yield drop

Symptoms. First-pass yield falls from its stable baseline, often on one ATP row, tester, lot, or shift.
Likely causes. Process drift, supplier-lot variation, assembly change, calibration or fixture drift, software revision, or a changed guardband.
Measurements. Pareto by test and lot, golden-unit history, gauge repeatability, station correlation, incoming data, and destructive analysis on selected failures.
Mitigations. Contain suspect material, clear the tester, identify the first changed input, correct it, and verify with a controlled lot before release.

8.12.4 How it is debugged

For a yield fall, freeze software, limits, and suspect material. Split by tester, shift, lot, supplier site, and ATP row. Run a golden unit across stations and a failing unit on a reference bench. If the failure follows the station, repair the measurement system. If it follows the unit, inspect upstream process data and choose failure analysis that can distinguish the top mechanisms. Contain first, find root cause second, change the process third, and verify the fix on fresh data.

Debug story

Observed. Module yield fell sharply after a supplier lot change.
Investigation. The failure Pareto pointed to one-lane TDECQ. Golden units passed all stations, and failed units kept the bad lane on the reference bench. Cross-sections showed a shifted fiber-array attach.
Finding. The electrical path and testers were stable.
Root cause. An assembly fixture change moved one fiber row outside its coupling window.
Resolution. The lot was held, the fixture was restored, first-article coupling checks were tightened, and the supplier control plan was revised.

8.13 Interview and design review questions

Concept.

- What distinguishes infant mortality, useful-life random, and wear-out failures?
- Why does qualification on a small sample not guarantee field reliability?
- What does FIT mean, and how does fleet size amplify it?

Design.

- Which failure mechanisms are accelerated by each qualification stress, and which mechanisms are not covered?
- What is tested on every unit, what is sampled, and why is that escape risk acceptable?
- Can yield and field returns be sliced by wafer, lot, supplier site, assembly line, tester, firmware, and date code?
- What spare, repair, and reroute plan limits the fleet cost of a failure that still escapes?

Debug.

- Yield dropped from 95% to 70% on one ATP row. What are your first three actions?
- A field return looks alive on the bench. What does a high NFF rate tell you?
- What measurement-system check runs before a supplier or process excursion is declared?

Manufacturing and operations.

- Which corrective action changes a control so the same defect cannot escape again?
- What data must a supplier provide at each NPI gate to pass?
- How do you set guardband limits that balance yield loss against DPPM escape?
- What is the cost of a wrong RMA classification (laser wear-out versus connector contamination)?

Key idea. Production readiness joins mechanism-based qualification, a stable measurement system, controlled supplier changes, and field data that preserve lot history. When yield or fleet health moves, contain the population, clear the test system, find the first changed input, and verify the corrective control with new data.

Chapter 9

AI datacenter networking

Optics only make sense once you see the fabric they sit in. Training and inference clusters are not one network; they are several overlapping networks with different bandwidth, latency, and reach needs. This chapter places the devices and validation methods from earlier chapters into that system context: how AI clusters are wired, why the interconnect sits on the inference critical path, and where optics dominates cost and power.

9.1 Scale-up versus scale-out

The industry borrowed two words from computer architecture and overloaded them for AI fabrics. Keeping them straight matters, because they buy different optics.

Scale-up tight coupling inside a node or rack: GPU-to-GPU over a memory-semantic fabric (for example NVLink). Very high bandwidth, very low latency. Increasingly optical as bandwidth outgrows copper reach.

Scale-out coupling across racks and pods over a switched network (Ethernet or InfiniBand). This is where pluggable and co-packaged optics have long lived.

9.1.1 Three networks, two that set the optics budget

The OIF CEI-448G framework names *three* distinct networks in a large AI cluster [14]. Figure 9.1 shows two; the third is easy to overlook:

Scale-up accelerator-to-accelerator inside a pod or rack. Highest bandwidth per link, lowest latency, often lossless. Dominates compute time if under- provisioned.

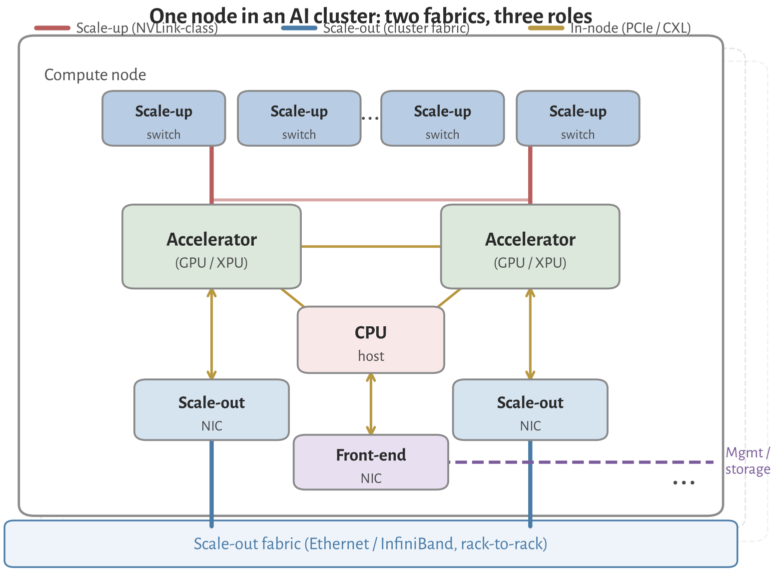

Figure 9.1: Schematic AI compute node (one tray in a larger cluster). *Accelerators* are the heavy compute engines (typically GPUs; the book also uses XPU for GPUs and custom ASICs). **Scale-up** links (red) tie accelerators through a high-bandwidth, low-latency switch fabric inside the node or rack (NVLink-class). **Scale-out** links (blue) leave via a dedicated NIC per accelerator into the datacenter Ethernet or InfiniBand fabric. The CPU and front-end NIC handle host, storage, and management traffic.

Scale-out accelerator-to-accelerator (or node-to-node) across the cluster over a switched fabric (InfiniBand, *UEC*/Ethernet with RoCE or Ultra Ethernet Transport). Lower per-link bandwidth than scale-up, but must reach 10^5+ endpoints.

UEC Ultra Ethernet Consortium: open Ethernet stack for AI/HPC scale-out (UET transport, PHY extensions).

Front-end CPU-attached traffic: management, control plane, checkpointing, storage, and training-data ingest. Analogous to a conventional cloud datacenter network; not the IM/DD bottleneck this book focuses on, but it shares the same rack power and cabling budget.

Scale-up carries the majority of *interconnect* bandwidth inside a training job; scale-out multiplies link count across the building. That is why both push 448G-class lane rates and why optics shows up first on scale-out, then on scale-up as copper reach shrinks [14].

9.1.2 Scale-up versus scale-out at a glance

Table 9.1 condenses OIF Table 1 (CEI-448G framework, §2.2): order-of- magnitude targets for node count, physical extent, and media type [14]. Numbers are industry snapshots, not hard limits, but they explain why "optics inside the rack" and "optics between racks" arrive on different timelines.

	Scale-up		Scale-out	
Metric	Today	Next gen	Today	Next gen
Accelerator nodes in domain	$\sim$100	$\sim$1 k	100 k+	$\gg$100 k
Physical extent	Rack	Rack to row	Datacenter	Datacenter
Network properties	Lossless, low latency		Large scale; multi-tier switching	
Media within rack	Electrical: passive PCB / twinax backplane		Electrical: twinax backplane	
Media between racks	AEC (adjacent racks)	Optical (within row)	Optical (pluggable or CPO on switch/NIC)	

Table 9.1. Scale-up vs. scale-out snapshots (adapted from OIF CEI-448G framework Table 1, 2025). Scale-up stays on copper as long as rack densification keeps channels short; scale-out is already optical at datacenter scale.

9.1.3 Standards bodies: who owns what

448G/lane signaling is not owned by one standards body, and that is by design. Electrical reach, Ethernet naming, connectors, and rack packaging evolved in different rooms; AI fabrics forced them to meet at the same lane rate. OIF's CEI-448G framework (§2.3) lists the groups that must align [14]. Table 9.2 maps each body to the layer an optical engineer actually touches. The short version: OIF sets the electrical baud and reach classes; IEEE names the Ethernet optical PMD; UALink and UEC own scale-up and scale-out protocol stacks; SNIA and OCP decide connectors and where the optics physically live. The *LPO MSA* is not a standards body, but it publishes the only open end-to-end spec that stitches CEI Linear electrical limits to IEEE optical PMD limits for DSP-less modules (Section 9.3.1). Prose below expands each row, OIF and non-OIF.

Body	Fabric	What matters for short-reach optics
OIF CEI	All reaches	Electrical PHY: XSR/VSR/MR/LR, 448G PAM4/6/8, LPO/LRO; CMIS for module management
UALink	Scale-up	Open load/store pod fabric; 200G/lane today, $\sim$400G/lane next; drives XSR-class PHY needs
UEC	Scale-out	Ultra Ethernet Transport, congestion control; PHY group surveying 400G/lane enhancements
SNIA SFF	Host / backplane	Cables, connectors, PCIe/EDSFF form factors; SFF 448G COM for backplane (complements CEI front-panel work)
OCP	Rack / system	Open rack designs, CPO/XPO placement, optical circuit switching (OCS) for AI clusters
IEEE 802.3	Scale-out (+ optics PMD)	Ethernet MAC rates (802.3dj, 400G/lane SG), KP4 FEC, IM/DD PMD clauses (DR/FR, TDECQ)
100G Lambda MSA	Scale-out optics	Originated the 100G/λ single-mode PMDs (100G-FR/LR, 400G-DR4/FR4/LR4) later folded into IEEE 802.3; RIN_xOMA transmitter method the DR/FR and LPO specs inherit
IBTA	Scale-out	InfiniBand Architecture; NDR 400G, XDR 800G/port at 200G/lane, 1.6T switch links; reuses QSFP/OSFP + MPO optics with Ethernet
IEEE 802.1	All fabrics	Link-layer security: MACsec (802.1AE) line-rate encryption; 802.1X port access control
PCI-SIG / CXL	Scale-up, in-node	PCIe 7.0 (128 GT/s PAM4) and CXL 4.0 memory fabric; PCI-SIG Optical Workgroup for optical PCIe
DMTF / OpenConfig / SONiC	Management	Box and fleet telemetry and config (Redfish, OpenConfig, SONiC); complement OIF CMIS module management

Table 9.2. SDO and consortium roles at 448G/lane. The top six rows follow OIF CEI-448G §2.3; the 100G Lambda MSA, IBTA, IEEE 802.1, PCI-SIG/CXL, and the management stack are added for coverage beyond OIF's own list.

OIF. Common Electrical I/O (*CEI*) Implementation Agreements are the modular electrical PHY recipes: die-to-die (*XSR*), chip-to-module (*VSR*), chip-to-chip (*MR*), and backplane (*LR*). CEI-448G is the current AI-driven push beyond 224G/lane [14].

Related OIF tracks include energy-efficient interfaces (EEI), CMIS management extensions at 448G, and coherent DCI (out of scope for this book). For optics, CEI tells you the *baud and reach* the module or CPO engine must meet; IEEE 802.3 tells you how Ethernet names the optical PMD.

UALink Consortium. *UALink* specifies an open *scale-up* stack (transaction, data link, and physical layers) optimized for direct accelerator memory access, not IP/Ethernet [1][15][14]. UALink 1.0 targets 200G/lane and pods up to roughly 1 k accelerators; the physical-layer working group is gathering requirements for ∼400G/lane. Optical engineers care because higher lane rate reduces port count on the accelerator package, and because scale-up eventually hits the same copper reach wall as CEI XSR (Section 9.5).

UALink Ultra Accelerator Link: open scale-up fabric for accelerator load/store (200G/lane class).

[1] UALink Consortium, UALink 200G 1.0 scale-up specification (April 2025).

Ultra Ethernet Consortium. *UEC* builds a complete Ethernet-based stack for AI/HPC *scale-out*: transport (UET), link layer, and PHY enhancements [2][16][14]. UEC 1.0 (June 2025) targets cluster-scale congestion control and RDMA-class performance over standard Ethernet hardware, including NICs, switches, optics, and cables. The PHY working group is surveying 400G/lane improvements. This is the protocol layer above the transceiver; IM/DD module specs still come from IEEE and MSAs.

[2] Ultra Ethernet Consortium, UEC Specification 1.0 (June 2025).

RDMA Remote direct memory access: zero-copy network transfer between accelerator memories; RoCE (Ethernet) or native IB.

NIC Network interface card: host adapter connecting an accelerator or CPU to the scale-out fabric (Ethernet or IB).

SNIA (SFF Technology Affiliate). SNIA's SFF working group defines connectors, cables, and form factors for storage and compute backplanes, not front-panel Ethernet [3][18][14]. The SFF 448G project (SFF-TA-1043) parallels CEI-448G but focuses on backplane COM, package insertion loss, and PAM choice on copper channels inside the box. Pair OIF VSR (module connector) with SFF (host PCB and PCIe cable) when debugging a link budget.

[3] SNIA SFF TA-1043 / OIF 448G workshop: backplane 448G scope (2025).

Open Compute Project. OCP publishes open rack, server, and networking designs for hyperscale [4][19][14]. Relevant work includes Open Systems for AI, short-reach optical interconnect guidelines, and the Optical Circuit Switching (OCS) subproject (2025) for reconfigurable cluster fabrics. OCP rarely specifies baud rate; it specifies *where* the optical engine lives (faceplate pluggable vs. CPO vs. XPO) and how it is cooled and serviced.

[4] OCP Open Systems for AI; OCS subproject for AI cluster fabrics (2025).

IEEE 802.3. Ethernet standards define MAC rates, FEC (KP4 in Clause 91), and interoperable optical PMDs [8][10][14]. 802.3dj (200G/lane) and the 400G/lane study group name the product generations that consume 224G and 448G-class optics. One naming trap: IEEE quotes **400 Gb/s per lane** (MAC/info rate); CEI quotes ∼**448 Gb/s** (line rate with coding overhead). Same physics, different accounting (Chapter 3 and table 3.6).

100G Lambda MSA. The single-mode PMDs this book leans on did not start at IEEE. The *100G Lambda MSA*, formed in 2017 by a broad group of suppliers and hyperscalers, wrote the first interoperable optical specifications built on one wavelength carrying 100 Gb/s PAM4 ($\approx$53 GBd): 100G-FR and 100G-LR for single-wavelength 100 GbE, and 400G-DR4/FR4/LR4 for four-lane 400 GbE over duplex single-mode fiber [5][54]. IEEE 802.3 then adopted the same 100 Gb/s-per-λ approach into its DR/FR/LR PMD clauses, and the LPO MSA 100G-DR-LPO profile (Section 9.3.1) inherits both that modulation and the RIN_xOMA transmitter method the MSA defined (Section 4.3). For a short-reach engineer this is the body behind the reach-class names on almost every single-mode datasheet: when a module is called "DR4" or "FR4," the per-wavelength recipe traces to this MSA even where the compliance point is now quoted against an IEEE clause.

[5] 100G Lambda MSA (2017): 100G/λ PAM4 single-mode PMDs (100G-FR/LR, 400G-DR4/FR4/LR4); basis for IEEE DR/FR/LR clauses and 100G-DR-LPO.

InfiniBand Trade Association. The *IBTA* maintains the InfiniBand Architecture Specification, the other dominant scale-out fabric for AI and HPC alongside Ethernet [6][17]. Data rates follow the same per-lane SerDes ladder: NDR reaches 400 Gb/s per port over four 100 Gb/s lanes, and XDR reaches 800 Gb/s per port over four 200 Gb/s lanes, with switch-to-switch links at 1.6 Tb/s (XDR figures are from the 2023 spec announcement and remain provisional). For the optical engineer the physical layer is close to the Ethernet case: the same QSFP and OSFP form factors, the same MPO fiber, and DR/FR-class IM/DD optics at matching lane rates. InfiniBand changes the transport and congestion model (credit-based flow control, RDMA), not the transceiver. It is absent from the OIF §2.3 list because IBTA runs its own specification, but it occupies the same scale-out slot as UEC.

IBTA InfiniBand Trade Association: owns the InfiniBand Architecture spec (NDR 400G, XDR 800G/port); reuses QSFP/OSFP optics.

[6] IBTA InfiniBand Architecture: NDR 400G, XDR 800G/port at 200G/lane; reuses QSFP/OSFP + MPO optics with Ethernet.

IEEE 802.1 (link-layer security). Encryption is a link-layer function, and at 800G and 1.6T it has to run at line rate. *MACsec* (IEEE 802.1AE) provides connectionless confidentiality, frame integrity, and data origin authenticity at the MAC layer, transparent to the MAC client; IEEE 802.1X handles port access control and the key agreement that establishes MACsec associations [7][12]. This lands on optics indirectly: line-rate MACsec costs latency and gate count in the switch ASIC or NIC, not in the transceiver, and an LPO or DSP-less module carries the ciphertext transparently. Encryption is therefore orthogonal to the optical PMD choice (Section 9.3.1), but it belongs in the standards map because a fabric-security requirement can still gate a module program at qualification.

[7] IEEE 802.1AE MACsec (line-rate L2 encryption) and 802.1X port access control.

PCI-SIG and CXL. Inside the node, the load-store fabric is PCI Express and *CXL*. PCI-SIG released PCIe 7.0 in 2025 at 128 GT/s per lane using PAM4 and flit encoding, and the CXL Consortium built CXL 4.0 on that same PCIe 7.0 electrical base for coherent memory pooling [8][25][9][26]. Both are 2025 releases, so treat the exact rates and the optical-interface scope as provisional. Both are copper-first and short today, but PCI-SIG runs an Optical Workgroup defining a technology-agnostic optical

CXL Compute Express Link: coherent memory interconnect built on PCIe PHY (CXL 4.0 on PCIe 7.0). Potential optical reach.

[8] PCI-SIG PCIe 7.0 (2025): 128 GT/s PAM4, flit encoding; Optical Workgroup for optical PCIe.

[9] CXL 4.0 (Nov 2025) on PCIe 7.0: coherent memory-pooling fabric; relevant to optical CXL.

PCIe interface, and CXL memory disaggregation across a rack is exactly the reach that pulls optics into a bus that was never optical before. For a short-reach optics team this is the emerging in-node and in-rack case to watch: PAM4 at PCIe rates over fiber, with the same modulator and detector problems as an Ethernet lane.

Management and telemetry. Module management is OIF CMIS (Section 7.8), but the box and fleet layers above it are not OIF. *DMTF Redfish* is the RESTful standard for server and switch management and telemetry; *OpenConfig* defines vendor-neutral data models and streaming telemetry (gNMI) for network devices; and *SONiC* is the open network operating system many hyperscalers run, with its own dataplane telemetry hooks [10][87]. These are where per-module CMIS monitors surface as fleet signals: optical power, case temperature, FEC symbol-error counts, and pre-FEC BER aggregated across 10^5 links (Chapter 8 and section 8.11). A module program that ships without a telemetry contract into one of these systems is effectively invisible at fleet scale.

[10] DMTF Redfish, OpenConfig (gNMI), and SONiC: box/fleet telemetry and config above OIF CMIS.

The frontier is optics entering the scale-up domain (optical NVLink-class links, co-packaged switches), because copper reach at 200G/lane is only about a meter.

9.2 Topologies and why optics count explodes

Cluster topology is where optical count stops being "a few modules per server" and becomes a fleet problem. Large AI fabrics mostly use fat-tree / Clos, rail-optimized, or dragonfly layouts. In a classic k-ary fat-tree, link count scales as $O(k^3)$ while endpoints scale as $O(k^2)$: optics multiply faster than compute. Rail-optimized designs (one NIC rail per accelerator row, all-to-all within a rail) rose with collective-heavy training and inference because they cut oversubscription on all-reduce paths, at the price of more parallel optical planes. Dragonfly and other hierarchical topologies trade some global bisection bandwidth for fewer long links.

The optical engineer cares because every topology choice sets link count, which sets laser count, module count, and FIT budget (Chapter 8); rail layouts drive fan-out from leaf to spine and push denser 800G/1.6T ports toward CPO/XPO (Sections 9.10 and 9.11); and scale-up inside the rack stays electrical longer while scale-out between racks is already optical (Table 9.1). That explosion in link count is the economic reason a hyperscaler builds an in-house optical engineering team.

9.3 Pluggable form factors and module styles

Faceplate modules differ in aggregate rate, lane count, electrical reach, and where the DSP lives. The form factor sets the passive channel the SerDes must survive, and the last decade of AI networking has mostly been a fight over that choice: keep a retimer in the module for interop, delete it for watts (LPO/LRO), or move the optics onto the package (CPO) and service the laser from the faceplate (ELSFP).

QSFP-DD / OSFP the incumbent datacom pluggables. QSFP-DD carries 800G/1.6T class products (eight lanes at 100G or 200G per lane). OSFP targets higher power and density (1.6T today, 3.2T roadmaps) with a larger cage and better thermal path. Both impose a VSR-class electrical channel from host PCB through the connector (Section 9.5 and section 9.5.1).

Retimed module DSP/CDR inside the module (Section 3.6). Default for interoperability; $\sim$15–20 W class at 800G/1.6T.

LPO / LRO linear or lightly retimed: delete or slim module DSP (Sections 3.14.3 and 9.5.1 and figures 3.2 and 3.6). Power drops to $\sim$7–9 W but host EQ and transmitter and dispersion eye closure quaternary (TDECQ) margin tighten.

ELSFP / external laser field-replaceable CW source for CPO (Section 9.10 and chapter 5): decouples laser FIT from switch FIT.

XPO liquid-cooled mega-pluggable (Section 9.11): 12.8 Tb/s per module, front-panel serviceability with CPO-class density.

CMIS (Section 7.8) is the management layer for module identity, monitors, and (at 224G/448G) link-training and host-side signal-integrity tuning extensions. Optical engineers touch it when debugging lock, FEC, and equalizer settings across vendors.

Half-retimed and asymmetric modules. Not every pluggable is fully retimed or fully linear. Common 2025–26 variants sit between those poles:

LPO linear drive and linear receive: no module DSP/CDR (Section 3.6 and section 3.14.3).

LRO / RTLR retimed transmit, linear receive (OIF RTLR): DSP on electrical $\rightarrow$ optical only; eases host TX while keeping a simpler RX path [72].

TRO / half-retimed symmetric opposite or partial retiming variants appear in vendor roadmaps; always check which direction carries the DSP.

Fully retimed DSP both directions; default for interoperability at 800G/1.6T.

Power scales with DSP content: retimed $\sim$15–20 W, LRO $\sim$9 W, LPO $\sim$7–9 W at 800G class (Section 9.13). The architecture choice is a trade between host margin (COM, TDECQ) and module watts.

9.3.1 The LPO MSA: stitching IEEE optics to CEI Linear

OIF CEI tells you the electrical recipe at the module cage. IEEE 802.3 tells you how Ethernet names the optical PMD and its TDECQ/OMA limits. Neither document, by itself, is a complete product spec for a linear pluggable that deletes the module DSP. The *LPO MSA* (Linear Pluggable Optics Multi-Source Agreement) fills that gap: one open specification for both sides of the module, with normative test points and host responsibilities spelled out [79][72].

The first published revision, *100G-DR-LPO* (v1.0, March 2025), targets 100 Gb/s per lane at 53.125 GBd PAM4 on single-mode fiber from 0.5 m to 500 m. It is explicitly a data-center profile: low power, low latency, high port density, RS(544,514) FEC on the host, and form-factor agnostic (QSFP, QSFP-DD, OSFP are examples, not the spec). The naming pattern generalizes to `n00G−DRn−LPO` for $n \in \{1, 2, 4, 8\}$ lanes. Optical reach and modulation track IEEE DR-class PMDs; the electrical interface tracks OIF CEI-112G-LINEAR-PAM4. That split is the template 224G linear modules follow: CEI-224G-Linear on the AUI, 802.3dj optical PMD limits on the fiber (Section 3.14.2 and section 3.13).

Who owns which block. In an LPO link the host is not a passive cable driver. It runs KP4 FEC (Section 3.12), full SerDes equalization (CTLE, FFE, DFE, CDR; Sections 3.6 and 3.7), and optional nonlinear compensation (NLC) and startup protocol functions. The module is analog: linear driver, modulator or laser, photodiode, TIA, and at most a fixed CTLE. No retiming, no FEC, no heavy DSP. CMIS (Section 7.8) is the management contract. SFF hardware specs (QSFP-DD, OSFP cages) set the mechanical envelope. The MSA's job is to define what must pass at each interface between those blocks so modules from different vendors close on the same host.

The test-point ladder. LPO MSA normative compliance is organized around six electrical/optical test points (Table 9.3), the concrete instance of the general TP0-to-TP5 planes in Section 3.9. Think of them as the validation script: host TX at TP1a, module optical TX at TP2, stressed optical RX at TP3, module electrical RX at TP4, and stressed host RX at TP4a. Section 10 of the MSA adds a host-to-host end-to-end BER test with FEC-encoded traffic, which is how you prove interop after the point tests pass.

Point	Location	Principal measurements
TP1a	Host SerDes output	EECQ (electrical eye closure), host TX quality
TP1	Module electrical input	Module input stressor calibration
TP2	Optical TX (2–5 m patch cord)	TDECQ, TECQ, $\text{OMA}_{\text{outer}}$, $\text{RIN}_{x\text{OMA}}$, ER
TP3	Optical RX input	Stressed receiver calibration (SECQ), sensitivity masks
TP4	Module electrical output	Module RX linear output (EECQ)
TP4a	Stressed host input	Host RX under worst-case module output

Table 9.3: LPO MSA test points (100G-DR-LPO).

Optical limits that matter. The MSA optical tables (Section 7.4 and chapter 7) inherit IEEE 802.3 measurement methods with LPO-specific reference equalizers. The numbers you will quote in a datasheet review:

- TDECQ and TECQ capped at 3.4 dB per lane, measured with a 9-tap T-spaced reference FFE and SER target 4.0×10^{-4}.
- $\text{OMA}_{\text{outer}}$ coupled to transmitter quality: launch power in OMA rises with max(TECQ, TDECQ) along a piecewise mask (for example $-3.2 + \max(\text{TECQ, TDECQ})$ dBm in the mid-TECQ region).
- $\text{RIN}_{\text{xOMA}} \leq -138$ dB/Hz at 17.2 dB ORL (the "x" subscript tracks return loss, same convention as IEEE DR PMDs).
- Illustrative link budget 6.7 dB total at 500 m (3 dB channel loss plus 0.3 dB MPI penalty plus 3.4 dB TDECQ allocation).
- Stressed receiver sensitivity -3.1 dBm $\text{OMA}_{\text{outer}}$ at TP3 with SECQ = 3.4 dB and aggressor lanes at 4.2 dBm OMA.

TECQ is TDECQ without the dispersion test fiber. The MSA uses both because outer OMA limits are written against max(TECQ, TDECQ), and a module can be fiber-limited even when the electrical eye looks clean.

Electrical limits and host COM. On the electrical side the MSA points to CEI-112G-LINEAR-PAM4 for the host-to-module interface and defines EECQ (electrical eye closure quaternary) at TP1a and TP4a. That is the electrical analogue of TDECQ: a host that passes optical TDECQ at TP2 but fails EECQ at TP1a still will not interoperate. Host PCB insertion loss, connector, and module input return loss sit in the channel reference model (Section 7). For 224G deployment, swap CEI-112G-LINEAR for CEI-224G-Linear and run the same two-ledger program (Sections 3.14.2 and 9.5.2).

What to read first. For a bring-up engineer the reading order is: Section 5 (system overview and host FEC requirements), Section 7 (electrical TPs and channel model), Section 8 (optical TX/RX tables), Section 9 (parameter definitions, especially 9.5 TDECQ/TECQ and 9.10 stressed RX), then Section 10 (host-to-host FEC BER). Cross-check every optical definition against the cited IEEE 802.3 clause; the MSA adds reference equalizer taps and SER targets, not new physics.

Key idea. OIF owns the electrical baud at the cage. IEEE owns the optical PMD metrics on the fiber. The LPO MSA is the product contract that binds them for DSP-less modules: normative test points, host FEC/EQ duties, and TDECQ/OMA masks you can test without a module retimer safety net.

9.3.2 The LPO supplier base and the adoption question

The MSA defines what a linear module must pass; it does not settle whether the market buys one. By 2025–26 the supplier base had formed around the analog parts that replace the DSP: high-linearity TIAs and linear drivers. Macom, Semtech, and Maxlinear are the named component proponents, and once the DSP is gone the TIA and driver become the make-or-break blocks [72].

The demonstrations track a fast rate climb. Eoptolink showed a 200G/λ four-channel LPO link with no DSP or CDR at OFC 2024 and moved a second-generation 100G/lane 800G and 400G single-mode line into volume, claiming full TP2 compliance at the transmit interface [75]. Macom exhibited its PURE DRIVE 200 Gb/s LPO parts at OFC 2024 and extended them toward 212 Gb/s/lane for a 1.6T module, with the TIA and driver as the headline [76]. Marvell, a DSP house, announced a 200G/lane TIA and laser-driver chipset for 800G and 1.6T LPO aimed at scale-up XPU fabrics [77]. Macom and Eoptolink are founding members of the LPO MSA (Section 9.3.1).

LPO reaches into PCIe scale-up. Alphawave and Innolight demonstrated a 64 Gb/s/lane PCIe 6.0 subsystem (controller plus PHY) over Innolight's LPO OSFP optics at OFC 2024, then a 128 Gb/s/lane platform pairing a PCIe 7.0-ready SerDes PHY with the same optics [78]. The pull is the switch-fabric pull again: larger, faster AI nodes need more PCIe reach than copper gives, and the linear module skips the DSP latency and power a retimed module would add (Sections 9.1 and 9.12).

Whether LPO wins is a separate question. Host support exists: Juniper's Broadcom-based QFX switches take LPO optics without hardware changes, and Arista has shown Broadcom TH5 compatibility for over two years [72]. The market read is still cautious. Cignal AI argues LPO stays a small share, at least at 800G, because the installed fabric was already designed around DSP-based modules; 100G/lane (800GbE) LPO looks late and likely to hold only a small slice long term [11][74]. At 200G/lane and 1.6T the balance tips toward LRO: early 1.6T LPO parts draw more than 30 W, a thermal problem at faceplate density (Section 9.13.1), while a transmit-DSP LRO design promises under 20 W with easier integration and interop [72]. Cignal notes that every vendor showing 1.6T LPO at OFC also showed an LRO part [74]. That matches the book's technical read: LPO and LRO win where the host electrical channel and TDECQ close without the module DSP, and lose where they do not (Sections 3.14.2 and 9.5.1).

[11] Cignal AI: LPO likely a small share (esp. at 800G) given DSP-based installed base; LRO favored at 1.6T. Analyst report, provisional.

9.4 Emerging link styles

The form-factor list above is the present. The industry argument for 2025–26 is where the DSP and the laser should live next. Retimed pluggables remain the interop default: a module DSP cleans the electrical channel, at the cost of watts. LPO and LRO delete or slim that DSP so the host SerDes carries more of the EQ burden, which is attractive for AI power budgets and painful for margin and interop. CPO moves the optical engine onto the switch or XPU substrate to cut electrical reach, and usually keeps the laser external and field-replaceable (ELSFP/CW-WDM). XPO appeared in 2026 as a middle path: a much larger, liquid-cooled pluggable that keeps front-panel serviceability while pushing density toward CPO territory (Section 9.11). The rest of this chapter treats electrical reach, eye budgets, and vendor CPO programs as consequences of that argument.

9.5 The electrical link: reach, conditioning, and the eye budget

Every form factor above is really an answer to one physical question: *how far can the electrical signal travel before it is cheaper (in power and dB) to convert to light?* As the per-lane rate climbs, that distance collapses, and it is what pushes the optics from the faceplate toward the die.

TRACE LOSS SCALES WITH FREQUENCY. A PCB stripline's insertion loss grows roughly with $\sqrt{f}$ (skin effect) plus a dielectric term linear in f, so it is quoted per inch *at the Nyquist frequency*. Going from 112G to 224G PAM4 moves Nyquist from 28 GHz to 56 GHz, and the loss per inch roughly doubles. Recent 224G board studies measure ≈ 2.8 dB/inch for regular stripline and ≈ 1.9 dB/inch with skip-layer routing at 56 GHz, against a next-generation *target* of 1 dB/inch that demands ultra-low-loss dielectric, HVLP copper, and short via stubs [68]. Figure 9.2 shows the consequence: at a fixed PCB-trace budget, doubling the baud rate roughly halves the copper reach.

THE CEI CHANNEL CLASSES NAME THE REACHES. OIF's Common Electrical I/O project defines the electrical link budgets the whole industry designs to. Table 3.8 is the CEI-224G lookup card (XSR / VSR / MR / LR, plus Linear for DSP-less modules); the reach map is Figure 3.5. At 56 GHz Nyquist the same names mean much shorter copper than at 112G [66]:

XSR *XSR*: die-to-die / die-to-engine: the shortest, lowest-power tier, the one CPO and chiplet optics live in ($\lesssim$50 mm package).

VSR *VSR*: chip-to-module: $\approx$200 mm of host plus 20 mm of module and one connector: the classic pluggable channel.

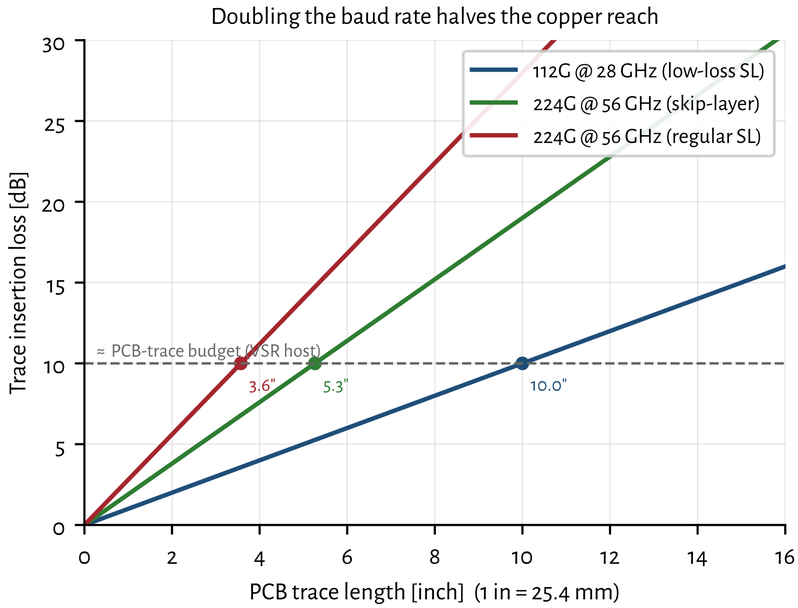

Figure 9.2: PCB trace insertion loss versus length at each rate's Nyquist. The reach to a fixed budget shrinks from ~ 10 inches (112G) to a few inches (224G), which is why the optical conversion must move closer to the ASIC.

MR *MR*: chip-to-chip across a board: ≈500 mm, one connector, ∼32–34 dB die-to-die.

LR *LR*: backplane or copper cable: ≈1000 mm of host and daughter cards, two connectors, up to ∼40 dB die-to-die (including a 1 m cable).

Linear *CEI-224G-Linear*: same faceplate-class ports without a module DSP/CDR; the electrical foundation for LPO (Section 3.14.2 and section 9.3).

Active copper: ACC, AEC, and DAC. Passive direct-attach copper (DAC) survives only where the reach is short: at 224G a passive DAC is good for roughly 0.5–1 m (strong 224G PHYs have demonstrated 2 m). *ACC* adds redrivers (CTLE + VGA) in the cable; *AEC* adds retimers (EQ + CDR) and stretches twinax to ∼2.5 m at higher power (Section 9.5.1 and section 3.6) [70]. Beyond that, and increasingly *within* the rack for scale-up, the answer is optics (Table 9.1).

ACC Active copper cable: redriver (CTLE+VGA) inside cable; extends DAC reach.

AEC Active electrical cable: retimer (EQ+CDR) inside cable; longer twinax.

Co-packaged and near-package copper: copper moves inward too. Optics is not the only thing the reach wall pushes toward the package. *CPC* (co-packaged copper) mates a copper cable or connector directly onto the ASIC package substrate, and *NPC* (near-package copper) places that connector just outside the package on the socket. Both leave the host at the package edge and skip the lossy PCB run to the faceplate, where most of the trace budget in Figure 9.2 is spent. The move mirrors CPO: shorten the electrical path until the channel closes, but keep the signal in copper instead of converting it to light.

On-substrate copper has been validated at 224G-PAM4 with a stated roadmap to 448G, and compression substrate connectors now target 224 Gb/s PAM4 and beyond [12] [105]. Inside its reach, copper keeps the properties that make it the default: a long-reach 224G SerDes into a CPC cable runs near 4 pJ/bit with no electro-optic conversion, adds less latency than a retimed optical hop, and costs less per lane. Passive co-packaged copper reaches about a meter at 448G-PAM4 under optimistic connector assumptions (Section 3.14.3), enough to cover many scale-up links inside a rack or between adjacent racks.

[12] Co-packaged copper (Molex): on-substrate routing validated at 224G-PAM4, roadmap to 448G, for short scale-up runs. Marvell 224G LR SerDes ∼4 pJ/bit. Provisional.

Past that wall the conversion to light pays for itself. When the reach exceeds what a clean substrate channel can carry, or the port count makes copper bulk and cabling weight unmanageable, optics takes the link (Table 9.1). CPC and NPC do not remove that crossover; they move it, buying copper one more rate generation before the optics win. Read the placement ladder in Table 9.4 from the copper side and this is the same trade seen in reverse.

So the optics march inward. Shortening the electrical path is exactly what trades power and reach for serviceability, the through-line of this chapter. Table 9.4 lays out the ladder from faceplate to interposer.

Placement	Host electrical reach	Energy/bit	Serviceability
Pluggable (faceplate)	full VSR host run (∼200 mm) + connector	highest (>30 pJ/bit w/ DSP; less for LPO)	hot-swap at faceplate (best)
On-board optics (OBO / COBO)	shorter mid-board trace	lower	board-level (solder/socket); largely bypassed for CPO
Near-packaged (NPO)	engine beside the ASIC on substrate	lower still	module on substrate; limited
Co-packaged (CPO)	mm-scale die-to-engine (XSR)	<5 pJ/bit	soldered; ELSFP lasers field-replaceable
Optical I/O on interposer	on-die / interposer	<2 pJ/bit	not field-serviceable (emerging)

Table 9.4. The optics-placement ladder. Moving the electro-optic conversion inward cuts trace loss and energy per bit but erodes field serviceability, the central tension behind pluggables, OBO/NPO, CPO (Section 9.10), and the XPO middle ground (Section 9.11). A mid-board fiber connector can shift breakage risk from the costly engine to a cheap jumper.

On-board optics: the step the industry mostly skipped. OBO (standardized by the *Consortium for On-Board Optics*, COBO) moves the optical engine off the faceplate and onto the main PCB, cutting the copper run without abandoning silicon photonics [COBO / OBO–NPO–CPO]. It works, but it gives up hot-plug serviceability while only partly closing the power gap, so with CPO maturing, most hyperscalers are leapfrogging OBO/NPO straight to co-packaging, keeping serviceability via field-replaceable lasers (Section 9.10) and the XPO pluggable hedge (Section 9.11).

THE DIE-TO-DIE INTERFACE IS A STANDARD TOO. At the innermost tier (Table 9.4), the electrical hand-off between chiplets is increasingly set by *UCIe* (Universal Chiplet Interconnect Express), an open die-to-die interconnect whose 2.0 revision (August 2024) added 3D packaging and in-field manageability, with a 3.0 revision roughly doubling bandwidth in 2025 [13][24]. UCIe is the parallel counterpart to CEI XSR: XSR is a serial die-to-OE link, while UCIe carries wide parallel lanes across a package or interposer. It reaches optics because co-packaged engines and optical-I/O chiplets attach to the compute die over a UCIe port, and optical-UCIe proposals aim to carry that traffic over fiber instead of a few millimeters of substrate. The chiplet-protocol and packaging detail sits outside this book's scope; the point for short reach is that the die-to-die interface, copper or optical, sets the shortest reach an optical engine must beat.

UCIe Universal Chiplet Interconnect Express: open die-to-die interconnect for chiplets on a package or interposer; parallel counterpart to CEI XSR.

[13] UCIe Consortium: open die-to-die chiplet interconnect; 2.0 (Aug 2024) adds 3D packaging + manageability; 3.0 (2025) doubles bandwidth.

9.5.1 Reshaping, retiming, and where the DSP lives

To survive that lossy channel the signal is conditioned at several points (Section 3.6), and it helps to keep two device classes distinct.[14]

[14] Rule of thumb: a redriver reshapes, a retimer rebuilds timing. Neither fixes reflections or a bad return path.

Redriver an analog *reshaper/amplifier*: a *CTLE* plus a VGA. It sharpens and boosts the eye but has *no* clock recovery, so it cannot reset accumulated jitter; it adds almost no latency. This is the active element in active copper cables (ACC) and on host boards used to stretch a marginal trace.

Retimer adaptive EQ *plus* a *CDR* that re-samples and regenerates the data. It breaks the channel into independent jitter/loss segments, at the cost of power and per-hop latency. Retimers sit in active electrical cables (AEC) and in fully retimed modules (Broadcom's Agera is a host retimer).

Inside the optical module the traditional "cleanup" is a full DSP (retimer + FEC engine). The retimed-to-linear spectrum trades exactly cleanup for power and latency:

Fully retimed DSPs in both directions: least sensitive to *ISI*/dispersion, but highest power ($\sim$14–18 W/module at 800G) and most latency ($\sim$8–10 ns/hop).

LPO (linear) *LPO*: *no* DSP or CDR in either direction: only a CTLE in the TIA and a linear driver, so the host SerDes must do all the *FFE*/CTLE equalization *and* carry the FEC. Lowest power ($\sim$7–9 W), lowest latency (<3 ns), but most ISI/dispersion-sensitive and shortest reach (<2 km) [72].

LRO / TRO (retimed TX, linear RX) DSP only on the electrical$\rightarrow$optical path, roughly half the power, sensitivity, and latency, with easier interop.

So yes: reshapers/amplifiers (redrivers) and retimers are routinely placed on the host path, and the module's own DSP is the last line of cleanup. The entire LPO bet (Section 9.13 and section 3.14.2) is to *delete* those active stages and let the host SerDes' CTLE/FFE carry the channel, trading electrical margin for power and latency.

9.5.2 The electrical eye: acceptable voltage and noise

What must the signal actually look like where it hands off to the module? The CEI-224G reference and draft numbers set the envelope [66]. Differential swing sits roughly in the 0.36–1.05 Vppd band (reference TX near 1 V peak-to-peak differential; lossier channels use the larger swing). Transmitter *SNDR* ($\mathrm{SNR_{TX}}$) is about 33 dB: the transmitter's own noise-plus-distortion ceiling. PAM4 level uniformity needs RLM ≥ 0.95 (1.0 = perfectly even eyes). Jitter budgets are tight: random ≈0.01 *UI* rms and bounded/uncorrelated ≈0.02 UI pk. The go/no-go metric that folds those pieces together is *COM* (channel operating margin) ≥ 3 dB: a statistical SNR of the fully equalized link (Section 9.5.2).

SNDR Signal-to-noise-and-distortion ratio: transmitter quality ceiling including nonlinearity.

UI Unit interval: one symbol period ($= 1/R_{\mathrm{sym}}$). Jitter specs are in UI.

Jitter and level uniformity. At 112 GBd, 1 UI ≈8.9 ps; 0.01 UI rms jitter is $\sim$90 fs rms. Random jitter adds vertical eye closure; deterministic jitter (ISI, crosstalk) shows up in bathtub curves and COM. *RLM* ≥ 0.95 keeps the three PAM4 eyes even; poor RLM wastes vertical margin the same way excess TDECQ does on the optical side (Section 7.4).

Retimers reset jitter per segment; redrivers do not. That is why long ACC chains accumulate timing budget stress while AEC segments stay independent (Section 9.5 and section 9.5.1).

Here is the number that reframes "acceptable noise." After the reference receiver's equalizer (an 8-tap *DFE*), the eye at the slicer is only about 4–10 mV tall and $\sim$0.06 UI wide, with $\sim$11–14 dB of vertical eye closure, at a *pre-FEC* error rate near 10^{-4}. The channel is deliberately driven deep into ISI and closed nearly shut; *KP4* FEC (pre-FEC 2.4×10^{-4} to post-FEC 10^{-15}, Chapter 4) is what turns that into a working link. "Acceptable," then, is not a clean open eye; it is whatever keeps COM ≥ 3 dB and the pre-FEC BER under the KP4 threshold. It is also why the optics must present a clean, highly linear interface, especially for LPO: with only a few mV of margin, any added noise or nonlinearity from the driver or TIA comes straight off COM.

Channel operating margin (COM) in one page. *COM* is the electrical go/no-go statistic for CEI-class channels: after the reference transmitter, channel, and receiver (CTLE + 8-tap DFE) are applied, COM is the SNR margin at the slicer, in dB. COM ≥ 3 dB is the usual pass line (Section 9.5.2 and section 3.6).

COM connects directly to optics because the module is the last analog segment before FEC. A retimed module can absorb some host-channel sin; LPO cannot (Section 9.3 and section 3.14.3). When COM is tight, debug in this order: host TX SNDR and RLM, connector return loss, equalizer tap saturation, then module input swing and TIA noise (Section 4.5 and chapter 4).

Optical-side analogs are not identical: TDECQ scores the transmitter with a reference equalizer; SECQ stresses the receiver (Sections 7.4 and 7.5). Think of COM as the *electrical* counterpart to those optical margin tests.

Key idea. Electrical reach is the hidden clock behind form factors. Trace loss per inch roughly doubles from 112G (28 GHz) to 224G (56 GHz), collapsing copper reach to a few inches on-board and $\sim$1 m in cable. Each step inward (pluggable, OBO, NPO, CPO, on-interposer optical I/O) buys power and reach and spends serviceability. The signal is held together by redrivers (reshape), retimers (reclock), and DSPs (both), which LPO deletes, inside a brutal budget: $\sim$1 Vppd in, 33 dB TX SNDR, COM ≥ 3 dB, and a post-equalizer eye of only a few mV rescued by FEC.

9.6 The two phases of inference, revisited

Chapter 1 introduced prefill and decode; here is why they matter for the network.

Prefill compute-bound and highly parallel: crunches the whole prompt through every layer at once.

Decode memory-bandwidth-bound and autoregressive: one token at a time, streaming all weights each step. The *KV cache* (scaling with batch $\times$ context length) is a major and growing memory consumer.

WHY DECODE IS MEMORY-BANDWIDTH-BOUND. It comes down to *arithmetic intensity*, FLOPs performed per byte read from memory. Each weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ costs $d_{\text{out}} d_{\text{in}}$ parameters to read. In decode you generate one token, so every layer is a matrix–*vector* product (GEMV): about $2\, d_{\text{out}} d_{\text{in}}$ FLOPs against $2\, d_{\text{out}} d_{\text{in}}$ bytes of FP16 weights, i.e. each weight is used once and discarded:

$$\text{AI}_{\text{decode}} \approx \frac{2\, d_{\text{out}} d_{\text{in}}}{2\, d_{\text{out}} d_{\text{in}}} = 1 \text{ FLOP/byte.}$$

Prefill instead pushes *N* prompt tokens through the *same* weights at once, a matrix–matrix product (GEMM) that reuses each loaded weight *N* times, so $\text{AI}_{\text{prefill}} \approx N$ FLOP/byte. On a roofline this is decisive: an H100 delivers $\sim$1000 TFLOP/s FP16 on $\sim$3.35 TB/s of HBM, so its balance point sits at

$$\frac{1000 \times 10^{12}}{3.35 \times 10^{12}} \approx 300 \text{ FLOP/byte.}$$

HBM High-bandwidth memory: stacked DRAM on interposer beside the accelerator die. Memory BW sets the decode roofline.

Decode at $\sim$1 FLOP/byte runs $\sim$300$\times$ *below* that ridge (under 1% of peak FLOPs) so token rate is set by HBM bandwidth/model bytes, not by compute [71]. Two effects compound it: the full weight set must be streamed from HBM *per token*, and attention re-reads the KV cache every step, with traffic that grows with context length.

BATCHING COUPLES LATENCY AND THROUGHPUT. Larger batches amortize the weight-streaming cost (better throughput and energy per token) but make each individual response wait (worse latency). Squeezing both at once is a central goal of an inference platform.

SHARDING PUTS THE NETWORK ON THE CRITICAL PATH. Because frontier models (especially mixture-of-experts) are sharded across many accelerators, each token triggers collectives: all-reduce for tensor parallelism, all-to-all for expert routing, point-to-point for pipeline stages. For MoE all-to-all in particular, interconnect bandwidth and tail latency can dominate. This is the concrete reason hyperscale inference platforms balance "compute, memory, and networking."

MoE Mixture of experts: transformer architecture that routes tokens to specialist subnetworks; all-to-all collective demand.

Table 9.5: Where optics fits the inference bottlenecks.

Phase	Bottleneck	Network role
Prefill	compute	moderate (parallel)
Decode	memory bandwidth	high for sharded models (collectives)
MoE routing	interconnect	dominant (all-to-all)

9.7 Collective communication and optics demand

Once a model is sharded across many accelerators, the network stops carrying only point-to-point streams. Training and large inference jobs spend a large fraction of wall time in *collective* patterns: all-reduce for tensor-parallel layers, all-to-all for MoE expert routing, and steadier point-to-point streams for pipeline stages. Those patterns set optical requirements even when the PHY still looks like ordinary Ethernet or InfiniBand.

All-reduce tensor-parallel layers sum partial results across a group; needs high bisection bandwidth and low latency; latency sensitive at small message sizes (decode).

All-to-all MoE expert routing sends tokens to remote experts; bandwidth dominates; tail latency spikes if any link in the pod is slow (Table 9.5).

Point-to-point pipeline parallelism moves activations between stages; steady streams rather than global sync.

Optical engineering maps to these patterns indirectly. More rails and higher per-lane rate cut time spent in collectives; CPO/XPO raise faceplate bandwidth so fewer hops are needed (Sections 9.2, 9.10 and 9.11). Protocol choice (UEC vs IB) sets lossless delivery and congestion behavior (Section 9.8), but the PHY job remains the same: deliver pre-FEC BER below KP4 at the lowest pJ/bit (Sections 3.12 and 9.13). The next section names the fabric stacks that carry those collectives; the sections after that show where the optics physically sit.

9.8 Fabric options

InfiniBand, Ethernet, and the Ultra Ethernet Consortium's AI-tuned Ethernet are the scale-out contenders. Momentum in 2025–26 favors open Ethernet for AI; large vertical AI stacks commonly use Broadcom Tomahawk switch silicon (Chapter 1 and section 9.10).

InfiniBand / NVLink fabric lossless, RDMA-native; NVIDIA Quantum switches; strong in closed NVIDIA stacks; CPO photonics on Quantum-X (Section 9.10).

RoCEv2 Ethernet RDMA over converged Ethernet; widely deployed; competes on congestion control and tail latency versus IB.

Ultra Ethernet (UEC) UET transport, enhanced PHY, congestion control aimed at AI collectives (Table 9.2). PHY work tracks 400G/lane class electrical and optical I/O.

Collectives (all-reduce, all-to-all for MoE) sit on this fabric (Section 9.6). The optical job is raw bandwidth and predictable latency at $\sim$200–400G per lane, not long-haul reach.

9.9 Optical circuit switching

The fabric options above are all *packet* switches: every link terminates in a switch ASIC that turns light into electrons, reads the header, and turns it back into light for the next hop. That O-E-O conversion is where much of a fabric's power and latency goes (Section 9.13), and at 10^5-plus endpoints it repeats at every tier. An *optical circuit switch* does a different job. It is a Layer-1 switch that steers light from an input fiber to an output fiber with no O-E-O conversion, so it is transparent to bit rate, modulation format, and wavelength. The same switch that passes 200 Gb/s PAM4 today passes 448G or a full WDM comb tomorrow with no change (Chapter 6 and section 3.14).

OCS Optical circuit switch: Layer-1 switch steering light with no O-E-O conversion; transparent to rate/format/wavelength. MEMS OCS: ms switching, $\sim$2 dB loss, topology reconfiguration.

Transparency is the point, and also the limit. Because an OCS never looks at a packet, it cannot buffer, arbitrate, or route per packet. It sets up a *circuit*: a fixed light path held open as long as the topology needs it. The mirrors or waveguides that steer the light reconfigure in milliseconds, far slower than a packet time, so an

OCS reshapes the *topology* between jobs or around failures, not the traffic inside a job. In return it deletes a whole tier of packet switches and their pluggable optics, along with the power, cost, and FIT that tier carried (Section 9.2 and chapter 8).

The device and its parameters. Most production OCS today is free-space: a fiber-collimator array launches beams onto a two-axis MEMS mirror array, and each mirror tilts to aim its beam at the chosen output collimator. Table 9.6 lists the main technologies. For an optics engineer the parameters that matter are the ones that land in the link budget and the fleet model.

MEMS Micro-electro-mechanical systems: tilting mirror arrays in OCS; millisecond switching, ∼2 dB loss, 136×136 radix.

Insertion loss a hop through the switch costs roughly 2 dB, straight off the optical budget (Section 7.7). Higher launch power or better receiver sensitivity has to cover it (Chapter 5 and section 4.4).

Radix how many fibers the switch connects. Production MEMS OCS runs about 136×136; circulators reuse each port in both directions and so double the effective radix (below).

Reconfiguration time milliseconds for MEMS, which fixes OCS as a topology switch. Silicon-photonic and SOA switches reach nanoseconds, but at low radix and higher loss (Table 9.6).

Crosstalk, return loss, polarization a mirror that leaks light into the wrong port is crosstalk; a reflective interface raises ORL and feeds laser RIN (Section 7.2.2 and section 4.3). Free-space paths are largely polarization-insensitive, which suits IM/DD.

Technology	Switching	Insertion loss	Radix	Where it fits
MEMS mirror array (free-space)	ms	∼1–3 dB	100s (136×136 shipping)	DC fabric and AI-pod topology reconfiguration
Piezoelectric beam steering	ms	low–moderate	10s–100s	free-space alternative to MEMS
Liquid crystal (LCoS)	ms	moderate	wavelength-selective	wavelength add/drop, WSS roles
3D robotic fiber	seconds–minutes	∼0.5 dB	1000s	automated patch and provisioning, not per-job
Silicon photonic (MZI / SOA)	ns–μs	higher (integrated)	10s	fast, low-radix; research and niche

Table 9.6. OCS technologies. Free-space MEMS switches dominate AI deployments today; robotic-fiber switches trade speed for radix and very low loss; integrated photonic switches trade radix and loss for nanosecond speed.

What it buys at fleet scale. Google's Jupiter datacenter fabric replaced a patch-panel Clos interconnect with a layer of MEMS OCS under software-defined control, and reported roughly 30% lower capex, 41% lower power, and 3x faster fabric reconfiguration while a direct-connect topology carried the same production traffic[15] [20]. The switch, Palomar, is a 136×136 MEMS OCS with about 2 dB insertion loss and millisecond switching, and circulators realize bidirectional links through

[15] Poutievski *et al.*, "Jupiter Evolving," ACM SIGCOMM 2022: MEMS OCS + SDN; Palomar 136×136, ∼2 dB, ms switching; ∼30% capex, ∼41% power, 3x reconfiguration.

it to double the effective radix [20]. The same building block reshapes AI pods: a TPU v4 pod wires 4096 accelerators through 48 OCS into a 3D torus that reconfigures per job and routes around failed racks, so a dead node becomes a topology the scheduler works around instead of a pod-wide outage [16][21]. That reconfiguration is the fabric-reliability lever Chapter 8 points at: component FIT still applies, but the fabric survives each failure by re-wiring optically rather than stalling the job.

[16] Jouppi *et al.*, TPU v4 optically reconfigurable supercomputer, ISCA 2023: 4096 chips, 48 OCS, reconfigurable 3D torus, route around failures.

What OCS asks of the transceivers. An OCS layer changes the module spec in ways this book cares about. Because the switch adds a fixed loss and is wavelength-transparent, single-fiber duplex reaches (FR, one fiber each way) fit an OCS better than parallel-fiber reaches (DR, many fibers), so an OCS deployment pulls the plant toward FR optics and toward circulators for bidirectional operation on one fiber [17][22]. The insertion loss argues for higher launch power and tighter ORL budgets, since every mated interface and mirror is a reflection the laser sees (Sections 4.3.1 and 7.2.2). Wavelength transparency means a WDM link passes through the switch unchanged, so CW-WDM and ring-based engines (Chapter 6 and section 6.6) compose with an OCS without a translation layer. None of this is exotic optics; it is the same DR/FR PMD and laser work from earlier chapters (Section 3.13 and chapter 5), specified against a channel that now includes a switch.

[17] Cignal AI, optical circuit switching market (2025): OCS technologies and transceiver impact (FR over DR, circulators). Analyst report, provisional.

Status and where it sits next to CPO. By 2025–26 OCS moved from a Google-specific technique to an industry theme, a headline topic at OFC 2026 with MEMS, piezoelectric, liquid-crystal, robotic-fiber, and silicon-photonic approaches competing on radix, loss, speed, and reliability [22]. It complements co-packaged optics rather than competing with it: CPO shortens the electrical path at the switch package (Section 9.10), while OCS removes packet-switch hops between packages and racks. A fabric can use both, CPO optics feeding an OCS layer, and the reliability question for each is the one Chapters 7 and 8 keeps returning to.

Key idea. An optical circuit switch reroutes light at Layer 1 with no O-E-O conversion, so it is transparent to rate, format, and wavelength and adds only insertion loss to the link budget. Millisecond MEMS switching makes it a topology and failure-reroute switch, not a packet switch. It buys fabric power, cost, and resilience (Google Jupiter and the TPU pods are the production proof), and it pulls the transceiver plant toward FR optics, circulators, and higher launch power. OCS and CPO are complementary bets on the same power and reliability problem.

9.10 Co-packaged optics: 2025–26 status

By 2025–26, co-packaged optics crossed from demonstrations into shipping products, pushed by the power and reliability limits of pluggables in AI scale-out. The programs converge on a common recipe: a photonic engine co-packaged with the switch ASIC, 200 Gb/s per channel, microring modulators (Section 3.14.3), *field-replaceable* lasers pulled out of the package, and TSMC COUPE packaging underneath.

9.10.1 Broadcom Tomahawk and CPO

Broadcom entered CPO earlier than most switch vendors and treated it as a product line, not a one-off demo. The current flagship is *Tomahawk 6*, a 102.4 Tb/s single-chip Ethernet switch (shipping 2025) offered with either copper or co-packaged optics, on 100G/200G SerDes.[18] The CPO variant, *TH6-Davisson*, began shipping in October 2025 as Broadcom's *third-generation* CPO switch. The public numbers sketch the architecture:

[18] Broadcom's broader AI-Ethernet stack: Tomahawk and Jericho switches, Thor NICs, Agera retimers, Sian optical DSPs, and CPO. Large hyperscale inference fabrics draw on this family (Chapter 1).

- 102.4 Tb/s optically enabled, built from sixteen 6.4 Tb/s "Davisson DR" optical engines at 200 Gb/s per channel.
- Photonic engines fabricated with *TSMC COUPE*; 64 Condor 3 nm SerDes cores (eight 212.5 Gb/s PAM4 lanes each).
- About 70% lower optical-interconnect power than pluggables.
- *Field-replaceable ELSFP laser modules*: lasers, the highest-failure component, made serviceable in the field.
- Scale-up to 512 XPUs; 100,000+ XPUs in a two-tier fabric at 200 Gb/s/link.

The lineage matters: CPO shipped on Tomahawk 4 and the second-generation *TH5-Bailly* (51.2 Tb/s), which logged "millions of hours" of reliability testing before Davisson. A fourth generation at 400 Gb/s per channel is already in development.

9.10.2 NVIDIA

NVIDIA's CPO story is the scale-up and scale-out fabric vendor converging on the same TSMC COUPE + microring recipe as Broadcom, announced as product families at GTC 2025: *Quantum-X* (InfiniBand) and *Spectrum-X* (Ethernet) Photonics switches, 200G SerDes, 1.6 Tb/s ports. Headline marketing claims include 3.5$\times$ power efficiency, 4$\times$ *fewer lasers*, and large resiliency gains versus pluggables; treat those as vendor orientation, and validate against your own FIT and power model.

- *Quantum-X InfiniBand*: 144 ports of 800G ($\approx$115 Tb/s), liquid-cooled; available late 2025. Each package integrates eighteen silicon-photonics engines fed by 36 laser inputs through six *detachable* optical sub-assemblies.
- *Spectrum-X Ethernet*: up to 409.6 Tb/s; available in 2H 2026.
- Ecosystem: lasers from Lumentum, silicon photonics with Coherent, packaging with TSMC.

9.10.3 TSMC COUPE (the shared foundation)

COUPE (Compact Universal Photonic Engine) stacks an electronic IC on a photonic IC via SoIC-X hybrid bonding (a 6 nm EIC on a 65 nm SOI PIC), giving a low-impedance die-to-die interface. The roadmap: pluggable qualification in 2025, CoWoS-based CPO integration and *mass production in 2026*, with 800G/1.6T engines now and 3.2T/6.4T (toward 12.8 Tb/s on-package) to follow. TSMC cites the energy-per-bit trajectory from >30 pJ/bit for copper toward <5 pJ/bit for CPO on substrate and <2 pJ/bit once optical I/O moves onto the interposer (Section 9.13). The hard problems it names (wafer-level test, fiber-array-unit integration, and high-speed optical packaging assembly) are exactly the validation and manufacturing challenges of Chapters 7 and 8.

COUPE Compact Universal Photonic Engine: TSMC's SoIC-X hybrid-bonded EIC-on-PIC packaging for CPO. Mass production 2026.

Program	Technology	Status
Broadcom TH6-Davisson	102.4 Tb/s, 200G/ch, COUPE, ELSFP lasers	3rd-gen CPO, shipping Oct 2025
Broadcom TH5-Bailly	51.2 Tb/s CPO	2nd-gen, extensively field-tested
NVIDIA Quantum-X (IB)	144 × 800G, MRM, COUPE, detachable lasers	available late 2025
NVIDIA Spectrum-X (Enet)	up to 409.6 Tb/s, MRM, COUPE	2H 2026
TSMC COUPE	SoIC-X EIC-on-PIC packaging	mass production 2026
Samsung (foundry)	optical engines / turnkey CPO	OE 2027, CPO 2029
Ayar Labs	TeraPHY optical I/O + SuperNova CW-WDM source	merchant scale-up optical I/O

Table 9.7. CPO programs, 2025–26.

Key idea. The 2025–26 CPO wave shares one architecture: 200G/lane microring engines on TSMC COUPE, with field-replaceable lasers because lasers fail first. Tomahawk-class CPO at 200G/lane makes IM/DD validation and ELSFP laser reliability direct gates on how many accelerators a pod can wire together dependably.

9.11 The serviceable-density middle ground: XPO (OFC 2026)

CPO buys density and power at a cost the operator feels: the optics are soldered to the switch, so a single failed engine can mean pulling the whole line card. That tension (pluggable serviceability versus co-packaged density) defined much of *OFC 2026*, where a third path drew the most attention.

Arista, with Coherent, Marvell, Lightmatter, and a broad partner list, launched the *XPO* (eXtra-dense Pluggable Optics) multi-source agreement. The bet is to keep the front-panel pluggable form factor (slide a module out, snap a new one in) while closing most of the density and power gap to CPO:

XPO eXtra-dense Pluggable Optics: liquid-cooled 12.8 Tb/s pluggable (Arista MSA, OFC 2026); front-panel serviceable at CPO density.

- **12.8 Tb/s per module**: 64 electrical lanes at 200 Gb/s PAM4 (with a roadmap to 400 Gb/s lanes for 25.6 Tb/s), roughly $4\times$ the density of a 1.6T-OSFP pluggable.
- **204.8 Tb/s per OCP rack unit**, from up to sixteen modules, front-panel density approaching co-packaged designs.
- **Integrated liquid-cooled cold plate** rated for 400 W+ per module, with blind-mate dripless quick-disconnects; this, not the connector, is what makes the high per-module power serviceable.
- **Universal reach and interface**: SR/DR/FR/LR plus ZR/ZR+, and fully-retimed, half-retimed, or linear (LPO/LRO) optics in one form factor.

Attribute	Retimed / LPO pluggable	XPO	CPO
Capacity	0.8–1.6 Tb/s/module	12.8 Tb/s/module	100+ Tb/s on-package
Density	baseline	$\sim 4\times$ (204.8 Tb/s per RU)	highest
Power path	full electrical run to faceplate	short run to dense faceplate	shortest (on substrate)
Cooling	air (or LPO savings)	integrated cold plate, 400 W+	switch-package liquid cooling
Serviceability	field-replaceable (best)	field-replaceable (slide-out)	soldered; ELSFP lasers replaceable
Energy/bit	highest	intermediate	lowest (<5, then <2 pJ/bit)
Maturity	shipping	MSA launched OFC 2026	shipping (Broadcom, NVIDIA)

Table 9.8. Where XPO sits between pluggables and co-packaged optics.

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[19] XPO is best read as a hedge against CPO's serviceability problem. If field-replaceable ELSFP lasers (Section 9.10) prove insufficient and CPO repair economics stay painful, a liquid-cooled pluggable that reaches CPO-class density lets operators keep the failure model they already trust. The reliability and validation questions of Chapters 7 and 8 are exactly what decides which path wins.

THE BROADER OFC 2026 PICTURE. XPO landed inside a clear consensus: 1.6T transceivers went mainstream and 3.2T (400G/lane) previews appeared, with initial demos expected around 2027; CPO moved from demo to imminent, with new MSAs (Open CPX, "socketed CPO") blurring the pluggable/co-packaged line; and hollow-core fiber (record loss now $\sim$0.091 dB/km) advanced toward low-latency intra-datacenter use (Section 9.12.1). The through-line is the one this book opened with: rising per-lane rate forcing optics closer to the silicon and squeezing every last pJ/bit.

9.12 The latency budget

The book has a link budget in dB (Section 7.7) and, next, an energy budget in pJ/bit. It needs a third ledger. Inference puts the optical link on the critical path (Section 9.6), so you should be able to add up a link's latency the way you add up its loss and its power. The contributors fall into two groups: fixed digital costs that do not care about distance, and a propagation term that does.

THE FIXED DIGITAL COST DOMINATES A SHORT LINK. FEC is the largest single term. KP4 RS(544,514) (Section 3.12) costs roughly 20 to 100 ns to encode

Contributor	Typical latency
PCS framing / serialization	a few ns
FEC encode + decode (KP4)	∼20–100 ns each
Module DSP / retimer (per hop)	∼8–10 ns (fully retimed)
LPO (no DSP)	<3 ns
Driver, modulator, PD, TIA	sub-ns
Fiber propagation (silica)	∼4.9 ns/m
hollow-core fiber	∼3.3 ns/m
Switch hop (O-E-O)	∼100–560 ns

Table 9.9: Approximate one-way latency contributors, 200G/lane class.

and again to decode, set by the codeword length and the implementation, not the fiber [20] [99]. The module DSP adds another 8 to 10 ns per hop when the link is fully retimed. The analog stages, driver, modulator, photodiode, and *TIA*, together contribute well under a nanosecond of group delay and are almost never the problem. On a 10 m in-rack link the fiber itself is about 50 ns, smaller than one pass through the FEC.

[20] KP4 RS(544,514) decode latency ≈20–100 ns in practice (arXiv:2402.09364; vendor notes). Provisional.

Propagation dominates only once the fabric is large. Light in standard single-mode fiber travels at roughly $c/1.47$, about 4.9 ns/m, because the glass has a group index near 1.47. A 100 m row-scale run is then about 490 ns, comparable to a switch hop and larger than the digital terms. Each switched tier adds an O-E-O conversion: a cut-through Ethernet switch forwards in a few hundred nanoseconds (∼560 ns for an 800GbE class device), while InfiniBand reaches under 100 ns per hop [21] [100]. Across a multi-tier scale-out fabric, the switch hops and their conversions, not the fiber, set the tail latency that stalls a collective (Section 9.6). This is the latency argument for optical circuit switching (Section 9.9) and for co-packaging: both remove conversions and electrical runs from the path.

[21] Per-hop switch latency: ∼560 ns for an 800GbE SONiC switch; sub-100 ns for InfiniBand XDR. Vendor figures, provisional.

Latency is where the module architecture shows up again. The retimed-to-linear choice you met in Section 9.13 for power repeats here. A fully retimed module spends ∼8–10 ns per hop; *LRO* roughly halves that; *LPO* deletes the module DSP and lands under 3 ns [72]. On a link with a few hops, deleting the DSP saves more time than shortening the fiber does. Latency and energy point the same way, which is why LPO and CPO are attractive for the scale-up domain where both matter most.

9.12.1 Hollow-core fiber

The one distance term you can attack is the group index. Hollow-core fiber (*HCF*) guides light mostly through air rather than glass, so its group index sits near 1.0 and light travels close to vacuum speed. A double nested antiresonant nodeless

HCF Hollow-core fiber: guides light in air ($n \approx 1$); 33% lower latency than silica SMF. Loss record 0.091 dB/km (2025).

fiber (*DNANF*) design reported 0.091 dB/km at 1550 nm, below the ∼0.14 dB/km floor that silica has held since the 1980s, and about 0.2 dB/km across a 66 THz window [22][101]. The latency payoff is the headline for short reach: propagation drops from ∼4.9 to ∼3.3 ns/m, roughly 45% faster or about a third lower latency, which on a 100 m run saves ∼150 ns, the size of a switch hop. Microsoft reports deploying cabled HCF in the Azure network.

[22] Chen *et al.*, DNANF hollow-core fiber, Nature Photonics 2025: 0.091 dB/km at 1550 nm, ∼45% faster than silica. Record, provisional.

For short-reach IM/DD the low nonlinearity and near-zero dispersion of air guiding matter less than they do for coherent long-haul; the draw here is purely latency and, secondarily, the wider low-loss window for more wavelengths. The open problems are practical: low-loss splicing and connectorization to solid-core plant, yield at volume, and cost. These figures are recent records, so treat them as provisional.

9.13 Energy per bit and the power wall

Frontier AI is *power-limited*: a site's usable megawatts, not just capital, caps how much compute it can host. Every watt spent moving bits is a watt not spent on compute, and interconnect energy multiplies across a fabric with millions of links, which is why energy per bit (*pJ/bit*) has become a headline design metric.

The trajectory that CPO is chasing, per TSMC's COUPE disclosures:

Table 9.10: Approximate interconnect energy per bit.

Link style	Energy per bit
Conventional copper / retimed	>30 pJ/bit
Co-packaged optics on substrate	<5 pJ/bit
Optical I/O on the interposer	<2 pJ/bit

This is the quantitative reason CPO exists: removing the power-hungry electrical run between a switch ASIC and a front-panel pluggable (and the module's retiming DSP) is roughly a 70% cut in optical-interconnect power in Broadcom's Davisson (Section 9.10). *LPO*/*LRO* attacks the same target from the pluggable side by deleting the DSP; CPO attacks it by shortening the electrical path.

Why it compounds. Multiply even a few pJ/bit by aggregate fabric bandwidth and link count and interconnect becomes a meaningful slice of cluster power. At a fabric moving petabits per second, a 5 pJ/bit versus 30 pJ/bit choice is megawatts, directly setting how many accelerators fit under a fixed power envelope. Laser wall-plug efficiency feeds the same budget: fewer, more efficient lasers (NVIDIA claims 4× fewer) cut both power and failure count at once.

The macro trend behind the metric. The pJ/bit fight has grid-scale stakes. US data centers drew about 176 TWh in 2023, roughly 4.4% of national electricity, up from 58 TWh in 2014; the Lawrence Berkeley National Laboratory projects 325 to 580 TWh, or 6.7 to 12% of US electricity, by 2028 [23][73]. Interconnect is a growing share of that draw as lane rates climb [72]. Every pJ/bit an LPO or CPO link removes is multiplied across that base, so interconnect power reads as an infrastructure problem, not only a per-module one.

[23] LBNL (2024): US data centers 176 TWh in 2023 (∼4.4% of US electricity), up from 58 TWh (2014), projected 325–580 TWh (6.7–12%) by 2028.

9.13.1 The thermal envelope

Every watt from the budget above turns into heat that has to leave the box, so interconnect power is also a cooling problem, and cooling sets a second ceiling beside the power wall. On a faceplate switch the optics are a large part of that heat: a 32-port 800G switch dissipates on the order of a kilowatt, and the pluggable modules account for roughly half of it [24][106]. Air cooling loses headroom as per-module power climbs past the mid-teens of watts, which is why dense high-rate switches are moving to liquid and immersion cooling, and why the LPO power cut (from ∼14–18 W to ∼8 W, Section 9.13) reads as a thermal cut as much as an electrical one [72].

[24] FiberMall (2026): 32-port 800G switch >1 kW, optics ∼50% of thermal load; power 15–25% over rated. Vendor, provisional.

Co-packaging changes the shape of the problem. Moving the optics onto the switch substrate (Section 9.10) puts heat-sensitive optical engines a few millimeters from a high-power ASIC. Absolute temperature still matters, but the steep on-package gradient becomes the performance-limiting term: rings drift off resonance and lock loops fight neighbor heaters (Section 6.5), and the laser is the least tolerant part of all [25][107]. This is the thermal half of the argument for external lasers (Section 5.14): holding the laser off the hot interposer at a controlled temperature protects both its wavelength and its life.

[25] Cao *et al.*, Front. Optoelectron. (2025): liquid-cooling thermal management for high-bandwidth CPO; power density and thermal crosstalk limit reliability.

Cooling is a reliability lever, not only a power one. Laser wear-out follows Arrhenius kinetics (Section 5.13): the acceleration factor is exponential in inverse junction temperature, so a few degrees of cooling buys a measurable drop in FIT and, across 10^5-plus lasers, fewer failures per day (Section 5.13). Power, cooling, and reliability are one constraint seen three ways. The link that fits under a fixed power and cooling envelope, and stays cool enough to last, is the one that scales.

Key idea. Energy per bit is a first-order lever on cluster size under a fixed power budget. The industry path, retimed pluggable (>30 pJ/bit) to LPO to CPO (<5, then <2 pJ/bit), is why "balance compute, memory, and networking" (Chapter 1) is a power statement as much as a performance one.

9.14 A first-order cost model

The book quantifies two of its three themes. Power has a ledger in pJ/bit (Section 9.13); reliability has one in FIT (Section 5.13). Cost is invoked everywhere but never counted. It deserves the same first-order treatment, kept deliberately relative and order-of-magnitude. What follows is an illustrative model, not a price sheet: absolute module prices move too fast and vary too much by volume to write down usefully, so the numbers here are assumptions you should replace with your own.

Total cost of ownership (TCO) for an optical link splits into three buckets. The first is acquisition, the *bill of materials* (BOM) and the yield of optical assembly and test: laser count, whether a DSP die is present, and the packaging and coupling steps that dominate transceiver cost. The second is lifetime energy, the module's power drawn over years and multiplied by a cooling overhead. The third is service: the failures per day implied by the fleet FIT (Section 5.13), each one costing a replacement part and a hands-on visit. Acquisition is capital; the other two are recurring.

TCO Total cost of ownership: acquisition (BOM, yield) plus lifetime energy plus service (FIT $\times$ replacement cost).

BOM Bill of materials: component and assembly cost of a module or system. DSP presence is a large BOM driver.

Energy is the bucket you can compute. Take an 800G module drawing $\sim$15 W fully retimed against $\sim$8 W for LPO (Section 9.13) [72]. Over a five-year life at $0.10/kWh, with a *power usage effectiveness* (PUE) of $\sim$1.3 to cover cooling, the retimed module burns about 850 kWh, near $85 of electricity; the LPO module about 460 kWh, near $46. The $\sim$$40 gap per module is a meaningful fraction of what the module itself costs, and it recurs every life cycle. Scaled up, the number stops being small: a vendor estimate puts the LPO saving on a 500,000-accelerator cluster on the order of 100 MW and roughly $100 million a year in electricity [26] [108]. Treat that figure as vendor orientation, but the order of magnitude is the point.

[26] Semtech (2025): LPO $\sim$50–60% less power than retimed; 500,000-GPU cluster $\sim$100 MW and $\sim$$100M/year electricity saving. Vendor, provisional.

Recurring cost is a large share of the total. A gigawatt-class AI site has been quoted near $38 billion of up-front capital and roughly $0.9 billion a year to run [27] [109]. Over a multi-year life the running cost is a real fraction of the capital, and interconnect power is a growing slice of it, so a pJ/bit cut reads directly as dollars saved. That estimate is an analyst breakdown, provisional, but it sets the scale: power and cost are the same argument seen through two units.

[27] Epoch AI (2025): 1 GW AI site $\sim$$38B up-front capex, $\sim$$0.9B/year opex. Analyst breakdown, provisional.

The architecture choice moves all three buckets at once. Per delivered Gb/s, a fully retimed pluggable carries the highest BOM (a DSP die and its packaging) and the highest power. LPO and LRO delete or relax that DSP, cutting both the BOM and $\sim$40–50% of the power [72], and move the cost that remains into host validation risk rather than the module. CPO and XPO remove the faceplate connector and the long electrical run and push energy under 5 pJ/bit (Table 9.10), but raise assembly and test cost and couple the optics to an expensive switch ASIC, which is why the lasers are field-replaceable: a laser failure must not scrap the package (Section 9.10). Co-packaged and near-package copper is cheaper still per Gb/s wherever it reaches, carrying only the connector and the host SerDes energy (Section 9.5); the crossover to optics is set by the reach wall, not by cost. At the fabric level, optical circuit switching is itself a cost lever: replacing a tier of O-E-O packet switches with a MEMS OCS cut capex $\sim$30% and power $\sim$41% in Google's Jupiter (Section 9.9) [20].

Key idea. Cost tracks power and reliability, not against them. The cheapest link per Gb/s is the one you do not light (copper within reach), then the one with the fewest active stages (LPO, then CPO), weighed against the validation risk and the failure blast radius each one adds. A pJ/bit saved is dollars saved every year the fabric runs, which is why the energy budget and the cost model point the same way.

9.15 Synthesis: from workload to fleet

This chapter sits at the end of the book because it joins the decisions from every earlier chapter into one system. The path below is not a waterfall; teams iterate. But each step produces evidence that either advances the design or sends it back.

1. **Workload and collective requirements.** What traffic pattern, latency target, and tail tolerance does the job demand? (Chapter 1 and sections 9.6 and 9.7)

2. **Reach and topology.** What physical extent, link count, and oversubscription does the cluster need? Does copper close, or does optics take the link? (Sections 9.2 and 9.5 and table 9.1)

3. **Electrical versus optical placement.** Pluggable, LPO, CPO, XPO, or co-packaged copper? The answer follows from reach, power envelope, and serviceability. (Table 9.4 and sections 9.5, 9.10 and 9.11)

4. **Laser, modulation, and WDM selection.** DFB, EML, CW+Si MZM, CW+ring, or CW+TFLN? Single-wavelength DR or dense WDM with locking? The choice sets the ATP, the supplier base, and the fleet FIT model. (Chapters 5 and 6 and tables 3.12 and 5.1)

5. **Link, power, and thermal budgets.** Close the optical ledger (OMA to sensitivity with penalties), the electrical ledger (COM), the energy ledger (pJ/bit), and the thermal envelope together. (Sections 7.7 and 9.13 and sections 9.5.2 and 9.13.1)

6. **Noise, sensitivity, and receiver design.** Match photodiode, TIA, and equalization to the power the laser and channel deliver. Budget RIN, shot, and thermal noise against the pre-FEC BER target. (Chapter 4 and sections 4.3 to 4.5)

7. **Validation evidence.** Walk the ladder: bring-up, characterization, margin corners, stress qualification, and production readiness. Name the instrument and reference plane for every number. (Chapter 7 and sections 7.1 and 7.9)

8. **Yield and supplier readiness.** Multi-lot yield, SPC, ATP coverage, NPI gates, first-article, and 8D discipline. Prove the part can be built at volume before committing the fleet. (Chapter 8 and sections 8.6 and 8.10)

9. **Fleet telemetry and corrective action.** CMIS monitors, FEC histograms, triage buckets, RMA codes, and the feedback loop from field to ATP. The system is not done until the fleet can detect, classify, and correct a failure without the design team. (Sections 7.12 and 8.11)

A design review should be able to point at evidence for every step. Where evidence is missing, the step is not done. Where two steps conflict (power budget versus serviceability, yield versus guardband), the trade must be stated and owned, not hidden behind a single-number spec.

9.16 Engineering lens

9.16.1 How it works

An AI fabric is judged by delivered workload time, not port rate, and the optical layer sits on that critical path. The chapter places every earlier device into scale-up and scale-out networks and shows where copper ends and optics wins the link.

9.16.2 How it is measured

Measure the network at workload, fabric, link, and module layers. At workload level, record step time, collective latency, accelerator idle time, and tail behavior. At fabric level, record delivered bandwidth, queue occupancy, route balance, retries, and failed links. At link level, record pre-FEC BER, FEC error distributions, equalizer state, and flaps. At module level, record optical power, temperature, wavelength or lock state, and electrical power. A peak-rate test does not replace an all-reduce or all-to-all run on the intended topology (Sections 7.12 and 9.7).

9.16.3 How it fails

Scaling can fail through oversubscription, poor route balance, head-of-line blocking, a straggling link, excess retries, switch-radix limits, fiber-count and service errors, or a power and cooling ceiling. Architecture adds distinct escape paths: pluggables can run out of faceplate power, linear optics can expose host margin, and co-packaged optics can turn package thermals, lock control, or fiber service into the limiting risk. A working link can still be a bad system choice if it cannot be monitored, replaced, or operated at fleet scale.

Failure mode: Collective slowdown

Symptoms. Accelerator count rises, but job throughput flattens or falls and tail step time grows.
Likely causes. Fabric oversubscription, uneven routing, one degraded rail, retransmits, switch congestion, or synchronization amplifying a straggler.
Measurements. Collective traces, route and queue telemetry, per-link FEC and retry counters, topology map, and a scale sweep.
Mitigations. Repair weak links, rebalance routes, remove oversubscription, change the collective or topology, and gate the next scale point on delivered workload performance.

9.16.4 How it is debugged

Start with the workload symptom and identify the slow collective, rail, or time window. Compare topology and route data with link counters. A single lane with rising FEC points to the optical path; many clean links with full queues point to fabric capacity or scheduling. Remove one rail, reroute one group, or replace one suspect module to test causality. Keep optics, switch, and workload timestamps aligned. Otherwise a link flap and a collective stall cannot be ordered reliably.

Debug story

Observed. All-reduce tail latency rose after a rack expansion, while average link utilization looked normal.
Investigation. Per-rail traces showed one path with FEC bursts and retries. A module swap moved the symptom with the module.
Finding. The fabric had capacity, but one marginal optical lane set the collective tail.
Root cause. A contaminated connector raised ORL and produced burst errors without a large average-power change.
Resolution. The connector was replaced, inspection was added to the expansion runbook, and collective-tail alarms were tied to link-level error bursts.

9.17 Interview and design review questions

Concept.

- Why does inference (not only training) put the optical link on the critical path?

- What is the difference between scale-up and scale-out, and why do they use different optics?
- Why does an optical circuit switch complement rather than compete with co-packaged optics?

Design.

- Which workload and collective set the latency, bandwidth, and tail targets?
- Where is the scale-up to scale-out boundary, and what measured reach or power limit places it there?
- How do pluggable, linear pluggable, and co-packaged choices trade host margin, power, cooling, fiber count, repair time, and supplier concentration?
- What happens to a running job when one lane, module, external laser, or switch fails?

Debug.

- Collective tail latency rose after a rack expansion. Average link utilization looks normal. Where do you look?
- A single lane has rising FEC errors but its Rx power is stable. Apply the debugging fork: is this a power problem or a signal-quality problem?
- Which telemetry reaches fleet software, and can it identify a weak link before the workload stalls?
- What data would show that the proposed optical architecture is not the cluster bottleneck?

Manufacturing and operations.

- How do laser choices, WDM architecture, and validation methodology from earlier chapters influence cluster-scale outcomes?
- What is the service-time budget for replacing a failed module before the scheduler must reroute?
- How does connector mating-cycle budget change with a weekly hot-swap maintenance schedule?

Key idea. An AI fabric is judged by delivered workload time, not aggregate port rate. Connect collective traces to queue, route, FEC, optical, thermal, and service data. Choose pluggables, linear optics, or co-packaging by the measured system constraint and by how the fleet detects, contains, and repairs each failure.

Chapter 10

Failure analysis handbook

This chapter is a symptom-first field guide. Start with what the bench, production line, or fleet reports. Preserve the failing state until its evidence has been captured, then use the same workflow for every incident:

$$\begin{aligned} \text{observe} &\longrightarrow \text{scope unit, lot, vendor, site, or fleet} \\ &\longrightarrow \text{classify sudden or gradual} \\ &\longrightarrow \text{identify the changing margin} \\ &\longrightarrow \text{measure, isolate, correct, and prevent recurrence.} \end{aligned}$$

Choose each measurement for its ability to separate competing hypotheses. The debugging pyramid in Section 1.8, the power-versus-signal fork in Section 4.8, and the fleet router in Table 7.5 provide the same method at different scales.

Power loss Split launch-power loss from coupling, connector, MUX, fiber, monitor, and receiver-plane errors (Section 10.1).

BER increase Separate a shifted waterfall from a BER floor, then check power, eye quality, noise, timing, and spectrum (Sections 10.2 and 10.3).

Eye closure or low ER Split bandwidth, drive, bias, reflection, dispersion, and resonance alignment (Sections 10.4 and 10.7).

Lane imbalance Split source, modulator or ring, filter or MUX, fiber attach, driver, and receiver (Section 10.5).

Temperature sensitivity Track power, wavelength, lock error, TEC or heater headroom, receiver margin, and package stress (Section 10.13).

Wavelength drift Split source movement from filter, ring, TEC, and control-loop movement (Section 10.6).

Intermittent bursts Preserve counters before reseating. Check connector contamination, ORL, supply noise, lock state, and weak attach (Section 10.9).

Yield drop Clear the tester, then split lot, site, process step, assembly, firmware, and calibration (Section 10.11).

The cases below provide the detailed measurements and corrective actions. Use the failure and debug callouts in Chapters 3–9 as shorter versions of the same method. Earlier chapters own the mechanism physics. This chapter owns the order of operations.

10.1 Power loss

Observed behavior. Received power or OMA falls at one or more lanes. BER may remain stable at first, then rise as receiver margin is consumed. The module monitor and an external power meter may disagree.

Likely hypotheses. Launch power can fall because the laser is disabled, thermally rolled over, or aged. Power can disappear after the source through modulator loss, coupling shift, MUX loss, fiber bend, connector contamination, or a wrong reference-plane calibration. A drifting monitor photodiode can report loss that does not exist.

Measurements and root-cause isolation.

1. Compare CMIS Tx and Rx power, external power at the faceplate, bias current, and case temperature. Disagreement identifies the first suspect plane.
2. Walk optical power plane by plane with a known source and calibrated meter. Do not change bias or equalization while locating the loss.
3. Rerun LIV at the failing temperature. Moved LIV points toward the source; stable LIV points toward the optical path or monitor.
4. Inspect and clean connectors, then measure insertion loss and ORL. Use a golden fiber and module swap to separate field plant from module.
5. For a weak lane, compare sibling lanes and per-lane coupling. Lot or lane clustering points toward assembly or MUX variation.

Corrective action and recurrence control. Repair the first plane where power diverges. Correct calibration or monitor coefficients before changing source bias. Add a power check at the earliest production plane that can catch the signature and retain golden-path baselines for fleet comparison.

10.2 BER increase: waterfall shift or floor

Observed behavior. Pre-FEC BER rises, but the first task is to determine whether the whole BER waterfall moved to higher received power or whether it stopped improving at a horizontal floor. A waterfall is BER versus received power from a VOA sweep at a named reference plane. One operating-point BER is not enough to classify the failure.

Likely hypotheses. A shifted waterfall points toward lost power, receiver sensitivity, eye closure, timing, or dispersion: the link still responds to more photons, but the power needed for a given BER has moved. A floor points toward signal-proportional noise, reflection, crosstalk, or bursty impairments: past a point, more power does not help because noise grows with the signal. The distinction prevents a power fix from being applied to a noise-limited link. FEC error timing further splits the floor and shift cases: randomly sprinkled errors fit Gaussian or steady RIN; clustered bursts fit MPI, connector intermittents, or shared supply and clock events.

Measurements and root-cause isolation.

1. Sweep received power and plot BER rather than recording one operating point. Confirm whether the curve shifted, floored, or both (a soft floor on top of a shift is common).
2. If the waterfall shifted, compare received OMA, TDECQ, receiver sensitivity, equalizer taps, wavelength, and temperature with the golden baseline. Golden-swap Tx and Rx to assign ownership.
3. If the curve floors, follow Section 10.3 and split intrinsic RIN, electrical noise, ORL, MPI, and crosstalk.
4. Use FEC error timing and lane correlation to separate random noise from bursts and shared disturbances. Preserve counters before reseating if the pattern is intermittent.

Corrective action and recurrence control. Restore the margin ledger that moved, then repeat the full BER sweep at loaded corners. Store waterfall shape, not only pass/fail BER, so later fleet changes can be classified without guessing. The interview study treatment of these shapes is in Section A.3.7.

10.3 BER floor

Observed behavior. Pre-FEC BER improves as you increase transmit or received power, then stops improving and flattens at a constant floor regardless of how much more power you add. The FEC histogram may still look random (steady RIN) or bursty (MPI, intermittents); the floor shape alone does not decide that split.

Likely hypotheses. A BER floor means a noise source that grows with signal power dominates the link. The classic cause is *relative intensity noise* (RIN): because $\sigma_{\mathrm{RIN}} \propto I$, once RIN dominates the noise budget, signal and noise grow together and Q saturates at $Q_{\max} = 1/\sqrt{\mathrm{RIN}_{\mathrm{lin}} \cdot \mathrm{BW}}$ (Section 4.3). Other causes: multipath interference (MPI) from dirty connectors or high back-reflection, or broadband noise on the laser bias rail that converts to equivalent RIN (Section 5.8).

Measurements and root-cause isolation.

1. Confirm the floor exists: sweep received power (or Tx OMA) and plot BER vs. power. A healthy link has a steep waterfall; a floor appears as a horizontal asymptote.
2. **Bisect optical vs. electrical RIN.** Measure RIN with a quiet SMU powering the laser (intrinsic RIN). Then repeat with the product bias board connected. If the floor moves, the electrical path is injecting noise (Section 5.8).
3. **Sweep ORL.** Add a controlled reflector. If BER floor worsens with lower ORL, the laser is feedback-sensitive. Check isolator, connector, and fiber-attach cleanliness.
4. **Check for MPI.** Look at the ESA (electrical spectrum analyzer) for discrete spurs at frequencies corresponding to round-trip delays of reflective interfaces. MPI creates bursty errors; the FEC histogram shows clustered symbol errors rather than random ones.
5. **Confirm RIN spec.** Compare measured $\mathrm{RIN}_x\mathrm{OMA}$ at the stated ORL against the PMD or ATP limit (e.g. -136 dB/Hz at 17.1 dB ORL for DR-class links [100G Lambda MSA / IEEE 802.3]). If the part exceeds spec, reject.

Corrective action and recurrence control. If intrinsic RIN is the limiter, the laser is marginal or aged; replace or derate. If electrical RIN, fix the bias supply (better PSRR, star ground, local decoupling). If ORL-driven, clean or replace connectors and verify isolator function. If MPI from multiple reflections, reduce the number of mated interfaces or improve their ORL.

10.4 Low extinction ratio

Observed behavior. Transmitter OMA looks low on the DCA even though average power is in range. TDECQ may or may not fail depending on how the reference equalizer compensates.

Likely hypotheses. Extinction ratio ER $= P_1/P_0$ sets how far apart the one and zero optical levels are for a given average power. Low ER means the zero level is too high (the modulator does not fully extinguish) or the one level is too low. In an EML, ER is set by the EAM reverse bias: insufficient bias leaves residual transmission in the off-state. In a DML, ER depends on modulation depth relative to threshold. The OMA penalty for finite ER is PP $= (\text{ER} + 1)/(\text{ER} - 1)$: at 10 dB ER the penalty is $\sim$0.87 dB; at 6 dB ER it rises to $\sim$2.2 dB (Section 4.4).

Measurements and root-cause isolation.

1. Measure ER on the DCA (outer OMA / average power, or directly from the histogram levels). Compare against the PMD limit.
2. **EML:** Sweep EAM bias. ER should peak at the optimal bias point; if the curve has shifted (aged EAM), the operating point needs recalibration. Check EAM bias DAC code vs. datasheet.
3. **DML:** Check bias current vs. LIV. If bias is close to threshold, modulation depth is limited. Increase bias (but watch thermal rollover and RIN).
4. **MZM:** Check quadrature bias. If the MZM has drifted off quadrature ($V_\pi/2$ point), extinction degrades. Log the bias-control loop error signal; a saturated loop indicates drift beyond correction range.
5. **Ring:** Check resonance alignment. If the ring is detuned from the laser wavelength, extinction drops. Monitor the thermal tuner current and wavelength-lock error.

Corrective action and recurrence control. Recalibrate the modulator operating point. For EML aging, update the EAM bias setpoint in firmware or flag the module for replacement if the absorption curve has shifted beyond the correctable range. For MZM drift, verify the bias controller and its monitor PD. For rings, retune or check for neighbor thermal crosstalk (Section 6.5).

10.5 Lane imbalance

Observed behavior. In a multi-lane module (DR4, FR4, DR8), one or more lanes show significantly different OMA, TDECQ, or pre-FEC BER compared with siblings in the same module. The weak lane may be marginal or failing while others are healthy.

Likely hypotheses. Multi-lane modules share a substrate, laser array (or CW-WDM source), and thermal environment. Lane-to-lane variation comes from: (1) die-level non-uniformity in the laser or modulator array (threshold, slope, V_π, coupling), (2) packaging variation in fiber-array alignment (one channel of the FAU slightly misaligned), (3) driver or TIA channel mismatch on the electronic IC, or (4) thermal gradient across the die (edge lanes hotter or cooler than center lanes).

Measurements and root-cause isolation.

1. Measure all lanes: OMA, ER, TDECQ, RIN, wavelength. Identify the outlier.
2. **Is it optical or electrical?** Swap the electrical lane assignment (if the host supports lane remapping) or test the suspect optical lane with a known-good driver channel. If the problem follows the optical path, it is laser/modulator/fiber-attach. If it follows the electrical channel, it is driver/TIA.
3. **Check coupling.** Measure per-lane fiber-coupled power with an integrating sphere or power meter array. A single weak lane with normal LIV suggests FAU misalignment.
4. **Thermal map.** Use an IR camera or CMIS per-lane monitors (if available) to check for hot spots. Edge lanes near the package wall or near a TEC boundary may run hotter.
5. **Laser array aging.** If the imbalance grows over time (HTOL or field life), one laser in the array is aging faster (threshold rise, slope drop). Compare LIV curves at t_0 and now.

Corrective action and recurrence control. FAU misalignment is a manufacturing escape; tighten incoming inspection or first-article coupling specs. Thermal gradient: redesign TEC zoning or derate the hot lane. Laser aging: flag the lot and check burn-in screening effectiveness. Driver mismatch: work with the IC supplier on channel-to-channel gain flatness.

10.6 Wavelength drift

Observed behavior. In a WDM system, one or more channels walk off the ITU grid or the ring/filter passband. BER degrades as the channel moves off the receiver filter or MUX/DEMUX passband. In ring-modulator CPO, this manifests as sudden unlock of the wavelength control loop.

Likely hypotheses. Laser wavelength moves with temperature and bias current. If the TEC or wavelength-locker servo cannot track, the channel walks off its assigned slot. In microring systems, resonance also moves strongly with temperature, and neighbor heaters create thermal crosstalk that pushes adjacent channels (Sections 6.4 and 6.5).

Measurements and root-cause isolation.

1. Measure wavelength on an OSA or wavemeter. Compare to the target grid.
2. **Laser-side:** Check TEC current. If saturated (at max or min drive), the thermal load exceeds TEC capacity. Check case temperature and airflow.
3. **Locker-side:** Read the wavelength-locker error signal. A healthy loop holds near zero; drift means the servo is losing lock. Check locker etalon alignment and PD balance.
4. **Ring CPO:** Check the ring thermal tuner DAC code. If it has railed (max heater power), the ring cannot reach the target wavelength. Check for neighbor heating (all adjacent lanes at full traffic and max case T).
5. **Aging:** If drift is progressive over weeks/months, suspect laser mode hop or gradual TEC degradation (reduced ΔT capacity).

Corrective action and recurrence control. TEC saturation: reduce case temperature (improve airflow or liquid cooling) or derate the laser operating current. Ring unlock: increase heater headroom in the design, reduce thermal crosstalk with layout changes, or shift the CW-WDM source grid to re-center the ring tuning range. Aging: schedule preventive replacement (ELSFP hot-swap, Section 5.14).

10.7 Eye closure (high TDECQ)

Observed behavior. TDECQ exceeds the PMD limit even though average power and ER look acceptable. The DCA eye appears compressed or distorted after the reference equalizer is applied.

Likely hypotheses. TDECQ measures how much noise the transmitter can tolerate before BER exceeds the FEC threshold, relative to an ideal transmitter (Section 7.4). High TDECQ means the equalized eye is poor. Common causes: (1) insufficient EO bandwidth (modulator or driver roll-off), (2) poor level linearity (RLM < 0.95; driver or modulator compression), (3) pattern-dependent effects (ISI from bandwidth limit, reflections, or impedance mismatch), (4) chromatic dispersion on FR-class fiber eating into the margin.

Measurements and root-cause isolation.

1. Inspect the raw (unequalized) eye on the DCA. Is it bandwidth-limited (rounded transitions), compressed (uneven levels), or noisy (RIN/jitter)?

2. **Bandwidth:** Measure EO S_{21} of the modulator (if accessible) or the combined Tx path. Compare to the Nyquist frequency. If 3-dB BW is below Nyquist, bandwidth is the limiter.

3. **RLM:** Compute relative level mismatch from the DCA histogram. If RLM < 0.95, the PAM4 levels are unevenly spaced. Check driver linearity (swept DAC code) and modulator transfer function (bias sweep).

4. **Reflections:** Check S_{11} of the RF path (driver to modulator) and the optical return loss. Reflections cause post-cursor ISI the FFE cannot fully cancel.

5. **Dispersion:** Measure TDECQ with and without the test fiber (TECQ vs. TDECQ). If TDECQ is significantly worse than TECQ, dispersion is the marginal contributor. Check Tx wavelength and chirp against the fiber length and dispersion coefficient.

Corrective action and recurrence control. Bandwidth: upgrade driver or modulator (higher-BW die or peaking network). RLM: tune driver pre-emphasis (DAC levels for each PAM4 symbol). Reflections: fix wirebond, impedance discontinuity, or connector. Dispersion: tighten wavelength tolerance or shorten the fiber (reroute).

10.8 Thermal runaway

Observed behavior. Module temperature rises continuously until shutdown (CMIS over-temperature alarm) or until the laser degrades catastrophically. May start subtly: BER creeps up as case T rises during traffic ramp or with neighbor-lane loading.

Likely hypotheses. In a faceplate pluggable, double-digit-watt module power must leave through the cage and heatsink [106]. If airflow is blocked, the cage is overloaded, or the TEC inside the module is fighting a losing battle against junction temperature, the thermal loop runs away. In CPO, the optical engine sits on the switch substrate beside the ASIC; cooling-path faults concentrate heat on the source and ring controls [107]. Both sources are vendor or research orientation, so the product thermal model remains authoritative.

Measurements and root-cause isolation.

1. Read CMIS module temperature and compare to the module's rated case T range. If case T exceeds the max, the system cooling is inadequate.

2. **Check TEC current.** A TEC at max drive current is saturated; it cannot pump more heat. The junction temperature is higher than the case T suggests.

3. **Measure LIV at temperature.** If threshold rises and slope drops steeply with T, the laser is near thermal rollover (Section 5.13). The operating point may be marginal.

4. **Neighbor loading.** Bring all lanes and neighbor modules to full traffic simultaneously. If the problem only appears under full-cage load, the thermal design margin is insufficient.

5. **Airflow audit.** Inspect the faceplate for blocked vents, incorrect fan speed, or missing blanking panels (bypass airflow). For liquid-cooled systems (CPO, XPO), check coolant flow rate and inlet temperature.

Corrective action and recurrence control. System-level: improve airflow, lower ambient, or reduce module count per cage. Module-level: derate the laser (lower bias current reduces self-heating) or switch to a lower-power module style (LPO instead of retimed, Section 9.5.1). CPO: ensure the cold-plate thermal interface material (TIM) is intact and the liquid loop meets flow-rate spec. Long-term: specify a tighter thermal class in the laser requirements (Section 5.6).

10.9 Intermittent failures

Observed behavior. Links flap, lose lock, or show bursts of FEC errors while average power and a short bench BER test look normal. The symptom may clear after reseating, cooling, or restarting firmware.

Likely hypotheses. Intermittent faults come from contacts, reflections, weak optical or electrical attach, supply noise, wavelength-lock loss, firmware state, or environmental stress. Reseating can remove the evidence by cleaning a contact, changing fiber stress, or resetting a state machine.

Measurements and root-cause isolation.

1. Freeze CMIS state, FEC error timing, LOS or LOL history, temperatures, rails, lock error, and neighbor activity before touching hardware.

2. Scope the pattern across lane, module, tray, rack, lot, and firmware revision. Shared timing points toward power, cooling, firmware, or a shared source.

3. Correlate bursts with Rx power, ORL, supply and clock spurs, lock-loop state, vibration, and temperature. Use trigger capture rather than averages.

4. Run controlled disturbance tests one at a time: connector motion, thermal ramp, neighbor load, rail load, and firmware state transition.

5. Repeat on a golden unit and fixture. If the fault follows the unit, inspect fiber attach, solder, contacts, and control logs before destructive analysis.

Corrective action and recurrence control. Fix the confirmed contact, attach, supply, lock, or firmware cause. Add event-triggered telemetry and a production stress that reproduces the fault. Keep intermittent and no-fault-found RMA codes separate from laser wear-out.

10.10 Connector contamination

Observed behavior. Intermittent link failures, burst errors, or elevated BER that clears after reseating or cleaning a connector. Rx power may fluctuate or show sudden drops. Often affects one direction of a duplex link.

Likely hypotheses. A particle of dust on an MT, LC, or MPO ferrule endface scatters and absorbs light, raising insertion loss and back-reflection (lowering ORL). Debris in the core zone can cause large loss even when most of the ferrule looks clean. Elevated ORL feeds back into the laser and raises RIN, causing burst errors even when average power looks acceptable. In high-power CW-WDM and ELSFP systems, trapped particles can burn onto the fiber endface and cause permanent damage.

Measurements and root-cause isolation.

1. **Inspect first.** Use a fiber-endface inspection scope (200–400×) on every mated connector before any other debug. Look for particles in the core zone, scratches, pits, and residue.

2. **Clean and re-inspect.** Use a dry-click cleaner or lint-free wipe with IPA. Re-inspect. If the endface still fails IEC 61300-3-35 zone criteria, replace the jumper [98].

3. **Measure IL and ORL.** After cleaning, measure insertion loss and ORL across the mated pair. Compare to the link-budget allocation (Section 7.7).

4. **Correlate with BER.** If BER clears after cleaning, the contamination was the root cause. Log the connector location and date code for trend analysis.

5. **Repeat offenders.** If the same location re-contaminates quickly, check for airborne particulates (raised-floor debris, construction dust) or improper cap/cover discipline during service events.

Corrective action and recurrence control. Immediate: clean and verify. Preventive: install dust caps on every unused port, enforce "inspect before connect" policy in the service runbook, use sealed cassettes or connectorized trunk cables that minimize open-ferrule exposure. For high-power paths (ELSFP, CW-WDM), any connector that shows burn damage must be replaced, not re-cleaned. Track contamination-related RMAs as a distinct failure code (not "laser failure") so FIT accounting stays honest (Section 7.12).

10.11 Yield drop

Observed behavior. First-pass yield falls below its stable baseline. The loss may cluster on one ATP row, lane, tester, shift, supplier lot, assembly site, or firmware revision.

Likely hypotheses. Process drift, incoming material variation, fiber-array alignment, die-attach or solder change, tester or fixture drift, stale calibration, a software limit change, or a guardband that no longer matches measurement spread.

Measurements and root-cause isolation.

1. Contain suspect work in process and freeze tester software, limits, fixtures, and calibration records.

2. Build a Pareto by ATP row, lot, date code, site, tester, and shift. Do not average away a one-lane or one-station signature.

3. Run a golden unit across stations and the same failed unit on the reference bench. This clears the measurement system before supplier action begins.

4. Compare upstream wafer, die, assembly, and incoming data at the first point where good and bad populations separate.

5. Select failure analysis that distinguishes the remaining mechanisms, then verify the correction on a controlled lot and watch the field code.

Corrective action and recurrence control. Restore the changed process or measurement input, add a statistical control at the first observable point, revise ATP only with correlation data, and require a new first-article check before releasing volume (Section 8.10).

10.12 Aging, thermal response, and margin erosion

Use time scale and recovery to route the incident. A reversible shift during a temperature or neighbor-load sweep is a thermal operating-point problem. A baseline that moves over stress hours or field months is aging. A sudden permanent step after a cycle points toward damage or an assembly defect, not ordinary thermal response.

1. Return the unit to its starting temperature and operating point. Record whether power, wavelength, bias, TDECQ, and BER recover.
2. Compare with ship and pre-stress data. Permanent LIV, spectrum, or bias-curve movement supports aging or damage.
3. Repeat the temperature sweep with source, wavelength-selective element, receiver, and neighbors isolated in turn.
4. Update the power, noise, timing, spectral, and control ledgers (Section 5.19). Several small shifts can explain a BER failure even when each component remains inside its stand-alone limit.
5. Route reversible thermal loss to cooling, control, calibration, or derating. Route cumulative change to HTOL and life-model review. Route lot-clustered permanent steps to manufacturing failure analysis.

The corrective action must restore margin at combined corners. A room-temperature retest does not close a high-temperature incident, and one clean HTOL readout does not explain a reversible lock failure.

10.13 Temperature sensitivity

Observed behavior. BER, TDECQ, optical power, or lock stability degrades during a case-temperature ramp or under loaded-neighbor conditions. The unit may recover when cooled.

Likely hypotheses. Laser threshold and slope drift, wavelength movement, ring-resonance drift, TEC or heater saturation, EAM or MZM bias error, receiver-noise rise, package stress, or a thermal gradient that affects one lane.

Measurements and root-cause isolation.

1. Record case temperature, external optical power, wavelength, bias, TEC or heater current, lock error, OMA, ER, TDECQ, and pre-FEC BER on one time axis.

2. Repeat with neighbors off and on. A shared shift points toward thermal or supply coupling; a lone lane points toward its local path.

3. Hold the source fixed and move the wavelength-selective element, then reverse the test. This splits laser drift from ring or filter drift.

4. Rerun LIV and receiver sensitivity hot and cold. Separate lost launch power from lost receiver margin.

5. Inspect package and cooling interfaces if optical and electrical blocks pass alone but fail in the assembled system.

Corrective action and recurrence control. Restore thermal headroom, correct calibration and control limits, reduce coupling, or derate the operating point. Add the loaded-neighbor temperature ramp to the ATP or design-validation plan that missed it.

10.14 Failure-analysis checklist

Step	Question	Required record
Preserve	What evidence will a reseat, reboot, clean, or retest destroy?	CMIS, BER and FEC history, rails, temperature, firmware, fixture, and time
Scope	One unit, lane, lot, vendor, site, or fleet?	Population and correlation plot
Classify	Sudden or gradual, constant or intermittent, thermal or cumulative?	Timeline and recovery test
Locate margin	Did power, noise, timing, spectrum, or control move first?	Golden comparison and margin ledger
Falsify	Which measurement best separates the leading hypotheses?	Expected result for each hypothesis before the test
Confirm	Does the fault follow the suspected block under a controlled swap or stress?	Repeated failure and passing control
Correct	Does the fix restore the original failing condition with margin?	Before and after data at loaded corners
Prevent	Where can production or fleet monitoring catch recurrence earliest?	ATP change, control limit, alarm, owner, and due date

Table 10.1. Failure-analysis checklist. The incident is not closed until the cause is reproduced, corrected, and covered by a recurrence control.

10.15 Interview and design review questions

Scoping and evidence.

- BER rose on one rack after a software rollout. What data do you preserve before reseating modules?
- How would you decide whether a symptom belongs to one unit, one lot, one vendor, one site, or the fleet?

Root-cause isolation.

- Received power is unchanged but BER worsened. Which measurement do you make next, and which hypotheses does it separate?
- How do you distinguish a BER waterfall shift from a BER floor?
- BER degrades after thermal cycling. How do you separate reversible thermal response, calibration error, assembly damage, and long-term aging?

Production and fleet action.

- A golden unit fails only on one production station. What must be cleared before opening supplier failure analysis?
- Which recurrence control belongs in ATP, and which belongs in fleet telemetry?
- When is a no-fault-found return evidence of weak triage rather than a good unit?

Key idea. A useful failure analysis starts with a symptom and ends with a new control. Preserve the failing state, split shared from local behavior, clear the measurement system, and choose one measurement that can disprove the leading hypothesis. The corrective action is incomplete until production or fleet data show that the same signature no longer escapes.

Chapter A

One-week optical systems interview review

Use this appendix as a standalone review sheet for about a week of focused prep. Its purpose is not to cover the most optics. Its purpose is to make your engineering process automatic under time pressure.

A.1 The philosophy: reduce uncertainty to decide

Key idea. The purpose of engineering is not to find certainty. It is to reduce uncertainty enough to make the next decision.

Validation, measurement, debugging, qualification, supplier choices, and production are the same activity under different names: uncertainty reduction.

$$\begin{aligned} \text{validation} &\longrightarrow \text{measurements} \longrightarrow \text{debugging} \\ &\longrightarrow \text{qualification} \longrightarrow \text{supplier decisions} \\ &\longrightarrow \text{production.} \end{aligned}$$

Key idea. The goal of an optical systems engineer is not to know every component. It is to make good engineering decisions under uncertainty using measurements, physics, and evidence.

This role owns laser direction inside an IM/DD interconnect effort: requirements, validation strategy, partner quality, and fleet risk. Lab measurement is how you decide, not the whole job. Hands-on fluency still matters. Senior people lose credibility if they cannot run or interpret LIV, RIN, ORL, TDECQ, and a BER waterfall. They also lose the level if they stop at plots and never close on a control.

Bayesian engineering, named. You already use this pattern. Name it so the interviewer hears the method:

$$\text{observation} \longrightarrow \text{hypotheses} \longrightarrow \text{measurement}$$
$$\longrightarrow \text{belief updated / hypothesis eliminated.}$$

Every measurement should change what you believe. A test that leaves the hypothesis list unchanged was the wrong test, or the wrong reference plane.

Ask the scale before the root cause. Root cause is not enough. Ask at what scale the problem exists:

$$\text{device} \longrightarrow \text{subsystem} \longrightarrow \text{module}$$
$$\longrightarrow \text{rack} \longrightarrow \text{datacenter} \longrightarrow \text{fleet.}$$

Scale picks the owner: design, process, supplier, manufacturing, or operations.

What decision does this measurement unlock? Make this a mantra. Instruments do not exist to produce plots. They exist to unlock a decision.

Power meter Should I chase the optical path or signal integrity?

OSA Is spectral alignment still plausible?

LIV Has the device itself changed?

DCA / TDECQ Is the eye still inside the budget at this corner?

BER waterfall Is this a sensitivity shift or a noise floor?

Work the day plan at the end of this appendix. Use each LLM practice box the day it is scheduled. Do not add new topics after Day 6.

A.2 The answer spine

Move every answer through the same sequence. The diagram is a memory aid, not the answer. In the interview you should be able to speak one clear paragraph for each arrow, naming what you know, what is still uncertain, and what measurement you would take next. Do not jump from a symptom to a component. The systems loop in Section 1.6, the debugging pyramid in Section 1.8, and the failure-analysis method in Chapter 10 are the full versions of this spine.

$$\text{requirements} \longrightarrow \text{architecture} \longrightarrow \text{measurements}$$
$$\longrightarrow \text{observations} \longrightarrow \text{hypotheses}$$
$$\longrightarrow \text{isolation} \longrightarrow \text{root cause}$$
$$\longrightarrow \text{corrective action} \longrightarrow \text{recurrence control.}$$

Requirements. Start by stating what the system must do before you name a part. Reach, lane rate, aggregate capacity, power envelope, lifetime, cost, and manufacturing volume are the constraints that decide everything downstream. A 500 m DR link at 100 G/lane with a tight power budget is a different problem from a 2 km FR link or a dense WDM co-packaged engine. Say the requirement out loud even when the interviewer hands you a failed module, because the requirement tells you which margins matter and which architectures were even eligible. If you cannot name the requirement, you cannot tell whether a measurement is relevant.

DR Datacenter reach: IEEE 802.3 single-mode class, typically 500 m at 1310 nm. Default for AI scale-out optics.

FR Far reach: IEEE 802.3 single-mode class, 2 km. Tighter chirp and dispersion budget than DR.

WDM Wavelength division multiplexing: multiple λ on one fiber. CWDM (20 nm spacing) or DWDM ($\leq$100 GHz).

Architecture. Translate the requirement into an architecture path: fiber plant, wavelength, source, modulator, receiver, and digital stack. A VCSEL path commits you to 850 nm multimode fiber and direct modulation. A DFB or EML path at 1310 nm commits you to single-mode fiber and leaves a choice among direct modulation, electro- absorption, a silicon MZM, or a ring. Each path then sets detector material, thermal control, test coverage, and service policy. Architecture is where you show that component choices are not preferences; they are consequences of the requirement (Table 5.1).

VCSEL Vertical-cavity surface-emitting laser: 850 nm class, MMF SR links.

DFB Distributed-feedback laser: grating in the active region for single-mode output. Dominant 1310 nm source for DR/FR and EML gain sections.

EML Electro-absorption modulated laser: DFB plus EAM on one InP chip; dominant 100–200G/lane pluggable Tx.

MZM Mach–Zehnder modulator: push-pull interferometer; broadband, low chirp. Si MZM, TFLN MZM, or hybrid platforms.

MRM Microring modulator: resonant Si modulator; compact, WDM-native, needs wavelength lock. CPO workhorse at 200G/ch.

Measurements. Name the instruments you would use and the uncertainty each one removes. Always state the reference plane, pattern, temperature, and pass criterion with the instrument. A number without a plane is not a measurement. Ask out loud: what decision does this measurement unlock?

A power meter does not "measure power" as an end in itself. It answers whether the optical path still owns the failure, or whether you should move to signal integrity. An LIV setup answers whether the laser itself changed threshold or slope. An OSA answers whether spectral alignment is still plausible (wavelength and SMSR).

A DCA answers whether the eye is still inside budget: OMA, ER, RLM, TDECQ, and eye shape. A BERT and FEC counters answer whether you have a sensitivity shift, a floor, or a burst pattern.

A VNA answers electrical or electro-optic bandwidth. An ORL meter and inspection scope answer whether the plant is reflecting or dirty.

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

OSA Optical spectrum analyzer: measures wavelength, SMSR, and side-mode structure of a laser source.

SMSR Side-mode suppression ratio: power difference (dB) between the lasing mode and the strongest side mode (OSA).

DCA Digital communication analyzer: sampling oscilloscope used for eye diagrams and TDECQ.

OMA Optical modulation amplitude: outer PAM4 level swing $P_1 - P_0$.

ER Extinction ratio: P_1/P_0 in dB. Higher ER widens the OMA for a given average power; trades against chirp and driver swing.

RLM Relative level mismatch: PAM4 level spacing uniformity. 1.0 is perfectly even; CEI often requires $\geq$0.95.

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

Observations. Report what the instruments actually showed, not what you hoped they would show. Separate facts from interpretation. "Received power held at -4 dBm while pre-FEC BER rose from 10^{-8} to 10^{-5} at $70°$C" is an observation. "The laser is dying" is not. Scope the observation: one lane or all lanes, one unit or a lot, sudden or gradual, recoverable on cool-down or permanent. Preserve telemetry, screenshots, and the failing state before you reseat, clean, reboot, or rewrite a calibration table. Observations that disappear under debugging are observations you never had.

BERT Bit-error-ratio tester: instrument or ASIC function that generates PRBS and counts errors for pre-/post-FEC BER.

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

VNA Vector network analyzer: measures S-

Hypotheses. Turn the observations into a short list of competing causes, ranked by how well they explain the full pattern. Use the power-versus-signal-quality fork as the first split. If average power moved, hypothesize source output, coupling, connector loss, multiplexer loss, or monitor calibration. If power held but BER worsened, hypothesize eye closure from ER, OMA, RLM, or TDECQ degradation; RIN under reflection; ISI or jitter; wavelength walking off a filter; receiver sensitivity loss; or a wrong calibration table. Keep the list short enough that the next measurement can kill more than one item. A hypothesis you cannot falsify with a bench step does not belong on the list yet.

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed $\mathrm{RIN_xOMA}$ under fixed ORL.

Isolation. Good engineers perform measurements. Great engineers perform the minimum measurement that eliminates the largest number of hypotheses. Choose the next cut for uncertainty removed per hour of bench time, not for the instrument you like. A golden swap of transmitter versus receiver splits Tx from Rx. An optical-versus-electrical lane swap in a multi-lane module splits the optical path from the driver or TIA. A bias sweep at the failing temperature asks whether the optimum moved away from the stored table. An LIV compared with ship data asks whether the device aged or only the setpoint drifted. A BER-versus-power waterfall asks whether the curve shifted or floored.

TIA Transimpedance amplifier: converts PD current to voltage; input capacitance sets input-referred noise.

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

Root cause. State the mechanism that survived isolation, with the evidence that killed the alternatives. "EAM bias table segment wrong above $60°$C" is a root cause. "Bad eye at high temperature" is still a symptom. Name the physical or process mechanism when you can: facet wear, monitor-PD corruption, FAU misalignment, MPI from a dirty connector pair, TEC railed under load, supplier lot with high threshold. If the evidence only reaches "calibration drift" and not a deeper mechanism, say so. Overclaiming the root cause is worse than stopping one layer early with honesty.

EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

FAU Fiber array unit: a precision V-groove assembly that mates multiple fibers to a PIC edge or grating coupler array.

MPI Multipath interference: coherent beating from two or more reflective paths. Produces a BER floor and clustered FEC errors that more launch power does not fix.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

Corrective action. Fix the mechanism you named, under the condition that failed. Retune the table, replace the lot, change the thermal design, clean and inspect the plant, filter the supply, or rewrite the ATP limit. Verify the fix by reproducing the original failing condition: same temperature, same host, same fiber, same pattern. A fix that only works on a quiet room- temperature bench is not a fix. Containment comes first when the fleet is already exposed: stop ship, quarantine lots, or derate while the permanent fix is built.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

Recurrence control. Close with the control that stops the same escape next month. That control is usually a new or tightened test, an SPC chart, a telemetry alarm, a process-control limit, a supplier screen, or a firmware guard that refuses to boot with a railed actuator. Name the owner and the measurable closure criterion. Root cause without recurrence control is a story, not an engineering result. In the in-

SPC Statistical process control: lot-by-lot control charts (I_{th}, SMSR, power) to catch drift between qual stress tests.

terview, ending on recurrence control is what separates a debug narrative from a product-engineering answer.

LLM practice

Summary. Every answer reduces uncertainty enough to decide. Walk requirements to architecture to measurements to observations to hypotheses to isolation to root cause to corrective action to recurrence control. Every measurement should change what you believe. Name the reference plane, the decision unlocked, and the ledger that moved. Do not jump from a symptom to a part number.

Paste to an LLM and answer aloud. Quiz me on the philosophy and the answer spine. Ask for the uncertainty- reduction line, then one paragraph per spine step. Give me a failed module at high temperature with stable average power and make me walk the full spine. Stop me if I skip scope, the power fork, the control ledger, or recurrence control.

A.3 Ten concepts to know cold

A.3.1 Start at the system

For a new link, walk downward from capacity to the service model. Capacity sets lane rate. Lane rate and reach together set the fiber plant and the source class. Power and cost then decide whether you can afford a TEC, a retimer, or a dense WDM engine. Only after those constraints are named do you choose the modulator and the receiver, then the DSP and FEC stack, then the validation and field-service model.

$$\begin{aligned}\text{capacity} &\longrightarrow \text{lane rate} \longrightarrow \text{reach} \longrightarrow \text{power}\\ &\longrightarrow \text{fiber plant} \longrightarrow \text{source} \longrightarrow \text{modulator}\\ &\longrightarrow \text{receiver} \longrightarrow \text{digital signal processing and FEC}\\ &\longrightarrow \text{validation and service model.}\end{aligned}$$

A good answer explains why each choice constrains the next one. A VCSEL path points toward 850 nm, multimode fiber, silicon detection, and direct modulation with a short reach. A DFB path at 1310 nm points toward single-mode fiber, germanium detection, and a choice among a directly modulated laser, an EML, or an external silicon-photonic modulator. Neither path is "better." Each closes a different reach, power, cost, and manufacturing problem. The decision matrix is in Table 5.1.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

WDM Wavelength division multiplexing: multiple λ on one fiber. CWDM (20 nm spacing) or DWDM ($\lesssim$100 GHz).

DSP SerDes DSP-based SerDes: ADC on the Rx lane, digital FFE/DFE and timing recovery, DAC on Tx. Standard above $\sim$50 GBd; the "DSP" that cancels ISI at 224G lives inside the SerDes.

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

VCSEL Vertical-cavity surface-emitting laser: 850 nm class, MMF SR links.

MMF Multimode fiber: 850–940 nm VCSEL links (OM3/OM4/OM5); reach and modal BW limit to SR distances.

DFB Distributed-feedback laser: grating in the active region for single-mode output. Dominant 1310 nm source for DR/FR and EML gain sections.

SMF Single-mode fiber: 9 μm core; O-band (1310 nm) or C-band (1550 nm). G.652.D and G.657 for datacenter plant.

DML Directly modulated laser: modulate bias current; high chirp, low cost.

EML Electro-absorption modulated laser: DFB plus EAM on one InP chip; dominant 100–200G/lane pluggable Tx.

A.3.2 Scope before root cause

Before you open an instrument, walk the failure up the scope ladder. Each rung changes the owner and the next action:

$$\begin{aligned}\text{failure observed} &\longrightarrow \text{single unit?} \longrightarrow \text{lot?}\\ &\longrightarrow \text{vendor?} \longrightarrow \text{rack?}\\ &\longrightarrow \text{datacenter?} \longrightarrow \text{fleet?}\end{aligned}$$

Also ask time and change: new failure or long-running trend; sudden, gradual, intermittent, or temperature-dependent; firmware, supplier, process, fixture, workload, or environment change just before the symptom. Scope often removes more hypotheses than the first bench measurement. A fleet-wide gradual drift cannot be a single dirty connector. A single-lane sudden failure after a rework cannot be a population aging problem. A vendor-lot signature points to supplier containment before you redesign the module.

Preserve the failing state and its telemetry before you reseat, clean, reboot, or change calibration. Capture CMIS monitors, pre- FEC BER, bias currents, temperatures, LOS and LOL flags, and firmware versions. An intermittent that disappears under debugging is still a real failure; you just destroyed the evidence (Section 7.12 and table 10.1).

CMIS Common Management Interface Specification: module management, monitors, link training at 224G/448G.

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

BER Bit error ratio: errored bits divided by bits received. Pre-FEC BER is the validation metric; post-FEC is the service metric.

LOS Loss of signal: receiver (or host) flag that optical input is below the detect threshold.

LOL Loss of lock: CDR or SerDes unlocked; electrical path not recovered even if light is present.

A.3.3 Use the power-versus-signal-quality fork

First ask whether received optical power changed. That one question splits the debug tree.

If power changed, the power ledger moved. Inspect source output and enable state, coupling and FAU alignment, connector cleanliness, ORL, fiber and multiplexer loss, and monitor-photodiode calibration. An APC loop that trusts a drifting monitor will hold a wrong launch power while reporting that everything is fine. Confirm with an external power meter at a named reference plane.

If power held but BER worsened, the loss is in signal quality or in the receiver. Inspect OMA, ER, RLM, and TDECQ on a DCA; RIN under controlled ORL; jitter and ISI; wavelength alignment to filters or rings; receiver sensitivity and equalization; and calibration tables that set modulator bias. Also ask the receiver-side equivalent: did the required optical power increase even though the power meter reading did not? A sensitivity shift can look exactly like transmitter degradation until you do a golden swap (Sections 4.8 and 7.11).

FAU Fiber array unit: a precision V-groove assembly that mates multiple fibers to a PIC edge or grating coupler array.

ORL Optical return loss: reflected power back toward the laser. Low ORL raises RIN and can seed burst errors.

APC Automatic power control: feedback loop from a monitor PD that holds average optical power; slow loop BW keeps it out of the RIN band.

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed RIN_xOMA under fixed ORL.

A.3.4 Track five margin ledgers

Links rarely fail from one dramatic excursion. They fail when several small shifts spend different ledgers at once. Keep this checklist whenever you narrate a failure. Name which ledger moved, by how much, and what spent it.

Power · Noise · Timing · Spectral · Control

Power Launch power, insertion loss, coupling, connector loss, multiplexer loss, and receiver sensitivity. Track average power and OMA separately: average power can look healthy while OMA collapses.

OMA Optical modulation amplitude: outer PAM4 level swing $P_1 - P_0$.

Noise RIN, receiver thermal noise, shot noise, crosstalk, supply noise through weak PSRR, and MPI from reflective interfaces. Noise that scales with the signal makes a BER floor that more launch power cannot fix.

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed $\mathrm{RIN}_x\mathrm{OMA}$ under fixed ORL.

Timing Bandwidth, dispersion, jitter, ISI, and equalization reserve. A SerDes that is already using most of its FFE and DFE taps has little timing margin left even if the eye still opens on a DCA.

SerDes Serializer/deserializer: high-speed I/O that maps parallel data to PAM4 on the wire.

Spectral Wavelength, SMSR, filter or ring passband, thermal drift, and lock range. A ring whose heater is near its DAC rail is spending spectral margin even while BER still passes.

SMSR Side-mode suppression ratio: power difference (dB) between the lasing mode and the strongest side mode (OSA).

Control Headroom in the loops that keep the optics healthy: APC, TEC, heaters, ring lock, bias DACs, and calibration tables. A railed actuator or a corrupted table can fail the link while the laser diode itself is still healthy. Control margin is what separates “optics broke” from “the product can no longer hold the operating point.”

APC Automatic power control: feedback loop from a monitor PD that holds average optical power; slow loop BW keeps it out of the RIN band.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

The device and link treatment is in Section 5.19. Always ask the control ledger when actuators and tables are in the product.

A.3.5 Use the validation ladder

Key idea. Validation is staged uncertainty reduction.

For every stage, name the question the stage answers and the uncertainty it removes. A test that answers no question is cost, not confidence.

Bring-up Does the hardware power, initialize, emit, lock, and pass data on a known-good host? Removes integration unknowns only.

Characterization How does performance vary with temperature, voltage, lane, wavelength, pattern, and unit? Maps the population, not a hero sample. Record LIV, spectrum, RIN, OMA, ER, RLM, and TDECQ as distributions.

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

Margin testing How far is the design from each failure boundary? Push power, temperature, ORL, and supplies to the cliff and measure the distance in dB, degrees, and volts.

ORL Optical return loss: reflected power back toward the laser. Low ORL raises RIN and can seed burst errors.

Interoperability Does it work with target hosts, production fiber, real modules, and real firmware? Removes combination failures that golden benches miss.

Environmental testing What changes under temperature cycling, humidity, vibration, shock, and connector mating?

Reliability qualification Which parameters drift with stress and time? HTOL, with a named mechanism and justified activation energy, projects wear-out as a FIT number.

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

FIT Failures in time: failures per 10^9 device-hours. Fleet FIT times link count sets expected failures per day.

Production readiness Can the ATP catch bad units and process drift at useful cost and cycle time, with station correlation and guard bands sized to repeatability?

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

Fleet monitoring Which telemetry catches margin erosion before service failure, and does the RMA Pareto match the life model?

RMA Return merchandise authorization: field-failed unit returned for FA. Distinct failure codes keep fleet FIT honest.

Table 7.1 maps these stages to activities and instruments. The worked interview answer in Section A.5.2 walks the same ladder as spoken prose.

A.3.6 Know what each instrument answers

Do not recite instrument names. Say what uncertainty each one removes, and what decision that unlocks.

Power meter Did optical power change at a named reference plane? Decision: chase the optical path, or move to signal integrity. External meters catch monitor-PD lies.

LIV setup Did laser threshold, slope efficiency, voltage, kink behavior, or thermal rollover change relative to ship data? Decision: has the device itself changed?

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

Optical spectrum analyzer or wavemeter Did wavelength, SMSR, mode structure, or spectral width change on an OSA? Decision: is spectral alignment still plausible for filters and rings?

SMSR Side-mode suppression ratio: power difference (dB) between the lasing mode and the strongest side mode (OSA).

RIN setup Is transmitter intensity noise limiting BER, and does it depend on ORL or on the product bias board rather than the laser itself? Decision: is the floor optical noise or something else?

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed RIN_xOMA under fixed ORL.

Digital communication analyzer Are OMA, ER, RLM, TDECQ, jitter, and eye shape acceptable on a DCA? Decision: is the eye still inside the transmitter budget at this corner?

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

Bit-error-ratio tester and FEC counters What is the BER waterfall, floor, burst pattern, lane correlation, and dwell-time behavior from a BERT? Decision: sensitivity shift, noise floor, or intermittent? FEC histograms separate Gaussian noise from clustered MPI.

BERT Bit-error-ratio tester: instrument or ASIC function that generates PRBS and counts errors for pre-/post-FEC BER.

Vector network analyzer Is electrical or electro-optic bandwidth, impedance, or reflection causing eye closure on a VNA? Decision: is the plant electrical rather than optical?

VNA Vector network analyzer: measures S-parameters versus frequency. Used for electrical and electro-optic bandwidth, return loss, and impedance match.

ORL meter and inspection scope Is the optical plant adding loss or reflection? Decision: clean and inspect before blaming the laser.

ORL Optical return loss: reflected power back toward the laser. Low ORL raises RIN and can seed burst errors.

Thermal chamber Is behavior reversible with temperature, and which actuator (TEC, heater, bias DAC) loses headroom? Decision: thermal operating point versus aging.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

Bias sweep Where is the optimum for the EAM, MZM, or ring, and did it move away from the stored table? Decision: retune the control ledger, or replace the device?

EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

Always name the reference plane, setup calibration, test pattern, bandwidth, temperature, sample identity, and pass criterion with the result (Tables 5.5 and 7.2).

A.3.7 Read a BER waterfall: shift, floor, and burst pattern

These three words show up in almost every debug answer. Know what each one looks like on the bench, what it rules in, and what it rules out. The operational procedures are in Sections 10.2 and 10.3.

What a BER waterfall is. A BER waterfall is a plot of bit error ratio versus received optical power. You build it by sweeping a calibrated VOA in the path, holding pattern, temperature, and host fixed, and counting errors long enough at each power that the BER is statistically meaningful. Plot received power on the horizontal axis (name the reference plane: usually TP3 at the receiver connector) and pre-FEC BER on a log vertical axis. A healthy link falls steeply as power rises: more photons, better signal-to-noise ratio, fewer errors. That falling curve is the waterfall. A single BER point at the operating power is not a waterfall. Without the sweep you cannot tell whether you are short on power or limited by noise that scales with the signal.

VOA Variable optical attenuator: calibrated loss element for sensitivity sweeps and stressed-receiver testing.

BER Bit error ratio: errored bits divided by bits received. Pre-FEC BER is the validation metric; post-FEC is the service metric.

What a shifted waterfall means. A parallel shift means the whole curve moved left or right while keeping a similar slope. The link still improves when you add power; it just needs a different power to hit the same BER. A rightward shift (worse sensitivity) means you now need more received power for the same pre-FEC BER. Common causes are lost launch or coupling (power ledger), a quieter or noisier receiver

(TIA noise, responsivity), eye closure that lowers effective OMA at constant average power (ER, RLM, TDECQ), wavelength walking onto a filter edge, or equalizer misadaptation. A leftward shift means the link got healthier. The diagnostic move after you see a shift is a golden swap: known-good transmitter, then known-good receiver, to decide which side owns the sensitivity change. Raising launch power can still help a shifted link, because the waterfall has not stopped responding to power.

TIA Transimpedance amplifier: converts PD current to voltage; input capacitance sets input-referred noise.

ER Extinction ratio: P_1/P_0 in dB. Higher ER widens the OMA for a given average power; trades against chirp and driver swing.

OMA Optical modulation amplitude: outer PAM4 level swing $P_1 - P_0$.

RLM Relative level mismatch: PAM4 level spacing uniformity. 1.0 is perfectly even; CEI often requires $\geq$0.95.

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

What a BER floor means. A floor is a horizontal asymptote: as you raise received power, BER improves for a while and then stops. More power no longer helps. That happens when the dominant noise scales with the signal itself. Relative intensity noise is the textbook case: intensity fluctuations grow in proportion to optical power, so the signal-to-noise ratio saturates and

$$Q_{\text{max}} = 1/\sqrt{\text{RIN} \cdot \text{BW}}.$$

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed RIN_xOMA under fixed ORL.

Multipath interference, bias-rail noise through weak PSRR, and aggressor crosstalk that tracks the aggressor power make the same shape. The interview trap is to keep raising launch power or cleaning for loss when the curve has already floored. The fix is to remove the multiplicative noise source (ORL and isolator hygiene, supply filtering, connector reflections, crosstalk isolation), not to buy more photons. Confirm the floor with a full sweep before you name the mechanism; then bisect with a quiet laser source versus the product bias board, an ORL reflector sweep, and the FEC error timing below.

MPI Multipath interference: coherent beating from two or more reflective paths. Produces a BER floor and clustered FEC errors that more launch power does not fix.

PSRR Power-supply rejection ratio: how well a bias board or driver rejects rail noise. Weak PSRR converts electrical noise into optical intensity noise through the laser.

ORL Optical return loss: reflected power back toward the laser. Low ORL raises RIN and can seed burst errors.

What a burst pattern means. Average BER alone hides how errors arrive in time. A burst pattern means errors cluster: many errored symbols in a short window, then quiet intervals, rather than a steady sprinkle of random bit flips. FEC histograms and pre-FEC error counters with timestamps make this visible. Gaussian thermal noise and well-behaved RIN tend to spread errors. MPI from a pair of reflective interfaces, connector intermittents, ESD events, supply glitches, and unlocked CDR intervals tend to cluster them. Lane- correlated bursts across a module point at a shared supply, clock, or thermal event. A single-lane burst pattern points at that lane's optical path or connector. In the fleet, bursty pre-FEC counters with stable average power are often intermittents: preserve the counters and CMIS event log before you reseat anything, or the evidence disappears (Section 10.9).

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

CDR Clock-data recovery: extracts bit clock from the data stream; present in retimed modules, absent in LPO.

CMIS Common Management Interface Specification: module management, monitors, link training at 224G/448G.

How to say the three together. "I sweep BER versus received power. A parallel shift is a sensitivity or power-ledger problem; more power can still help. A floor is multiplicative noise; more power cannot help. A bursty FEC histogram means clustered impairments such as MPI or intermittents, not plain Gaussian noise." Offer to draw the shifted curve next to the floored curve on the whiteboard.

A.3.8 Separate thermal response from aging

Thermal effects are usually reversible when you return to the starting temperature: wavelength shift, ring detuning, EAM or MZM bias movement, TEC loading, and receiver-noise increase. Aging is cumulative and does not reverse on cool-down: threshold rise, slope loss, contact degradation, defect growth, and permanent absorption or spectral change.

The practical test is simple. Return to the starting temperature and compare with pre-stress data at the same junction temperature. Full recovery supports a thermal or control hypothesis, often a calibration-table segment or an actuator that ran out of range. A permanently moved LIV baseline supports aging, damage, or assembly change. Mixing the two owners wastes weeks: a reliability team cannot fix a wrong table, and a firmware team cannot fix a worn facet (Section 5.10).

EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

MZM Mach–Zehnder modulator: push-pull interferometer; broadband, low chirp. Si MZM, TFLN MZM, or hybrid platforms.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

A.3.9 Accelerated life tests need a mechanism

HTOL predicts field life only if elevated temperature and bias accelerate the same physical mechanism the fleet will see. The stress must not introduce a different mechanism, such as solder creep at a temperature the product never reaches, and it must not omit a dominant field stress, such as thermal cycling or connector wear.

In the interview, state the assumed mechanism, the activation energy and why it applies to this process, the sample size and confidence bound, the stress and use temperatures, the measured drift parameter (threshold, slope, wavelength), and the field-data check. Treat the projected FIT result as a hypothesis that field returns must confirm or falsify. A FIT number without those pieces is a spreadsheet, not a prediction. The life acceleration model is Arrhenius (Section 5.13).

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

FIT Failures in time: failures per 10^9 device-hours. Fleet FIT times link count sets expected failures per day.

Arrhenius / E_a Life-acceleration model: temperature stress multiplies wear-out by $\exp[(E_a/k)(1/T_{\text{use}} - 1/T_{\text{stress}})]$.

A.3.10 Calibration is part of the product

The operating points a product actually stores are as much the product as the die. Know them by name: laser bias or APC target; EAM bias versus temperature; MZM quadrature; ring heater or wavelength-lock point; monitor-photodiode calibration; and receiver and power-monitor offsets. Tables are usually segmented by temperature, and segment boundaries are a real failure mode.

Recalibrate when evidence demands it, not by habit: loop residual stops converging, an actuator approaches its rail, telemetry disagrees with an external reference, temperature leaves the table range, or repair, rework, or firmware changes invalidate coefficients. ATP must verify calibration at the temperature corners the fleet will see, not only at station ambient (Section 5.11).

APC Automatic power control: feedback loop from a monitor PD that holds average optical power; slow loop BW keeps it out of the RIN band.

EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

MZM Mach–Zehnder modulator: push-pull interferometer; broadband, low chirp. Si MZM, TFLN MZM, or hybrid platforms.

PD Photodiode: converts photons to photocurrent. Ge-on-Si PIN is mainstream; APD adds gain; UTC for high saturation.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

A.3.11 Corrective action must prevent recurrence

Root cause is not the end of the answer. Close with immediate containment; the design, process, calibration, firmware, or supplier correction; verification under the original failing condition; an acceptance test, process control, alarm, or telemetry control that catches recurrence; and an owner with a measurable closure criterion. An interview answer that stops at "we replaced the laser" sounds like repair. An answer that ends on the new ATP corner or the new telemetry alarm sounds like product engineering.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

LLM practice

Summary. Ten cold concepts: start at the system; scope before root cause; power versus signal-quality fork; five margin ledgers; validation ladder; instruments named by uncertainty removed; waterfall shift versus floor versus burst; thermal response versus aging; HTOL needs a mechanism; calibration is part of the product; corrective action must prevent recurrence.

Paste to an LLM and answer aloud. Act as my interviewer for optical systems. Ask me one question from each cold concept, in random order. Push for a named reference plane, a next measurement that kills hypotheses, and a recurrence control. Grade me on process, not on jargon volume.

A.4 Two stories to prepare

Each story is a spoken version of the answer spine. Prepare one component or bench story and one system, production, or fleet story. Use only work you personally performed, and separate your contribution from the team's. Walk the same eight beats every time:

1. Requirement and system context: what the link or product had to do.
2. Observed symptom and scope: what failed, how wide, sudden or gradual.
3. Competing hypotheses: the short list after the power fork.
4. Measurements chosen and why: which instrument, which uncertainty.
5. Evidence that eliminated each hypothesis.
6. Root cause: the mechanism that survived, with evidence.
7. Corrective action: the fix, verified under the failing condition.
8. Recurrence control and measured result: the test, alarm, or process change, and the metric that proved it worked.

LLM practice

Summary. Prepare two true stories, one bench or component and one system, production, or fleet. Each story must hit the eight beats of the answer spine and end on a measured recurrence control. Separate your work from the team's.

Paste to an LLM and answer aloud. I will tell you my two story titles only. Interview me on each for five minutes. Force the eight beats in order. Cut me off if I invent metrics, skip scope, or end without a control that would catch the next escape.

A.5 Questions to rehearse aloud, with model answers

Rehearse these until the structure is automatic. Each answer starts with scope or requirements, names measurements and the hypotheses they separate, and ends with a control. Do not recite; adapt the skeleton to the follow-up questions. For a Staff-level conversation, end one beat past the fix: the decision and the owner of the recurrence control.

Three lengths for every long answer. You will not deliver six pages in an interview. Practice each worked answer at three depths and stop when the interviewer redirects:

Executive (30 seconds) Frame, decision, and one control.

Engineering (3 minutes) Scope, fork or ladder, key measurements, root cause class, recurrence control.

Deep dive (10 minutes) Full spine with numbers, reference planes, and tradeoffs. Offer this; do not dump it unprompted.

A.5.1 How would you set laser requirements for a new IM/DD link?

This is the ownership question. Start from the system, not from a laser datasheet. The output is a requirements slice a supplier and an ATP can both test against (Tables 5.1 and 5.4).

Step 1: freeze the system constraints. Name reach, lane rate, fiber plant (MMF or SMF), wavelength class, power envelope, lifetime FIT target, operating temperature, and manufacturing volume. Those constraints choose the architecture path before any part number appears.

FIT Failures in time: failures per 10^9 device-hours. Fleet FIT times link count sets expected failures per day.

A short multimode path points toward a VCSEL. A 500 m or 2 km single-mode path points toward a DFB or EML at 1310 nm. The modulator choice among direct

VCSEL Vertical-cavity surface-emitting laser: 850 nm class, MMF SR links.

EML Electro-absorption modulated laser: DFB plus EAM on one InP chip; dominant 100–200G/lane pluggable Tx.

modulation, EAM, silicon MZM, or ring comes after that path is fixed.

Step 2: write the optical budget the laser must close. Translate the link into numbers the source must support: minimum OMA at the launch reference plane, maximum RIN at a stated ORL, SMSR if filters or WDM sit downstream, chirp or dispersion allowance for the reach, and the bias and extinction window the modulator path needs.

OMA Optical modulation amplitude: outer PAM4 level swing $P_1 - P_0$.

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed $\mathrm{RIN_x OMA}$ under fixed ORL.

Say the reference plane with every number. A launch spec without TP2 or the module connector is not a requirement.

Step 3: write the thermal, life, and control requirements. Case temperature range, whether a TEC is allowed, wavelength drift over temperature and life, threshold and slope aging limits, and the calibration tables the product will store (APC target, EAM bias versus temperature, or ring lock range).

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

HTOL and burn-in belong here as named mechanisms with justified activation energy, not as a ritual checklist. If the laser cannot hold lock or APC headroom at the hot corner, the link does not close even if room-temperature OMA looks fine.

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

Step 4: write what production and partners must prove. Every requirement needs a measurement method, a limit, and a reaction plan. Map OMA, RIN, SMSR, LIV, and eye metrics into the ATP. State lot traceability, FAIR triggers, and how RMA codes stay split by supplier so field data can falsify the qual.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

RMA Return merchandise authorization: field-failed unit returned for FA. Distinct failure codes keep fleet FIT honest.

The decision you are making is not “which laser looks best on a bench.” It is “which requirements let us ship, second-source, and operate the fleet without guessing.”

How to say it aloud. “System constraints first, then OMA and RIN at a named plane and ORL, then thermal and life with a named HTOL mechanism, then ATP and supplier reaction plan.” Offer to walk one number, usually RIN at ORL or hot-corner APC headroom, if they want depth.

Executive answer (30 seconds). Freeze reach, lane rate, fiber, power, lifetime, and volume. Those choose the source path. Then write OMA and RIN at a named plane and ORL, thermal and control headroom, a named HTOL mechanism, and an ATP with supplier reaction plan. The decision is which requirements let us ship and second-source, not which laser looks best on a bench.

Engineering answer (3 minutes). Walk the four steps above: system constraints, optical budget with reference planes, thermal/life/control requirements, then production proof. Name one hard number you would fight for (RIN under ORL, or APC headroom at hot) and what fails if it is missing.

Deep dive (10 minutes). Expand one constraint into the full budget table and show how it lands in ATP limits and partner FAIR triggers.

A.5.2 How would you validate a new optical transmitter from bring-up through production?

This is the question most likely to open the interview, so it gets the full treatment. The frame to state before any detail: validation is staged uncertainty reduction. Each stage answers a question the previous stage could not, and a test that answers no question is cost, not confidence. Walk the stages in order, and at every one name the reference plane where the measurement is taken and the criterion that decides pass (Table 7.1).

Stage 1: bring-up. For bring-up, get one unit alive on a known-good bench and remove the integration unknowns. Power the device through its specified sequence and confirm rails, inrush, and power state transitions. Initialize the firmware and read identity and status over the management interface, because a part that cannot report its own state cannot be debugged later. Enable the laser and confirm first light on a calibrated power meter at the specified output reference plane, not at whatever connector is convenient. Then close a real loop: lock the receiver, run a PRBS pattern from a BERT or host SerDes, and count errors. The uncertainty removed is basic: the design functions, the chips talk to each other, and the test setup itself is sane. The trap is that everything here is golden: pristine connectors, one temperature, one host, one unit. Passing bring-up says nothing about the population, the corners, or the lifetime. Its only job is to make the next stage's failures meaningful.

CMIS Common Management Interface Specification: module management, monitors, link training at 224G/448G.

PRBS Pseudo-random binary sequence: deterministic test pattern that exercises all bit transitions for BER and eye measurements.

BERT Bit-error-ratio tester: instrument or ASIC function that generates PRBS and counts errors for pre-/post-FEC BER.

Stage 2: characterization. For characterization, map what the design actually does across its operating range, on enough units to see variation. Sweep temperature, supply voltage, bias, and test pattern, and at each point record the transmitter's vital signs: the LIV curve for threshold, slope efficiency, and thermal rollover; the optical spectrum on an OSA for center wavelength and SMSR; RIN measured at the stated ORL, because a quiet-bench RIN number is flattery; and the eye on a DCA for ER, OMA, and TDECQ. The key discipline is to record distributions, not a golden sample: ten units across two lots tell you what the process gives you, one hero unit tells you what marketing wants. The uncertainty removed is where the population sits relative to the spec at every corner. The output is not a pass stamp; it is the dataset that will later set production test limits and telemetry alarm thresholds.

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

OSA Optical spectrum analyzer: measures wavelength, SMSR, and side-mode structure of a laser source.

SMSR Side-mode suppression ratio: power difference (dB) between the lasing mode and the strongest side mode (OSA).

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed $\mathrm{RIN_xOMA}$ under fixed ORL.

ORL Optical return loss: reflected power back toward the laser. Low ORL raises RIN and can seed burst errors.

DCA Digital communication analyzer: sampling oscilloscope used for eye diagrams and TDECQ.

ER Extinction ratio: P_1/P_0 in dB. Higher ER widens the OMA for a given average power; trades against chirp and driver swing.

Stage 3: margin testing. For margin, stop asking whether the part passes and ask how far it sits from the cliff. Push each parameter to its failure boundary: sweep received power down with a calibrated VOA until the pre-FEC BER waterfall crosses the FEC threshold; take case temperature to the limit and past it; pull supply rails to their corners; degrade ORL with a controlled reflector and watch the error floor. Measure the distance to failure in dB, degrees, and volts, then compare it against the variation found in characterization. The uncertainty removed is the one that kills fleets: a design can pass every nominal test while sitting half a decibel from its limit, and normal lot spread plus aging will spend that half decibel in the first year. Margin testing also reveals which of the five margins, power, noise, timing, spectral, or control, runs out first, which tells you what telemetry must watch (Section A.3.4 and section 5.19).

VOA Variable optical attenuator: calibrated loss element for sensitivity sweeps and stressed-receiver testing.

pre-FEC BER measured before the FEC decoder. Standards and validation sweeps use this number, not post-FEC.

Stage 4: interoperability. For interoperability, take away everything golden. Replace the BERT with the target hosts, plural, because SerDes equalizers from different vendors adapt differently to the same transmitter. Replace the lab jumper with production fiber plant: real connector losses, real MPO polarity, realistic ORL from the connectors the datacenter will actually use. Run real link bring-up sequences, firmware interactions, and traffic patterns rather than one endless PRBS. The uncertainty removed lives in combinations rather than components: a transmitter and a receiver that each meet spec can still fail together, because the standard's transmitter eye and stressed-receiver tests only guarantee interoperability if both sides were tested against the specified reference, and reference receivers are idealizations. Interop failures found here cost a lab week; found in production they cost a fleet rollback.

MPO Multi-fiber push-on: standard multi-fiber connector for parallel optics (8, 12, 16, 24, or 32 fibers per ferrule).

SECQ Stressed eye closure quaternary: receiver-side stressed test, analog of TDECQ. Applies calibrated optical stress before BER test.

Stage 5: environmental stress and reliability qualification. For qualification, ask which mechanisms move with stress and time. Each stress is chosen for the failure mechanism it accelerates, not run as a ritual: temperature cycling exercises solder joints, die attach, and fiber attach; damp heat exercises corrosion and delamination; vibration and shock exercise the mechanical path; HTOL accelerates the wear-out of the active region so that weeks of stress project years of field life through an Arrhenius model. State the projection's fine print without being asked: the activation energy must be justified for this mechanism on this process, the sample size sets the confidence interval, and the projection holds only if the stress accelerates the same mechanism the field will see. The uncertainty removed is the shape of lifetime: infant mortality rate for the burn-in decision, and wear-out rate as a FIT number the fleet math can consume.

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

Arrhenius / E_a Life-acceleration model: temperature stress multiplies wear-out by $\exp[(E_a/k)(1/T_{\text{use}} - 1/T_{\text{stress}})]$.

FIT Failures in time: failures per 10^9 device-hours. Fleet FIT times link count sets expected failures per day.

Stage 6: production readiness. For production, the question changes from is the design good to can the factory tell a good unit from a bad one, at rate, at acceptable cost. Write the ATP so that every limit traces to the link budget or to the charac-

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

terization data; a limit nobody can trace will be waived under ship pressure. Correlate stations by running the same golden units on every station and site, so a limit means the same thing everywhere. Size guard bands from measurement repeatability, because a test that cannot repeat tighter than its limit ships escapes and scraps good units in equal measure. Verify calibration at the temperature corners the fleet will see, not only at station ambient. Archive raw data, instrument identity, and calibration-table versions so any shipped result can be replayed. Then keep SPC charts on threshold current, power, and wavelength, because process drift between qualification lots is caught here or not at all.

SPC Statistical process control: lot-by-lot control charts (I_{th}, SMSR, power) to catch drift between qual stress tests.

Stage 7: fleet telemetry and the feedback loop. Deployment is the last validation stage, not the end of validation. Fleet telemetry watches the margins that stage 3 identified: per-lane transmit power, bias current, pre-FEC BER, temperature, and actuator drive such as TEC current, with alarms on trends and disagreements rather than only on hard thresholds. Field returns close the loop: the RMA Pareto is compared against the qualification projection, and divergence means the life model was wrong, not that the fleet is unlucky. A high NFF rate is itself a finding, usually pointing at intermittents or triage gaps. The uncertainty removed is the honest one: what every earlier stage missed.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

RMA Return merchandise authorization: field-failed unit returned for FA. Distinct failure codes keep fleet FIT honest.

NFF No fault found: RMA unit passes all tests on return. High NFF rate points at triage or intermittent connector faults.

How to say it aloud. In the interview, give the frame in one sentence, then walk the seven stages in order, one sentence each, naming for each stage one instrument and the uncertainty it removes. Then offer to go deep on whichever stage the interviewer cares about. That structure shows judgment before detail, and it turns a memorized list into a conversation.

Executive answer (30 seconds). Validation is staged uncertainty reduction. Bring-up proves the setup is sane. Characterization maps the population. Margin finds the cliff. Interop removes combination risk. Qual projects life with a named mechanism. Production proves the ATP catches escapes. Fleet telemetry closes the loop. Every stage answers a question the previous stage could not.

Engineering answer (3 minutes). Walk the seven stages in order. For each, name one instrument, the uncertainty removed, and the decision unlocked (continue, redesign, tighten ATP, stop ship). End on which ledger (power, noise, timing, spectral, control) the telemetry must watch.

Deep dive (10 minutes). Expand the stage the interviewer picks. Bring numbers, reference planes, sample sizes, and the failure mode that stage exists to catch.

A.5.3 BER worsens at high temperature but average power is stable. What do you do?

This is a classic fork question. The frame to state first: average power held means the power ledger is intact, so the loss lives in signal quality or in a receiver that got noisier. Scope before instruments, measure at the failing temperature, and end with the control that stops recurrence (Section 10.13).

Step 1: scope the failure. Ask how wide the problem is before touching a knob. One unit or many? One lane or all lanes? One host or every host? One lot or the population? Then ask whether the BER recovers when the unit is cooled. Recovery makes it a thermal operating-point problem, not aging. Aging does not reverse on a cool-down. A single-unit, single-lane, recoverable failure points at a local bias table, heater, or alignment. A multi-unit, multi-lane, recoverable failure points at a system thermal design or a shared calibration method. Scope cuts the candidate list before the first instrument is opened.

Step 2: apply the power-versus-signal-quality fork. Average power held by the monitor or an external meter means the APC loop is still hitting its setpoint. The candidates that remain are the ones that close the eye without moving average power: extinction ratio collapse, modulator bias parked on the wrong part of its curve, wavelength walking off a filter or ring passband, a TEC or heater running out of range, and receiver noise rising with temperature. Do not chase laser threshold or fiber loss first. Those would have moved the power meter.

APC Automatic power control: feedback loop from a monitor PD that holds average optical power; slow loop BW keeps it out of the RIN band.

ER Extinction ratio: P_1/P_0 in dB. Higher ER widens the OMA for a given average power; trades against chirp and driver swing.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

Step 3: measure at the failing temperature. Take the unit to the temperature where it fails and stay there. On a DCA, read the eye for ER, OMA, and TDECQ. Sweep the EAM or MZM bias and ask whether the optimum moved away from the stored table value. Read wavelength on an OSA and compare it with the filter or ring passband. Read TEC current and heater DAC codes for actuator headroom: an actuator near its rail is consumed margin even when performance still passes. If a bias sweep restores a clean eye, the laser is healthy and the calibration table is wrong. If the eye stays closed across the full bias range, look at wavelength lock, thermal crosstalk, or the receiver.

DCA Digital communication analyzer: sampling oscilloscope used for eye diagrams and TDECQ.

OMA Optical modulation amplitude: outer PAM4 level swing $P_1 - P_0$.

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

MZM Mach–Zehnder modulator: push-pull interferometer; broadband, low chirp. Si MZM, TFLN MZM, or hybrid platforms.

OSA Optical spectrum analyzer: measures wavelength, SMSR, and side-mode structure of a laser source.

Step 4: fix the root cause and close the escape. A collapsed eye that a bias sweep restores is usually a temperature-segment boundary in the calibration table, or a table that was taken at station ambient and never verified at the loaded high-temperature corner. Fix the table or the thermal design, then add that loaded corner to the ATP that missed it. If the actuator is railed, the fix is thermal design or a wider tuning range, not a tighter BER limit. Recurrence control is the new test condition, not a hope that the next lot will behave.

How to say it aloud. Scope, fork, measure at the failing temperature, name the hypothesis the measurement eliminates, then name the control. Offer to walk the bias-sweep or the calibration-table story if the interviewer wants depth.

Executive answer (30 seconds). Power held, so leave the power ledger. Scope unit/lane/lot and whether cool- down recovers. At the failing temperature, read eye, bias sweep, wavelength, and actuator headroom. Fix the table or thermal design; put that corner in ATP.

Engineering answer (3 minutes). Walk the four steps: scope ladder, power fork, hot-corner measurements that kill hypotheses, then recurrence control. Name control margin (TEC/heater DAC rails) as a first-class ledger.

A.5.4 How do you distinguish laser aging from calibration drift?

This question tests whether you separate device physics from control-loop bookkeeping. Aging changes the device. Calibration drift changes the operating point applied to a healthy device. The frame is an external reference plus a recovery test (Sections 5.10 and 5.11).

Step 1: state the two hypotheses cleanly. Laser aging raises threshold and lowers slope efficiency at a fixed junction temperature. The APC loop then raises bias to hold power, and eventually the eye or the RIN budget fails even though average power still looks fine. Calibration drift leaves the LIV curve unchanged but parks the product at the wrong bias, wrong EAM voltage, wrong MZM quadrature, or wrong ring heater code. The symptoms can look identical in telemetry. The separation requires an external reference the product does not control.

APC Automatic power control: feedback loop from a monitor PD that holds average optical power; slow loop BW keeps it out of the RIN band.

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed RIN_xOMA under fixed ORL.

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

MZM Mach–Zehnder modulator: push-pull interferometer; broadband, low chirp. Si MZM, TFLN MZM, or hybrid platforms.

PD Photodiode: converts photons to photocurrent. Ge-on-Si PIN is mainstream; APD adds gain; UTC for high saturation.

Step 2: remeasure the device against ship data. Bring the unit to a controlled temperature and rerun LIV. Compare threshold and slope with the shipping record at the same junction temperature. A threshold rise or slope drop is device aging. If LIV matches ship data, the device is still healthy and the problem is in the operating point or the monitor path. Next compare the monitor photodiode current against an external power meter at the same launch. A monitor that disagrees with the external meter will drive the APC loop to the wrong place silently. Then sweep modulator bias and ask whether the optimum moved away from the stored table. A healthy LIV with a moved optimum is calibration drift.

Step 3: use time behavior as a second separator. Aging accumulates monotonically across stress intervals. Plot threshold or bias against cumulative stress hours and look for a steady climb. Calibration error appears as a step after a thermal excursion, a repair, a rework, a firmware flash, or a table rewrite, and it is fully corrected by recalibration. A unit that returns to ship performance after a table reload was never aged. A unit whose LIV still fails after a fresh calibration is aged, and recalibration only masks the wear until the next corner.

Step 4: choose the corrective action by the mechanism. Aging routes to life modeling, derating, burn-in, or field replacement. Calibration drift routes to table version control, temperature-segment verification, monitor-PD integrity checks, and an ATP corner that exercises the table under load. Mixing the two owners wastes weeks: a reliability team cannot fix a wrong table, and a firmware team cannot fix a worn facet.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

How to say it aloud. "Aging changes the device; calibration drift changes the setpoint. I separate them with an external LIV and a recovery test, then use time behavior to confirm." Offer the monitor-PD corruption story as the silent failure mode.

Executive answer (30 seconds). Aging changes the LIV. Drift changes the setpoint on a healthy LIV. External LIV plus recovery after recalibration separates them. Time behavior confirms: monotonic climb versus a step after a table or firmware change. Owner follows the mechanism.

A.5.5 How would you qualify a second laser or photonic-integrated-circuit supplier?

This question tests supplier judgment, not vendor names. The frame: the first supplier's failure distribution does not transfer. Qualify against the requirements slice, not against the incumbent's datasheet (Table 5.4 and section 8.10).

Step 1: freeze the requirements, not the part number. Write what the new part must close: link budget, RIN at the stated ORL, bias window, wavelength class, thermal class, and lifetime FIT target. Those are the acceptance criteria. The incumbent's datasheet is evidence that one process can meet them, not a template the second source must copy. A second source that matches the datasheet but fails the link-budget corners is not qualified.

RIN Relative intensity noise: laser amplitude noise spectral density (dB/Hz). Specs often quote stressed RIN_xOMA under fixed ORL.

ORL Optical return loss: reflected power back toward the laser. Low ORL raises RIN and can seed burst errors.

FIT Failures in time: failures per 10^9 device-hours. Fleet FIT times link count sets expected failures per day.

Step 2: characterize distributions, not samples. Measure threshold, slope, wavelength, SMSR, and RIN across wafers, lots, and temperature. Compare spreads against the incumbent, not only means. A hero sample from a new supplier proves nothing about the process. Ask for wafer maps and lot genealogy so edge-of-wafer

SMSR Side-mode suppression ratio: power difference (dB) between the lasing mode and the strongest side mode (OSA).

outliers are visible before they enter your module line. Record the same reference planes and fixtures you use on the incumbent, or the comparison is fiction.

Step 3: run the full qualification on this process. HTOL with the supplier's own activation-energy justification for the named mechanism, temperature cycling, damp heat, and burn-in. Wear- out mechanisms and infant-mortality rates are process-specific: epi, facet coat, attach, and hermeticity all differ. Borrowing the incumbent's E_a is the most common quiet mistake. State the projection's fine print: sample size, confidence bounds, and evidence that the stress accelerates the field mechanism without inventing a new one.

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

Step 4: correlate ATPs and plan the ramp. Run the same units on the supplier's line and yours so an ATP limit means the same thing at both sites. Size guard bands from the combined repeatability. Ramp with lot traceability, a FAIR, and an agreed reaction plan for excursions. Keep fleet RMA codes split by supplier so field data can falsify the qualification. A second source that cannot be traced in the fleet is not a second source; it is a blind risk.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

FAIR First-article inspection report: re-qualify after tooling, site, epi, or CMIS firmware change before open volume.

RMA Return merchandise authorization: field-failed unit returned for FA. Distinct failure codes keep fleet FIT honest.

How to say it aloud. "Requirements first, distributions second, process-specific qual third, ATP correlation and split field codes fourth." Offer to go deep on HTOL validity or on what you would put in the reaction plan.

A.5.6 A single lane is weak in a multi-lane module. How do you isolate optical, electrical, thermal, and assembly causes?

Multi-lane modules give you a free control group: the sibling lanes. The frame is pattern recognition across lanes before any single-lane deep dive (Section 10.5).

Step 1: confirm one outlier, not a gradient. Measure every lane for OMA, ER, TDECQ, and wavelength at the same temperature and pattern. A single-lane cliff is a local defect. A smooth edge-to-center gradient is thermal or FAU alignment. A checkerboard pattern often points at electrical routing or a shared supply. Write the pattern down before changing anything. The siblings tell you what "normal" is for this unit.

Step 2: bisect optical versus electrical. Swap the electrical lane assignment, or drive the suspect optical path with a known-good driver channel. If the weakness follows the optical path, the candidates are the laser or EML element, the modulator, the attach, or the fiber. If it follows the electrical channel, the candidates are the driver, the TIA, the package routing, or the host SerDes lane. This one swap often cuts the tree in half.

OMA Optical modulation amplitude: outer PAM4 level swing $P_1 - P_0$.

ER Extinction ratio: P_1/P_0 in dB. Higher ER widens the OMA for a given average power; trades against chirp and driver swing.

TDECQ Transmitter and dispersion eye closure quaternary: headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE, including a test fiber. Lower is better; LPO MSA 100G-DR caps near 3.4 dB. Replaced simple eye-mask tests. Stressed counterpart: SECQ.

FAU Fiber array unit: a precision V-groove assembly that mates multiple fibers to a PIC edge or grating coupler array.

EML Electro-absorption modulated laser: DFB plus EAM on one InP chip; dominant 100–200G/lane pluggable Tx.

TIA Transimpedance amplifier: converts PD current to voltage; input capacitance sets input-referred noise.

SerDes Serializer/deserializer: high-speed I/O that maps parallel data to PAM4 on the wire.

Step 3: separate assembly from device and thermal. Per-lane coupled power with a normal LIV on the weak lane points at fiber-array alignment or a contaminated ferrule: the device is fine, the light is not leaving. A thermal map or per-lane temperature monitors separate an edge-lane thermal gradient from a device defect. If the imbalance grows over HTOL or burn-in time, one array element is aging faster, which is a lot or screening question, not a design question.

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

Step 4: pick the corrective action from the pattern. Across lanes, units, and lots, the pattern chooses the fix: a rework instruction for alignment, a coupling-spec change for the FAU, a thermal- design change for edge lanes, a driver or TIA lot screen, or a supplier corrective action on one array element. Do not rewrite the whole module spec for a single-lane assembly escape.

How to say it aloud. "Siblings first, then optical-versus-electrical swap, then assembly versus device versus thermal, then the pattern picks the control." Offer the FAU-alignment story as the most common production escape.

A.5.7 Received power is unchanged but required receiver power increased. What hypotheses remain?

This is the other face of the power-versus-signal-quality fork. Average received power held means the loss is not in the link-budget power ledger. The frame: eye quality or the receiver itself (Section 10.2).

Step 1: restate what the observation rules out. Unchanged received power rules out average launch drop, connector loss, and fiber attenuation as the primary cause. What remains is signal-quality erosion on the transmitter or channel, or a receiver whose sensitivity got worse. Say that out loud so the interviewer hears the fork.

Step 2: list the remaining hypotheses by location. Transmitter: eye closure at constant average power from modulator bias drift, falling ER, rising RIN under reflection, or increased jitter and chirp. Channel: a new reflection raising MPI, or wavelength moving relative to a filter edge, both of which degrade quality without moving the power meter. Receiver: TIA noise, photodiode responsivity, or equalizer misadaptation. Keep the list short and location-tagged.

MPI Multipath interference: coherent beating from two or more reflective paths. Produces a BER floor and clustered FEC errors that more launch power does not fix.

Step 3: isolate with golden swaps and a waterfall. Swap a known-good transmitter into the link, then a known-good receiver. Whichever side restores the old sensitivity owns the fault. Sweep BER versus received power with a VOA. A parallel shift of the waterfall means the sensitivity moved. A floor that will not go down with more

VOA Variable optical attenuator: calibrated loss element for sensitivity sweeps and stressed-receiver testing.

power points at multiplicative noise: RIN, MPI, or crosstalk. Finish with a stressed-eye check of the receiver against a known transmitter if the golden swap indicts the receive path.

SECQ Stressed eye closure quaternary: receiver-side stressed test, analog of TDECQ. Applies calibrated optical stress before BER test.

Step 4: close with the mechanism, not the symptom. A shifted waterfall with a clean golden transmitter is a receiver problem: noise, responsivity, or adaptation. A floor is a noise problem; chase ORL, supply PSRR, and aggressor lanes before replacing a laser. The recurrence control depends on the mechanism: a calibration table, a connector hygiene rule, a supply filter, or a receiver lot screen.

PSRR Power-supply rejection ratio: how well a bias board or driver rejects rail noise. Weak PSRR converts electrical noise into optical intensity noise through the laser.

How to say it aloud. “Power ledger intact, so quality or receiver. Golden swap, then waterfall shape, then mechanism.” Offer to draw the shifted-versus-floored waterfall on the whiteboard.

A.5.8 Why can a link show a BER floor that more launch power does not fix?

This question has a short physics answer and a longer debug answer. Give both (Sections 4.3 and 10.3).

Step 1: state the physics in one sentence. A BER floor means the dominant noise scales with the signal. Raising launch power raises that noise in proportion, so the signal-to-noise ratio saturates. For relative intensity noise the ceiling is $Q_{\mathrm{max}} = 1/\sqrt{\mathrm{RIN} \cdot \mathrm{BW}}$. More power cannot buy you past that ceiling. Additive noise (thermal, dark current) does not make a floor; multiplicative noise does.

Step 2: name the usual multiplicative sources. Laser RIN, often feedback-driven through poor ORL. Multipath interference from a pair of reflective interfaces. Bias-rail noise converting to intensity noise through the laser when PSRR is weak. Crosstalk that scales with the aggressor. Each of these raises the error rate in a way that looks like “the link just will not clean up,” and each is fixed by removing the noise source, not by raising OMA.

ORL Optical return loss: reflected power back toward the laser. Low ORL raises RIN and can seed burst errors.

Step 3: diagnose by waterfall shape and bisect. Sweep received power and plot the BER waterfall. Confirm it floors rather than shifts. Then bisect: measure RIN with a quiet lab source versus the product bias board to separate intrinsic laser RIN from board-induced noise. Sweep ORL with a controlled reflector and watch the floor rise and fall. Read the FEC error histogram: MPI and burst noise cluster errors in time, while Gaussian noise spreads them. That histogram often picks the mechanism before a scope does.

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

Step 4: fix the mechanism, then prevent recurrence. Isolator or connector hygiene for feedback. Supply filtering and PSRR improvement for electrical noise. Aggressor silencing or better isolation for crosstalk. Replace the laser only if intrinsic RIN is truly out of spec after the board and ORL have been cleared. Add the ORL or supply stress to the qualification and to the ATP if that path was never tested.

How to say it aloud. "Floor means multiplicative noise; more power cannot help. I confirm with a waterfall, then bisect RIN, ORL, and the FEC histogram." Offer the $Q_{\max}$ line if they want the equation.

A.5.9 What makes an HTOL projection credible?

A FIT number without a mechanism is a spreadsheet. The frame is four conditions, said in order (Section 5.13). HTOL is only as credible as those four.

FIT Failures in time: failures per 10^9 device-hours. Fleet FIT times link count sets expected failures per day.

HTOL High-temperature operating life: accelerated stress test (GR-468). Arrhenius projects field wear-out from HTOL lots.

Condition 1: a named failure mechanism. The projection applies to a specific drift parameter: threshold current, slope efficiency, wavelength, or a defined hard failure such as COD. "The laser" is not a mechanism. Name the parameter, the end-of-life criterion, and the distribution you are projecting.

COD Catastrophic optical damage: sudden, irreversible facet failure in a laser under thermal or optical overstress.

Condition 2: a justified activation energy. E_a must be measured or justified for that mechanism on that process, with sample size and confidence bounds. Copying an E_a from another product, another vendor, or a textbook table is the most common way a projection lies. State the Arrhenius form and the temperatures used, and be ready to say what happens if E_a is wrong by 0.1 eV.

Arrhenius / E_a Life-acceleration model: temperature stress multiplies wear-out by $\exp[(E_a/k)(1/T_{\text{use}} - 1/T_{\text{stress}})]$.

Condition 3: the stress matches the field mechanism. The stress must accelerate the mechanism the fleet will see, without adding a new one. A stress temperature that triggers solder creep the field never sees produces a pessimistic fiction. A bench HTOL that omits thermal cycling, connector wear, or bias-rail transients produces an optimistic fiction. Say which field mechanisms the HTOL does and does not cover.

Condition 4: a field feedback loop. Compare the RMA Pareto against the projection. Divergence is evidence the model is wrong, not that the fleet is unlucky. Update E_a, the mechanism list, or the use condition when the field disagrees. A projection that never meets field data is theater.

RMA Return merchandise authorization: field-failed unit returned for FA. Distinct failure codes keep fleet FIT honest.

How to say it aloud. "Named mechanism, justified E_a, stress matches field, field feedback. Without those four, FIT is a spreadsheet." Offer one example of a stress that invents a false mechanism.

A.5.10 Which data must an automated test save so a failure can be replayed?

This question is about whether you have ever had to defend a ship decision months later. The frame: enough data that a unit failing today can be re- analyzed without the station or the person.

What identity and setup must be saved. Sample identity down to serial, lot, and date code. Instrument identity and calibration state, including the path-loss table version. Fixture, cable, and reference-plane description, because a power number without a plane is not a measurement. Software, firmware, and calibration-table versions on the device under test. Test conditions: temperature, supplies, pattern, dwell, and timestamps that let results correlate with environmental logs.

What measurement content must be saved. Raw measured data, not only pass/fail. Eyes, LIV points, spectra, and BER sweeps that can be re-plotted. Station-to-station golden-unit correlation results, so measurement drift can be separated from product drift. Without the raw data, a later engineer can only argue about the limit, not about the unit.

LIV Light–current–voltage curve: threshold, slope efficiency, kink-free range, and thermal rollover versus bias.
BER Bit error ratio: errored bits divided by bits received. Pre-FEC BER is the validation metric; post-FEC is the service metric.

The replay test of sufficiency. Given a field return, can you reproduce the exact shipping measurement and decide whether the unit changed or the measurement did? If the answer is no, the archive is incomplete. That question is the acceptance criterion for the data system, not a nice-to-have.

How to say it aloud. "Identity, instrument state, reference plane, raw data, and versions. The test is whether I can replay ship data on a return." Offer one war story about a missing path-loss table if you have one.

A.5.11 When would you choose an EML, silicon MZM, or ring modulator?

Choose by constraint, not preference. State the requirement slice first, then name what each path buys and costs (Tables 3.12 and 5.1).

When an EML wins. An EML wins for single-wavelength DR or FR at 100–200G/lane when cost, maturity, and one-chip integration dominate. You get low chirp relative to a DML, a mature supply chain, and a bias-versus-temperature calibration that the industry already knows how to qualify. You pay for an InP process, EAM aging as a second wear mechanism, and a calibration table that must be verified at temperature corners.

EML Electro-absorption modulated laser: DFB plus EAM on one InP chip; dominant 100–200G/lane pluggable Tx.
DR Datacenter reach: IEEE 802.3 single-mode class, typically 500 m at 1310 nm. Default for AI scale-out optics.
FR Far reach: IEEE 802.3 single-mode class, 2 km. Tighter chirp and dispersion budget than DR.
DML Directly modulated laser: modulate bias current; high chirp, low cost.
InP Indium phosphide: III-V semiconductor for lasers, EAMs, SOAs, and high-speed photodiodes in the 1310/1550 nm bands.
EAM Electro-absorption modulator: voltage-controlled absorption on InP; low chirp vs DML; paired with DFB in EML.

When a silicon MZM wins. A silicon MZM wins when the modulator must sit on the PIC beside other functions and the link wants a broad, flat passband with no wavelength lock. You get CMOS-compatible integration and a wide optical bandwidth. You pay for die area, drive swing, RF co-design of the traveling-wave electrode, and a bias-control loop that holds quadrature. Validation must cover driver matching and bias-rail behavior across temperature.

MZM Mach–Zehnder modulator: push-pull interferometer; broadband, low chirp. Si MZM, TFLN MZM, or hybrid platforms.

PIC Photonic integrated circuit: waveguides, modulators, detectors, and couplers on one chip (typically SOI at 220 nm Si).

CMOS Complementary metal-oxide-semiconductor: mainstream IC process; TIAs and SerDes at 16–3 nm nodes.

When a ring wins. A ring wins when many wavelengths must fit on one die, in dense WDM and co-packaged engines. It is small and WDM-native. You pay for a wavelength-lock loop, heater power, thermal-crosstalk validation, and a resonance-alignment failure mode the others do not have. The validation burden is the decision: if you cannot afford lock-range and crosstalk testing, you cannot afford the ring.

MRM Microring modulator: resonant Si modulator; compact, WDM-native, needs wavelength lock. CPO workhorse at 200G/ch.

WDM Wavelength division multiplexing: multiple λ on one fiber. CWDM (20 nm spacing) or DWDM ($\leq$100 GHz).

Thermal crosstalk Heater or self-heating on one ring shifts neighbors' resonances; worst case is full-bank traffic at max case T.

How to say it aloud. "Constraint first. EML for mature single-λ DR/FR. Silicon MZM for broadband PIC integration. Ring for dense WDM with the lock and heater cost accepted." Offer the validation-burden line as the closer.

A.5.12 How do you decide whether a field issue is performance, reliability, or manufacturability?

Wrong bucket, wrong owner, wasted weeks. Classify on the ticket before failure analysis starts (Section 7.12).

Bucket 1: performance. Ask: did the unit ever meet spec under the field condition? If the design or operating point never closed the budget at that corner, it is performance. The fix is retune, derate, a thermal redesign, or a honest spec change. Shipping more of the same design will not help.

Bucket 2: reliability. Ask: did it meet spec at ship and then degrade with time or stress? That is reliability. The fix is a life model, a screen, derating, burn-in, or field replacement. Telemetry trends (bias rising at constant power, actuator heading toward rail) usually show this before the hard failure.

Bucket 3: manufacturability. Ask: does the failure cluster by lot, date code, site, or process step rather than by time? That is manufacturability. The fix is containment, supplier corrective action (8D), and an ATP or SPC change. A lot-correlated escape is not a physics mystery; it is a process-control gap.

8D Eight disciplines (8D) corrective action: contain, root-cause, correct, and prevent with the supplier; close only with verification evidence.

ATP Acceptance test plan: measurable ship tests that prove the requirements contract with a supplier (often mapped to GR-468 stresses).

SPC Statistical process control: lot-by-lot control charts (I_{th}, SMSR, power) to catch drift between qual stress tests.

How telemetry answers all three before a pull. Install age, lot correlation, and trend shape (sudden, gradual, or corner- dependent) usually pick the bucket before a unit is removed. Classify first, then pull hardware with a plan that matches the bucket. Performance issues need a design owner. Reliability issues need a life-model owner. Manufacturability issues need a supplier and process owner.

How to say it aloud. "Ever meet spec? Then degrade with time? Or cluster by lot? Those three questions pick the owner." Offer a one-line example of each bucket from your experience if you have one.

A.5.13 What would you put in fleet telemetry, and why?

Telemetry exists to catch margin erosion early and to triage without pulling hardware. Log what discriminates hypotheses, not every register on the chip (Sections 7.8 and 7.12).

Per-lane observables. Transmit and receive optical power, laser bias current, and pre-FEC BER with FEC error histograms. Bias rising at constant power is aging. Power dropping at constant bias is the optical path. Clustered errors mean bursts (MPI, connector, ESD event) rather than Gaussian noise. Per-lane data is what makes the sibling-lane method possible in the field.

Module observables. Temperature, supply rails, TEC or heater drive, and lock-loop error. Actuators near their rails are consumed margin even while performance still passes. A railed TEC today is a BER fail next summer. Alarm on actuator headroom, not only on hard optical thresholds. Read state over CMIS.

Events and context. LOS and LOL history with timestamps, resets, and firmware versions, preserved through reboots so intermittents leave evidence. Lot and date code, install age, and rack position, so one query separates unit, lot, and environment. Without context, every incident looks unique.

Alarm philosophy. Alarm on trends and disagreements, such as monitor versus expected power or bias versus a peer lane, not only on hard thresholds. Hard thresholds catch dead units. Trends catch dying ones. The point of fleet telemetry is the dying ones.

How to say it aloud. "Per-lane power, bias, and pre-FEC BER; module temperature and actuator drive; events with context; alarms on trends." Justify each item with the hypothesis it separates, then stop.

BER Bit error ratio: errored bits divided by bits received. Pre-FEC BER is the validation metric; post-FEC is the service metric.

FEC Forward error correction: parity symbols let the receiver fix bit errors without retransmit. Datacom links specify a *pre-FEC* BER the code must correct.

MPI Multipath interference: coherent beating from two or more reflective paths. Produces a BER floor and clustered FEC errors that more launch power does not fix.

TEC Thermoelectric cooler: Peltier device that holds laser junction or ring at a controlled temperature.

CMIS Common Management Interface Specification: module management, monitors, link training at 224G/448G.

LOS Loss of signal: receiver (or host) flag that optical input is below the detect threshold.

LOL Loss of lock: CDR or SerDes unlocked; electrical path not recovered even if light is present.

LLM practice

Summary. The worked answers cover laser requirements, the validation ladder, hot BER with stable power, aging versus calibration, second-source qual, weak lane isolation, sensitivity shift, BER floor, HTOL credibility, replayable test data, modulator choice, field triage buckets, and fleet telemetry. Speak the frame, walk the steps, close on the decision and control.

Paste to an LLM and answer aloud. Run a 45-minute mock interview. Pick six of the rehearsal questions at random. After each answer, ask one follow-up that forces a measurement choice or a supplier/fleet decision. Score me as Staff-level only if I name the decision unlocked, not only the instrument used.

A.6 Must-know abbreviations

A fuller glossary follows this appendix. By the end of the week, know these without notes. Entries are alphabetical by the leading abbreviation.

8D / CAPA Eight-discipline problem solving and corrective and preventive action. Structured containment, root cause, correction, and recurrence control for supplier or production failures.

APC Automatic power control. Feedback from a monitor photodiode that holds average launch power; a drifting monitor corrupts it silently.

Arrhenius / E_a Life-acceleration model. Temperature stress multiplies wear-out by $\exp[(E_a/k)(1/T_{\text{use}} - 1/T_{\text{stress}})]$. E_a must be justified for the named mechanism on that process.

ATP Acceptance test plan. Production test limits, methods, reference planes, and reaction rules.

Bathtub curve Infant mortality (falling rate), useful life (roughly constant FIT), then wear-out (rising rate). Burn-in targets the left; Arrhenius life targets the right.

BER Bit error ratio. Pre-FEC BER is the validation metric; post-FEC is the service metric. Always say which.

BERT Bit-error-ratio tester. Generates PRBS and counts errors for waterfalls, floors, and dwell tests.

Burn-in Production or sample screen that removes infant-mortality parts before ship. Distinct from HTOL life projection.

CDR Clock-data recovery. Extracts bit clock from the data stream; LOL means unlock even if light is present.

CMIS Common Management Interface Specification. Module state, controls, alarms, and telemetry.

COD Catastrophic optical damage. Sudden, irreversible laser-facet failure.

COM Channel operating margin. Statistical electrical-link margin after loss, noise, crosstalk, and equalization.

DCA Digital communication analyzer. Sampling oscilloscope for eyes, OMA, ER, RLM, and TDECQ.

DFB Distributed-feedback laser. Single-mode 1310 nm class source; also the gain section in an EML.

DML Directly modulated laser. Simple and efficient, but chirp limits reach.

DPA Destructive physical analysis. Cross-section or EDX on failed units to confirm facet, solder, FAU, or die failure modes.

DPPM Defective parts per million. Incoming or outgoing quality rate.

DR / FR Datacenter reach ($\sim$500 m) and far reach (2 km). IEEE single-mode classes at 1310 nm.

DVT / PVT Design validation test and production validation test.

EOL End of life. The defined wear-out criterion (threshold rise, slope drop, or hard fail) used in HTOL projection and derating.

ESD Electrostatic discharge. Handling or assembly damage to drivers or TIAs; sudden hard fail, not Arrhenius wear-out. Qual uses HBM/CDM models.

EAM Electro-absorption modulator. Voltage-controlled absorption; pairs with a DFB in an EML.

ELSFP External Laser Small Form-Factor Pluggable. Replaceable continuous-wave source for co-packaged optics.

EML Electro-absorption modulated laser. DFB plus EAM on one InP chip; dominant 100–200G/lane pluggable transmitter.

ER Extinction ratio. P_1/P_0 in dB. Higher ER widens OMA at fixed average power; trades against chirp and swing.

FAIR First-article inspection report. Re-qualify after tooling, site, epi, or firmware change before open volume.

FAU Fiber array unit. Precision multi-fiber attachment to a photonic integrated circuit.

FEC / KP4 Forward error correction; KP4 is Reed–Solomon RS(544,514). Pre-FEC BER and error histograms show margin before FEC failure.

FIT Failures in time. Failures per 10^9 device-hours.

Golden unit / host Known-good reference used to bisect station, fixture, host, module, and fiber. Essential in debug and ATP correlation.

GR-468 Telcordia optoelectronic qualification framework (HTOL, environmental, mechanical).

HAST Highly accelerated stress test. Humidity plus temperature and bias; exercises corrosion and delamination.

HTOL High-temperature operating life. Accelerated stress used with an Arrhenius model to project field wear-out; credible only with a named mechanism.

HTSL High-temperature storage life. Unbiased bake; separates storage mechanisms from biased HTOL wear-out.

JESD47 JEDEC IC qualification stress suite. Silicon-side counterpart to GR-468 for drivers and TIAs.

LIV Light–current–voltage curve. Threshold, slope efficiency, kink-free range, and thermal rollover versus bias.

LOS / LOL Loss of signal and loss of lock. LOS points first toward power; LOL can occur with adequate power but poor timing or signal quality.

MPI Multipath interference. Coherent beating from reflective paths; often floors BER and clusters FEC errors.

MPO Multi-fiber push-on connector. Parallel-optics plant (8–32 fibers).

MRM Microring modulator. Compact, WDM-native silicon modulator; needs wavelength lock and heater budget.

MSA Multi-source agreement. An industry specification for interoperable products.

MTBF Mean time between failures. For constant failure rate, MTBF $= 10^9/$FIT hours. Fleet math usually uses FIT times population.

MZM Mach–Zehnder modulator. Broadband interferometric modulator (Si or TFLN); needs quadrature bias control.

NFF / RMA No fault found and return merchandise authorization. High NFF rates often indicate weak triage or intermittent faults.

OMA Optical modulation amplitude. Outer level swing $P_1 - P_0$.

ORL Optical return loss. Reflected power toward the laser; low ORL raises RIN and can seed burst errors.

OSA Optical spectrum analyzer. Wavelength, SMSR, and side-mode structure.

OSFP / QSFP-DD High-density pluggable module form factors with different mechanical and thermal limits.

PIC / SOI Photonic integrated circuit on silicon-on-insulator. The common silicon-photonics chip and substrate.

PRBS Pseudo-random binary sequence. Repeatable pattern for eye and BER measurements.

PSRR Power-supply rejection ratio. Weak PSRR can turn electrical rail noise into optical intensity noise.

RIN Relative intensity noise. Laser amplitude noise (dB/Hz); often measured under a stated ORL.

RLM Relative level mismatch. PAM4 level-spacing quality.

SECQ Stressed eye closure quaternary. Receiver-side margin measured with a calibrated stressed optical signal.

SerDes Serializer/deserializer. Host high-speed I/O; equalization reserve is a timing-margin ledger.

SMSR Side-mode suppression ratio. Power difference between the lasing mode and the strongest side mode.

SPC Statistical process control. Tracks process distributions and trends.

TDECQ Transmitter and dispersion eye closure quaternary. Headline PAM4 transmitter-quality metric after a reference receiver and bounded FFE.

TEC Thermoelectric cooler. A TEC near its current limit has little thermal control margin left.

TIA Transimpedance amplifier. Converts photodiode current to voltage.

VCSEL Vertical-cavity surface-emitting laser. 850 nm class for multimode short-reach links.

VNA Vector network analyzer. Electrical or electro-optic *S*-parameters versus frequency.

VOA Variable optical attenuator. Calibrated loss for sensitivity and BER-waterfall sweeps.

WDM Wavelength division multiplexing. Multiple wavelengths on one fiber.

LLM practice

Summary. Know the alphabetical list cold. Optical/debug core: ATP, APC, BER/BERT, CMIS, DCA, EML, ER, FEC/KP4, FIT, HTOL, LIV, LOS/LOL, MPI, OMA, ORL, RIN, RLM, SECQ, TDECQ, TEC, TIA, VOA. Reliability/manufacturing core: Arrhenius/E_a, bathtub, burn-in, DPA, ESD, FAIR, golden unit, GR-468, HAST, HTSL, JESD47, MTBF, NFF/RMA, SPC, 8D/CAPA. Expand the term, then say why it matters in a debug or ship decision.

Paste to an LLM and answer aloud. Drill abbreviations. Give me ten random terms mixing optical debug and reliability/manufacturing. For each, I must expand it in one sentence and give one measurement, stress, or failure mode it connects to. Fail me if I only expand the letters or confuse burn-in with HTOL.

A.7 One-week study plan

Assume seven days, with the interview near the end of Day 7. Protect sleep. Each day has one primary drill and one LLM practice box. If a day slips, cut new reading first, not story rehearsal or the mock interview.

Day 1: philosophy, answer spine, and ownership frame. Read the philosophy section and the answer spine. Memorize the uncertainty- reduction line, the Bayesian update rule (every measurement changes belief), the scope ladder, the five ledgers, and the nine spine steps. Speak one paragraph per spine step. Use the answer-spine LLM practice box. Write down the decision language you will reuse: ship, derate, second-source, ATP change, partner action.

Day 2: requirements and validation ladder. Rehearse Section A.5.1 and Section A.5.2 aloud until the structure is automatic. Name a reference plane in every measurement sentence. Skim Tables 5.1, 5.4 and 7.1 for numbers you already believe; do not hunt new optics.

Day 3: two real stories. Draft and rehearse one bench or component story and one system, production, or fleet story. Use only work you did. Hit all eight beats and end on a measured recurrence control. Use the stories LLM practice box. Record yourself once and cut invented metrics.

Day 4: instruments, waterfall, and debug forks. Review what each instrument removes as uncertainty. Drill waterfall shift versus floor versus burst pattern (Section A.3.7 and sections 10.2 and 10.3). Rehearse hot BER with stable power, aging versus calibration, and weak-lane isolation. Use the ten-concepts LLM practice box.

Day 5: reliability, manufacturing, and suppliers. Drill HTOL credibility, Arrhenius/E_a, bathtub (burn-in versus wear-out), second-source qualification, FAIR, DPA, ESD versus wear-out, and field triage buckets (performance / reliability / manufacturability). Rehearse the HTOL, second-source, and triage worked answers. Skim Chapter 8 only where a story needs a fact.

Day 6: abbreviations, telemetry, and modulator choice. Run the abbreviations LLM practice box until expansions are fast. Rehearse fleet telemetry, modulator choice (EML / Si MZM / ring), replayable test data, and BER-floor physics. Do one short mixed quiz: three debug questions and three reliability questions.

Day 7: mock interview and stop. Use the 45-minute mock-interview LLM practice box. No new chapters. Light review of your two stories and the laser-requirements opener only. Stop two to three hours before the call. Sleep.

If you have less than seven days. Compress in this order: Day 1 spine, Day 3 stories, Day 2 requirements and validation, Day 5 HTOL and triage, Day 7 mock. Cut Day 6 reading; keep only the abbreviation drill.

Key idea. The goal of an optical systems engineer is not to know every component. It is to make good engineering decisions under uncertainty using measurements, physics, and evidence. Show that process under time pressure: requirements and scope, a measurement that changes belief at a named reference plane, hypothesis elimination, root cause verified under the failing condition, and a control that prevents recurrence. Lab fluency proves you can decide; the control proves you can ship. A week is for making that spine automatic, not for learning a new device family.

Abbreviations

This glossary is adapted from the OIF *Next Generation CEI-448G Framework* glossary (pages 7–11, OIF-FD-CEI-448G-01.0, September 2025) [14], extended with optical, reliability, and manufacturing terms used in the book (including the must-know list in Chapter A). Some entries include material from public reference sources noted in the original.

ADC An analog-to-digital converter is a system that converts an analog signal into a digital signal.

AI Artificial intelligence.

ACC Active copper cable. ACCs are a type of active copper cable used in data centers. ACCs primarily use Redriver chips and Continuous Time Linear Equalization (CTLE) to amplify and equalize the signal and provide longer reach than DACs. ACCs generally are lower power and cost than AECs (another type of active copper cable), but provide shorter reach than AECs.

AEC Active electrical cable. AECs are a type of active copper cable used in data centers. AECs utilize Retimer chips with Clock and Data Recovery (CDR) to reshape and retransmit signals which generally offer longer reach and better signal integrity compared to ACCs, but at a higher cost.

Application spaces Portions of equipment or network architecture that could benefit from having a defined set of interconnection parameters.

ASIC An application-specific integrated circuit is an integrated circuit (IC) customized for a particular use, rather than intended for general-purpose use.

Backplane A group of electrical connections used as a backbone to connect several printed circuit boards together to make up a switch, computing or storage system.

BER Bit error ratio is a measure of the number of bit errors that occur during data transmission, expressed as a ratio of erroneous bits to the total number of bits transmitted.

BoW Bunch of wires is a die-to-die interface specification. It is part of the open compute project (OCP) and, like UCIe, is used to connect chiplets within a package.

Cabled host An implementation where twinaxial cable is used instead of PCB traces expressly for the insertion loss benefits or 3D architecture.

CDR Clock and data recovery, a component that re-establishes the timing of a signal that may have degraded due to impairments on a transmission line, the retimed signal is able to continue further to its destination.

CEI Common Electrical IO, an OIF Implementation Agreement containing clauses defining electrical interface specifications, each optimized for various reaches at minimal power.

Coded modulation Coded modulation is a technique in digital communication that combines error-control coding with modulation to enhance reliability and efficiency. Its main goal is to balance bandwidth efficiency, power efficiency, and error probability.

CMIS Common management information specification.

CMIS-LT CMIS-based link training provides a message set and exchange mechanism for out-of-band link training or tuning between a pluggable module and a host.

CMIS-VCS CMIS versatile control set extends the base CMIS standard to allow for more advanced and flexible signal integrity (SI) capabilities.

CPC Co-packaged copper is an emerging interconnect technology where copper cables are directly attached to the top of an Application-Specific Integrated Circuit (ASIC) or other high-speed integrated circuit within a single package. This design minimizes the length of high-speed electrical signals on the printed circuit board (PCB) and package, addressing the limitations of traditional copper traces at increasingly high data rates (e.g., 224G and above).

CPO Co-packaged optics. An electrical to optical device intended to be mounted on the host package.

CTLE Continuous time linear equalizer.

DAC Direct attach copper cable. A high-speed cable assembly made of copper twinaxial cable with fixed passive transceiver modules on each end. DAC cables enable direct, electrical connections between networking devices over short distances.

DER Detector error ratio.

DFE Decision feedback equalizer. An equalizer by adding a filtered version of previous symbol estimates to the original filter output.

DSP Digital signal processing.

ENOB Effective number of bits. A measure of the dynamic performance of an analog-to-digital converter (ADC).

Faceplate A plate, cover, or bezel on the front of a device which may contain I/O ports.

FEC Forward error correction gives a receiver the ability to correct errors without needing a reverse channel to request retransmission of data.

FFE Feed forward equalizer.

Gbps Gigabits per second. The throughput or data rate of a port or piece of equipment. Gbps is 1×10^9 bits per second.

GBd The baud rate is the number of electrical transitions per second, also called symbol rate. Giga Baud is 1×10^9 symbols per second.

Gearbox A component used for managing and manipulating data streams, primarily by converting multiple serial data streams at one rate to multiple streams at another rate,

Hamming codes A type of linear error-correcting code used in digital communication and data storage systems. It enhances data integrity by detecting and correcting single-bit errors that may occur during transmission or storage.

HPC High performance compute.

HPDC High-performance data center. A general term for data centers designed for high-compute workloads like AI/HPC.

IA Implementation Agreements, what the OIF names their defined interface specifications.

IC Integrated circuit.

IMDD Intensity modulation direct detection is a method where the intensity of a light source is modulated by an electrical signal. This modulated light then travels through an optical medium (like a fiber optic cable) and is directly detected by a photo detector at the receiving end. This is a common and relatively straightforward technique for transmitting information over optical links.

I/O Input Output, a common name for describing a port or ports on equipment.

ISI Intersymbol interference.

KP4 FEC A specific Reed-Solomon FEC (544,514) defined in IEEE 802.3 Clause 91, commonly used in Ethernet standards.

LDPC A low-density parity-check code is a linear error correcting code, a method of transmitting a message over a noisy transmission channel. An LDPC is constructed using a sparse Tanner graph. LDPC codes are capacity-approaching codes.

LR Long reach. CEI LR specifies backplane/midplane and copper cable electrical interfaces.

LPO Linear pluggable optic is a technology used in optical transceivers that simplifies the design of pluggable optical modules by removing the traditional Digital Signal Processor (DSP) and Clock Data Recovery (CDR) chips. Instead, LPO utilizes a direct-drive linear approach where the signal path is considered linear, relying on the capabilities of the Application Specific Integrated Circuit (ASIC) in the host system (like a switch or Network Interface Card) to perform signal conditioning and equalization.

MCM Multi chip module, a specialized electronic package where multiple integrated circuits (ICs), semiconductor dies or other discrete components are packaged onto a unifying substrate, facilitating their use as a single component (as though a larger IC).

Mid-board optics an optical transceiver that is mounted on a PCBA away from the PCBA edge, close to a switch ASIC to reduce the amount of PCBA trace loss between an ASIC and the optical transceiver. This is in contrast to the common practice today of locating optical transceivers at the PCBA edge.

Midplane Some backplanes are constructed with slots for connecting to devices on both sides and are referred to as midplanes.

MLSD Maximum likelihood sequence detection is a mathematical algorithm to extract useful data out of a noisy data stream.

MR Medium reach. CEI MR specifies chip-to-chip electrical interfaces.

NG Next generation.

NRZ (PAM2) Non return to zero, a binary code in which 1s are represented by one significant condition (usually a positive voltage) and 0s are represented by some other significant condition (usually a negative voltage), with no other neutral or rest condition.

NPC Near-package copper. NPC uses a copper cable to bring the front panel signal to a location close to the host silicon to minimize the host PCB losses. It reduces PCB losses by bringing the signals to a connector on the PCB close to the ASIC whereas CPC (Co-packaged copper) brings the signal to a connector on the ASIC package.

NPO Near-package optics. Similar to CPO (Co-packaged optics) and NPC (Near-package copper), NPO is an electrical to optical device intended to be mounted on the host PCB at a location adjacent to the host silicon to minimize host PCB traces to minimize electrical signaling requirements.

OE Optical engine.

O-to-E and E-to-O Optical to electrical interface and Electrical to optical interface, a component that converts an optical signal to an electrical signal or vice versa.

PAM Pulse amplitude modulation, a form of signal modulation where the message information is encoded in the amplitude of a series of signal pulses. For optical links it refers to intensity modulation.

PAM4 Pulse amplitude modulation-4 is a two-bit modulation that takes two bits at a time and maps the signal amplitude to one of four possible levels.

PAM6 A digital signal modulation scheme that encodes information by varying the amplitude of a pulse across six distinct voltage levels. Each of these six levels can represent approximately 2.5 bits of data.

PAM8 A digital signal modulation scheme that encodes information by varying the amplitude of a pulse across eight distinct voltage levels. Each of these eight levels can represent 3 bits of data.

PCB/PCBA Printed circuit board (PCB) assembly, an assembly of electrical components built on a rigid glass-reinforced epoxy-based board.

Repeater A low-latency electronic device that receives a signal and retransmits it. Repeaters are used to extend transmissions so that the signal can cover a longer distance. Besides signal equalization, clock and data recovery (CDR) functions could be also added to remove jitter from received signals effectively.

RoCE RDMA over Converged Ethernet (CE) is a network protocol which allows remote direct memory access (RDMA) over an Ethernet network.

RS Reed Solomon FEC coding is a type of block code. Block codes work on fixed-size blocks (packets) of bits or symbols of predetermined size. It can detect and correct multiple random and burst errors.

RTLR (Retimed Transmit, Linear Receive) also generically referred to as Linear receive optic (LRO), is a type of optical transceiver technology used primarily in high-speed data center and networking applications, especially within AI clusters. The RTLR naming convention is within OIF. LRO is characterized by the presence of a Digital Signal Processor (DSP) solely on the transmit path, while the receive path operates with a linear, non-retimed architecture. This differs from fully retimed optical modules that utilize DSPs on both transmit and receive paths, and also from Linear Pluggable Optics (LPO) that eliminate DSPs entirely.

Scale out also known as horizontal scaling, refers to the process of increasing capacity and performance by adding more individual machines or nodes to a distributed system.

Scale up also known as vertical scaling, refers to the process of increasing the capacity or performance of a single server or system within an AI data center by adding more resources.

SDO standard development organizations.

SerDes A Serializer/Deserializer is a pair of functional blocks commonly used in high-speed communications to transfer data over a relatively low number of lanes.

SFP Small form-factor pluggable connector is a modular, hot-pluggable interface used in networking devices to connect to various types of fiber optic or copper cables. It uses a PCB card edge interface.

SI Signal integrity is a set of measures of the quality of an electrical signal.

SNDR Signal-to-noise-and-distortion ratio is a measurement of the purity of a signal.

SNR Signal-to-noise ratio.

TBD To be determined.

Tbps Terabits per second. The throughput or data rate of a port or piece of equipment. Tbps is 1×10^{12} bits per second.

TCM Trellis coded modulation is a technique that combines convolutional coding and modulation to improve data transmission efficiency over bandwidth-limited channels, like telephone lines. It achieves this by intelligently integrating the encoding and modulation processes, increasing the distance between signal points in the constellation to enhance error correction without expanding bandwidth.

TME Test and measurement equipment.

Twinax copper cable A type of copper cable similar to coaxial cable, but with two inner conductors instead of one.

UCIe Universal chiplet interconnect express is an open industry standard that defines the interconnect between chiplets, or small component dies, within a single package.

VLC Vertical line card. A new line card design in which vertical I/O connectors and ASIC are mounted side by side, reducing the signal trace distance.

VSR Very short reach. CEI VSR specifies chip-to-module electrical interfaces.

XSR Extra short reach. CEI XSR specifies die-to-optical engine (D2OE) and die-to-die (D2D) electrical interfaces.

448G A generic name for an expected technology enabling data rates (including overhead) of approximately 448 Gbps per lane.

8D / CAPA Eight-discipline problem solving and corrective and preventive action. Structured containment, root cause, correction, and recurrence control for supplier or production failures.

ALS Automatic laser shutdown. A safety mechanism that cuts laser output when fiber continuity is lost. Modern systems prefer APR with automatic restart over full shutdown.

APC Automatic power control. A feedback loop from a monitor photodiode that holds average optical output power constant against temperature drift and aging.

APD Avalanche photodiode. A photodiode with internal multiplication gain (5–9 dB sensitivity improvement over PIN) at the cost of excess noise factor and bias voltage.

APR Automatic power reduction. Holds laser output at or below Hazard Level 1M on fiber break and probes for re-mate with safe low-power pulses (ITU-T G.664).

Arrhenius / E_a Life-acceleration model. Temperature stress multiplies wear-out by $\exp[(E_a/k)(1/T_{\text{use}} - 1/T_{\text{stress}})]$. E_a must be justified for the named mechanism on that process.

ASE Amplified spontaneous emission. Broadband optical noise generated by optical amplifiers (SOA, EDFA); adds to the receiver noise floor.

ATP Acceptance test plan. Production test limits, methods, reference planes, and reaction rules for ship or hold decisions.

AUI Attachment unit interface. The electrical serial lane set between a host ASIC and the module cage connector (e.g. 400GAUI-4).

Bathtub curve Reliability failure-rate shape: infant mortality (falling), useful life (roughly constant FIT), then wear-out (rising). Burn-in targets the left; Arrhenius life targets the right.

BERT Bit-error-ratio tester. An instrument or ASIC function that generates PRBS patterns and counts bit errors for pre-FEC and post-FEC BER measurement.

BiCMOS A semiconductor process combining bipolar transistors (high f_T) with CMOS on one die; used for high-speed TIAs and modulator drivers.

BOM Bill of materials. The component and assembly cost of a module or system; DSP presence is a large BOM driver.

Burn-in Production or sample screen that removes infant-mortality parts before ship. Distinct from HTOL life projection.

CMOS Complementary metal-oxide-semiconductor. The mainstream IC fabrication process; TIAs and SerDes are built at 16 nm to 3 nm nodes.

COBO Consortium for On-Board Optics. Standardized mid-board optical engine placement; largely leapfrogged by CPO in hyperscale deployments.

COD Catastrophic optical damage. A sudden, irreversible failure of a laser facet under thermal or optical overstress.

COM Channel operating margin. A statistical SNR metric for the fully equalized electrical link; CEI go/no-go is typically COM $\geq$ 3 dB.

COUPE Compact Universal Photonic Engine. TSMC's SoIC-X hybrid-bonded EIC-on-PIC packaging platform for co-packaged optics; mass production in 2026.

CW Continuous wave. Unmodulated laser output; external modulators (MZM, ring, EAM) encode data onto a CW source.

CW-WDM Continuous-wave wavelength division multiplexing. A multi-source agreement for multi-wavelength CW laser sources feeding WDM photonic engines.

CXL Compute Express Link. A coherent memory interconnect protocol built on the PCIe PHY (CXL 4.0 on PCIe 7.0); potential optical-reach application.

DCA Digital communication analyzer. A sampling oscilloscope used for PAM4 eye diagrams, TDECQ, OMA, and RLM measurements.

DBR Distributed Bragg reflector. A laser architecture where the grating sits outside the gain region, enabling tunable single-mode output.

DDM Digital diagnostic monitoring. Per-lane telemetry in CMIS: Tx/Rx optical power, laser bias, module temperature, LOS/LOL flags.

DFB Distributed feedback laser. A laser with a grating along the active region that selects one longitudinal mode; the workhorse CW or directly modulated source.

DML Directly modulated laser. Modulates bias current to encode data; simple and efficient but chirp-limited over dispersive fiber.

DPA Destructive physical analysis. Cross-section, SEM, or EDX on failed units to confirm facet, solder, FAU, or die failure modes.

DPPM Defective parts per million. Incoming or outgoing quality rate used in supplier quality and ATP escape tracking.

DR Datacenter reach. An IEEE 802.3 single-mode fiber class, typically 500 m at 1310 nm; the default for AI scale-out optics.

DVT / PVT Design validation test and production validation test. DVT proves the design; PVT proves the factory can build it at rate.

DWDM Dense wavelength division multiplexing. WDM with channel spacing $\leq$100 GHz; used in metro/long-haul and dense CPO architectures.

EAM Electro-absorption modulator. A voltage-controlled absorption region on InP; low chirp compared with direct modulation; paired with a DFB in an EML.

EECQ Electrical eye closure quaternary. The electrical analog of TDECQ, quoted at host-referenced test points TP1a and TP4a.

ELS External laser source. A CW laser module feeding a co-packaged optical engine; superset of ELSFP and CW-WDM module forms.

ELSFP External Laser Small Form-Factor Pluggable. An OIF-defined faceplate-pluggable CW laser module for CPO with a 24-pin card edge, CMIS management, and blind-mate MT optics.

EML Externally modulated laser. A DFB laser integrated with an EAM on one InP chip; the dominant 100–200G/lane pluggable transmitter.

EO Electro-optic. Conversion from voltage to optical phase or intensity; EO bandwidth is the primary speed metric for a modulator.

EOL End of life. The defined wear-out criterion (threshold rise, slope drop, or hard fail) used in HTOL projection and derating.

ER Extinction ratio. P_1/P_0 in dB; higher ER widens OMA for a given average power and trades against chirp and driver swing.

ESD Electrostatic discharge. Handling or assembly damage to drivers or TIAs; sudden hard fail, not Arrhenius wear-out. Qual uses HBM/CDM models.

FAIR First-article inspection report. Re-qualify after tooling, site, epi, or firmware change before open volume.

FAU Fiber array unit. A precision V-groove assembly that mates multiple fibers to a PIC edge or grating coupler array.

FIT Failures in time. Failures per 10^9 device-hours; fleet FIT times link count sets expected failures per day.

FR Far reach. An IEEE 802.3 single-mode fiber class, typically 2 km; tighter chirp and dispersion budget than DR.

FSR Free spectral range. The wavelength or frequency span between adjacent resonances of a ring resonator or etalon.

Golden unit / host Known-good reference used to bisect station, fixture, host, module, and fiber. Essential in debug and ATP correlation.

GR-468 Telcordia optoelectronic device qualification framework covering HTOL, environmental, and mechanical stresses.

HAST Highly accelerated stress test. Humidity plus temperature and bias; exercises corrosion and delamination.

HBM High-bandwidth memory. Stacked DRAM on an interposer beside the accelerator die; memory bandwidth sets the decode roofline.

HCF Hollow-core fiber. Guides light primarily in air ($n \approx 1$), giving roughly 33% lower latency than solid silica SMF.

HTOL High-temperature operating life. An accelerated reliability stress test (GR-468); Arrhenius modeling projects field wear-out from HTOL results.

HTSL High-temperature storage life. Unbiased bake; separates storage mechanisms from biased HTOL wear-out.

IBTA InfiniBand Trade Association. Maintains the InfiniBand Architecture spec (NDR 400G, XDR 800G/port); reuses QSFP/OSFP optics with Ethernet.

IEEE Institute of Electrical and Electronics Engineers. IEEE 802.3 owns Ethernet PHY/MAC rates, KP4 FEC, and IM/DD optical PMD clauses.

IL Insertion loss. Optical power lost traversing a component (connector, fiber, MUX); quoted in dB.

InP Indium phosphide. A III-V semiconductor material for lasers, EAMs, SOAs, and high-speed photodiodes in the 1310/1550 nm bands.

JESD47 JEDEC IC qualification stress suite. Silicon-side counterpart to GR-468 for drivers and TIAs.

LIV Light–current–voltage. The fundamental laser characterization curve: plots optical power and voltage versus bias current to read threshold, slope, kinks, and rollover.

LOL Loss of lock. A CDR or SerDes flag indicating the clock recovery circuit has lost symbol timing.

LOS Loss of signal. A receiver or host flag indicating optical input power has fallen below the detect threshold.

LRO Linear receive optic (also called RTLR). A module with a retimer/DSP only on the transmit path; the receive path is linear into the host SerDes.

MAC Media access control. The Ethernet data-link sublayer that builds frames, handles addressing and CRC. MAC rate is payload throughput.

MACsec IEEE 802.1AE media access control security. Provides line-rate encryption, frame integrity, and data origin authentication at Layer 2; transparent to the optical PMD.

MDI Medium dependent interface. The optical connector where a module meets the fiber; test points TP2 (Tx launch) and TP3 (Rx input) sit at the MDI.

MEMS Micro-electro-mechanical systems. Tilting mirror arrays used in optical circuit switches; millisecond switching at $\sim$2 dB loss.

MLC Multi-level coding. Coded modulation paired with PAM6/PAM8; OIF 448G workshop models often add a strong outer RS code at $\sim$12% overhead.

MMF Multimode fiber. Fiber supporting many spatial modes (50/125 μm); used with VCSELs at 850–940 nm for SR links (OM3/OM4/OM5 grades).

MoE Mixture of experts. A transformer architecture that routes tokens to specialist subnetworks; drives all-to-all collective traffic on the fabric.

MPI Multipath interference. Coherent beating from reflective paths; often floors BER and clusters FEC errors.

MPO Multi-fiber push-on. A standard multi-fiber connector for parallel optics (8, 12, 16, 24, or 32 fibers per ferrule).

MRM Microring modulator. A resonant silicon modulator; compact and WDM-native but requires wavelength lock. The CPO workhorse at 200G/channel.

MSA Multi-source agreement. An industry specification for interoperable products (e.g. LPO MSA, CW-WDM MSA, 100G Lambda MSA).

MTBF Mean time between failures. For constant failure rate, MTBF $= 10^9/$FIT hours. Fleet math usually uses FIT times population.

MZM Mach–Zehnder modulator. A push-pull interferometer; broadband, low chirp. Built in silicon, TFLN, or III-V platforms.

NFF No fault found. An RMA unit that passes all tests on return; high NFF rate points at triage or intermittent connector faults.

NIC Network interface card. A host adapter connecting an accelerator or CPU to the scale-out fabric (Ethernet or InfiniBand).

OBO On-board optics. Optical engines mounted mid-board on the host PCB (COBO standard); mostly bypassed for CPO in hyperscale.

OCS Optical circuit switch. A Layer-1 switch that steers light from input to output fiber with no O-E-O conversion; transparent to rate, format, and wavelength.

OMA Optical modulation amplitude. The outer PAM4 signal swing ($P_1 - P_0$); the primary power metric for IM/DD transmitters.

ORL Optical return loss. The ratio of reflected to incident power at a fiber interface; low ORL raises laser RIN and can cause burst errors.

OSA Optical spectrum analyzer. An instrument that measures wavelength, SMSR, side-mode structure, and spectral width of a laser source.

OSFP Octal Small Form-factor Pluggable. A high-power faceplate module cage for 800G/1.6T/3.2T; larger thermal envelope than QSFP-DD.

PCS Physical coding sublayer. The IEEE 802.3 sublayer that performs 64B/66B line coding, 256B/257B transcoding, scrambling, and RS-FEC encoding.

PD Photodiode. A semiconductor device that converts photons to photocurrent (PIN, APD, or UTC variants).

PHY Physical layer. The combined PCS, PMA, and PMD that maps MAC frames to signals on the wire or fiber.

PIC Photonic integrated circuit. A chip integrating waveguides, modulators, detectors, and couplers (typically on SOI at 220 nm silicon).

PIN A p-intrinsic-n photodiode with no internal gain and the lowest excess noise; Ge-on-Si waveguide PIN is the mainstream short-reach detector.

PMA Physical medium attachment. The IEEE 802.3 sublayer that serializes, lane-multiplexes, and recovers clock. The SerDes lives here.

PMD Physical medium dependent. The IEEE 802.3 sublayer that modulates and detects light: the optical transceiver (laser, driver, PD, TIA).

PRBS Pseudo-random binary sequence. A deterministic test pattern that exercises all bit transitions; used for BER and eye measurements.

PSRR Power supply rejection ratio. A circuit's ability to suppress supply-rail noise; critical for laser bias drivers to avoid injecting RIN.

PUE Power usage effectiveness. Total facility power divided by IT equipment power; typical hyperscale values 1.1–1.3.

QSFP-DD Quad Small Form-factor Pluggable Double Density. A faceplate module cage carrying eight electrical lanes for 400G/800G/1.6T.

RDMA Remote direct memory access. Zero-copy network transfer between accelerator memories; carried over RoCE (Ethernet) or native InfiniBand.

RIN Relative intensity noise. The laser's own amplitude-noise spectral density (dB/Hz); sets a BER floor that power cannot overcome.

RLM Relative level mismatch. A measure of PAM4 level spacing uniformity (1.0 = perfectly even); CEI typically requires $\geq$0.95.

RMA Return merchandise authorization. A field-failed unit returned to the supplier for failure analysis. Distinct RMA codes keep FIT accounting honest.

SECQ Stressed eye closure quaternary. A receiver-side metric that applies calibrated optical stress and reports remaining margin before BER threshold.

SiGe Silicon-germanium BiCMOS. A high-f_T semiconductor process for TIAs and modulator drivers operating at 100–200+ GHz bandwidth.

SiPh Silicon photonics. Waveguides and modulators on silicon-on-insulator; CMOS fab-compatible and the mainstream for DR/FR and CPO.

SMF Single-mode fiber. Fiber with a $\sim$9 μm core carrying one spatial mode; G.652.D (standard) and G.657 (bend-insensitive) for datacenter plant.

SMSR Side-mode suppression ratio. The power difference (dB) between the dominant lasing mode and the strongest side mode on an OSA.

SOA Semiconductor optical amplifier. A III-V inline optical gain block; adds ASE noise. Used as a receiver preamplifier for sensitivity.

SOI Silicon on insulator. A wafer substrate for silicon photonics: a thin ($\sim$220 nm) silicon layer on buried oxide confines the optical mode.

SPC Statistical process control. Tracks process distributions and trends to catch drift before escapes ship.

SR Short reach. An IEEE 802.3 multimode fiber class, typically $\leq$100 m over OM4; VCSEL at 850 nm.

TCO Total cost of ownership. Acquisition (BOM, yield) plus lifetime energy plus service cost (FIT $\times$ replacement + labor).

TDECQ Transmitter and dispersion eye closure quaternary. The headline PAM4 transmitter-quality metric; measured after a reference receiver and bounded FFE, including a test fiber.

TEC Thermoelectric cooler. A Peltier device that holds a laser junction or ring at a controlled temperature against case thermal variations.

TECQ Transmitter eye closure quaternary. Same measurement as TDECQ without the test fiber; used alongside TDECQ in LPO MSA OMA tables.

TFLN Thin-film lithium niobate. A sub-micrometer LN layer on a silicon or silica handle; >110 GHz EO BW and the leading path to native 224 GBd PAM4.

TIA Transimpedance amplifier. Converts photodiode current to voltage with low input-referred noise; co-packaging with the PD minimizes capacitance and noise.

TP Test point. A defined reference plane where a link measurement is taken (TP0 through TP5, plus TP1a/TP4a).

TRO Transmit retimed optic. A module with DSP only on the transmit (electrical-to-optical) path; the receive path is linear.

TWE Traveling-wave electrode. An RF transmission line on a modulator that velocity-matches the electrical and optical group indices for high EO bandwidth.

TWI Two-wire interface. An I2C-like serial bus (SCL + SDA) used for CMIS module management communication.

UALink Ultra Accelerator Link. An open scale-up fabric specification for direct accelerator memory access (200G/lane class; $\sim$400G/lane next).

UEC Ultra Ethernet Consortium. Builds an open Ethernet-based stack (UET transport, PHY) optimized for AI/HPC scale-out with cluster-scale congestion control.

UI Unit interval. One symbol period ($= 1/R_{\mathrm{sym}}$); jitter budgets and eye diagrams are referenced in UI.

UTC / MUTC Uni-traveling-carrier (modified UTC) photodiode. Uses electron-only transport for >200 GHz bandwidth and high saturation current; linear/LPO niche.

VCSEL Vertical-cavity surface-emitting laser. 850 nm class source for multimode short-reach links.

VNA Vector network analyzer. Measures electrical or electro-optic *S*-parameters versus frequency for bandwidth and reflection.

VOA Variable optical attenuator. A calibrated loss element used for sensitivity sweeps and stressed-receiver testing.

WDM Wavelength division multiplexing. Sending multiple wavelengths on one fiber; CWDM (20 nm spacing) or DWDM ($\leq$100 GHz spacing).

XPO eXtra-dense Pluggable Optics. A liquid-cooled, 12.8 Tb/s faceplate pluggable module (Arista MSA, OFC 2026); front-panel serviceable at CPO-class density.

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[35] HyperLight Corp. 110-GHz packaged TFLN IQ modulators for 240-GBaud-class signaling. September 2025 (vendor announcement; provisional).

[36] MACOM Technology Solutions. 448G per-lane PAM4 modulator drivers (MAOM-025408 MZM, MAOM-022404 EML). March 2026 (OFC). >120 GHz RF bandwidth; SiPh, EML, and TFLN platforms; wire-bond or bumped die.

[37] Semtech Corporation. 224 Gb/s/lane MZM drivers (GN1877/GN1887) and TIA family for LPO/LRO/CPO. March 2026 (OFC); CEI-224G-Linear / LPO-MSA positioning (vendor announcement).

[38] L. Da Silva *et al*. A 460 Gb/s PAM-4 linear distributed driver with 105 GHz BW for TFLN modulators in 130 nm SiGe BiCMOS. RFIC 2026: 105.7 GHz BW, 2.25 V_{pp}, 232 GBd PAM4 with offline DSP (research).

[39] OFC 2026, paper W4J.4. A 420 Gb/s/lane O-band PAM-4 TOSA based on thin-film lithium niobate. Co-designed TFLN modulator + driver; 210 GBd / 420 Gb/s PAM4.

[40] OFC 2026, paper W3E.6. 180 GBaud PAM4 driver–modulator engine for IM/DD in the O-band. Co-packaged 76 GHz InP MZM with 224 GBd-class linear differential EML driver.

[41] A. Aimone *et al*. Truly-differential drive of TFLN TWE-MZM by linear SiGe driver in a codesigned hybrid integrated assembly. OFC 2026, M4D.4: $V_\pi L = 1.1$ V·cm; 140 GBd PAM8; 1.4 pJ/bit.

[42] M.-W. Lin *et al*. Record-high 90-GHz silicon microring modulator and 224-Gb/s PAM4 operation. OFC 2026, paper Th2A.14. Inductive and wavelength tuning; compact RLC model; 90-GHz EO BW; CPO target.

[43] C.-W. Peng *et al*. Ultra-high bandwidth silicon microring modulator with T-coil peaking for >400 Gbps transmission. OFC 2026, paper M2A.2. >110 GHz EO BW; 416 Gb/s PAM4, TDECQ 2.88 dB at 1 Vpp.

[44] OFC 2026, paper M2A.1. 256 Gb/s silicon Euler microring modulator with 3 THz FSR and >67 GHz bandwidth. O-band Euler ring; inductive peaking; 256 Gb/s PAM4.

[45] P. Dong *et al*. 400G/lane PAM4 modulation using silicon Mach–Zehnder modulators. OFC 2026 postdeadline, paper Th4A.4. Si MZM with commercial SiGe driver (2.5 V swing); 400 Gb/s/lane PAM4 IM/DD.

[46] OFC 2026, paper M2A.6. Compact Mach–Zehnder modulator in 300-mm silicon photonic platform toward 400Gbps/lane. 500 μm MZM; 94.7 GHz median EO BW; 2.4 dB median IL at 1310 nm.

[47] OFC 2026, paper W3E.5. Toward 400 G/lane silicon differential-drive Mach–Zehnder modulator with >80 GHz bandwidth. Differential-drive Si MZM; 81.8 GHz 3-dB EO BW; 100 GBd PAM8 eyes (lab setup).

[48] CW-WDM MSA. *Continuous-Wave WDM Multi-Source Agreement* Technical Specifications Rev 1.0 (4 June 2021). O-band grids: 8/16/32-line sets in 9/18/36 nm spans; modular (one λ/fiber) and integrated (full comb/fiber) configs; measurement methods for power, RIN, SMSR, linewidth. Form factor and management left to suppliers / ELSFP. Informative 18 nm examples often cite ≈30 dB SMSR, ≈ -135 dB/Hz RIN, ≈20 MHz linewidth.

[49] Optical Internetworking Forum. External Laser Small Form-Factor Pluggable (ELSFP) Implementation Agreement. OIF-ELSFP-02.0, 2023. 24-pin card-edge electrical interface; CMIS management; blind-mate MT optical connector; optical/thermal/electrical power classes.

[50] Ayar Labs. *SuperNova light source and TeraPHY optical I/O chiplet*. ayarlabs.com.

[51] Arista Networks (with Coherent, Marvell, Lightmatter, et al.). *XPO: eXtra-dense Pluggable Optics multi-source agreement* (12.8 Tb/s liquid-cooled pluggable). Announced OFC 2026; arista.com white paper, March 2026.

[52] E. Säckinger. *Analysis and Design of Transimpedance Amplifiers for Optical Receivers*. Wiley, 2018 (Chapter 4: Receiver Fundamentals).

[100G Lambda MSA / IEEE 802.3] 100G Lambda MSA and IEEE 802.3. *400G-FR4 and 100GBASE-BR technical specifications* ($\text{RIN}_{17.1}$OMA ≤ -136 dB/Hz; RIN method per IEEE Std 802.3 clause 52.9.6).

[54] 100G Lambda MSA Group. *100G Lambda MSA specifications* (100G-FR/LR, 400G-DR4/FR4/LR4). Formed 2017 (22 promoter members). Specifications built on 100 Gb/s per wavelength PAM4 (≈53 GBd) over duplex single-mode fiber, 2–40 km reaches; defined the single-λ 100G approach IEEE 802.3 adopted into its DR/FR/LR PMD clauses and that the LPO MSA 100G-DR-LPO profile inherits, including the RIN_xOMA transmitter method.

[Koheron DRV200] Koheron. *DRV200 low-noise laser diode driver datasheet* (current noise density at 1 kHz: 55 / 270 / 480 pA/$\sqrt{\text{Hz}}$ for 40 / 200 / 400 mA variants).

[56] Intensity-noise studies of epitaxial/heterogeneous quantum-dot lasers on silicon. *Opt. Lett.* 45(17):4887 (2020, p-doped QD RIN ≈ -150 dB/Hz); *J. Photonics* (2024, quiet-pump/injection-locked RIN to ~ -168 dB/Hz); *JLT* 43(4):1855 (2025, feedback-tolerant QD on 300 mm Si).

[RIO PLANEX] RIO Lasers. *PLANEX 1550 nm external-cavity laser datasheet*. Planar gain-chip and Bragg-grating external cavity with narrow linewidth, low phase noise, and wavelength stability; sensing and metrology product rather than a short-reach IM/DD default. Vendor orientation, provisional.

[58] Waveguide germanium-on-silicon photodiodes for 200 GBd and beyond. Shahin et al., *OECC/PSC* (2025), recessed Ge/Si PIN: 106 GHz, 0.93 A/W, < 10 nA dark; *ACS Photonics* (2025), SiN-coupled lateral Ge PD > 110 GHz at 1 mA (O-band).

[59] A 112-Gb/s -8.2-dBm-sensitivity 4-PAM linear TIA in 16-nm CMOS with co-packaged photodiodes. *IEEE J. Solid-State Circuits* 58(4), 2023 (32 GHz BW, 16.9 pA/$\sqrt{\text{Hz}}$).

[60] A 4×112-GBaud linear PAM-4 TIA with 65-GHz bandwidth and 13.2-pA/$\sqrt{\text{Hz}}$ input noise for 1.6-T links. *IEEE Trans. Microw. Theory Tech.*, 2026 (55-nm SiGe BiCMOS, 224 Gb/s, 1.22 pJ/bit).

[61] Ultrafast avalanche photodiode exceeding 100 GHz bandwidth. *Nat. Commun.* (2025): Ge/Si UMC-APD, 105 GHz at gain $\approx$7; GBP $\sim$4800 GHz; 224/260 Gb/s PAM4 at -10.9/-10.1 dBm ($\sim$9 dB over PIN).

[62] A. Shahin *et al*. Towards 100 GHz waveguide Ge/Si avalanche photodiodes for high-speed optical interconnects. *JLT* 2026: >100 GHz peak BW; 70 GHz at 2 A/W; $\sim$7 V; 300 mm SOI.

[63] OFC 2026, paper Th3F.2. 360 Gbps Ge-on-Si avalanche photodiodes operating in the O- and C-band. Up to 180 GBd PAM4 below HD-FEC; 70/100 GHz BW; 2/1.5 A/W.

[64] Semtech Corporation. OFC 2026: 224G optical demos and 448G/lane PMD ICs (TN622 driver, TN14740 TIA). Vendor demonstration; provisional until datasheet limits are published.

[65] Wideband uni-traveling-carrier photodiodes for near-300 Gbps communications. Type-II GaInAsSb/InP UTC-PD > 110 GHz; InP MUTC-PD 206 GHz (arXiv:2501.02812, 2025).

[66] Optical Internetworking Forum, Common Electrical I/O (CEI-224G) framework and implementation agreements; CEI interoperability demos, OFC/ECOC 2025. Reach classes: XSR (die-to-die), VSR (200 mm host + module), MR ($\sim$500 mm, $\sim$32–34 dB), LR ($\sim$1 m, up to $\sim$40 dB die-to-die). CEI-224G-Linear extends the CEI-112G-Linear methodology to 224G full-linear modules (LPO/CPO/NPO): TP1/TP1a and TP4/TP4a electrical specs without module DSP/CDR.

[67] Optical Internetworking Forum, *448G, 224G, 112G CEI Interoperability Demo*, OFC 2025. Project map: CEI-224G-XSR ($\lesssim$50 mm), VSR (200+20 mm), MR ($\sim$500 mm), LR ($\sim$1000 mm), and CEI-224G-Linear (no module DSP); BER 10^{-15} with FEC allowed. Live Linear/RTLR/VSR/LR demos.

[68] M. Li et al., *224G Package and PCB Investigations and COM Reference Model*, IEEE 802.3df, 2022. M7N stripline loss at 56 GHz: $\approx$2.8 dB/inch (regular), $\approx$1.9 dB/inch (skip-layer); next-gen target 1 dB/inch at Nyquist.

[COBO / OBO–NPO–CPO] Consortium for On-Board Optics, *On-Board Optical Module Specification*; and comparative analyses of pluggable, on-board (OBO), near-packaged (NPO), and co-packaged (CPO) optics architectures.

[70] Reliable 2 m+ passive DAC connectivity for AI-scale networks with 224G PHY IP (BER $\sim$2 $\times$10^{-9} at 2 m); passive DAC $\sim$0.5–1 m nominal, AEC to $\sim$2.5 m at 224G.

[71] S. Williams, A. Waterman, and D. Patterson. *Roofline: An Insightful Visual Performance Model for Multicore Architectures*. Communications of the ACM, 2009. Arithmetic intensity (FLOP/byte) vs. the compute/bandwidth balance point; the framework behind the decode memory-bound argument.

[72] Linear Pluggable Optics (LPO) and Linear-Receive Optics (LRO/TRO): DSP-less and half-retimed module architectures. IEEE EPS overview (2026); OIF/MSA interop demos, OFC 2025. Typical 800G: DSP module $\sim$14–18 W, $\sim$8–10 ns/hop; LPO $\sim$7–9 W, <3 ns, <2 km; LRO $\sim$9 W. Redriver = CTLE+VGA (no CDR); retimer = EQ+CDR. Names component proponents (Macom, Semtech, Maxlinear) and OFC 2024–25 demos from Eoptolink, Macom, Marvell, Alphawave, and Innolight.

[73] A. Shehabi et al. *2024 United States Data Center Energy Usage Report*. Lawrence Berkeley National Laboratory, December 2024. US data centers: 58 TWh (2014) $\rightarrow$ 176 TWh (2023, $\sim$4.4% of US electricity) $\rightarrow$ projected 325–580 TWh (6.7–12%) by 2028.

[74] Cignal AI, *The Linear Drive Market Opportunity* (2023) and OFC 2025 show report. Market view: LPO stays a small share, especially at 800G, because the installed fabric is DSP-based; 100G/lane 800GbE LPO late to market; at 1.6T, LRO favored over LPO on power/thermal. Provisional (analyst reports).

[75] Eoptolink, *Industry-first 200G-per-lane LPO; 100G/lane 800G LPO entering mass production*, March 2024. 200G/λ four-channel LPO, no DSP/CDR; Gen2 100G/lane 800G and 400G SMF (OSFP, QSFP-DD, QSFP112), full TP2 compliance claimed. Vendor, provisional.

[76] Macom, *Macom to showcase 200G-per-lane products at OFC*, March 2024. PURE DRIVE 200 Gb/s LPO, extensible to 212 Gb/s/lane for 1.6T; linear TIA and driver as the key blocks. Vendor, provisional.

[77] Marvell, *Marvell introduces 1.6 Tbps LPO chipset*, December 2024. 200G/lane TIA and laser-driver chipset for 800G and 1.6T LPO, targeting scale-up XPU compute fabrics. Vendor, provisional.

[78] Alphawave Semi and InnoLight, *Low-latency LPO with a PCIe 6.0 subsystem for AI infrastructure*, March 2024. 64 Gb/s/lane PCIe 6.0 controller+PHY over InnoLight LPO OSFP optics (OFC 2024); later a 128 Gb/s/lane PCIe 7.0-ready platform. Vendor, provisional.

[79] LPO MSA, *100G-DR-LPO Specification*, rev. 1.0, 19 March 2025. 100 Gb/s/lane PAM4 SMF to 500 m; builds on IEEE 802.3 and OIF CEI-112G-LINEAR. TDECQ/TECQ $\leq$3.4 dB; OMA coupled to max(TECQ,TDECQ); host RS(544,514) FEC. Template for linear-module optical limits; 224G linear modules follow CEI-224G-Linear electrically while optical PMDs track 802.3dj.

[80] Telcordia. *GR-468-CORE: Generic Reliability Assurance Requirements for Optoelectronic Devices*.

[81] Telcordia. *GR-1221-CORE: Generic Reliability Assurance Requirements for Passive Optical Components*, Issue 2 (January 1999). Passive-component companion to GR-468: connectors, couplers, WDM filters, splitters, and isolators. Same stress families (damp heat, temperature cycle, mechanical, aging) scored on insertion and return loss rather than LIV.

[82] Broadcom Inc. *Tomahawk 6 – Davisson: 102.4 Tb/s co-packaged-optics Ethernet switch* (press release and briefs). October 2025.

[83] NVIDIA. *Spectrum-X and Quantum-X silicon photonics co-packaged-optics switches* (GTC 2025; Hot Chips 2025, Shainer).

[84] TSMC. *Compact Universal Photonic Engine (COUPE): SoIC-X electronic/photonic integration; mass-production milestone*. 2026.

[85] Optical Internetworking Forum. OIF Implementation Agreement for 400ZR. 400 Gb/s coherent DWDM pluggable; 15 W module power budget; DCI reach to $\sim$120 km; interoperable QSFP-DD/OSFP coherent optics.

[86] Optical Internetworking Forum. CMIS 5.3 Implementation Agreement (OIF-CMIS-05.3). Host-to-module management over two-wire bus; module and data-path state machines; DDM, VDM, CDB; link training; ELSFP and CPO extensions.

[87] DMTF, OpenConfig, and the SONiC community. DMTF Redfish (DSP0266), OpenConfig models and gNMI, and SONiC. Box- and fleet-level management and telemetry above OIF CMIS. Redfish: RESTful server/switch management API (v1.0, August 2015). OpenConfig: vendor-neutral YANG models and streaming telemetry. SONiC: open network OS with dataplane telemetry. Where per-module optical monitors surface at cluster scale.

[88] IEC. IEC 60825-1:2014/AMD1:2021 and IEC 60825-2:2021. Laser product classes (1, 1M, 3R, 3B, 4) and OFCS hazard levels at accessible fiber ports.

[89] ITU-T. Recommendation ITU-T G.664 (02/2012). Optical safety procedures; APR with automatic restart; reduce to Hazard Level 1M within 3 s of continuity loss.

[90] ITU-T. Recommendation ITU-T G.694.1 (10/2020), Spectral grids for WDM applications: DWDM frequency grid and ITU-T G.694.2 (12/2003), CWDM wavelength grid. G.694.1 anchors the DWDM grid at 193.1 THz with 12.5/25/50/100 GHz spacings and integer multiples; G.694.2 defines the 20 nm CWDM grid, 18 channels from 1271 to 1611 nm. Datacom DWDM locks to G.694.1; CWDM4 uses four O-band lines of G.694.2.

[91] ITU-T. Recommendation ITU-T G.652 (11/2016), Characteristics of a single-mode optical fibre and cable and ITU-T G.657 (08/2024), Characteristics of a bending-loss insensitive single-mode optical fibre and cable. G.652.D sets the standard single-mode fiber used in short-reach links: attenuation $\approx$0.2 dB/km at 1550 nm, $\approx$0.32 dB/km at 1310 nm, and zero-dispersion wavelength near 1310 nm. G.657 defines bend-insensitive fiber for tight-radius routing in patch panels and shelves; category A stays G.652-compatible, category B trades some standard-fiber interoperability for tighter bend radii.

[92] ISO/IEC and TIA. ISO/IEC 11801:2017, Information technology: Generic cabling for customer premises and the TIA-492 series, Detail Specifications for Optical Fibers. ISO/IEC 11801 defines the cabled fiber performance classes used in datacenter patch plant: OM1–OM5 for multimode and OS1/OS2 for single-mode. TIA-492 gives the matching detailed fiber specifications (for example TIA-492AAAD corresponds to OM4); TIA-568 adopts the same OM/OS class names for structured cabling.

[93] JEDEC Solid State Technology Association. JESD47M, Stress-Test-Driven Qualification of Integrated Circuits. Published August 2025. Baseline acceptance-test flow (temperature cycle, HTOL/HTSL, autoclave/HAST, mechanical shock and vibration) for qualifying an IC as a new product, a product family, or after a process change. The silicon-side counterpart to Telcordia GR-468-CORE for optoelectronics.

[94] JEDEC Solid State Technology Association and ESD Association. ANSI/ESDA/JEDEC JS-001-2024, Human Body Model (HBM) ESD Sensitivity Testing and JS-002-2025, Charged Device Model (CDM) ESD Sensitivity Testing. Component-level ESD classification for driver, TIA, and DSP silicon. Supersede the older JEDEC-only JESD22-A114 (HBM) and JESD22-C101 (CDM) test methods.

[95] JEDEC Solid State Technology Association. JESD78F.02, IC Latch-Up Test. Published November 2023. Overvoltage and ± 100 mA current-injection test method for parasitic-thyristor latch-up susceptibility in CMOS, bipolar, and BiCMOS ICs; the failure mode COD is to a laser facet.

[96] Automotive Electronics Council. AEC-Q100 Rev-J, Failure Mechanism Based Stress Test Qualification for Integrated Circuits. Automotive-grade IC qualification built on the

JEDEC JESD47/JESD22 methods with tightened ESD targets and temperature grades (Grade 3: -40 to 85°C through Grade 0: -40 to 150°C). Not required for datacenter optics, but a useful borrow when a driver or TIA die also ships in an automotive part number and the datasheet cites a Q100 grade.

[97] IEC. IEC 61754-7-1:2014, Fibre optic connector interfaces – Type MPO connector family: one fibre row and IEC 61754-7-2:2017, Type MPO connector family: two fibre rows. Mechanical interface standard for the MPO/MT connector family: 6.4 mm $\times$ 2.5 mm rectangular ferrule, guide-pin alignment, 8/12/16/24-fiber counts. Superseded the single-part IEC 61754-7:2008 to add active-device receptacles and up-angled plugs.

[98] IEC. IEC 61300-2-2:2009, Basic test and measurement procedures – Mating durability and IEC 61300-3-35:2022, Visual inspection of fibre optic connectors and fibre-stub transceivers. IEC 61300-2-2 defines the mate/unmate cycling test method that connector datasheets rate against; TIA-568.3 sets 500 cycles as the structured-cabling durability floor, and MPO/MTP-class connectors are commonly rated well above 1000. IEC 61300-3-35 grades endface scratches, pits, and debris against zones on the fiber core and cladding for pass/fail inspection.

[99] Concatenated RS-BCH coded modulation latency study (incl. KP4). Performance-complexity-latency trade-offs of concatenated codes, arXiv:2402.09364, 2024; and vendor RS(544,514) implementation notes. KP4 RS(544,514) decode latency in practice $\approx$20–100 ns depending on implementation and clock; encode is comparable. Academic/vendor figures, provisional.

[100] Vendor switch datasheets (Ethernet and InfiniBand). 800GbE 51.2 Tb/s SONiC switch, $\sim$560 ns port-to-port; NVIDIA Quantum-X800 InfiniBand advertises sub-100 ns port-to-port with in-network compute. Per-hop O-E-O switch latency: a few hundred ns for cut-through Ethernet, under $\sim$100 ns for InfiniBand. Vendor orientation, provisional.

[101] Y. Chen, M. N. Petrovich, E. Numkam Fokoua, *et al*. (Southampton/Microsoft). Record-low-loss hollow-core fibre (double nested antiresonant nodeless fibre, DNANF). *Nature Photonics*, September 2025: 0.091 dB/km at 1550 nm (below the $\sim$0.14 dB/km silica floor), $\approx$0.2 dB/km over a 66 THz window; air guiding gives group index near 1.0, $\sim$45% faster propagation (about a third lower latency) than silica. Microsoft reports Azure network deployment. Record result, provisional.

[102] Quantum-dot mode-locked laser (QD-MLL) frequency combs for datacom. High-efficiency QD lasers as comb sources for DWDM, *Appl. Sci.* 12(4):1836 (2022): among DFB arrays, nonlinear microcombs, and semiconductor MLLs, QD-MLLs currently offer the best efficiency, simplicity, and size for a single-device comb. O-band IM/DD demos: a 1.3 µm InAs/InGaAs QD-MLL comb carrying 14$\times$100 Gb/s PAM4 over 10 km SMF at $\sim$284 fJ/bit; and an isolator-free QD-MLL comb reported for interconnects beyond 3.2 Tb/s (arXiv:2506.02402, 2025). Research demos, provisional.

[103] Integrated Kerr/soliton microcomb multi-wavelength sources. Integrated multi-port multi-wavelength coherent optical source, *Nat. Commun.* 16 (2025): a Kerr microcomb with a monolithically integrated demultiplexer that autonomously locks to and tracks the comb lines, aimed at beyond-Tb/s links. A microcomb turns one pump into hundreds of evenly spaced lines on a chip, but per-line power is low (pump-conversion efficiency is modest), so a booster or per-line SOA is usually required (Section 6.6.2). Research demos, provisional.

[104] Semiconductor optical amplifiers for datacom booster/distribution use. Commercial O-band SOA modules quote $\sim$15 dB gain, noise figure $\leq$7 dB, and $\leq$1.5 dB polarization-dependent gain (Anritsu SOA datasheet); the quantum NF floor is 3 dB. O-band quantum-dot SOAs grown on a CMOS-compatible silicon substrate (*ACS Photonics*,

2019) give wide O-band gain with low noise for co-packaged integration. An SOA restores launch power after comb generation and fan-out, at the cost of added ASE. Vendor/academic figures.

[105] Co-packaged and near-package copper, copper's answer to the same reach wall that drives CPO. Molex co-packaged copper (CPC): on-substrate routing validated at 224 Gb/s PAM4 with a stated roadmap to 448G, aimed at short scale-up runs where power, latency, and cost favor copper over optics. The Impress compression substrate connector and mating cable support 224 Gb/s PAM4 and beyond. Marvell 224G long-reach SerDes reports $\sim$4 pJ/bit and targets CPC cables, near-packaged optics, and CPO. Near-package copper (NPC) mates the connector near the ASIC rather than onto the package substrate; shared NPC/NPO sockets have been proposed. Vendor orientation, provisional.

[106] FiberMall. OSFP thermal management: data center cooling guide. 2026. A 32-port 800G switch dissipates >1 kW, with optical modules $\sim$50% of the thermal load; observed power runs 15–25% above rated. Vendor orientation, provisional.

[107] Cao, *et al*. Simulation and experimental investigation of liquid-cooling thermal management for high-bandwidth co-packaged optics. Front. Optoelectron., 2025. Power density and on-package thermal crosstalk limit CPO reliability; liquid cooling investigated to hold optical-engine and laser temperature.

[108] Semtech. Low-power 1.6T datacom transceivers and the path to 3.2T. Webinar recap, 2025. Linear modules cited at $\sim$50–60% less power than fully retimed; a 500,000-GPU cluster estimated to save >100 MW and $\sim$\$100 million/year in electricity. Vendor orientation, provisional.

[109] Epoch AI. Total cost of ownership of a one-gigawatt AI data center. Data insight, 2025. A 1 GW AI site estimated near \$38 billion up-front capex and $\sim$\$0.9 billion/year opex. Analyst breakdown, provisional.